

KUMMER DIVISION, POLYNOMIAL SECANTS, AND TRACE RIGIDITY

ABSTRACT. We develop an algebraic and arithmetic theory of division correspondences and polynomial secants in semisimple groups. On the regular semisimple locus, division by m is a finite Kummer torsor under the m -torsion of the universal centralizer. In type A_1 , Hilbert 90 turns the projective square-root correspondence into the rational formula $[1 + g]$. We prove that this phenomenon is exceptional: a rational section of the generic projective m -th-root correspondence exists only when $m = 2$ and the group is a product of groups of type A_1 . In higher type A , projective polynomial secants $[f(g)]$ take the place of roots. Their fields of definition detect the Galois action on the Dynkin diagram and thereby distinguish inner from outer forms.

Combining secants with double-coset dynamics and arithmetic descent gives uniform gaps for signed trace sumsets, a trace-gap classification for lattices in $SL_d(\mathbb{C})$, and a reduced-trace classification for lattices in $SL_m(\mathbb{H})$. For compact Fuchsian trace sparsity we identify the canonical role of the subgroup generated by squares and prove that, on word balls of radius R , every fiber of two trace coordinates has size $\exp(o(R))$ over every finitely generated characteristic-zero field. We then deduce the sharp universal bounds $D_H(X) \geq X^{1-o(1)}$ and $N_H^*(L) \geq e^{(1/2-o(1))L}$ for every closed hyperbolic surface. The same $X^{1-o(1)}$ lower bound holds in every component indexed by every finite quotient. Apart from the inverse identity $A + A^{-1} = (\text{tr } A)I$, the additive representation multiplicities inside an exact trace fiber are $\exp(o(R))$ on a word ball of radius R , once three fixed trace coordinates are used to recover A . Consequently, the usual small-doubling arguments do not by themselves prove arithmeticity.

To prove arithmeticity, we fix an index N in the Chebyshev recurrence and vary the group element to which the recurrence is applied. A reduced-divisor form of Roth's theorem gives a lower bound with coefficient $2N - 4 - o(1)$. Expansion modulo p^2 is enough to choose elements $b^m h_m$ for which every prime occurs to uniformly bounded order in the product of two consecutive recurrence values. A uniform fiber estimate for two trace coordinates, followed by an explicit congruence counting argument, gives an upper bound with coefficient $(2N - 1)/2$. Letting N tend to infinity shows that the global recurrence height attached to every element of the subgroup generated by squares is at most half its original translation length. At an algebraic character, the distinguished real place already contributes exactly this half-length. All other nonnegative local terms must therefore vanish. It follows that the traces of the subgroup generated by squares are algebraic integers, their trace field is totally real, and every nondistinguished conjugate lies in $[-2, 2]$. Takeuchi's criterion [47] then gives arithmeticity. If the original character were transcendental, a nearby algebraic Fuchsian specialization in the same component would satisfy $\ell_y(\gamma) \leq \ell_x(\gamma)$ for every $\gamma \neq 1$. Thurston's asymmetric metric then forces the two characters to be equal, contradicting the choice of an algebraic specialization of a transcendental character. Thus a cocompact Fuchsian group with linearly growing signed trace set is arithmetic. Hence the compact Schmutz linear-growth and compact Sarnak bounded-clustering conjectures both hold.

The proof does not settle several stronger statements about the recurrence of a single fixed element. For every algebraic Fuchsian character satisfying the linear trace-growth hypothesis and every hyperbolic a , expansion modulo rational squares gives the one-term estimate

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log N \text{rad}_S(q_n(\text{tr } a)) \leq \frac{\ell(a)}{2}.$$

The matching universal lower bound is implied by number-field abc , but at $s = 3/\sqrt{2}$ it is equivalent to full exponential growth of $\text{rad}(2^n - 1)$, and implies infinitely many base-2 non-Wieferich primes. Thus the universal pointwise assertion remains open. So does superapproximation for arbitrary ideals in general twisted restriction-of-scalars groups. Neither statement is used in the proof of the compact conjectures. The paper also proves a Lucas–Wieferich localization theorem and an exact entropy–pole-degree inequality for algebraic curves on which prescribed trace equalities persist.

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1. INTRODUCTION AND MAIN RESULTS

Power maps on reductive groups are naturally correspondences rather than single-valued rational operations. Their branches are controlled by regular centralizer tori, their fields of definition by Kummer cohomology, and their monodromy by Weyl and Galois actions. Polynomial secants $[f(g)]$ are rational maps obtained from these correspondences without choosing a root. When paired with matrix traces, they express certain products in terms of finitely many additive trace

differences. The same construction therefore applies to rational root formulas, descent questions for algebraic group forms, and arithmetic rigidity of sparse trace sets.

The first results concern algebraic groups and their representations. We then apply polynomial secants and commensurator rigidity to trace gaps. In the rank-one application, let $\Gamma < \mathrm{PSL}_2(\mathbb{F})$, $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, be a lattice, let $\tilde{\Gamma}$ be its full inverse image in $\mathrm{SL}_2(\mathbb{F})$, and write

$$T(\Gamma) = \{\mathrm{tr} \gamma : \gamma \in \tilde{\Gamma}\}.$$

Thus $T(\Gamma) = -T(\Gamma)$; it is a set of values, so repeated traces are counted only once. For a subset S of $\mathbb{R}$ or $\mathbb{C}$, define

$$\mathrm{Gap}(S) = \inf\{|x - y| : x, y \in S, x \neq y\}.$$

The set is *uniformly discrete* when $\mathrm{Gap}(S) > 0$. We say that S has *bounded clustering*, and write $\mathrm{BC}(S)$, if

$$\sup_z \#(S \cap \{w : |w - z| \leq 1\}) < \infty.$$

The radius 1 is immaterial: the same property results from any fixed positive radius, because a ball of one fixed radius can be covered by finitely many balls of another. For $S \subset \mathbb{R}$, *linear growth* means

$$\#(S \cap [-X, X]) = O(X) \quad (X \rightarrow \infty).$$

Uniform discreteness implies bounded clustering, and bounded clustering implies linear growth, but neither converse holds for arbitrary subsets of $\mathbb{R}$.

We use the following arithmetic conventions throughout. Two subgroups are *commensurable* if their intersection has finite index in both, and

$$\mathrm{Comm}_G(\Lambda) = \{g \in G : g\Lambda g^{-1} \text{ is commensurable with } \Lambda\}.$$

Here a *lattice* in a locally compact group means a discrete subgroup whose quotient has finite invariant volume. For a linear algebraic k -group $\mathbf{G}$, an arithmetic subgroup is one commensurable with the integral points of a fixed $\mathcal{O}_k$ -model of $\mathbf{G}$; changing the faithful representation or integral model does not change this commensurability class.

In rank one, an *admissible quaternion algebra* B/k is one whose chosen archimedean realization is $M_2(\mathbb{F})$ and whose other archimedean factors are compact. Thus, in the Fuchsian case, k is totally real and B is ramified at every real place except the chosen one; in the Kleinian case, k has one complex place up to conjugation and B is ramified at every real place. An order $\mathcal{O} \subset B$ is a subring containing 1 which is a full $\mathcal{O}_k$ -lattice, and $\mathcal{O}^1 = \{b \in \mathcal{O} : \mathrm{Nrd}(b) = 1\}$. A lattice is *arithmetic* if, after conjugation and projectivization, it is commensurable with the image of some $\mathcal{O}^1$; it is *derived from the order* if it is actually contained in that image (and hence has finite index there). Whenever a normal subgroup inside the integral norm-one group is needed, we take the normal core of its intersection with the arithmetic lattice. This produces a finite-index normal subgroup, denoted Λ_0 or Γ_0 , that is contained in the relevant integral norm-one group.

We also fix references for several standard results used repeatedly below. Northcott's finiteness theorem is [4, Theorem 1.6.8]. Borel density is used in the form [5]; the finite-covolume theorem for arithmetic groups is [6]; and Skolem–Noether is used in its usual central-simple-algebra form [31, Section 12.6]. For a non-elementary convex-cocompact Fuchsian group H , orbital counting gives

$$\#\{h \in H : d(o, ho) \leq R\} = \exp((\delta_H + o(1))R),$$

and primitive closed geodesics have the corresponding prime-geodesic asymptotic; see [34] (and [13, Chapter 9] in the compact surface case). These citations cover later invocations of “orbit counting” and the “prime geodesic theorem.”

With these conventions, the positive trace-gap classification [3], which is used in Sections 5 and 9, is

$$\Gamma \text{ is derived from an admissible quaternion algebra,} \quad \text{Gap } T(\Gamma) > 0, \quad \overline{T(\Gamma) - T(\Gamma)} \neq \mathbb{F}, \quad (1.1)$$

where the three conditions are equivalent. The closure in the last condition is essential.

The principal results are as follows.

- (i) The projective square root $[1 + g]$ is Hilbert–90 division by two on the regular centralizer torus. The obstruction to an actual rational root is a Kummer class.
- (ii) A rational generic projective m -th-root section exists for a semisimple simply connected group only in the square map on products of SL_2 .
- (iii) For type A_{d-1} , $d \geq 3$, a polynomial secant with at least two terms has target descent to the adjoint group precisely for an inner form. In an outer form its unitary norm is generically noncentral.
- (iv) If Γ is derived, then its entire difference set, and more generally every fixed integral signed trace sumset, is uniformly discrete with an explicit norm bound. Otherwise the first difference set is dense.
- (v) For $d \geq 3$, non-density of trace differences for a lattice in $\text{SL}_d(\mathbb{C})$ is equivalent to containment with finite index in the norm-one group of an order in a degree- d central simple algebra over an imaginary quadratic field.
- (vi) For $m \geq 2$, the analogous reduced-trace classification holds for lattices in $\text{SL}_m(\mathbb{H})$; the defining field is $\mathbb{Q}$.
- (vii) The compact Sarnak and Schmutz conjectures are equivalent to a trace-gap implication for the subgroup $\Gamma^{(2)}$ generated by squares. For a fixed regular semisimple matrix B , every fiber of the coordinate pair $A \mapsto (\text{tr } A, \text{tr}(AB))$ has size $\exp(o(R))$ on a word ball of radius R , over every finitely generated characteristic-zero field. Consequently every closed hyperbolic surface satisfies $D_H(X) \geq X^{1-o(1)}$, or equivalently has at least $e^{(1/2-o(1))L}$ distinct primitive lengths up to L ; the exponent is sharp.
- (viii) The lower bound $X^{1-o(1)}$ holds separately in every component indexed by a finite quotient, and linear trace growth forces mixed energy $X^{3-o(1)}$ between all pairs of components in odd-order quotients. On the other hand, in an exact trace fiber the only matrix sum with large multiplicity is the forced inverse identity $A + A^{-1} = (\text{tr } A)I$; every other sum has representation multiplicity $\exp(o(R))$ on a word ball of radius R .
- (ix) Invariant lattices at all prime divisors of a proper normal model descend an absolutely irreducible representation to a central simple algebra over its constant field; the vertical divisors produce an order. Within an admissible form, arithmeticity also follows from a non-elementary normal subgroup in an order, one bounded spanning adjoint orbit, or nonexpanding adjoint directions on growing word balls. At any divisor where the adjoint image is unbounded, every direction instead expands exponentially.
- (x) On a character locus preserving all trace collisions at a Fuchsian point, two evaluations of $A = \rho(a)$ give trace fibers of size $\exp(o(R))$ in a word ball of radius R , as soon as their relative matrix has distinct eigenlines. Hence restriction to each of Hao’s complementary subgroups is birational on every positive-dimensional irreducible locus through the point on which these collisions persist, and has injective differential on every reduced subvariety of such a locus. At an algebraic point of the exceptional trace-collision locus, the full adjoint trace field is already generated by squared traces on each such subgroup. Finite covers strengthen this to an exponential family of globally distinct trace functions that coincide on the rational closure of the point.

- (xi) At a divisorial valuation, powers of a loxodromic element $k \in \ker \phi$ separate the two spectral coefficients of each trace equality that holds identically on the locus. This gives

$$h_{\text{tr}}(x) \geq \frac{\delta_H}{1 + \ell_x(k) \mathfrak{c}_v(a, k; H) / \ell_v(k)}.$$

Every unbounded divisorial realization contains such an element k . Thus all divisorial degenerations with sufficiently large local translation are excluded; the remaining case is the explicit opposite inequality between local translation and cancellation.

- (xii) Linear trace growth produces a set of $e^{(1/2-o(1))R}$ traces on which all Chebyshev recurrence dilates have simultaneously controlled sumsets. On a function field of a curve, the pairwise coprime Chebyshev zero divisors satisfy a Chinese-remainder uncertainty theorem, yielding the exact entropy–pole-degree inequality (13.10bh10).
- (xiii) An explicit injective map converts uniform trace counts modulo $(q_{n-1}(s)q_n(s))$ into a comparison between global recurrence height and geometric translation length. Fixing the recurrence index, Roth’s theorem gives a lower bound for almost the entire reduced divisor of two consecutive terms. A sieve over varying group elements, using expansion only modulo p^2 , chooses $a_m = b^m h_m$ for which this divisor has bounded prime exponents. Superapproximation for moduli whose prime exponents are uniformly bounded, together with the uniform estimate

$$\#\{g : d_x(o, go) \leq Cm, \text{tr}_y g = \xi, \text{tr}_y(a_m g) = \zeta\} = \exp(o_C(m)),$$

then gives

$$\mathcal{J}_{k_y}(\text{tr}_y b) \leq \frac{\ell_x(b)}{2}$$

for every algebraic faithful cocompact Fuchsian character y in the rational closure of x .

- (xiv) At an algebraic Fuchsian point the distinguished real place gives the reverse inequality, and Takeuchi’s criterion [47] gives arithmeticity. If the original point were transcendental, its rational closure would contain a nearby algebraic Fuchsian point in the same component. The preceding inequality implies that every marked length at the specialization is at most the corresponding original marked length. Thurston’s asymmetric metric [49] forces the two marked hyperbolic structures to coincide, contradicting transcendence. Thus linear trace growth forces arithmeticity unconditionally, proving both compact Schmutz and compact Sarnak.
- (xv) Independently, superapproximation for moduli m^E , with m square-free and E fixed, proves the stronger one-term bound

$$\limsup_n \frac{1}{n} \log N \text{rad}_S(q_n(s)) \leq \frac{\ell(a)}{2}.$$

It also shows that any failure of the radical bound for a fixed recurrence must occur in the initial Lucas–Wieferich multiplicities.

- (xvi) The stronger universal one-term Chebyshev radical hypothesis

$$\limsup_n n^{-1} \log N \text{rad}_S(q_n(s)) = \mathcal{J}_k(s)$$

is implied by number-field abc . At $s = 3/\sqrt{2}$, however, it is exactly the assertion that $\text{rad}(2^n - 1)$ has exponential rate 2^n , which implies infinitely many base-2 non-Wieferich primes. This explains why the unconditional proof varies the group element rather than proving the stronger pointwise statement.

The main unconditional conclusion is Theorem 13.59. The earlier arbitrary-ideal and number-field abc implications are retained because they prove stronger statements about a single recurrence and show why that approach encounters familiar open problems. Proposition 13.47 gives a

finite-ring example in which uniform estimates modulo separate prime powers do not combine into a uniform estimate modulo their product. The proof using varying group elements does not require this invalid inference. Section 16 records the logical dependencies among these results, and Section 17 compares them with the literature.

2. THE A_1 SQUARE ROOT IS HILBERT 90

2.1. Quadratic centralizer algebra. Recall that a finite-dimensional associative algebra D/k is *central simple* if its center is k and it has no nonzero proper two-sided ideals. Its degree is $\deg D = \sqrt{\dim_k D}$, and it is *split* over an extension K/k when $D \otimes_k K \simeq M_{\deg D}(K)$. The reduced characteristic polynomial, trace Trd , and norm Nrd are the ordinary characteristic polynomial, trace, and determinant after such a splitting; Galois descent makes them independent of the splitting. A quaternion algebra is exactly a central simple algebra of degree two.

An *involution* on a central simple algebra is an anti-automorphism $*$ of order two: $(xy)^* = y^*x^*$ and $(x^*)^* = x$. It is of the *first kind* if it fixes the center pointwise. It is of the *second kind*, or *unitary*, if the center is a quadratic extension of the fixed field and $*$ induces the nontrivial automorphism of that extension. Thus an involution of the second kind is not linear over the center; it is linear only over its fixed subfield. If k is a number field, an *order* in a central simple k -algebra D is a subring containing 1 which is also a full $\mathcal{O}_k$ -lattice in D , namely a finitely generated $\mathcal{O}_k$ -submodule spanning D over k .

Let k have characteristic different from two, let B/k be a quaternion algebra, and let $g \in \text{SL}_1(B)$ be regular semisimple; equivalently, $\text{Trd}(g)^2 - 4 \neq 0$. Indeed, the reduced characteristic polynomial is

$$X^2 - \text{Trd}(g)X + 1,$$

because $\text{Nrd}(g) = 1$. Its discriminant is $\text{Trd}(g)^2 - 4$, so the nonzero-discriminant hypothesis makes

$$E = k[g]$$

a quadratic étale k -algebra. Its nontrivial k -involution sends g to the other root, $\bar{g} = \text{Trd}(g) - g = g^{-1}$. Regularity excludes $\text{Trd}(g) = -2$, and

$$\text{Nrd}(1 + g) = (1 + g)(1 + g^{-1}) = 2 + \text{Trd}(g) \neq 0.$$

Thus $u = 1 + g$ is invertible. Then

$$\bar{u} = 1 + g^{-1} = g^{-1}(1 + g), \quad \frac{u}{\bar{u}} = g, \tag{2.1}$$

and

$$u\bar{u} = 2 + g + g^{-1} = 2 + \text{Trd}(g). \tag{2.2}$$

Equation (2.1) is Hilbert 90 in its most elementary form. After adjoining a square root of (2.2), normalization gives the claimed root. Explicitly, $u/\bar{u} = g$ implies $u^2 = (u\bar{u})g$; division by a chosen square root of the scalar $u\bar{u}$ therefore gives

$$R(g) = \frac{1 + g}{\sqrt{2 + \text{Trd}(g)}}, \quad R(g)^2 = g. \tag{2.3}$$

The two choices of the scalar square root differ by a sign, so the normalized object is a two-sheeted algebraic correspondence, or equivalently a map on the quadratic Kummer cover on which that square root has been chosen. Projectivization removes exactly that scalar ambiguity, making the following class rational over k :

$$R_{\text{ad}}(g) = [1 + g] \in \text{PGL}_1(B), \quad R_{\text{ad}}(g)^2 = [g]. \tag{2.4}$$

Proposition 2.1 (Kummer obstruction in type A_1). *For $g \in \mathrm{SL}_1(B)(k)$ with $2 + \mathrm{Trd}(g) \neq 0$,*

$$g \text{ has a square root in } \mathrm{SL}_1(B)(k) \iff 2 + \mathrm{Trd}(g) \in k^{\times 2}.$$

Thus the obstruction is the class of $2 + \mathrm{Trd}(g)$ in $H^1(k, \mu_2) = k^\times / k^{\times 2}$.

Proof. If $2 + \mathrm{Trd}(g)$ is a square, formula (2.3) gives a k -rational root. Conversely, suppose $h^2 = g$. Cayley–Hamilton for h gives

$$h^2 - \mathrm{Trd}(h)h + 1 = 0,$$

so $g + 1 = \mathrm{Trd}(h)h$. Taking reduced norms yields

$$\mathrm{Nrd}(g + 1) = 2 + \mathrm{Trd}(g) = \mathrm{Nrd}(\mathrm{Trd}(h)h) = \mathrm{Trd}(h)^2 \mathrm{Nrd}(h) = \mathrm{Trd}(h)^2.$$

□

The reduced Cayley–Hamilton equation $g^2 - \mathrm{Trd}(g)g + 1 = 0$ also gives the secant identity

$$(g - 1)^2 = g^2 - 2g + 1 = (\mathrm{Trd}(g) - 2)g. \quad (2.5)$$

Thus $[1 + g]$, when $2 + \mathrm{Trd}(g) \neq 0$, and $[g - 1]$, when $\mathrm{Trd}(g) - 2 \neq 0$, are projective square roots. The advantage of the second is its linear trace derivative. In any split matrix realization, for every fixed matrix γ , trace linearity gives

$$\mathrm{tr}(\gamma(g - 1)) = \mathrm{tr}(\gamma g) - \mathrm{tr}(\gamma). \quad (2.6)$$

2.2. Geometric meaning. For a loxodromic element of $\mathrm{PSL}_2(\mathbb{C})$, write its eigenvalues as $e^{\pm\lambda/2}$, where $\lambda = \ell + i\theta$ is complex translation length. A branch of $R(g)$ has eigenvalues $e^{\pm\lambda/4}$. It fixes the same oriented axis because it is a rational function of g , hence commutes with g and has the same eigenspaces. It translates by $\ell/2$ and rotates by $\theta/2$; changing the square-root branch multiplies the lift by $-I$, which is invisible in PSL_2 . In $\mathrm{PSL}_2(\mathbb{R})$ this is the half-translation along the same geodesic. Equation (2.6) gives the arithmetic link: when γ and γg are lattice elements, $\mathrm{tr}(\gamma(g - 1))$ is literally the difference of two lattice trace values. A density theorem for the relevant double coset can therefore be read as a density theorem for additive trace differences.

3. UNIVERSAL DIVISION TORSORS AND RATIONAL SECTIONS

3.1. Construction from the universal centralizer. Let G be connected reductive, let G^{rs} be its regular semisimple locus, and let

$$\mathcal{J} \longrightarrow G^{\mathrm{rs}}, \quad \mathcal{J}_g = Z_G(g),$$

be the regular centralizer group scheme. Equivalently, it is the pullback to G^{rs} of the universal centralizer on the regular conjugacy stack. Concretely,

$$\mathcal{J} = \{(g, h) \in G^{\mathrm{rs}} \times G : gh = hg\},$$

with group law in the second coordinate. The tautological section is $\iota(g) = (g, g)$. It is a torus of relative dimension $\mathrm{rank} G$. If the characteristic does not divide m , define

$$\mathcal{R}_m = [m]^{-1}(\iota) \longrightarrow G^{\mathrm{rs}}. \quad (3.1)$$

Thus, more concretely,

$$\mathcal{R}_m = \{(g, h) \in G^{\mathrm{rs}} \times G : h^m = g\};$$

the commutation relation is automatic because every element commutes with its own powers.

Proposition 3.1. *The morphism $\mathcal{R}_m \rightarrow G^{\text{rs}}$ is a finite $\mathcal{J}[m]$ -torsor of degree $m^{\text{rank } G}$. At $g \in G^{\text{rs}}(k)$, the obstruction to a k -rational root is the Kummer class*

$$\delta_m(g) \in H^1(k, \mathcal{J}_g[m]).$$

Étale-locally on G^{rs} , (3.1) is ordinary division by m on a split torus. When G is split, one may use the usual cameral cover; in general one first makes a splitting base change. Weyl and Galois monodromy permute the branches.

Proof. Multiplication by m on a torus is finite, faithfully flat, and, under the characteristic hypothesis, finite étale. Its kernel is $\mathcal{J}[m]$. Pulling it back along ι proves the torsor statement. After a finite étale splitting base change, $\mathcal{J}_g \simeq \mathbb{G}_m^r$, where $r = \text{rank } G$, and its m -torsion is μ_m^r . This explains the degree m^r .

At a k -point g , the exact Kummer sequence

$$1 \longrightarrow \mathcal{J}_g[m] \longrightarrow \mathcal{J}_g \xrightarrow{[m]} \mathcal{J}_g \longrightarrow 1$$

has a connecting map $\mathcal{J}_g(k) \rightarrow H^1(k, \mathcal{J}_g[m])$. The image of the element g is $\delta_m(g)$. By exactness, this class vanishes precisely when $g = [m](h) = h^m$ for some $h \in \mathcal{J}_g(k)$. For split G , the cameral cover is the finite cover which chooses and labels the maximal torus containing a regular element—the higher-rank analogue of choosing its eigenvalues. It removes the Weyl-group ambiguity and splits the preceding torus locally. $\square$

Thus roots do generalize geometrically, but normally as a finite correspondence rather than a rational single-valued map.

The correspondence has an intrinsic torus presentation. Let T be a maximal torus after a splitting base change, let $N = N_G(T)$, and put

$$\tilde{\mathcal{R}}_m(T) = \{(t, u) \in T^{\text{rs}} \times T : u^m = t\}.$$

Proposition 3.2 (torus and quotient-stack model). *After the chosen splitting base change there is a canonical equivalence*

$$[\mathcal{R}_m/G] \simeq [\tilde{\mathcal{R}}_m(T)/N]. \quad (3.2)$$

In particular, the geometric branches form a torsor under $T[m]$, Weyl monodromy acts through N/T , and over a nonsplit field this action is combined with the Galois action on the based root datum.

Proof. The quotient stack notation remembers not only a conjugacy class but also the conjugating elements and stabilizer; this is why the normalizer N , rather than only the abstract Weyl group N/T , appears. Every regular semisimple element is conjugate to an element of T^{rs} , and two such elements are conjugate precisely through N . If $u^m = t$ and t is regular, then u is regular as well: for every root α , $\alpha(u)^m = \alpha(t) \neq 1$. Thus a commuting root pair is conjugate to a pair in $\tilde{\mathcal{R}}_m(T)$: once t is in T , its centralizer is exactly T , so the commuting root u lies in the same torus. The pair has exactly the indicated normalizer ambiguity. This is the standard regular-semisimple equivalence $[G^{\text{rs}}/G] \simeq [T^{\text{rs}}/N]$ with the division correspondence pulled back along it. $\square$

There is also a uniform geometric meaning for every real division branch.

Proposition 3.3 (Jordan displacement under division). *Let G be a connected real semisimple linear group, and let $g \in G$ be regular semisimple. If $h^m = g$, then the Jordan projections satisfy*

$$\lambda(h) = \frac{1}{m} \lambda(g). \quad (3.3)$$

For the associated symmetric space, the translation lengths therefore obey $\ell(h) = \ell(g)/m$.

Proof. Recall the definition. Every element x of a real linear semisimple group has a unique multiplicative Jordan decomposition

$$x = x_e x_h x_u$$

into commuting elliptic, hyperbolic, and unipotent factors. The hyperbolic factor is conjugate to a unique element $\exp H$ with H in a fixed closed positive Weyl chamber $\mathfrak{a}^+$. This element H is the Jordan projection $\lambda(x)$.

Apply uniqueness to $h^m = g$. If $h_u = \exp N$, then the unipotent part of g is $h_u^m = \exp(mN)$. The element g is semisimple, so this part is the identity; characteristic zero gives $N = 0$. Thus h is semisimple. Conjugate its hyperbolic part to $\exp H$, $H \in \mathfrak{a}^+$. Its m -th power is $\exp(mH)$, and mH remains in $\mathfrak{a}^+$. The exponential map is injective on the positive split torus $A = \exp \mathfrak{a}$. More explicitly, choose the simple-root characters $\alpha_1, \dots, \alpha_r$. The map

$$A \longrightarrow \mathbb{R}^r, \quad a \longmapsto (\log |\alpha_1(a)|, \dots, \log |\alpha_r(a)|)$$

is injective on the positive split torus, and the chosen closed Weyl chamber is described by linear inequalities in these logarithmic coordinates. The absolute values remove the signs which can occur on other components of a real torus; on $A = \exp \mathfrak{a}$ the character values are already positive. For every real character χ one has

$$\log |\chi((\exp H)^m)| = m \log |\chi(\exp H)|.$$

Thus taking the m -th power is literally multiplication by m in logarithmic character coordinates. The simple-root characters determine H , so the unique chamber representative for the hyperbolic part of g is mH . Therefore $\lambda(g) = m\lambda(h)$, which is (3.3).

Finally, for the standard G -invariant metric on the symmetric space, the minimum displacement of a semisimple element x is $\|\lambda(x)\|$, where the norm comes from the invariant inner product on $\mathfrak{a}$. Homogeneity of that norm gives $\ell(g) = m\ell(h)$. $\square$

3.2. Classification of rational projective root sections. The next result classifies when the finite division correspondence can collapse to a rational projective branch. Notice that no conjugation-equivariance hypothesis is needed.

Theorem 3.4 (classification of rational projective root sections). *Let G_{sc} be a nontrivial semisimple simply connected group over an algebraically closed field of characteristic zero, and let $\pi : G_{\text{sc}} \rightarrow G_{\text{ad}}$ be the adjoint quotient. Suppose that, for some $m \geq 2$, a rational map on a nonempty open set satisfies*

$$s : G_{\text{sc}} \dashrightarrow G_{\text{ad}}, \quad s(g)^m = \pi(g). \quad (3.4)$$

Then $m = 2$ and every simple factor of G_{sc} is of type A_1 . Conversely, for a product of type- A_1 factors the componentwise map $g \mapsto [1 + g]$ satisfies (3.4) for $m = 2$.

Proof. Choose a regular semisimple point g in the domain and put $T_{\text{sc}} = Z_{G_{\text{sc}}}(g)$. The domain of the rational map is open, so its intersection with T_{sc} contains a nonempty open neighbourhood of a suitable such g . For generic t in this intersection, equation (3.4) implies that $s(t)$ commutes with $s(t)^m = \pi(t)$. Since $\pi(t)$ is regular, its centralizer is T_{ad} . Hence restriction gives a rational map $T_{\text{sc}} \dashrightarrow T_{\text{ad}}$.

Write

$$P = X^*(T_{\text{sc}}), \quad Q = X^*(T_{\text{ad}});$$

these are the weight and root lattices. Pullback by the central isogeny π identifies Q with the root sublattice of P . For every $\alpha \in Q$,

$$(s^* \alpha)^m = \pi^* \alpha = \chi^\alpha \quad \text{in } k(T_{\text{sc}}). \quad (3.5)$$

In the fraction field of the Laurent UFD $k[P]$, a monomial χ^α is an m -th power only if $\alpha \in mP$. To see this, write a putative root as a coprime quotient a/b . Valuation at every irreducible divisor

shows that all exponents in a^m and b^m must cancel; hence both a and b are Laurent monomials times scalars. Comparing their character exponents then gives $\alpha = m\beta$ for some $\beta \in P$. Therefore

$$Q \subset mP. \quad (3.6)$$

Write a simple root in the fundamental-weight basis as

$$\alpha_i = \sum_j a_{ji} \omega_j,$$

where (a_{ji}) is the Cartan matrix. In every irreducible root system of rank at least two, some off-diagonal entry is -1 . The corresponding simple root therefore has coefficient -1 in this basis and cannot be divisible by any $m \geq 2$ in P . Apply this argument to each simple factor. In type A_1 , by contrast, $Q = 2\mathbb{Z}\omega \subset P = \mathbb{Z}\omega$, and (3.6) forces $m = 2$. Formula (2.4) proves the converse. $\square$

Remark 3.5. The theorem concerns a map $G_{\text{sc}} \dashrightarrow G_{\text{ad}}$. It does not give a rational square-root map on PGL_2 itself. It is also distinct from surjectivity of power maps, rational sections of the conjugacy quotient, and Cayley birationality; see Steinberg [43], Popov [33], and Lemire–Popov–Reichstein [25].

4. PROJECTIVE SECANTS AND FORM DETECTION

Let $\rho : G \rightarrow \text{GL}(V)$ be a linear representation and let $0 \neq f \in k[z, z^{-1}]$. Laurent evaluation is legitimate because $\rho(g)$ is invertible. On the open set where $f(\rho(g))$ is invertible, define the projective secant

$$\Sigma_{f,\rho}(g) = [f(\rho(g))] \in \text{PGL}(V). \quad (4.1)$$

This construction commutes with base change and conjugation of the representation. Its target need not lie in the projective image of G . That failure records the tensors which cut out $\rho(G)$ inside $\text{GL}(V)$.

For self-dual classical representations this target condition is explicit. Let $\dagger$ be the adjoint involution of a nondegenerate bilinear or Hermitian form and assume $\rho(g)^\dagger = \rho(g)^{-1}$. The coefficients of f lie in the fixed field, so

$$f(\rho(g))^\dagger = f(\rho(g)^{-1}).$$

An invertible operator a is a similitude of the form exactly when $a^\dagger a$ is a scalar. Put

$$q_f(z) = f(z^{-1})f(z). \quad (4.2)$$

Then $[f(\rho(g))]$ is a projective similitude precisely when

$$f(\rho(g))^\dagger f(\rho(g)) = q_f(\rho(g)) \quad \text{is scalar.} \quad (4.3)$$

Theorem 4.1 (classical secant obstruction). *Let k have characteristic zero and let $f \neq 0$. Consider either the standard representation, over an algebraically closed field, of*

$$\text{Sp}_{2r} \ (r \geq 2), \quad \text{SO}_N \ (N \geq 3),$$

or the standard representation of an outer unitary k -form of type A_{d-1} , $d \geq 3$, tested after base change to a splitting field. The generic projective secant $\Sigma_{f,\rho}$ lies in the corresponding projective similitude group—and in the unitary case has target descent to the adjoint k -group—if and only if f is a Laurent monomial.

Proof. If the smallest and largest exponents occurring in f are $a < b$, the coefficient of z^{b-a} in q_f is the nonzero product of the two extreme coefficients. This term comes only from pairing exponent a in $f(z^{-1})$ with exponent b in $f(z)$, so it cannot cancel. Thus q_f is constant exactly when f is a Laurent monomial cz^a , $c \neq 0$.

On a standard symplectic or even orthogonal torus one may write

$$\text{diag}(z_1, \dots, z_r, z_1^{-1}, \dots, z_r^{-1}).$$

Equation (4.3) then has the eigenvalues $q_f(z_i) = q_f(z_i^{-1})$. In rank at least two, scalarity forces the nonconstant Laurent polynomial q_f to take the same value on two independent variables, which is impossible. An odd orthogonal torus has in addition the eigenvalue 1; already $q_f(z_1) = q_f(1)$ forces q_f to be constant. After splitting a unitary form, a diagonal torus has eigenvalues $z_1, \dots, z_d$, subject only to $\prod z_i = 1$. When $d \geq 3$, z_1, z_2 vary independently and the remaining coordinates absorb the determinant relation, with the same conclusion. Conversely, for a monomial $f(z) = cz^a$, equation (4.3) is scalar. $\square$

The excluded symplectic rank-one case is $\text{Sp}_2 = \text{SL}_2$, whose projective similitude group is all of PGL_2 . This tensor-theoretic exception is the same A_1 exception found in the rational root-section theorem.

4.1. Polynomial secants in type A. Let k have characteristic zero and let G be a simply connected k -form of type A_{d-1} . For a nonzero Laurent polynomial $f \in k[z, z^{-1}]$, consider the projective polynomial secant

$$s_f(g) = [f(g)] \tag{4.4}$$

on the open set where $f(g)$ is invertible.

Here “descent” means *target descent*: after scalar extension, the projective class lands in the canonical adjoint k -subgroup. It does not mean that the map factors through the central quotient of its source; indeed $[f(\zeta g)]$ need not equal $[f(g)]$.

Here a k -form of SL_d means a k -group which becomes isomorphic to SL_d over an algebraic closure. It is *inner* when its Galois descent maps use only conjugations, and *outer* when the transpose-inverse diagram automorphism occurs. Equivalently, the image of its descent cocycle in the order-two automorphism group of the A_{d-1} Dynkin diagram is trivial in the inner case and nontrivial in the outer case.

4.2. Inner forms. The simply connected inner forms of type A_{d-1} are precisely $\text{SL}_1(D)$ for degree- d central simple k -algebras D , and their adjoint groups are $\text{PGL}_1(D)$. Every positive or negative Laurent power of g lies in D . Hence $f(g) \in D$, and invertibility cuts out a Zariski-open set on which (4.4) is automatically the k -rational map

$$s_f : G \dashrightarrow \text{PGL}_1(D) = G_{\text{ad}}. \tag{4.5}$$

This is the higher-rank remnant of the rationality of $[1 + g]$. In this inner case, “the defining algebra” is merely shorthand for the central simple algebra D/k from which the group $\text{SL}_1(D)$ is constructed; the word “defining” imposes no additional condition on D .

4.3. Outer forms. Suppose instead that the diagram action is nontrivial and

$$G = \text{SU}(E, *).$$

The *defining unitary datum* $(L/k, E, *)$ consists of the quadratic field L/k , a degree- d central simple L -algebra E , and an *involution of the second kind* $*$. Explicitly, $*$ is a k -linear anti-automorphism of E of order two and $\ell^* = \bar{\ell}$ for $\ell \in L$, where the bar is the nontrivial element of $\text{Gal}(L/k)$. The special unitary group is the k -group whose k -points are

$$\text{SU}(E, *)(k) = \{g \in E^\times : g^*g = 1, \text{Nrd}_{E/L}(g) = 1\}.$$

For example, if $E = \text{End}_L(V)$ and $*$ is adjoint with respect to a nondegenerate Hermitian form h on V , this is the usual special unitary group $\text{SU}(h)$. Thus $G = \text{SU}(E, *)$ is the simply connected outer form attached to this datum. The restriction of scalars $\text{Res}_{L/k} \text{PGL}_1(E)$ is the k -group whose k -points are $\text{PGL}_1(E)(L)$. In this manuscript, the occasional shorthand “the defining algebra” of

an outer form always means this *full unitary datum* $(L/k, E, *)$, not the central simple L -algebra E by itself. Indeed, E alone does not record either the descent field k or the semilinear involution that defines the k -form. Inside $\text{Res}_{L/k} \text{PGL}_1(E)$, a projective class $[a]$ lies in the projective unitary group precisely when $a^*a \in L^\times$ (then this scalar is automatically fixed by $*$, because $(a^*a)^* = a^*a$, and hence lies in $k^\times$). On G one has $g^* = g^{-1}$, so

$$f(g)^* f(g) = f(g^{-1}) f(g). \quad (4.6)$$

We also fix the meaning of a distinguished split factor. If v is the archimedean place used to realize the lattice and $L \otimes_k k_v \simeq k_v \times k_v$, the two factors correspond to the two places of L above v , and $*$ exchanges them. Choosing one place $w \mid v$ selects the factor E_w . When $E_w \simeq M_d(\mathbb{F})$, projection to that factor identifies the local special unitary group with $\text{SL}_d(\mathbb{F})$; this chosen projection is the *distinguished realization*. The arithmetic datum is assumed compact at every other archimedean factor whenever the resulting image is called a lattice in a single factor.

Theorem 4.2 (detection of inner and outer forms by polynomial secants). *Let $d \geq 3$, and let $f \in k[z, z^{-1}]$ have at least two nonzero coefficients. The generic projective secant $[f(g)]$ has target descent to the adjoint k -group of a type- A_{d-1} form if and only if that form is inner.*

Proof. The inner assertion is (4.5). In the outer case put

$$q_f(z) = f(z^{-1})f(z).$$

If the smallest and largest exponents occurring in f are $a < b$, then the coefficient of z^{b-a} in q_f is the product of the two extreme coefficients and is nonzero. Hence q_f is nonconstant; equivalently, q_f is constant only when f is a unit of $k[z, z^{-1}]$, that is, a monomial.

After base change to an algebraic closure, the involution exchanges the standard module with its dual; on a split diagonal torus this sends $\text{diag}(z_1, \dots, z_d)$ to $\text{diag}(z_1^{-1}, \dots, z_d^{-1})$. Take a generic such element with $\prod z_i = 1$. Equation (4.6) is diagonal with entries $q_f(z_i)$. For $d \geq 3$, two eigenvalues can be varied independently subject to the determinant relation, so these entries are not generically equal when q_f is nonconstant. Thus (4.6) is generically noncentral, and $[f(g)]$ does not descend to the projective unitary group. $\square$

For the basic secant $f(z) = z - \lambda$, $\lambda \in k^\times$, the scalar λ is fixed by $*$, and direct multiplication gives

$$(g - \lambda)^*(g - \lambda) = (g^{-1} - \lambda)(g - \lambda) = (1 + \lambda^2)I - \lambda(g + g^{-1}). \quad (4.7)$$

In particular, $[g - I]$ descends in an outer form exactly on the proper locus where $g + g^{-1}$ is scalar. When $d = 2$, Cayley–Hamilton says $g^2 - \text{tr}(g)g + I = 0$. Multiplying by g^{-1} gives $g + g^{-1} = \text{tr}(g)I$, so the obstruction disappears; indeed every symmetric Laurent polynomial in g is scalar. This is the same exceptional rank in which (2.5) makes the secant a projective square root.

4.4. The Galois action being detected. For $d \geq 3$,

$$\text{Aut}(\text{SL}_d) = \text{PGL}_d \rtimes \langle \iota \rangle, \quad \iota(g) = (g^{-1})^t.$$

A k -form is inner when its descent cocycle has trivial image in the diagram automorphism group. More explicitly, composing the descent cocycle with $\text{Aut}(\text{SL}_d) \rightarrow \langle \iota \rangle \simeq \mathbb{Z}/2$ gives the diagram character. A trivial character gives an inner form $\text{SL}_1(D)$. A nontrivial character cuts out a quadratic extension L/k ; over L , Galois conjugation identifies the standard representation with its dual and produces the unitary datum $(L/k, E, *)$. Formula (4.7) detects precisely that exchange.

This should not be confused with the distinction between an arithmetic lattice and a lattice *derived* from an order. Type A_1 has no nontrivial diagram automorphism, so all of its forms are inner; nevertheless an arithmetic A_1 -lattice can fail to be derived. In rank one, the square-root argument detects the latter Kummer and integrality obstruction. Thus the diagram character

is an invariant of the algebraic k -form, whereas being derived from an order is an integral and commensurability property of a particular lattice.

We record the two dynamical facts used in the next corollary. First, if an invertible matrix commensurates the rational subgroup of an outer arithmetic lattice, its distinguished realization is projectively a global unitary similitude; this is the outer case of Lemma 6.1. Second, the double-coset trace theorem [3] says that, for an invertible matrix outside that commensurator, the values $\{\mathrm{tr}(\gamma A) : \gamma \in \Lambda_0\}$ are dense in the distinguished field, component by component.

The finite-index rational subgroup used here always exists. Indeed, an outer arithmetic lattice is, by definition, commensurable with the integral points of a unitary $\mathcal{O}_k$ -model. Intersect the two groups and, if a normal subgroup is wanted, take the intersection of its finitely many conjugates in the lattice. The resulting subgroup is still finite index and consists of genuine unitary norm-one points. This is the same normal-core construction used in the introduction.

Corollary 4.3 (trace density for outer forms). *Let Λ_0 be the finite-index subgroup consisting of rational unitary norm-one points of an outer arithmetic lattice. Assume that at the unique noncompact archimedean place v , its unitary datum has a split factor $E_w \simeq M_d(\mathbb{F})$, for a chosen $w \mid v$, giving the distinguished realization $\Lambda_0 < \mathrm{SL}_d(\mathbb{F})$, $d \geq 3$. Let $f(z) = \sum_j c_j z^j$ have at least two nonzero coefficients. For generic $g \in \Lambda_0$, $[f(g)]$ does not commensurate the lattice, and hence*

$$\left\{ \sum_j c_j \mathrm{tr}(\gamma g^j) : \gamma \in \Lambda_0 \right\} \quad (4.8)$$

is dense in $\mathbb{F}$.

Here “generic” means every lattice point in a fixed nonempty Zariski-open set on which $f(g)$ is invertible and $f(g^{-1})f(g)$ is noncentral. Borel density [5] guarantees lattice points in that open set.

Proof. If $[f(g)]$ commensurates, the unitary version of Lemma 6.1 writes its distinguished realization as sb , where $b \in E^\times$ and $b^*b \in L^\times$. Both $f(g)$ and b are global elements of E ; since their quotient realizes the scalar sI , faithfulness of the chosen realization makes that quotient central in E . Hence $[f(g)] = [b]$ is a projective unitary similitude, contrary to Theorem 4.2 for generic g . The double-coset trace theorem [3] applies to arbitrary invertible matrices in $\mathrm{GL}_d(\mathbb{F})$, on every nonzero-determinant component, and therefore makes $\{\mathrm{tr}(\gamma f(g))\}$ dense, and trace linearity gives (4.8). $\square$

Warning 4.4. The element g in this fixed-element statement is initially chosen in the rational unitary subgroup, where $g^* = g^{-1}$ is justified. For an arbitrary element of a larger arithmetic overgroup, a determinant-one lift may only be a unitary similitude. Borel density in the finite-index rational subgroup is enough for all form-exclusion arguments.

5. EXACT ADDITIVE CONSEQUENCES OF THE POSITIVE-GAP THEOREM

Suppose Γ is derived from an admissible quaternion algebra over a number field k of degree n . Every trace is an algebraic integer; at each nondistinguished archimedean embedding its absolute value is at most two. The same product-formula argument used for T applies to much larger fixed subsets.

For an integral vector $c = (c_1, \dots, c_q)$, set

$$S_c(T) = \left\{ \sum_{i=1}^q c_i t_i : t_i \in T(\Gamma) \right\}, \quad C(c) = \sum_{i=1}^q |c_i|.$$

Theorem 5.1 (uniform gaps for fixed trace sumsets). *If Γ is derived over k , then $S_c(T)$ is uniformly discrete. More precisely,*

$$\text{Gap } S_c(T) \geq \begin{cases} (4C(c))^{1-n}, & \mathbb{F} = \mathbb{R}, \\ (4C(c))^{1-n/2}, & \mathbb{F} = \mathbb{C}. \end{cases} \quad (5.1)$$

For the zero vector the assertion is interpreted trivially.

Proof. Let $x \neq y$ belong to $S_c(T)$. Their difference is a nonzero algebraic integer

$$z = \sum_i c_i(t_i - u_i).$$

At any nondistinguished archimedean embedding, $|\sigma z| \leq 4C(c)$. In the Fuchsian case the norm inequality is

$$1 \leq |N_{k/\mathbb{Q}}(z)| \leq |z|(4C(c))^{n-1}.$$

In the Kleinian case the distinguished conjugate pair contributes $|z|^2$ and the other $n-2$ embeddings contribute at most $(4C(c))^{n-2}$. Solving the two inequalities for $|z|$ proves (5.1). $\square$

Taking $c = (1, -1)$ gives the promised sharpening.

Corollary 5.2 (exact difference-set dichotomy). *Put $D = T(\Gamma) - T(\Gamma)$. If Γ is not derived, then D is dense in $\mathbb{F}$. If Γ is derived over a degree- n field, then*

$$\text{Gap } D \geq \begin{cases} 8^{1-n}, & \mathbb{F} = \mathbb{R}, \\ 8^{1-n/2}, & \mathbb{F} = \mathbb{C}. \end{cases}$$

The first assertion is Bogachev's theorem; the second is Theorem 5.1. Thus the main theorem is not merely a dichotomy between positive and zero gap of T : it is a dichotomy between a uniformly discrete first additive derivative and a dense one.

6. A COMPLEX HIGHER-RANK TRACE-GAP THEOREM

We now give the complex analogue of the corresponding classification for $\text{SL}_d(\mathbb{R})$. We use Margulis arithmeticity [28, Chapter IX], the standard classification of arithmetic type- A forms [32], and the following commensurator-normalizer fact. The Skolem–Noether step in that fact is the classical theorem for central simple algebras [31, Section 12.6].

Lemma 6.1 (projective normalizer). *Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, let K be a number field with a chosen archimedean embedding into $\mathbb{F}$, and let D/K be a degree- d central simple algebra satisfying $D \otimes_K \mathbb{F} \simeq M_d(\mathbb{F})$. Let $\Lambda_0 < D^\times \cap \text{SL}_d(\mathbb{F})$ be Zariski dense. If $A \in \text{GL}_d(\mathbb{F})$ commensurates Λ_0 , then*

$$A = sb \quad (s \in \mathbb{F}^\times, b \in D^\times). \quad (6.1)$$

*For an outer unitary form $(E/L, *)$, where L/K is quadratic, let $\Lambda_0 < \text{SU}(E, *) (K)$ be Zariski dense and use a chosen embedding, or a chosen factor when $L \otimes_K \mathbb{R}$ is split, to identify the distinguished component of E with $M_d(\mathbb{F})$. If A commensurates its image, then*

$$A = sb \quad (s \in \mathbb{F}^\times, b \in E^\times, b^*b \in L^\times).$$

Proof. Put $H = \Lambda_0 \cap A^{-1}\Lambda_0A$. It is finite index and Zariski dense, and $AHA^{-1} \subset \Lambda_0$. Choose d^2 elements in H which are $\mathbb{F}$ -linearly independent. They form a K -basis of D . Conjugation by A therefore preserves D and fixes its center. By Skolem–Noether [31, Section 12.6] it agrees on D with conjugation by some $b \in D^\times$. The matrix $b^{-1}A$ centralizes $D \otimes_K \mathbb{F} = M_d(\mathbb{F})$, and is scalar. In the outer case Zariski density implies that the L -span of H is E ; the same basis argument and Skolem–Noether over L give $A = sb$, $b \in E^\times$. Applying the unitary equation to a Zariski-dense finite-index subgroup shows that b^*b centralizes E , hence lies in $L^\times$. $\square$

Theorem 6.2 (complex higher-rank classification). *Let $d \geq 3$, let $\Lambda < \mathrm{SL}_d(\mathbb{C})$ be a lattice, and put $T_\Lambda = \{\mathrm{tr} g : g \in \Lambda\}$. The following are equivalent.*

- (1) *After conjugation, Λ has finite index in $\mathcal{O}^1$, where $\mathcal{O}$ is an order in a degree- d central simple algebra D/K and K is imaginary quadratic.*
- (2) *For some imaginary quadratic field $K \subset \mathbb{C}$, $T_\Lambda \subset \mathcal{O}_K$.*
- (3) *$\mathrm{Gap} T_\Lambda > 0$ in the Euclidean metric on $\mathbb{C}$.*
- (4) *$\overline{T_\Lambda - T_\Lambda} \neq \mathbb{C}$.*

When these conditions hold, $\mathrm{Gap} T_\Lambda \geq 1$.

Proof. The implications (1) $\Rightarrow$ (2) $\Rightarrow$ (3) $\Rightarrow$ (4) are immediate. Indeed, reduced traces of order elements lie in $\mathcal{O}_K$, and a nonzero $z \in \mathcal{O}_K$ satisfies $|z|^2 = |N_{K/\mathbb{Q}} z| \geq 1$.

Assume (4). If $g \in \Lambda$ and $\det(g - I) \neq 0$, then

$$\{\mathrm{tr}(\gamma(g - I)) : \gamma \in \Lambda\} \subset T_\Lambda - T_\Lambda.$$

Bogachev's double-coset trace theorem applies to every invertible $A \in \mathrm{GL}_d(\mathbb{C})$: scalar normalization places A in $\mathrm{SL}_d(\mathbb{C})$, and trace maps the corresponding nonzero-determinant hypersurface onto $\mathbb{C}$. It therefore forces $g - I$ to commute with Λ . Margulis arithmeticity [28, Chapter IX] applies to $\mathrm{SL}_d(\mathbb{C})$, viewed as a real group of real rank $d - 1 \geq 2$.

We first exclude an outer arithmetic form. Pass to its finite-index rational unitary subgroup and choose, by Borel density [5], an element g satisfying

$$\det(g - I) \neq 0, \quad g + g^{-1} \text{ is not scalar.}$$

The secant $A = g - I$ commensurates. Lemma 6.1 writes $A = sb$, with $b \in E^\times$ and $b^*b \in L^\times$. Since also $A \in E$, injectivity of the distinguished realization shows that $Ab^{-1} = sI$ is central in E , and hence $s \in L$. Consequently

$$(g - I)^*(g - I) = 2I - g - g^{-1}$$

would be central, a contradiction. Hence the arithmetic form is inner, say $\mathrm{SL}_1(D)$ over a number field K .

There is only one noncompact archimedean factor. Every complex place of an inner type- A_{d-1} group gives $\mathrm{SL}_d(\mathbb{C})$. At a real place the local group is either $\mathrm{SL}_d(\mathbb{R})$, or, when d is even, $\mathrm{SL}_{d/2}(\mathbb{H})$; both are noncompact for $d \geq 3$. Thus K has no real places and exactly one complex place. It is imaginary quadratic.

It remains to show that the entire lattice, not merely a finite-index subgroup, lies in D^1 . Every $g \in \Lambda$ commensurates the initial arithmetic subgroup, so Lemma 6.1 writes $g = sb$ with $s \in \mathbb{C}^\times$, $b \in D^\times$. If, in addition, g is nonscalar and $\det(g - I) \neq 0$, write similarly $g - I = tc$. Then

$$c = \frac{s}{t}b - \frac{1}{t}I \in D.$$

The elements I, b are K -linearly independent and remain $\mathbb{C}$ -linearly independent after scalar extension. Comparing coordinates in a K -basis of D gives $s/t, 1/t \in K$, hence $s, t \in K$, and $g \in D$. Its reduced norm is its determinant at the distinguished embedding, so $\mathrm{Nrd}(g) = \det(g) = 1$ and $g \in D^1$.

Let U be this nonempty good Zariski-open set. For an arbitrary $h \in \Lambda$, the two open conditions $x \in U$ and $hx \in U$ have a common point of Λ by Zariski density. Both x and hx lie in D^1 , and therefore $h = (hx)x^{-1} \in D^1$. This two-open-set argument is needed: the countable subset $D \subset M_d(\mathbb{C})$ is not complex-Zariski closed.

Choose an initial order $\mathcal{O}_0 \subset D$ whose norm-one group is commensurable with Λ . For every $g \in \Lambda$, some positive power g^r lies in $\mathcal{O}_0^1$. Each eigenvalue α of g then has α^r integral, hence is itself integral. Since $\mathrm{tr} g \in K$, we obtain $\mathrm{tr} g \in \mathcal{O}_K$.

Finally choose d^2 elements $\gamma_i \in \Lambda$ forming a K -basis of D . The reduced-trace Gram determinant

$$m = \det(\text{Trd}(\gamma_i \gamma_j))$$

is a nonzero element of $\mathcal{O}_K$. Cramer's rule puts all coordinates of all elements of Λ in the fixed fractional ideal $m^{-1}\mathcal{O}_K$. Therefore $\mathcal{O} = \text{span}_{\mathcal{O}_K} \Lambda$ is a finite $\mathcal{O}_K$ -module; it contains 1, spans D , and is closed under multiplication. It is an order with $\Lambda < \mathcal{O}^1$. Since a discrete overgroup of a lattice has finite index, $[\mathcal{O}^1 : \Lambda] < \infty$. This proves (1). $\square$

Remark 6.3. We have not found the precise trace-difference criterion in Theorem 6.2 in the literature. The proof combines standard results, and we make no claim here about priority for this particular formulation.

7. SECANT ACCUMULATION AND FURTHER LOCAL FORMS

In absolute degree at least three, repeated secants can prove arithmeticity without using the special root formula available in type A_1 .

Lemma 7.1 (secant accumulation). *Let $g \in \text{GL}_N(\mathbb{C})$, $N \geq 3$, be regular semisimple, with every eigenvalue off the unit circle. If a subgroup $C < \text{PGL}_N(\mathbb{C})$ contains $[g^n - I]$ for every $n \geq 1$, then C is nondiscrete.*

Proof. All $g^n - I$ are invertible. Put

$$S_n = (g^n - I)(g - I)^{-1}, \quad Q_n = S_{n+1}S_n^{-1} = (g^{n+1} - I)(g^n - I)^{-1}.$$

Their projective classes lie in C . On the g -eigenline with eigenvalue z , the operator Q_n acts by

$$q_n(z) = \frac{z^{n+1} - 1}{z^n - 1} \rightarrow \begin{cases} z, & |z| > 1, \\ 1, & |z| < 1. \end{cases} \quad (7.1)$$

Thus Q_n converges to an invertible operator. Its projective classes are not eventually constant. Indeed, if $q_n(z) = cq_n(w)$ for every large n , cross-multiplication gives

$$(z - cw)(zw)^n + (c - z)z^n + (cw - 1)w^n + (1 - c) = 0.$$

Linear independence of exponential sequences with distinct nonzero bases (and $z, w \neq 1$) shows that the ratio can be constant only for $z = w$ or $zw = 1$; in the latter case $q_n(z^{-1}) = z^{-1}q_n(z)$. A regular spectrum of cardinality at least three cannot be a single inversion orbit. Hence C contains an infinite convergent sequence, and is nondiscrete. $\square$

Remark 7.2. For spectrum $\{z, z^{-1}\}$, the classes $[Q_n]$ are constant. Thus this proof fails in absolute degree two, precisely the degree in which the projective square root is rational.

Suppose a lattice in an inner local type- A group has non-dense reduced trace differences. Whenever $g^n - I$ is invertible, the identity

$$\text{Trd}(\gamma(g^n - I)) = \text{Trd}(\gamma g^n) - \text{Trd}(\gamma)$$

and the double-coset theorem force $[g^n - I]$ into the commensurator. A regular element with the spectrum required in Lemma 7.1 then makes the commensurator nondiscrete. In higher real rank this recovers Margulis arithmeticity. For suitable rank-one inner forms it gives a proof by a different argument.

The same argument gives a quaternionic theorem. In rank one, its proof uses secant accumulation; in higher rank, the conclusion already follows from Margulis arithmeticity.

We shall use the following standard consequence of Shah's orbit-closure theorem [41].

Lemma 7.3 (double-coset dichotomy). *Let G be a connected noncompact almost simple real Lie group generated by one-parameter unipotent subgroups, and let $\Gamma < G$ be a lattice. For every $a \in G$, either $a \in \text{Comm}_G(\Gamma)$, or $\Gamma a \Gamma$ is dense in G .*

Proof. Let the diagonal copy $H = \Delta G$ act on the finite-volume homogeneous space

$$(G \times G)/(\Gamma \times \Gamma)$$

through the point $x = (e, a)(\Gamma \times \Gamma)$. Shah's theorem makes the orbit closure homogeneous. A Lie subalgebra of $\mathfrak{g} \oplus \mathfrak{g}$ containing $\Delta \mathfrak{g}$ is either $\Delta \mathfrak{g}$ or all of $\mathfrak{g} \oplus \mathfrak{g}$: subtracting a diagonal element from any nondiagonal element produces a nonzero ideal in one factor. Connectedness of the orbit closure therefore gives the following dichotomy. Either Hx is closed of finite volume, in which case

$$\text{Stab}_H(x) \simeq \Gamma \cap a\Gamma a^{-1}$$

is a lattice in G , so the two lattices are commensurable; or Hx is dense. In the second case, approximate an arbitrary $(e\Gamma, y\Gamma)$ by $(h_i\Gamma, h_i a\Gamma)$. Choosing $\gamma_i, \delta_i \in \Gamma$ with $h_i\gamma_i \rightarrow e$ and $h_i a\delta_i \rightarrow y$ gives $\gamma_i^{-1} a\delta_i \rightarrow y$, proving density of the double coset. $\square$

Theorem 7.4 (quaternionic trace-gap classification). *Let $m \geq 2$, let*

$$G_m = \text{SL}_m(\mathbb{H}) = \text{SL}_1(M_m(\mathbb{H})),$$

and use reduced trace. For a lattice $\Lambda < G_m$, the following are equivalent:

- (1) *There are a degree- $2m$ central simple algebra $D/\mathbb{Q}$, an order $\mathcal{O} \subset D$, and an $\mathbb{R}$ -algebra isomorphism $D \otimes_{\mathbb{Q}} \mathbb{R} \simeq M_m(\mathbb{H})$ such that, after conjugation in G_m , Λ has finite index in the image of $\mathcal{O}^1$.*
- (2) $\text{Trd}(\Lambda) \subset \mathbb{Z}$;
- (3) *the reduced-trace set has positive gap;*
- (4) *its difference set is not dense in $\mathbb{R}$.*

When these conditions hold, the gap is at least one.

Proof. The first three implications are immediate. We prove (4) $\Rightarrow$ (1). First note the quaternionic version of the double-coset trace corollary. If $A \in M_m(\mathbb{H})^\times$, multiplication by a positive real scalar normalizes its reduced norm to one without changing whether it commensurates Λ . If it does not commensurate, Lemma 7.3 makes $\Lambda A \Lambda$, after that normalization, dense in G_m . Reduced trace is cyclic, and it maps G_m onto $\mathbb{R}$: for $x > 0$,

$$\text{diag}(x, -x^{-1}, 1, \dots, 1) \in G_m, \quad \text{Trd} = 2(x - x^{-1} + m - 2). \quad (7.2)$$

Consequently $\{\text{Trd}(\gamma A) : \gamma \in \Lambda\}$ is dense in $\mathbb{R}$.

Assume now that the reduced-trace difference set is not dense. For every $g \in \Lambda$ and every $n \geq 1$ for which $g^n - I$ is invertible,

$$\text{Trd}(\gamma(g^n - I)) = \text{Trd}(\gamma g^n) - \text{Trd}(\gamma)$$

shows that $g^n - I$ commensurates Λ .

For $m \geq 3$, Margulis arithmeticity [28, Chapter IX] applies because $\text{rank}_{\mathbb{R}} G_m = m - 1 \geq 2$. For $m = 2$, pass to a torsion-free finite-index subgroup. The complex regular-semisimple (simple-spectrum) locus in the standard four-dimensional complex realization is a nonempty real Zariski-open set, so Borel density [5] gives such a g in that subgroup. A nontrivial semisimple element of a torsion-free discrete rank-one group cannot be elliptic, so g is axial. Its four standard complex eigenvalues have moduli r, r, r^{-1}, r^{-1} for some $r \neq 1$, and in particular all lie off the unit circle. Thus every $[g^n - I]$ lies in the projective commensurator, and Lemma 7.1 makes that commensurator nondiscrete. Margulis's commensurator criterion [28, Chapter IX] again gives arithmeticity.

The standard classification of arithmetic type- A_{2m-1} forms [32] now gives either an inner central simple algebra or an outer unitary form. We first rule out the latter. (A quadratic extension can split at the distinguished real place, so local quaternionic type alone does not do this.) Pass to the finite-index rational unitary subgroup. Borel density [5] gives an element g for which $g - I$ is

invertible and $g + g^{-1}$ is not central. The secant $g - I$ commensurates. The outer version of the normalizer argument in Lemma 6.1, with the distinguished simple factor now $M_m(\mathbb{H})$, writes

$$g - I = sb,$$

where b belongs to the central simple L -algebra E occurring in the unitary datum and $b^*b \in L^\times$. Because $g - I$ and b are global elements, the identity $(g - I)b^{-1} = sI$ in the faithful distinguished factor forces $s \in L^\times$. It follows that

$$(g - I)^*(g - I) = 2I - g - g^{-1}$$

is central, a contradiction. The arithmetic form is therefore inner, given by a central simple algebra D/K .

Every complex completion of D gives a noncompact $\mathrm{SL}_{2m}(\mathbb{C})$, while at a real completion its norm-one group is $\mathrm{SL}_{2m}(\mathbb{R})$ or $\mathrm{SL}_m(\mathbb{H})$, both noncompact because $m \geq 2$. The arithmetic construction has only the distinguished noncompact factor. Hence K has exactly one archimedean place, which is real, and therefore $K = \mathbb{Q}$.

It remains to pass from commensurability to actual containment in an order. We use the local normalizer argument from Lemma 6.1: if $A \in M_m(\mathbb{H})^\times$ commensurates a Zariski-dense subgroup of $D^\times$, then

$$A = sb \quad (s \in \mathbb{R}^\times, b \in D^\times). \quad (7.3)$$

Indeed, a basis chosen in a finite-index intersection shows that conjugation by A preserves D ; Skolem–Noether [31, Section 12.6] gives b , and the centralizer of $D \otimes_{\mathbb{Q}} \mathbb{R} = M_m(\mathbb{H})$ in that algebra is $\mathbb{R}$.

For a nonscalar $g \in \Lambda$ with $g - I$ invertible, apply (7.3) to both $g = sb$ and $g - I = tc$. From

$$c = (s/t)b - (1/t)I \in D$$

and linear independence of I, b , coordinate comparison gives $s/t, 1/t \in \mathbb{Q}$. Thus $g \in D$, and its reduced norm is one. The two-open-set argument used in Theorem 6.2 propagates this from the good Zariski-open set to every element of Λ , so $\Lambda < D^1$.

Choose an order $\mathcal{O}_0 \subset D$ whose norm-one group is commensurable with Λ . Every $g \in \Lambda$ has a positive power in $\mathcal{O}_0^1$, so its reduced eigenvalues, and hence $\mathrm{Trd}(g) \in \mathbb{Q}$, are integral. Therefore $\mathrm{Trd}(\Lambda) \subset \mathbb{Z}$. Finally choose $(2m)^2$ elements of Λ forming a $\mathbb{Q}$ -basis of D . The nondegenerate reduced-trace pairing and Cramer's rule put every element of Λ in one fixed fractional $\mathbb{Z}$ -lattice. Hence $\mathcal{O} = \mathrm{span}_{\mathbb{Z}} \Lambda$ is an order, contains Λ , and is closed under multiplication. Since $\mathcal{O}^1$ is a discrete overgroup of the lattice Λ , the index is finite. This is (1), and integer traces give the stated gap. $\square$

7.1. A quantitative root-commensurator criterion. The root idea does give an unconditional arithmeticity criterion in every rank-one group.

Theorem 7.5. *Let G be a connected center-free simple rank-one isometry group, $\Gamma < G$ a lattice, h its volume entropy, and $m \geq 2$. Let $P_m(L)$ count primitive axial Γ -conjugacy classes $[\gamma]$ of length at most L for which some $r \in \mathrm{Comm}_G(\Gamma)$ satisfies $r^m = \gamma$. If Γ is nonarithmetic, then*

$$P_m(L) \ll e^{hL/m}. \quad (7.4)$$

Consequently, an exponential rate strictly greater than h/m , and in particular the existence of such a root for every primitive axial element, forces arithmeticity.

Proof. For nonarithmetic Γ , its commensurator Δ is discrete and contains Γ with finite index. If $r^m = \gamma$, then $\ell(r) = \ell(\gamma)/m$. A Δ -conjugacy class meets Γ in at most $[\Delta : \Gamma]$ Γ -classes. The prime-geodesic upper bound for Δ counts $O(e^{hR})$ axial classes of length at most R . Taking $R = L/m$ gives (7.4). $\square$

Counting axial classes is important in the nonuniform case: parabolic classes have zero translation length and cannot be included in this estimate.

8. DECOMPOSITION OF ARITHMETIC TRACES BY SQUARE CLASSES

An arithmetic lattice need not be derived, and its full trace set need not have positive gap. What survives is a finite decomposition indexed by square classes, compatible with the Galois action.

Proposition 8.1 (finite decomposition of an arithmetic trace set). *Let $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ be arithmetic. There is a finite-index normal subgroup $H \triangleleft \Gamma$, derived from an order, such that, for every coset aH , the signed trace set*

$$T(aH) = \{\mathrm{tr} \widetilde{ah} : h \in H\}$$

is uniformly discrete. Consequently $T(\Gamma)$ is a finite union of uniformly discrete sets and has bounded clustering. Differences between two components may nevertheless be dense.

Proof. Take a normal finite-index subgroup H contained in the norm-one group of an order in the defining quaternion algebra B/k . A lift of a coset representative commensurates that order group. Skolem–Noether [31, Section 12.6] writes it as $s_a b_a$, with $b_a \in B^\times$, while the determinant condition gives $s_a^2 \in k^\times$. Hence

$$\mathrm{tr}(s_a b_a h) = s_a \mathrm{Trd}(b_a h). \quad (8.1)$$

The second factor lies in a fixed fractional $\mathcal{O}_k$ -ideal. Put $L_a = k(s_a)$, of degree at most two over k . After one fixed denominator is cleared, all differences in (8.1) are algebraic integers of L_a . At every extension of a nondistinguished archimedean embedding of k , they are bounded: the corresponding norm-one group is compact, and the factors s_a, b_a are fixed. Over the distinguished real place there are either one or two embeddings of L_a ; in the quadratic case they send s_a to $\pm s_a$, so both absolute values equal the one at the distinguished place. The norm inequality in L_a therefore gives a positive separation constant for (8.1). There are only finitely many cosets. $\square$

For the Fricke extension in Bogachev’s manuscript,

$$T(\Gamma) = 2\mathbb{Z} \cup \sqrt{2}\mathbb{Z}. \quad (8.2)$$

Each of the two displayed subsets is separated, while Pell approximants using one value from each subset give zero total gap and $2\mathbb{Z} + \sqrt{2}\mathbb{Z} \subset T - T$, a dense subgroup of $\mathbb{R}$.

8.1. Ordinary versus invariant trace field. Let

$$K = \mathbb{Q}(\mathrm{tr} \widetilde{\Gamma}), \quad k = \mathbb{Q}(\mathrm{tr} \widetilde{\Gamma}^{(2)}) = \mathbb{Q}((\mathrm{tr} \widetilde{\gamma})^2 : \gamma \in \Gamma).$$

This is the ordinary/invariant trace-field convention of [26, Chapter 3]; in particular, the invariant trace field is unchanged on passage to a finite-index non-elementary subgroup. Because Γ is finitely generated, the standard finite trace-generation theorem for SL_2 characters expresses all word traces as integral polynomials in finitely many basic word traces [21]. Together with

$$(\mathrm{tr} \widetilde{\gamma})^2 = \mathrm{tr}(\widetilde{\gamma}^2) + 2 \in k, \quad (8.3)$$

this shows that K/k is a finite multiquadratic extension. There is a more precise compatibility with the quotient by the subgroup generated by squares.

Proposition 8.2 (Galois signs are mod-two characters). *Assume Γ is nonelementary. For every $\sigma \in \mathrm{Gal}(K/k)$, there is a character*

$$\varepsilon_\sigma : \Gamma \longrightarrow \{\pm 1\}$$

such that

$$\sigma(\mathrm{tr} \widetilde{\gamma}) = \varepsilon_\sigma(\gamma) \mathrm{tr} \widetilde{\gamma}. \quad (8.4)$$

Consequently $\text{Gal}(K/k)$ embeds in $H^1(\Gamma, \mathbb{F}_2)$. For a fixed class in $\Gamma/\Gamma^{(2)}$, all its traces lie in a one-dimensional k -eigenspace $s_v k \subset K$.

Proof. Work first on the full inverse image $\tilde{\Gamma}$, with its standard representation ρ , viewed over $\mathbb{C}$. Extend σ to a field automorphism of $\mathbb{C}$ and apply it entrywise (equivalently, realize the irreducible character over a finite algebraic extension of its trace field and extend σ there). After semisimplification this gives a representation ρ^σ whose character is $\sigma(\text{tr } \rho)$. The adjoint characters agree, since

$$\text{tr Ad}(\tilde{\gamma}) = (\text{tr } \tilde{\gamma})^2 - 1 \in k.$$

Because Γ is nonelementary, its projective image is Zariski dense; the adjoint representation is irreducible. Brauer–Nesbitt therefore identifies the two adjoint representations. Since every automorphism of PGL_2 is inner, after conjugating ρ^σ their projectivizations agree. Hence

$$\rho^\sigma(\delta) = \tilde{\varepsilon}_\sigma(\delta)\rho(\delta) \quad (\delta \in \tilde{\Gamma}),$$

where multiplicativity and determinant one make $\tilde{\varepsilon}_\sigma : \tilde{\Gamma} \rightarrow \{\pm 1\}$ a character. It takes the value $+1$ on the central element $-I$, because its trace -2 is fixed by σ , and therefore descends to Γ . This gives (8.4), and it is trivial on squares.

The twisting character is unique. Indeed, if $\rho \simeq \rho \otimes \chi$ for a nontrivial quadratic character χ , then $\text{tr } \rho$ vanishes on the nontrivial coset of $\ker \chi$. That coset is Zariski dense because $\rho(\tilde{\Gamma})$ is Zariski dense and $\ker \chi$ has finite index, which is impossible for the nonzero trace function. Applying σ to the twist for τ , and then using the twist for σ , shows that $\varepsilon_\sigma \varepsilon_\tau$ is the twisting character for $\sigma\tau$. Uniqueness therefore makes $\sigma \mapsto \varepsilon_\sigma$ a homomorphism, including on trace-zero elements. Its kernel fixes the trace-generated field K , so it is injective. A multiquadratic extension decomposes into one-dimensional eigenspaces over k , proving the last assertion. $\square$

The two Galois actions occur at different levels. The diagram action distinguishes inner and outer forms of higher type A , whereas the characters in (8.4) distinguish the components indexed by square classes inside one rank-one arithmetic commensurability class.

9. REDUCTION TO THE SUBGROUP GENERATED BY SQUARES

Let

$$H = \Gamma^{(2)}$$

be the subgroup generated by the elements γ^2 , $\gamma \in \Gamma$. Every automorphism carries a square to a square, so H is characteristic. Every element of Γ/H has order at most two; a group of exponent two is abelian because $(xy)^2 = 1 = x^2 y^2$ implies $xy = yx$. Thus Γ/H is an elementary abelian two-group. It is finite because Γ is finitely generated, and it is the largest quotient of Γ with these properties. For a closed surface, it is $H_1(\Gamma, \mathbb{F}_2)$, so H is the fundamental group of the mod-two homology cover.

Theorem 9.1 (arithmeticity and the square subgroup). *For a Fuchsian lattice Γ , the following are equivalent:*

- (1) Γ is arithmetic;
- (2) $\Gamma^{(2)}$ is derived from a quaternion algebra;
- (3) $\text{Gap } T(\Gamma^{(2)}) > 0$;
- (4) $T(\Gamma^{(2)}) - T(\Gamma^{(2)})$ is not dense in $\mathbb{R}$.

Consequently, the compact bounded-clustering conjecture is equivalent to

$$\text{BC}(T(\Gamma)) \implies \text{Gap } T(\Gamma^{(2)}) > 0, \tag{9.1}$$

and the compact linear-growth conjecture is equivalent to

$$\#(T(\Gamma) \cap [-X, X]) = O(X) \implies \text{Gap } T(\Gamma^{(2)}) > 0. \tag{9.2}$$

Proof. The classical invariant-trace-field characterization says that Γ is arithmetic exactly when $\Gamma^{(2)}$ is derived from the invariant quaternion algebra; see [47] and [26, Chapter 8]. Bogachev's positive trace-gap theorem applied to the finite-index lattice H makes (2), (3), and (4) equivalent. Bounded clustering and linear growth do pass from $T(\Gamma)$ to its subset $T(H)$, but this alone does not give a gap. The exact reformulations follow instead by replacing, pointwise, the conclusion “ Γ is arithmetic” in the Sarnak or Schmutz implication by its already proved equivalent “ $\text{Gap } T(H) > 0$.” Thus each implication in (9.1) and (9.2) entails the original conjecture, and conversely the corresponding original conjecture, together with $(1) \Leftrightarrow (3)$, entails that implication. $\square$

Thus (9.1) is not a weaker substitute for Sarnak's conjecture. It is an equivalent formulation in which the finite ambiguity coming from square classes has been removed. The reformulation is useful because $\Gamma^{(2)}$ is characteristic and $\Gamma/\Gamma^{(2)}$ is the maximal elementary abelian two-quotient of Γ .

Corollary 9.2 (finite-cover formulation). *A Fuchsian lattice is arithmetic if and only if it has a finite-index subgroup H whose signed trace set is uniformly discrete. If it is arithmetic, one may moreover choose H normal so that every coset trace set is uniformly discrete.*

Proof. The forward statement is Proposition 8.1. Conversely, positive trace gap makes H derived by Bogachev's theorem, and arithmeticity passes to the finite-index overgroup Γ . $\square$

10. THE TRACE ALGEBRA GRADED BY SQUARE CLASSES

Put $V = \Gamma/\Gamma^{(2)}$, written additively. For $v \in V$, let T_v be the signed traces of lifts of elements in that coset. The Fricke identity gives a coherent multiplication law which an arbitrary coloring does not possess.

Theorem 10.1 (trace identities respecting square classes). *For $a, b \in V$,*

$$T_a T_b \subset T_{a+b} + T_{a+b}. \quad (10.1)$$

For $t \in T_a$,

$$t^2 - 2 \in T_0. \quad (10.2)$$

If $R_\Gamma = \mathbb{Z}[T(\Gamma)]$, then

$$R_\Gamma = \mathbb{Z} + \text{span}_{\mathbb{Z}} T(\Gamma). \quad (10.3)$$

More precisely, with $M_a = \text{span}_{\mathbb{Z}} T_a$,

$$M_a M_b \subset M_{a+b}. \quad (10.4)$$

The sum of the M_a need not be direct.

Proof. Choose lifts A, B with traces $x \in T_a, y \in T_b$. Then

$$xy = \text{tr}(AB) + \text{tr}(AB^{-1}). \quad (10.5)$$

Because V has exponent two, both AB and AB^{-1} have class $a+b$, proving (10.1). Cayley–Hamilton gives $t^2 - 2 = \text{tr}(A^2)$, which proves (10.2). Finally (10.5) reduces every product of trace generators to an integral sum of traces, inductively proving (10.3) and (10.4). $\square$

Corollary 10.2 (the trace algebra over its neutral component). *Put $A_0 = \mathbb{Z} + M_0$ and $A = \mathbb{Z}[T(\Gamma)]$. Then A_0 is a ring and A is a finite integral A_0 -algebra generated by elements satisfying quadratic equations over A_0 .*

Proof. Equations (10.2) and (10.4) make A_0 a ring. The trace algebra of a finitely generated SL_2 -group is generated by finitely many word traces, say $t_1, \dots, t_r$. Each satisfies

$$X^2 - (\text{tr}(\gamma_i^2) + 2) = 0$$

over A_0 . Hence $A = A_0[t_1, \dots, t_r]$ is integral and finite over A_0 . $\square$

These identities show why an approximate-ring theorem is relevant. They also show why it cannot yet be applied: an approximate subring requires both $X + X$ and XX to be covered by finitely many additive translates of X . Formula (10.1) only puts products in a two-fold sumset.

10.1. An abstract graded-ring counterexample.

Proposition 10.3 (a scalar model that does not come from group traces). *Let*

$$S = 2\mathbb{Z} \cup 2\sqrt{2}\mathbb{Z}, \quad S_0 = S, \quad S_1 = 2S. \quad (10.6)$$

Thus $S_0 \cup S_1 = S$ and $\pm 2 \in S_0$. Then:

- (i) *S has bounded clustering and linear growth, but $\text{Gap } S = 0$ by Pell approximation;*
- (ii) *$S_i S_j \subset S_{i+j}$ for $i, j \in \mathbb{Z}/2$, and $s^2 - 2 \in S_0$ for every $s \in S$;*
- (iii) *every Chebyshev trace polynomial P_n , defined by $P_0 = 2, P_1 = x, P_{n+1} = xP_n - P_{n-1}$, respects the grading: $P_n(S_i) \subset S_{ni}$;*
- (iv) *$\mathbb{Z}[S] = \mathbb{Z} + \text{span}_{\mathbb{Z}}(S) = \mathbb{Z}[2\sqrt{2}]$.*

The component declared neutral is $S_0 = S$, and hence has zero gap. Moreover, $S + S$ contains the dense group $2\mathbb{Z} + 2\sqrt{2}\mathbb{Z}$, while a finite union of translates of S is locally finite. Thus S is not an approximate subring.

Proof. Each unit interval meets each of the two lattices in a uniformly bounded number of points, and their counts in $[-X, X]$ are linear. Continued fractions for $\sqrt{2}$ give $|2p - 2\sqrt{2}q| \rightarrow 0$, proving zero gap. Put $B = \mathbb{Z} \cup \sqrt{2}\mathbb{Z}$, so $S_0 = 2B$, $S_1 = 4B$, and $BB \subset B$. It follows immediately that $S_0 S_0 \subset S_0$, $S_0 S_1 \subset S_1$, and $S_1 S_1 \subset S_0$. Also $s^2 - 2 \in 2\mathbb{Z} \subset S_0$.

The polynomial P_n has the parity of n . If its argument lies in $2B$, its even powers and constant term give an element of $2\mathbb{Z}$, while its odd powers remain in the same one of the two lattices making up $2B$. Thus $P_n(S_0) \subset S_0$. If its argument lies in $4B$, the same observation gives an element of $4B$ for odd n , and of $2\mathbb{Z}$ for even n . This proves the graded Chebyshev assertion. The ring identity is direct. Finally, $2\mathbb{Z} + 2\sqrt{2}\mathbb{Z}$ is dense in $\mathbb{R}$, whereas every finite union of translates of S is locally finite. Thus no finite collection of additive translates of S covers $S + S$. $\square$

This model satisfies every purely scalar operation in Theorem 10.1, including the displayed two-component multiplication law, but its neutral component has zero gap and its whole set is not an approximate subring. Therefore closure under the Fricke identity, Chebyshev iteration, finite generation of the trace ring, and a grading by square classes cannot prove (9.1) without information from noncommutative multiplication or hyperbolic geometry. The arithmetic example (8.2) shows that the whole-set obstruction is not an artificial pathology.

There is, however, a sharp distinction between assigning degrees to a scalar model and realizing those degrees by traces of group elements. The full Fricke relation forces purity in any direct grading.

Theorem 10.4 (Fricke purity for a direct grading). *Let Q be an elementary abelian two-group, and let*

$$R = \bigoplus_{q \in Q} R_q \subset \mathbb{R}$$

be a direct sum of additive subgroups such that $R_p R_q \subset R_{p+q}$ and $2 \in R_0$. Let $\Delta < \text{SL}_2(\mathbb{R})$ be a group for which every trace is nonzero and homogeneous:

$$\text{tr}(\Delta) \subset \bigcup_{q \in Q} R_q.$$

For $g \in \Delta$, let $\varepsilon(g)$ be the unique degree of $\text{tr } g$. Then

$$\varepsilon : \Delta \longrightarrow Q$$

is a homomorphism. In particular,

$$\mathrm{tr}(\Delta^{(2)}) \subset R_0. \quad (10.7)$$

If Δ is the full lift of a torsion-free cocompact Fuchsian lattice and R_0 is uniformly discrete, then the lattice is arithmetic.

Proof. The identity has trace $2 \in R_0$, and $\mathrm{tr}(g^{-1}) = \mathrm{tr} g$, so inversion preserves degree. Put $p = \varepsilon(g)$ and $q = \varepsilon(h)$. The Fricke identity says

$$\mathrm{tr} g \mathrm{tr} h = \mathrm{tr}(gh) + \mathrm{tr}(gh^{-1}). \quad (10.8)$$

Its left side is a nonzero element of R_{p+q} . Each term on the right is a nonzero homogeneous element. Because the sum of the R_r is direct, neither right-hand term can have degree different from $p + q$: a component in any other degree would have to vanish, and if both terms had the same wrong degree their sum could not equal a nonzero element of R_{p+q} . Consequently

$$\varepsilon(gh) = \varepsilon(gh^{-1}) = p + q.$$

Thus ε is a homomorphism. Since Q has exponent two, it kills every square and hence the subgroup generated by squares, proving (10.7).

For a torsion-free cocompact Fuchsian group there are no trace-zero elements. The central element $-I$ has degree zero, so ε descends from the full lift. If R_0 is uniformly discrete, (10.7) gives a positive trace gap on the square subgroup. Theorem 9.1 then gives arithmeticity. $\square$

For example, fix any integer $q \geq 1$ and put

$$R_0 = 2\mathbb{Z}, \quad R_1 = 2q\sqrt{2}\mathbb{Z}.$$

These groups form a direct $\mathbb{Z}/2$ -grading: their products have the required degrees. Hence, if all traces of a torsion-free cocompact Fuchsian lift lay in $R_0 \cup R_1$, Theorem 10.4 would force every trace on the square subgroup into $2\mathbb{Z}$, and the group would be arithmetic. The scalar example (10.6) obtains its zero neutral gap only by declaring the whole union to have degree zero; that declaration cannot agree with the canonical quotient by square classes.

11. WHY THE TRACE IDENTITIES DO NOT YET GIVE AN APPROXIMATE RING

A set X in a ring is *symmetric* if $0 \in X$ and $X = -X$. It is an *approximate subring* if there is a finite set F such that

$$X + X \cup XX \subset F + X.$$

Two subsets of an additive group are *additively commensurable* if each is contained in finitely many additive translates of the other. Krupiński and Machado prove the following 2026 structure theorem [22].

Theorem 11.1 (Krupiński–Machado, specialized form). *Let X be an infinite uniformly discrete approximate subring spanning a finite-dimensional simple real algebra A . Then there are a number field K , an archimedean place v , a K_v -algebra structure on A , and a K_v -basis $e_1, \dots, e_r$ such that X is additively commensurable with*

$$\sum_{i=1}^r \mathcal{O}_{K,v} e_i, \quad \mathcal{O}_{K,v} = \{a \in K : |a|_w \leq 1 \text{ for every } w \neq v\}. \quad (11.1)$$

For $A = \mathbb{R}$, these are the classical Pisot–Salem model sets.

The model set on the right has the three features that occur in Takeuchi’s criterion [47]: a number field, integrality at the finite places, and boundedness at the nondistinguished archimedean places. Additive commensurability, however, does not imply that the original set or the finitely many translating elements lie in that number field. One must also relate the model set to the

trace field and to the components indexed by square classes. In addition, the approximate-ring hypothesis is strictly stronger than (10.1).

Lemma 11.2 (covering close differences already gives gap). *Let $S \subset \mathbb{R}$ be locally finite. If*

$$(S - S) \cap (-\eta, \eta) \subset F + S$$

for a finite set F and some $\eta > 0$, then $\text{Gap } S > 0$.

Proof. The finite union $F + S$ is locally finite, so zero is isolated in it. Every nonzero difference smaller than η belongs to this union; all other differences are already at least η . Hence the nonzero differences are bounded away from zero. $\square$

Thus a finite-translate cover of the sufficiently small differences in $T(\Gamma^{(2)})$ would already prove the required trace gap, without an application of Theorem 11.1. By contrast, finding a small-doubling subset at selected scales leaves the other trace values uncontrolled, and those values may still contain arbitrarily close pairs.

11.1. Finite-place boundedness and arithmeticity. We shall use the following elementary consequence of the product formula.

Lemma 11.3 (separation by the product formula). *Let k be a number field with a distinguished real embedding v_0 , and let I be a fractional $\mathcal{O}_k$ -ideal, that is, a nonzero finitely generated $\mathcal{O}_k$ -submodule of k whose k -linear span is k . Equivalently, some nonzero $a \in \mathcal{O}_k$ satisfies $aI \subset \mathcal{O}_k$. Suppose $S \subset I$ and, for every other archimedean embedding σ , there is a constant C_σ such that*

$$|\sigma(s)| \leq C_\sigma \quad (s \in S). \quad (11.2)$$

Then $v_0(S) \subset \mathbb{R}$ is uniformly discrete.

Proof. Choose $a \in \mathcal{O}_k \setminus \{0\}$ with $aI \subset \mathcal{O}_k$. For distinct $s, t \in S$, put $z = a(s - t)$. Then z is a nonzero algebraic integer, so $N_{k/\mathbb{Q}}(z)$ is a nonzero rational integer. If complex embeddings are counted with their usual multiplicities, then

$$1 \leq |N_{k/\mathbb{Q}}(z)| = |v_0(z)| \prod_{\sigma \neq v_0} |\sigma(z)|.$$

For $\sigma \neq v_0$, the triangle inequality and (11.2) give

$$|\sigma(z)| \leq 2|\sigma(a)|C_\sigma.$$

There are only finitely many archimedean embeddings, so the product of these upper bounds is a fixed constant M . It follows that

$$|v_0(s - t)| = |v_0(z)|/|v_0(a)| \geq (|v_0(a)|M)^{-1}.$$

This lower bound is independent of s and t , which is precisely uniform discreteness. $\square$

The hypothesis of Lemma 11.3 controls an entire scalar set simultaneously. For arithmeticity, a weaker matrix-valued condition is enough: one adjoint orbit which spans the Lie algebra may be controlled instead.

Theorem 11.4 (arithmeticity inherited from a normal subgroup). *Let $\Gamma < \text{SL}_2(\mathbb{R})$ be a lattice and let $\Delta \triangleleft \Gamma$ be non-elementary. Suppose that, under a fixed splitting at its distinguished real place, Δ is contained in the norm-one group of an order in an admissible quaternion algebra B/k . Then Γ is arithmetic.*

It is enough, in particular, to find one noncentral $g \in \Gamma$ whose entire Γ -conjugacy orbit is contained in one such order.

Proof. The non-elementary group Δ is absolutely irreducible. Hence, by Burnside's theorem,

$$k[\Delta] \otimes_{k, v_0} \mathbb{R} = M_2(\mathbb{R}),$$

and therefore $k[\Delta] = B$. The algebra

$$\mathcal{O}_\Delta = \mathcal{O}_k[\Delta]$$

is an $\mathcal{O}_k$ -submodule of the given order. Since $\mathcal{O}_k$ is Noetherian, it is a finite $\mathcal{O}_k$ -module; because it spans B , it is itself an order. Normality says that conjugation by every $\gamma \in \Gamma$ preserves Δ , and therefore preserves both $B = k[\Delta]$ and $\mathcal{O}_\Delta$. By Skolem–Noether [31, Section 12.6], this conjugation on B is induced by an element $b \in B^\times$. After passage to the distinguished split factor, $b^{-1}\gamma$ centralizes $B \otimes_{k, v_0} \mathbb{R} = M_2(\mathbb{R})$, hence is scalar, and $b\mathcal{O}_\Delta b^{-1} = \mathcal{O}_\Delta$. Thus the projective image of Γ lies in the orientation-preserving projective normalizer of $\mathcal{O}_\Delta$.

For an admissible quaternion algebra, the projective order normalizer is an arithmetic lattice and is commensurable with the norm-one group of the order [26, Chapter 8]. It contains the projective image of Γ . Since an inclusion between two lattices has finite index, Γ is arithmetic.

For the last assertion, take Δ to be the normal closure of g . Every conjugate of g has reduced norm one and lies in the given order. Its inverse also lies in the order (for a norm-one quaternion, $u^{-1} = \text{Trd}(u) - u$), and the order is closed under multiplication. Consequently the entire normal closure is contained in the norm-one group of that order. It remains only to check that this normal closure is non-elementary. Every elementary normal subgroup of a non-elementary Fuchsian group has trivial projective image, equivalently it is contained in the center of $\text{SL}_2(\mathbb{R})$. Since g is noncentral, Δ is non-elementary. $\square$

The preceding criterion assumes that the number-field algebra is already known. The next two results construct such an algebra, and then an order in it, from boundedness at divisorial valuations. They apply over arbitrary finitely generated coefficient fields and treat rational descent and integrality in a single argument.

We first record the codimension-one patching fact used in both results.

Lemma 11.5 (patching lattices on a normal scheme). *Let X be a normal noetherian integral scheme with function field F , and let V be a finite-dimensional F -vector space. For every prime divisor $D \in X^{(1)}$, prescribe an $\mathcal{O}_{X, D}$ -lattice $L_D \subset V$. Suppose that, on each affine open subset of X , these lattices agree with the localizations of one fixed free lattice except at finitely many prime divisors. Then*

$$\mathcal{E}(U) = \bigcap_{D \in U^{(1)}} L_D \subset V$$

defines a coherent reflexive sheaf whose stalk at every prime divisor D is exactly L_D .

Proof. The assertion is local, so write $X = \text{Spec } A$, identify the fixed lattice with A^d , and let S be the finite set of exceptional height-one primes. Choose $0 \neq f \in A$ so that, for every $\mathfrak{p} \in S$,

$$fA_{\mathfrak{p}}^d \subset L_{\mathfrak{p}} \subset f^{-1}A_{\mathfrak{p}}^d.$$

For nonexceptional height-one primes put $L_{\mathfrak{p}} = A_{\mathfrak{p}}^d$, and set

$$M = \bigcap_{\text{ht } \mathfrak{p}=1} L_{\mathfrak{p}} \subset F^d.$$

Then $fA^d \subset M \subset f^{-1}A^d$. Hence M is a finite A -module. We verify that localization recovers the prescribed lattice. If $x \in L_{\mathfrak{p}}$, only finitely many height-one valuations prevent x from lying in the other prescribed lattices. By prime avoidance there is an element $s \in A \setminus \mathfrak{p}$ with sufficiently large valuation at each of them. Then $sx \in M$, so $M_{\mathfrak{p}} = L_{\mathfrak{p}}$.

A finite torsion-free module over a normal domain is reflexive precisely when it is the intersection, inside its generic fibre, of its localizations at the height-one primes. The displayed definition of M ,

together with the equality of its height-one localizations just proved, therefore gives $M = M^{**}$. These modules agree on overlaps because all were defined by the same valuation conditions, and hence glue to the asserted coherent reflexive sheaf. $\square$

Theorem 11.6 (divisorial descent to the constant field). *Let K be a field of characteristic zero, let F/K be a finitely generated extension in which K is relatively algebraically closed, and let X/K be a normal projective integral variety with function field F . Let V be a d -dimensional F -vector space, let Γ be finitely generated, and let $\rho: \Gamma \rightarrow \mathrm{SL}_F(V)$ be absolutely irreducible. Suppose that, for every prime divisor $D \in X^{(1)}$, there is an $\mathcal{O}_{X,D}$ -lattice $L_D \subset V$ satisfying*

$$\rho(\gamma)L_D = L_D \quad (\gamma \in \Gamma).$$

Then there are a central simple K -algebra B of degree d and an F -algebra isomorphism

$$B \otimes_K F \simeq \mathrm{End}_F(V)$$

under which

$$\rho(\Gamma) \subset B^1 := \{b \in B^\times : \mathrm{Nrd}_{B/K}(b) = 1\}.$$

Here K is the *constant field* in F : relative algebraic closedness means that every element of F algebraic over K already lies in K . A lattice over the discrete valuation ring $\mathcal{O}_{X,D}$ is a free $\mathcal{O}_{X,D}$ -submodule of rank d that spans V over F .

Proof. Fix an F -basis of V . Away from the finitely many divisors that support a pole of a matrix entry of a chosen finite generating set or of its inverse, the standard lattice is invariant. At each exceptional divisor, choose one of the invariant lattices in the hypothesis. The resulting codimension-one lattices differ from one fixed rational lattice at only finitely many divisors. Lemma 11.5 patches them to the coherent reflexive sheaf

$$\mathcal{E}(U) = \{s \in V : s \in L_D \text{ for every } D \in U^{(1)}\};$$

whose stalk at D is the prescribed lattice L_D . Each $\rho(\gamma)$ preserves every L_D , and the codimension-one criterion for reflexive sheaves therefore gives

$$\rho(\Gamma) \subset \mathrm{Aut}_X(\mathcal{E}).$$

Because X is proper over K ,

$$M := H^0(X, \mathcal{E}nd(\mathcal{E}))$$

is a finite-dimensional K -algebra. Put $B = K[\rho(\Gamma)] \subset M$. Absolute irreducibility and Burnside's theorem say that the natural map

$$B \otimes_K F \longrightarrow \mathrm{End}_F(V)$$

is surjective. We include the descent argument that makes this map an isomorphism. The image of $\mathrm{rad}(B) \otimes_K F$ is a nilpotent two-sided ideal of the full matrix algebra, hence is zero. Since B itself embeds in $\mathrm{End}_F(V)$, this gives $\mathrm{rad}(B) = 0$. A central idempotent of B becomes a central idempotent of the full matrix algebra and is therefore zero or one; consequently B has a single simple factor. Its center is a finite field extension of K whose image in $\mathrm{End}_F(V)$ consists of scalar matrices, so it embeds in F . Relative algebraic closedness forces $Z(B) = K$. Thus B is central simple. Then $B \otimes_K F$ is simple, and its displayed nonzero surjection is an isomorphism. In particular $\deg B = d$. Reduced norm becomes determinant after extension to F , which proves $\rho(\Gamma) \subset B^1$. $\square$

Theorem 11.7 (integral descent on a proper model). *Let K be a number field, let F/K be finitely generated with K relatively algebraically closed in F , and let $\mathcal{X}$ be a normal integral scheme, projective and flat over $\mathrm{Spec} \mathcal{O}_K$, with function field F . Let V , Γ , and ρ be as in Theorem 11.6. If*

$\rho(\Gamma)$ preserves an $\mathcal{O}_{\mathcal{X},D}$ -lattice for every prime divisor $D \in \mathcal{X}^{(1)}$, then there are a degree- d central simple K -algebra B , an $\mathcal{O}_K$ -order $\mathcal{O} \subset B$, and an isomorphism

$$B \otimes_K F \simeq \text{End}_F(V)$$

for which $\rho(\Gamma) \subset \mathcal{O}^1$.

Proof. Choose the standard lattice away from the finitely many pole divisors and patch the chosen invariant lattices to a reflexive sheaf $\mathcal{E}$ on $\mathcal{X}$, exactly as above. Properness now makes

$$H^0(\mathcal{X}, \text{End}(\mathcal{E}))$$

a finite $\mathcal{O}_K$ -module. Applying Theorem 11.6 on the generic fibre shows that $B := K[\rho(\Gamma)]$ is central simple of degree d and splits over F . The algebra

$$\mathcal{O} := \mathcal{O}_K[\rho(\Gamma)]$$

is a submodule of that finite module of global endomorphisms, hence is finite over the noetherian ring $\mathcal{O}_K$. It is torsion-free, contains one, is closed under multiplication, and spans B ; it is therefore an order. Determinant one again becomes reduced norm one, so $\rho(\Gamma) \subset \mathcal{O}^1$. $\square$

For a finitely generated field $F/\mathbb{Q}$, the relative algebraic closure of $\mathbb{Q}$ in F is a number field. Thus Theorem 11.7 does not presuppose a number-field form. Boundedness at horizontal divisors descends the generic representation to its constant field, while boundedness at vertical divisors produces the order.

Corollary 11.8 (Fuchsian divisorial descent). *In Theorem 11.7, let $d = 2$, and suppose that one real realization of the projective image of Γ is a cocompact Fuchsian lattice. Assume moreover that $K = \mathbb{Q}(\text{Trd } \Gamma)$. For each embedding $\sigma: K \hookrightarrow \mathbb{C}$, form the realization of $\Gamma \subset B^1$ in $(B \otimes_{K,\sigma} \mathbb{C})^1$. Assume that its image is relatively compact whenever σ is not the distinguished real embedding. Then K is totally real, B splits at the distinguished real place and ramifies at every other real place, and Γ is arithmetic.*

Proof. The distinguished realization identifies B with $M_2(\mathbb{R})$. A compact subgroup of $\text{SL}_2(\mathbb{C})$ is conjugate into $\text{SU}(2)$, so all its traces are real. Since the reduced traces of Γ generate K , every embedding of K has real image; hence K is totally real. At another real place, if B split, the compact conjugate of Γ would lie in a compact subgroup of $\text{SL}_2(\mathbb{R})$. Its real Zariski closure would then be proper, contrary to Zariski density, which follows by transport from the distinguished Fuchsian realization. Hence B ramifies there. The norm-one group of the order constructed in the theorem is consequently an arithmetic Fuchsian lattice containing Γ . An inclusion of lattices has finite index, so Γ is arithmetic. $\square$

We write $\mathbb{A}_{k,f}$ for the finite adele ring of k , namely the restricted product of the completions k_v over the nonarchimedean places with respect to their integer rings $\mathcal{O}_{k_v}$.

Theorem 11.9 (arithmeticity from one bounded adjoint orbit). *Let k be a number field, let G/k be an absolutely simple adjoint group, and let v_0 be an archimedean place. Assume that $G(k_v)$ is compact at every archimedean $v \neq v_0$. Let $\Gamma < G(k)$ be Zariski dense and suppose that its image in $G(k_{v_0})$ is a lattice. Choose $0 \neq X \in \mathfrak{g}(k)$.*

Suppose that the orbit

$$\mathcal{O}_X = \{\text{Ad}(\gamma)X : \gamma \in \Gamma\} \tag{11.3}$$

is bounded in $\mathfrak{g}(k_v)$ at every finite place v , and lies in a fixed integral lattice for all but finitely many v . Then Γ is arithmetic. Equivalently, it is enough that $\mathcal{O}_X$ have compact closure in the finite-adelic Lie algebra $\mathfrak{g}(\mathbb{A}_{k,f})$.

Proof. The k -linear span of $\mathcal{O}_X$ is invariant under Γ , hence under its Zariski closure. It is therefore a nonzero ideal in the simple Lie algebra $\mathfrak{g}$, and so equals $\mathfrak{g}$. Choose $\gamma_1, \dots, \gamma_N \in \Gamma$ such that $X_i = \text{Ad}(\gamma_i)X$ is a k -basis.

Relative to this basis, the i -th column of $\text{Ad}(\gamma)$ is the coordinate vector of

$$\text{Ad}(\gamma)X_i = \text{Ad}(\gamma\gamma_i)X,$$

which belongs to the same orbit. Thus $\text{Ad}(\Gamma)$ is bounded at every finite place and integral outside a fixed finite set. The same statement for γ^{-1} shows that, relative to the basis X_i , both $\text{Ad}(\gamma)$ and its inverse are bounded at every finite place and preserve one fixed global integral lattice outside a finite set S . For $v \notin S$, $\text{Ad}(\Gamma)$ therefore lies in the compact-open stabilizer of that lattice. For $v \in S$, its compact closure preserves some $\mathcal{O}_{k_v}$ -lattice and is contained in a compact-open subgroup $K_v < G(k_v)$. Enlarging S to accommodate an integral model of G , the product of these stabilizers is a compact-open subgroup $K_f < G(\mathbb{A}_{k,f})$ containing Γ . Here adjointness matters: it makes Ad a faithful closed immersion.

At every nondistinguished archimedean place the entire local group is compact. The diagonal intersection

$$G(k) \cap \left(G(k_{v_0}) \times \prod_{v|\infty, v \neq v_0} G(k_v) \times K_f \right)$$

is an arithmetic lattice in the full archimedean product, by the Borel–Harish-Chandra theorem [6]. Since all factors other than v_0 are compact, its projection is a lattice $\Lambda < G(k_{v_0})$. It contains Γ . An inclusion of one lattice in another has finite index, so $[\Lambda : \Gamma] < \infty$. $\square$

In type A_1 the orbit criterion can be checked using only three families of actual word traces, once the rational form is fixed. More precisely, assume that the projective image of Γ lies in a fixed admissible adjoint k -form $G(k)$. Let $g \in \Gamma < \text{SL}_2(\mathbb{R})$ be noncentral, put $t = \text{tr } g$, and set

$$X = g - \frac{t}{2}I.$$

Assume $X \in \mathfrak{g}(k)$, and choose $h_1, h_2, h_3 \in \Gamma$ so that $X_i = h_i X h_i^{-1} \in \mathfrak{g}(k)$ is a basis of $\mathfrak{g}$; Zariski density guarantees such a choice. The trace pairing is nondegenerate, and

$$\text{tr}((\gamma X \gamma^{-1})X_i) = \text{tr}(\gamma g \gamma^{-1} h_i g h_i^{-1}) - \frac{t^2}{2}. \quad (11.4)$$

The three explicitly displayed traces on the right are therefore k -linear coordinates of the orbit (11.3). If they are contained in fixed fractional ideals at the finite places and are uniformly bounded at every nondistinguished archimedean place, then Theorem 11.9 proves arithmeticity. The assumption that the ambient admissible k -form is already fixed is essential: membership of three scalar families in a number field does not by itself construct that rational form. Subject to this necessary hypothesis, a finite collection of trace coordinates which faithfully records the matrix orbit suffices; no simultaneous bound for every scalar trace is logically necessary.

There is a still weaker criterion which uses only one exact trace fiber. It is stated using the coordinate map $z(A) = (\text{Trd}(AB_1), \text{Trd}(AB_2), \text{Trd}(AB_3))$.

Theorem 11.10 (arithmeticity from coordinate sums in one trace fiber). *Let B/k be an admissible quaternion algebra with distinguished split real place v_0 , and let $\Lambda < \text{SL}_1(B)(k)$ have Zariski-dense projective image which is a lattice in $\text{PGL}_1(B)(k_{v_0})$. Choose a noncentral $g \in \Lambda$, put*

$$B^0 = \{Y \in B : \text{Trd}(Y) = 0\}, \quad t = \text{Trd}(g), \quad X = g - \frac{t}{2}1 \in B^0,$$

and choose $h_1, h_2, h_3 \in \Lambda$ so that $X_i = \text{Ad}(h_i)X$ is a k -basis of B^0 . Set

$$B_i = h_i g h_i^{-1}, \quad z(A) = (\text{Trd}(AB_1), \text{Trd}(AB_2), \text{Trd}(AB_3)),$$

and

$$Z_g^{\text{orb}} = \{z(\gamma g \gamma^{-1}) : \gamma \in \Lambda\}.$$

If the following translated set of coordinatewise trace sums

$$\mathcal{W}_g = z(g) + Z_g^{\text{orb}} - (t^2, t^2, t^2)$$

has compact closure in $\mathbb{A}_{k,f}^3$, then Λ is arithmetic. Equivalently, it is enough that the three coordinates of $\mathcal{W}_g$ be contained in fixed fractional ideals of k . This is a criterion inside the specified quaternion algebra B/k ; it does not construct k or B from the real trace set.

Proof. For $C = \gamma g \gamma^{-1}$, define

$$P_\gamma = g + C - t1 = X + \text{Ad}(\gamma)X \in B^0.$$

For context, Cayley–Hamilton gives the following intertwining identities:

$$(A + C - t1)A = C(A + C - t1), \quad (A + C - t1)C = A(A + C - t1)$$

whenever $A, C \in \text{SL}_1(B)$ have reduced trace t . Thus P_γ , when invertible, conjugates g to C projectively. The proof below only needs its linear expression $P_\gamma = X + \text{Ad}(\gamma)X$, so no invertibility assumption is required. Since $B_i = X_i + \frac{t}{2}1$ and $\text{Trd}(P_\gamma) = 0$,

$$\begin{aligned} \text{Trd}(P_\gamma X_i) &= \text{Trd}(P_\gamma B_i) \\ &= \text{Trd}(g B_i) + \text{Trd}(C B_i) - t \text{Trd}(B_i) \\ &= z_i(g) + z_i(C) - t^2. \end{aligned}$$

Consequently $\mathcal{W}_g$ is exactly the image of $\{P_\gamma : \gamma \in \Lambda\}$ under the k -linear isomorphism

$$B^0 \longrightarrow k^3, \quad P \longmapsto (\text{Trd}(P X_1), \text{Trd}(P X_2), \text{Trd}(P X_3)).$$

This is an isomorphism because the reduced-trace pairing on B^0 is nondegenerate and the X_i form a basis. Hence the hypothesis is equivalent to finite-adelic precompactness of the set of matrices $\{P_\gamma\}$. To justify the equivalent formulation in the statement, recall that a diagonally embedded subset of k^3 has compact closure in $\mathbb{A}_{k,f}^3$ exactly when it is bounded at every finite place and integral at all but finitely many places. Clearing the finitely many remaining denominators puts each coordinate in one fractional $\mathcal{O}_k$ -ideal. Translation by the fixed rational vector X preserves precompactness, so

$$\{\text{Ad}(\gamma)X : \gamma \in \Lambda\} = \{P_\gamma - X : \gamma \in \Lambda\}$$

has compact closure in $B^0(\mathbb{A}_{k,f})$. Apply Theorem 11.9 to $G = \text{PGL}_1(B)$. Admissibility gives the compact nondistinguished archimedean factors, and the theorem gives arithmeticity. $\square$

The full fiber of the coordinate map z , $Z_t = \{z(A) : A \in \Lambda, \text{Trd}(A) = t\}$ contains Z_g^{orb} . Finite-place boundedness of $z(g) + Z_t - t^2(1, 1, 1)$ is therefore sufficient, but unnecessarily strong: the theorem needs only the subset arising from the conjugacy orbit of g . This makes the remaining extraction problem precise. It is enough to bound, at every finite place, the matrices $g + \gamma g \gamma^{-1} - t1$; one need not first construct an integral model for the entire scalar trace set.

For the high-multiplicity application even the basepoint need not be fixed. The following asymptotic version is strictly weaker than a bounded full orbit.

Theorem 11.11 (arithmeticity from nonexpanding adjoint directions). *Let G/k be absolutely simple and adjoint, with $G(k_v)$ compact at every archimedean place $v \neq v_0$. Let $\Gamma < G(k)$ be finitely generated and Zariski dense, and suppose that its image at v_0 is a lattice. Fix a word metric and a finite set S of finite places such that $\Gamma \subset \mathcal{G}(\mathcal{O}_{k,S})$ for an integral model $\mathcal{G}$. Suppose that for every n there is $0 \neq X_n \in \mathfrak{g}(k)$ such that, for every $v \in S$,*

$$\max_{|\gamma| \leq n} \|\text{Ad}(\gamma)X_n\|_v \leq C_v \|X_n\|_v,$$

where C_v is independent of n . Then Γ is arithmetic. As in Theorem 11.9, the hypotheses already include a fixed k -form of G ; the conclusion is not a method for recovering that form from a representation over $\mathbb{R}$.

Proof. Fix $v \in S$. We regard the same rational vector X_n in $\mathfrak{g}(k_v)$, but we are free to multiply it by a scalar in $k_v^\times$, since the displayed ratio of norms is homogeneous in X_n . Normalize it to lie in a fixed compact annulus which is bounded away from zero. Such an annulus is compact over a nonarchimedean local field. Since S is finite, successive extraction gives one subsequence which converges at every $v \in S$, say to $0 \neq X_v \in \mathfrak{g}(k_v)$. Fix $\gamma \in \Gamma$. For every sufficiently large index in this subsequence one has $|\gamma| \leq n$; continuity of $\text{Ad}(\gamma)$ and the assumed inequality then give

$$\|\text{Ad}(\gamma)X_v\|_v \leq C_v \|X_v\|_v.$$

The constant is independent of γ , so the full orbit $\text{Ad}(\Gamma)X_v$ is bounded.

Its k_v -span is a nonzero $\text{Ad}(\Gamma)$ -invariant subspace. Zariski density makes it invariant under G_{k_v} , and absolute simplicity makes the adjoint representation irreducible; the span is therefore all of $\mathfrak{g}(k_v)$. Choose $\delta_1, \dots, \delta_N \in \Gamma$ such that $\text{Ad}(\delta_i)X_v$ is a basis. The i -th column of $\text{Ad}(\gamma)$ in this basis is the coordinate vector of $\text{Ad}(\gamma\delta_i)X_v$, which belongs to the bounded orbit. Thus all matrix entries of $\text{Ad}(\Gamma)$ are bounded. The same statement applies to inverses because Γ is a group. Hence $\text{Ad}(\Gamma)$ has compact closure in $\text{GL}(\mathfrak{g}(k_v))$. Since G is adjoint, $\text{Ad} : G \rightarrow \text{GL}(\mathfrak{g})$ is a faithful closed immersion, and Γ is relatively compact in $G(k_v)$.

This holds for each $v \in S$. Outside S , the assumed integral model places Γ in the compact-open subgroup $\mathcal{G}(\mathcal{O}_{k_v})$. At each $v \in S$, the compact closure of Γ is contained in some compact-open subgroup of $G(k_v)$. Taking their restricted product gives a compact-open $K_f < G(\mathbb{A}_{k,f})$ containing Γ . The Borel–Harish-Chandra argument in Theorem 11.9 now proves arithmeticity. $\square$

In type A_1 , take varying elements g_n , put $t_n = \text{Trd}(g_n)$, $X_n = g_n - \frac{t_n}{2}1$, and define

$$P_{n,\gamma} = g_n + \gamma g_n \gamma^{-1} - t_n 1 = X_n + \text{Ad}(\gamma)X_n.$$

The hypothesis of Theorem 11.11 is equivalent, up to fixed constants, to uniform bounds $\|P_{n,\gamma}\|_v \ll_v \|X_n\|_v$ on the growing conjugacy balls. Choose three fixed traceless matrices B_i whose trace-pairing functionals form a basis of the dual space. The preceding matrix bounds are then equivalent to bounds for the three scalar quantities

$$\text{Trd}(g_n B_i) + \text{Trd}(\gamma g_n \gamma^{-1} B_i) - t_n \text{Trd}(B_i).$$

Thus the trace fiber of high multiplicity may vary with the scale. What is needed is a nonzero direction in each such fiber whose adjoint orbit, up to the corresponding word radius, remains bounded by a fixed multiple of the size of that direction at every relevant finite place.

This condition has an exact quantitative converse.

Lemma 11.12 (uniform expansion over archimedean and divisorial fields). *Let K be $\mathbb{R}$, $\mathbb{C}$, or a complete discretely valued field, and let $\rho : \Gamma \rightarrow \text{GL}(V)$ be an irreducible finite-dimensional representation of a finitely generated group. Fix a norm on V ; in the discretely valued case take the norm defined by a lattice. Put*

$$m_n = \inf_{0 \neq x \in V} \max_{|\gamma| \leq n} \frac{\|\rho(\gamma)x\|}{\|x\|}.$$

Then $\rho(\Gamma)$ is bounded if and only if $\sup_n m_n < \infty$. If it is unbounded, there are $c > 0$ and $\lambda > 1$ such that $m_n \geq c\lambda^n$ for every n . Over $\mathbb{R}$, $\mathbb{C}$, or a nonarchimedean local field, boundedness is equivalent to relative compactness. Over an arbitrary complete discretely valued field, boundedness is equivalent to preservation of a lattice.

Proof. The word balls give the supermultiplicativity $m_{n+r} \geq m_n m_r$. Indeed, for a fixed $x \neq 0$, choose a word δ of length at most r which realizes the maximum over the r -ball. Applied to $\rho(\delta)x$, the definition of m_n provides a word γ of length at most n with

$$\|\rho(\gamma\delta)x\| \geq m_n \|\rho(\delta)x\| \geq m_n m_r \|x\|.$$

Taking first the maximum and then the infimum proves the inequality.

Suppose first that $K = \mathbb{R}$ or $\mathbb{C}$ and that $\sup_n m_n = M < \infty$. Choose unit vectors x_n for which the maximum over the n -ball is at most $M + 1/n$. Compactness of the unit sphere gives a subsequence converging to a unit vector x . For fixed $\gamma \in \Gamma$, passage to the limit after $n \geq |\gamma|$ gives $\|\rho(\gamma)x\| \leq M$. Thus the full orbit of x is bounded.

Now let K be complete discretely valued, with valuation ring R and uniformizer π , and identify the chosen lattice with R^d . After normalizing x to be primitive, every ratio occurring in the finite maximum defining m_n lies in the discrete value group. The identity is in every word ball, so these ratios are at least one. Consequently the infimum is attained by a primitive vector. Choose an integer a with $M \leq |\pi|^{-a}$, and define the descending R -submodules

$$L_n = \{x \in R^d : \rho(\gamma)x \in \pi^{-a}R^d \text{ for every } |\gamma| \leq n\}.$$

Every L_n contains a primitive vector. For $r \geq 1$, let

$$I_{n,r} = (L_n + \pi^r R^d) / \pi^r R^d.$$

For fixed r , this descending chain stabilizes because $(R/\pi^r R)^d$ has finite module length; finiteness of the residue field is not needed. Write its stable value as N_r . We have $N_1 \neq 0$, and the reduction maps $N_{r+1} \rightarrow N_r$ are surjective: choose n beyond the two stabilization indices, and use the surjection from the image of L_n modulo π^{r+1} onto its image modulo π^r . Starting with a nonzero element of N_1 , choose compatible lifts through all N_r . Completeness identifies their inverse limit with a primitive vector $x \in R^d$. For fixed n , the vector x belongs to $L_n + \pi^r R^d$ for every r . The module L_n , being a finite intersection of inverse images of lattices, is closed, so $x \in L_n$. Thus x again has a bounded full orbit.

In either case, irreducibility gives orbit vectors $\rho(\delta_1)x, \dots, \rho(\delta_{\dim V})x$ which form a basis. In that basis every column of $\rho(\gamma)$ is the coordinate vector of another point of the bounded orbit. Hence $\rho(\Gamma)$ is bounded; the same assertion for inverses is automatic because Γ is a group. Conversely, a bounded group plainly makes the m_n uniformly bounded. Over a complete discrete valuation ring, a bounded group preserves the lattice $\sum_{\gamma \in \Gamma} \rho(\gamma)R^d$; boundedness sandwiches this module between two homothetic lattices, and noetherianity makes it a lattice. The converse is immediate. Over a locally compact field, bounded matrices together with bounded inverses have compact closure.

If $\rho(\Gamma)$ is unbounded, the preceding equivalence says that m_n is unbounded, so $m_N > 1$ for some N . Write $n = qN + r$, with $0 \leq r < N$. Since the identity belongs to every word ball, $m_r \geq 1$, and supermultiplicativity gives

$$m_n \geq m_N^q \geq m_N^{-1} (m_N^{1/N})^n.$$

Taking $c = m_N^{-1}$ and $\lambda = m_N^{1/N} > 1$ proves the final claim. $\square$

Corollary 11.13 (descent from uniformly nonexpanding vectors in word balls). *Use the notation of Theorem 11.7. For a prime divisor $D \in \mathcal{X}^{(1)}$, let F_D be the completion of F at its discrete valuation. Suppose that, for every D and every $n \geq 1$, there is a nonzero $x_{D,n} \in V \otimes_F F_D$ such that*

$$\max_{|\gamma| \leq n} \|\rho(\gamma)x_{D,n}\|_D \leq C_D \|x_{D,n}\|_D, \quad (11.5)$$

where C_D is independent of n . Then $\rho(\Gamma)$ descends to the norm-one group of an order in a degree- d central simple algebra over K , as in Theorem 11.7.

If $d = 2$, the distinguished real projective image is a cocompact Fuchsian lattice, the reduced traces generate K , and every nondistinguished archimedean realization is relatively compact, then Γ is arithmetic.

Proof. Global absolute irreducibility remains absolute after extension to F_D . Lemma 11.12 applied over F_D turns (11.5) into boundedness of $\rho(\Gamma)$ at D , hence into an invariant $\mathcal{O}_{X,D}$ -lattice. Theorem 11.7 applies simultaneously to all divisors. The last assertion is Corollary 11.8. $\square$

The same conclusion follows from nonexpansion in the adjoint representation. Indeed, boundedness in $\mathrm{PGL}_d(F_D)$ lifts to boundedness in $\mathrm{SL}_d(F_D)$: in Cartan coordinates the adjoint bound controls all differences of exponents, while determinant one says that their sum is zero.

At any divisorial or finite place where a proposed nonarithmetic group has unbounded adjoint image, Lemma 11.12 says that *every* choice of X_n has some $|\gamma| \leq n$ with

$$\|\mathrm{Ad}(\gamma)X_n\|_v \geq c_v \lambda_v^n \|X_n\|_v.$$

For large n , the factor $c_v \lambda_v^n$ is greater than one. The two summands in $P_{n,\gamma} = X_n + \mathrm{Ad}(\gamma)X_n$ then have different norms, so the ultrametric equality gives $\|P_{n,\gamma}\|_v = \|\mathrm{Ad}(\gamma)X_n\|_v$. Since the three chosen trace-pairing coordinates form a fixed linear coordinate system, at least one of them also grows exponentially, up to constants depending only on that system. This gives a valuative sufficient condition for the compact conjectures that does not assume a number-field form in advance. It is enough to deduce (11.5) for the adjoint representation from sparse real traces, at every horizontal and vertical divisor of one proper model, and then to prove that the nondistinguished archimedean realizations are compact. Corollary 11.13 then implies arithmeticity. If (11.5) fails at a divisor, the failure is quantitative: every nonzero vector is expanded by at least $c_v \lambda_v^n$ somewhere in the word ball of radius n .

The following stronger scalar statement would also settle the conjectures. It combines two logically separate tasks: constructing a common number field from the trace set and proving integrality and boundedness at all of its nondistinguished places.

Conjecture 11.14 (adelic trace decomposition by square classes). Let Γ be a cocompact Fuchsian lattice whose trace set has bounded clustering, or merely linear growth, and put $V = \Gamma/\Gamma^{(2)}$. There exist a number field k with a distinguished real place, Kummer representatives s_v , fractional ideals I_v , and constants $C_{v,\sigma}$ such that every trace in the component indexed by v has the form

$$\mathrm{tr} \tilde{\gamma} = s_v a_\gamma, \quad a_\gamma \in I_v, \quad |\sigma(a_\gamma)| \leq C_{v,\sigma} \quad (11.6)$$

for every nondistinguished archimedean σ . For the identity component one may take $s_0 \in k^\times$.

For the trace gap on $\Gamma^{(2)}$, (11.6) is stronger than logically necessary: it would suffice to place all differences in the neutral component in one fractional ideal with uniformly bounded nondistinguished conjugates. The component-by-component form is stated because it also recovers the canonical decomposition of arithmetic traces by square classes.

Proposition 11.15. *Conjecture 11.14 implies both the compact Sarnak and compact Schmutz conjectures.*

Proof. Apply Lemma 11.3 to the identity component $T(\Gamma^{(2)})$, after absorbing s_0 into its fractional ideal. This gives positive gap. Theorem 9.1 gives arithmeticity. $\square$

The content of (11.6) is precisely what arbitrary graph coloring omits. Any bounded-clustering subset of $\mathbb{R}$ can be finitely colored into separated sets: connect two values at distance less than one and greedily color the resulting bounded-degree graph. Such colors need not factor through $\Gamma/\Gamma^{(2)}$, respect (10.1), lie in one number field, or have bounded conjugates.

11.2. Why algebraicity alone is insufficient. Even a fixed number field and algebraic integrality do not give a gap at the distinguished embedding. The two-color set (10.6) already proves this. A one-point-per-unit-interval variant, with $\alpha = \sqrt{2}$, is the set

$$S = \{n + \{\alpha n^2\} : n \in \mathbb{Z}\} \subset \mathbb{Q}(\sqrt{2}).$$

This set has one point in each unit interval, and therefore has bounded clustering. Weyl equidistribution of $(\{\alpha n^2\}, \{\alpha(n+1)^2\})$ gives adjacent pairs whose differences tend to zero, while the second embedding is unbounded. Variants over $\mathbb{Q}$ with denominators tending through primes show the independent finite-place obstruction. The trace ring can also be misleading: under one real embedding, $\mathcal{O}_k$ is dense whenever $[k : \mathbb{Q}] > 1$. Derived trace sets are separated not because their additive span is discrete, but because their images under every nondistinguished embedding remain uniformly bounded.

12. TRACE MULTIPLICITY, FIXED TRACE COORDINATES, AND ADDITIVE ENERGY

12.1. A finite-type divisor theorem. We first prove a divisor bound over an arbitrary finitely generated field of characteristic zero. This generality is necessary because the matrix entries of a finitely generated group need not be algebraic numbers.

Let $L/\mathbb{Q}$ be finitely generated, and let K be the algebraic closure of $\mathbb{Q}$ inside L ; then K is a number field. Choose a transcendence basis $t = (t_1, \dots, t_r)$, so $L/K(t)$ is finite. After enlarging a finite set S of places of K and inverting one nonzero polynomial $a \in \mathcal{O}_{K,S}[t]$, one may choose a normal domain $R \subset L$, with fraction field L , which is finite free over

$$A = \mathcal{O}_{K,S}[t_1, \dots, t_r, a^{-1}].$$

Here S contains all archimedean places and

$$\mathcal{O}_{K,S} = \{x \in K : v_{\mathfrak{p}}(x) \geq 0 \text{ for every finite place } \mathfrak{p} \notin S\}$$

is the ring of S -integers. Thus its elements may have denominators only at the finitely many finite places belonging to S . Fix an A -basis $e_1, \dots, e_D$ of R . For $C > 0$, let $R_C[n]$ consist of the elements which can be written

$$\sum_{j=1}^D a^{-m} P_j(t) e_j, \quad 0 \leq m \leq Cn, \quad \deg P_j \leq Cn, \quad h_{\text{aff}}(P_j) \leq Cn, \quad (12.1)$$

where $P_j \in \mathcal{O}_{K,S}[t]$ and

$$h_{\text{aff}}(P_j) = h_K([1 : (\text{coefficients of } P_j)]) \leq Cn.$$

Here h_K is the normalized logarithmic projective Weil height: it is the sum, over all places of K , of the logarithm of the largest absolute value of the homogeneous coordinates, with the usual local-degree weights. Thus the height is *affine*, not merely the projective height of the coefficient vector: in particular, the constant polynomial $N \in \mathbb{Z}$ has height $\log N$, rather than zero. This normalization is essential for controlling vertical prime content. Changing the chosen model, basis, or finite generating set only changes C .

Theorem 12.1 (finite-type divisor bound). *For every fixed C ,*

$$\sup_{0 \neq q \in R_C[n]} \#\{(b, c) \in R_C[n]^2 : bc = q\} = \exp(o_C(n)). \quad (12.2)$$

The $o_C(n)$ is uniform in q .

Proof. We first count divisors in the polynomial ring before passing to its finite normal extension. Enlarge S so that $\mathcal{O}_{K,S}$ is a principal ideal domain; this is possible because the ideal class group of K is finite. Thus $\mathcal{O}_{K,S}[t]$ is a unique factorization domain.

We put a positive weight on each of its prime elements. A vertical prime is a prime $\mathfrak{p} \notin S$ of $\mathcal{O}_K$; give it weight $\log N\mathfrak{p}/\log 2$, so every vertical weight is at least one. A horizontal prime is represented, up to multiplication by $K^\times$, by a primitive irreducible polynomial $f \in K[t]$. At a nonarchimedean place v , let $M_v(f)$ be the Gauss norm, the maximum v -absolute value of its coefficients. At an archimedean place put

$$\log M_v(f) = \int_{(S^1)^r} \log |f(z_1, \dots, z_r)|_v d\mu.$$

The global multivariable Mahler height is the normalized sum

$$m_K(f) = \frac{1}{[K : \mathbb{Q}]} \sum_v [K_v : \mathbb{Q}_v] \log M_v(f).$$

The product formula makes m_K unchanged when f is multiplied by a nonzero scalar. Gauss norms are multiplicative, and logarithms turn the archimedean identity $M_v(fg) = M_v(f)M_v(g)$ into addition. Hence $m_K(fg) = m_K(f) + m_K(g)$.

The standard Mahler–coefficient inequalities compare $m_K(f)$ with the projective height of the primitive coefficient vector up to a constant times $\deg f$. Choose a constant C_r , depending only on the number of variables, so that

$$w(f) = C_r \deg f + m_K(f)$$

is at least one for every nonconstant irreducible f , and controls both $\deg f$ and its projective coefficient height. This weight is additive under multiplication. Moreover, only finitely many horizontal primes have $w(f) \leq T$: their degrees and coefficient heights are bounded, so this is Northcott’s finiteness theorem [4, Theorem 1.6.8]. A vertical prime is indexed by a prime ideal of the fixed ring of integers, so only finitely many have norm, and hence weight, at most a given bound.

We use the following elementary weighted divisor observation. Suppose prime weights satisfy $w(p) \geq 1$, have finite sublevel sets, and

$$\sum_p e_p w(p) \leq C_0 n.$$

Then

$$\log \prod_p (e_p + 1) = o(n). \tag{12.3}$$

Indeed, fix T . The finitely many primes of weight at most T contribute $O_T(\log n) = o(n)$. For the remaining primes, $\log(e + 1) \leq e \log 2$, and hence their contribution is at most $C_0 n \log 2/T$. First let $n \rightarrow \infty$, and then let $T \rightarrow \infty$.

If $Q \in \mathcal{O}_{K,S}[t]$ has total degree and affine coefficient height $O(n)$, the sum of the weights of its horizontal prime factors, with multiplicity, is $O(n)$ by additivity and the coefficient comparison. The weighted multiplicity of its vertical content is also $O(n)$: after choosing a primitive polynomial, the norm of the removed coefficient content is bounded by the affine height. Equation (12.3) therefore says that Q has only $\exp(o(n))$ divisors. Inverting the fixed polynomial a does not change this estimate: its finitely many prime factors become units, and clearing a power $a^{O(n)}$ adds only polynomially many choices.

Now take $0 \neq q \in R_C[n]$. Multiplication by q on the fixed free A -module R has a $D \times D$ matrix whose entries, after clearing $a^{O(n)}$, have degree and coefficient height $O(n)$. Its determinant is the field norm

$$Q = N_{L/K(t)}(q).$$

It has the same complexity bounds. Let P be a height-one prime of the normal domain R , away from the already inverted boundary, and put $p = P \cap A$. The norm valuation formula is

$$v_p(Q) = \sum_{P|p} f(P/p) v_P(q), \quad (12.4)$$

where the residue degrees $f(P/p)$ are positive and the number of primes above p is at most D .

If $bc = q$ in R , normality makes every height-one localization R_P a discrete valuation ring. Since $b, c \in R$, it follows that

$$0 \leq v_P(b) \leq v_P(q)$$

at every such P . Formula (12.4) shows that the possible collections of all valuations $v_P(b)$ are bounded by a fixed power, depending only on D , of the number of divisors of Q . There are therefore $\exp(o(n))$ possible divisor patterns.

It remains to count elements having one fixed pattern. If b, b' have the same valuations at every height-one prime, then $u = b'/b$ and u^{-1} belong to every localization R_P . Normality gives the Krull intersection identity

$$R = \bigcap_{\text{ht } P=1} R_P,$$

so $u \in R^\times$. Take a fixed normal projective compactification of the generic fiber $\text{Spec}(R \otimes_{\mathcal{O}_{K,S}} K)$. Orders along its finitely many boundary divisors map $R^\times$, modulo constant units, into a fixed lattice $\mathbb{Z}^s$; this is the Rosenlicht–Samuel unit theorem [35, 40]. The fixed filtration bounds the *absolute* valuation of b and b' by $O(n)$ along every boundary divisor. Indeed, their displayed coordinate expressions give the lower bound. Applying the field norm gives a rational function of degree and affine coefficient height $O(n)$, hence of boundary valuation $O(n)$; the norm-valuation formula and the lower bounds at all other divisors above the same base divisor give the corresponding upper bound. Thus every boundary order of $u = b'/b$ is $O(n)$, and there are only $n^{O(1)}$ possible order vectors. Choose once and for all generators for their image lattice. A representative u_0 made from these generators with exponents $O(n)$ has logarithmic height $O(n)$. In $b' = ku_0b$, clear the common denominator $a^{O(n)}$ and compare one nonzero coefficient in the fixed basis; the affine coefficient bounds give $h(k) = O(n)$. The residual constant k is an S' -unit of K , for one fixed finite S' . Dirichlet's S' -unit theorem [30, Chapter I, Sections 7 and 11] gives only $n^{O(1)}$ such constants. Thus each divisor pattern contributes only polynomially many factors, which proves (12.2).

For completeness, the model used in the statement always exists. Noether normalization makes L finite over $K(t)$. Start with the integral closure of $\mathcal{O}_{K,S}[t]$ in L ; after initially enlarging S , this normalization is finite. Invert the finitely many denominators needed to contain all prescribed entries and their inverses. Generic freeness then provides one further polynomial localization after which the resulting normal algebra is finite free over A . After one final finite enlargement of S , the ring $\mathcal{O}_{K,S}$ is a principal ideal domain, as required above. $\square$

Theorem 12.2 (subexponential fibers for two trace coordinates). *Let $F/\mathbb{Q}$ be finitely generated, let $\Delta < \text{SL}_2(F)$ be finitely generated, and let $B \in \Delta$ be regular semisimple. For any word metric,*

$$\sup_{x,z} \#\{A \in \Delta : |A| \leq n, \text{tr } A = x, \text{tr}(AB) = z, \langle A, B \rangle \text{ absolutely irreducible}\} = \exp(o(n)). \quad (12.5)$$

Proof. Pass to one finite extension of F which splits B , and conjugate

$$B = \text{diag}(\lambda, \lambda^{-1}), \quad \lambda \neq \lambda^{-1}.$$

Choose the normal arithmetic model R above to contain the entries of the conjugated finite generating set and of its inverses. Repeated multiplication by these fixed matrices shows that every entry of a word of length at most n belongs to $R_C[n]$ for one fixed C : total degrees, denominator orders, and logarithmic coefficient heights all grow at most linearly with the number of factors.

Write $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The two specified values $\text{tr } A = x$ and $\text{tr}(AB) = z$ determine the diagonal entries:

$$a = \frac{z - \lambda^{-1}x}{\lambda - \lambda^{-1}}, \quad d = \frac{\lambda x - z}{\lambda - \lambda^{-1}}. \quad (12.6)$$

The determinant equation then fixes

$$bc = q := ad - 1. \quad (12.7)$$

The pair $\langle A, B \rangle$ is absolutely reducible exactly when it has a common B -eigenline, which in these coordinates is equivalent to $bc = 0$. Thus absolute irreducibility is precisely $q \neq 0$.

If a fiber is nonempty, choose one matrix in it. Its entries show that the fixed a, d, q have complexity $O(n)$. Theorem 12.1 gives only $\exp(o(n))$ pairs (b, c) of the required complexity, after enlarging the fixed filtration constant C , satisfying (12.7). Once a, b, c, d are fixed, so is A . The estimate is uniform in x, z . $\square$

The same divisor argument applies uniformly to a family of quadratic norm equations. We can therefore study addition inside an exact trace fiber without separately diagonalizing over each varying quadratic field.

Lemma 12.3 (a uniform divisor bound for quadratic norm equations). *Use the arithmetic model and filtration $R_C[n]$ of (12.1). For every fixed $C > 0$, uniformly in $d, N \in R_C[n]$ with $dN \neq 0$,*

$$\#\{(y, v) \in R_C[n]^2 : y^2 - dv^2 = N\} = \exp(o_C(n)).$$

The conclusion remains true if the four filtration constants are different but fixed.

Proof. Put $E_d = \text{Frac}(R)[z]/(z^2 - d)$, a quadratic étale algebra which may be a product, and let R_d be the normalization of R in E_d . A solution determines $\alpha = y + vz \in R_d$ with

$$N_{E_d/\text{Frac}(R)}(\alpha) = N.$$

The map $(y, v) \mapsto \alpha$ is injective because $1, z$ is a basis of the rank-two algebra.

The divisor-pattern argument in Theorem 12.1 is uniform in this varying quadratic algebra. Indeed, R_d has degree at most $2D$ over A . Applying the norm-valuation formula down to A bounds the possible height-one valuations of α , by a fixed power depending only on $2D$, by the divisor count of

$$N_{\text{Frac}(R)/K(t)}(N).$$

The latter has degree and affine coefficient height $O_C(n)$. Thus there are $\exp(o_C(n))$ possible divisor patterns, uniformly in d, N .

It remains to count associates uniformly as d varies. Let $X = \text{Spec}(R \otimes_{\mathcal{O}_{K,S}} K)$, fix a normal projective compactification $\bar{X}$, and write its boundary as $D_1 \cup \dots \cup D_s$. Let $\bar{Y}_d$ be the normalization of $\bar{X}$ in E_d . Above each D_i there are at most two prime divisors. Their orders embed $R_d^\times$, modulo its constant units, in $\mathbb{Z}^{2s}$. For quotients of two admissible α 's every such order is $O_C(n)$. Indeed,

$$\alpha + \bar{\alpha} = 2y, \quad \alpha\bar{\alpha} = N.$$

At a boundary valuation the orders of y and N are $O_C(n)$. If the two root valuations differ, their minimum is the valuation of $2y$; if they agree, each is half the valuation of N . Thus both are $O_C(n)$, giving only $n^{O(1)}$ boundary vectors.

For one boundary vector, the quotient of two units is a constant unit in the constant étale algebra of E_d , of degree at most $2[K : \mathbb{Q}]$ over $\mathbb{Q}$. Its height is $O_C(n)$: use $\alpha^{-1} = \bar{\alpha}/N$ and express its trace and norm as rational functions of degree and affine coefficient height $O_C(n)$. These constants are S' -units for one fixed finite set of rational places. In the full archimedean-plus- S' logarithmic embedding their rank is bounded in the fixed degree. Northcott's theorem [4, Theorem 1.6.8] gives a uniform positive lower height for a non-torsion algebraic number of that degree, and Minkowski's second theorem then shows that an $O(n)$ box contains only $n^{O(1)}$ such units. The split case is

identical componentwise. There are $\exp(o(n))$ possible divisor patterns, so summing over them proves the lemma. $\square$

Theorem 12.4 (additive representations inside a fixed trace fiber). *Let $F/\mathbb{Q}$ be finitely generated and let $\Delta < \mathrm{SL}_2(F)$ be finitely generated. Assume that the identity is the only element of Δ having trace 2; this holds, in particular, for the full lift of a torsion-free cocompact Fuchsian lattice. For a word metric put*

$$\mathcal{F}_t(n) = \{A \in \Delta : |A| \leq n, \operatorname{tr} A = t\}.$$

Uniformly in $t \in F$ and $S \in M_2(F)$ with $S \neq tI$,

$$\#\{(A, C) \in \mathcal{F}_t(n)^2 : A + C = S\} = \exp(o(n)).$$

For the excluded sum $S = tI$, the pairs are exactly (A, A^{-1}) , so the fiber has size $|\mathcal{F}_t(n)|$.

Proof. For a nonempty fiber choose one pair. In one fixed normal arithmetic model, all entries of A, C , and hence t, S , have complexity $O(n)$. Put

$$P = S - tI = A + C - tI = \begin{pmatrix} r & s \\ u & -r \end{pmatrix}, \quad d = r^2 + su = -\det P,$$

and write

$$A = \frac{t}{2}I + \begin{pmatrix} x & b \\ c & -x \end{pmatrix}, \quad q_0 = \frac{t^2}{4} - 1.$$

After inverting 2 in the fixed model, the equations $\det A = \det C = 1$ are exactly

$$x^2 + bc = q_0, \quad 2rx + sc + ub = d.$$

First suppose $d = 0$. Cayley–Hamilton gives $P = A - C^{-1}$, and

$$\det(A - C^{-1}) = 2 - \operatorname{tr}(AC).$$

Thus $\operatorname{tr}(AC) = 2$. The hypothesis forces $AC = I$, whence $P = A + A^{-1} - tI = 0$, contrary to $S \neq tI$. Hence this case is empty.

Assume $d \neq 0$ and set $q = q_0 - d/4$. Polarizing the determinant in the preceding system gives $\det(A - C) = -4q$. If $q = 0$, then $2 - \operatorname{tr}(AC^{-1}) = \det(A - C) = 0$, so again the trace-two hypothesis gives $A = C = S/2$. There is at most one pair.

It remains to consider $q \neq 0$. If $s \neq 0$, put

$$y = 2(sx - rb), \quad v = 2b - s.$$

A direct calculation from the two determinant equations yields

$$y^2 - dv^2 = s^2(4q_0 - d) = 4s^2q \neq 0.$$

The map $(x, b, c) \mapsto (y, v)$ is injective: recover b from v , then x from y , and finally c from the second determinant equation. Lemma 12.3 gives $\exp(o(n))$ possibilities. If $s = 0$ but $u \neq 0$, transpose the calculation, using

$$y = 2(ux - rc), \quad v = 2c - u.$$

Finally, if $s = u = 0$, then $r \neq 0$, the second determinant equation gives $x = r/2$, and

$$bc = q_0 - r^2/4 = q \neq 0.$$

Theorem 12.1 gives $\exp(o(n))$ possibilities. These cases exhaust the noncentral fiber. If $S = tI$, Cayley–Hamilton gives $C = tI - A = A^{-1}$, proving the last assertion. $\square$

If $r_{t,n}(S)$ denotes the representation function in the additive group of $M_2(F)$, the theorem gives

$$\sum_S r_{t,n}(S)^2 = |\mathcal{F}_t(n)|^2 \exp(o(n)), \quad |\mathcal{F}_t(n) + \mathcal{F}_t(n)| \geq |\mathcal{F}_t(n)|^2 \exp(-o(n)).$$

Indeed, the sum tI , arising from the inverse pairs (A, A^{-1}) , contributes $|\mathcal{F}_t(n)|^2$ to the energy. Every other multiplicity is $\exp(o(n))$, and the total number of ordered pairs is $|\mathcal{F}_t(n)|^2$; Cauchy–Schwarz gives the sumset bound. The same statements hold after applying an injective trace-coordinate map. Choose $B_1, B_2, B_3 \in \Delta$ whose traceless parts form a basis and put $z(A) = (\text{tr}(AB_i))_{i=1}^3$. On $\text{tr } A = t$, equality of two sums of vectors $z(A)$ is equivalent, by nondegeneracy of the trace pairing, to equality of the corresponding matrix sums. The unique coordinate-vector sum of large multiplicity is $t(\text{tr } B_1, \text{tr } B_2, \text{tr } B_3)$; when the B_i are conjugate and have trace t , it is $t^2(1, 1, 1)$, corresponding to the zero matrix $A + A^{-1} - tI$.

12.2. Balanced torus orbits in arithmetic word balls. We begin with a counting theorem which is independent of hyperbolic surfaces. It explains the arithmetic content of these fixed trace-coordinate maps in any rank.

For an algebraic number ξ , write $H(\xi)$ for its absolute multiplicative Weil height. If $T = \mathbb{G}_m^r$ acts diagonally on K^m , its weights are the vectors $w_i \in \mathbb{Z}^r$ determined by

$$(t_1, \dots, t_r) \cdot (x_1, \dots, x_m) = (t^{w_1} x_1, \dots, t^{w_m} x_m).$$

The weights are called *balanced* if they span $\mathbb{Q}^r$ and 0 lies in the interior of their convex hull. The second condition means that every direction in the torus expands at least one active coordinate and contracts at least one other; it is the elementary diagonal-torus form of GIT stability.

Theorem 12.5 (uniform counting in balanced algebraic torus orbits). *Let K be a number field, let S be a finite set of places containing the archimedean places, and let $T = \mathbb{G}_m^r$ act diagonally on K^m with balanced weights $w_1, \dots, w_m$. Fix $C > 0$. For $x \in (\mathcal{O}_{K,S} \setminus \{0\})^m$, put*

$$\mathcal{O}_x(K) = (T(\overline{K}) \cdot x) \cap K^m.$$

Uniformly for $n \geq 2$ and $H(x_i) \leq e^{Cn}$,

$$\# \left\{ y \in \mathcal{O}_x(K) \cap \mathcal{O}_{K,S}^m : H(y_i) \leq e^{Cn} \text{ for every } i \right\} \leq \exp \left(O_{K,S,C,w_i} \left(\frac{n}{\log n} \right) \right). \quad (12.8)$$

The same conclusion holds when some coordinates vanish, provided the weights on their common nonzero support remain balanced.

Proof. First suppose $y = t \cdot x$ for $t = (t_1, \dots, t_r) \in T(K)$. At a nonarchimedean place $\mathfrak{p} \notin S$, put

$$a_{i,\mathfrak{p}} = v_{\mathfrak{p}}(x_i), \quad \nu_{\mathfrak{p}} = (v_{\mathfrak{p}}(t_1), \dots, v_{\mathfrak{p}}(t_r)).$$

Both x and y are S -integral. Therefore

$$a_{i,\mathfrak{p}} \geq 0, \quad a_{i,\mathfrak{p}} + \langle w_i, \nu_{\mathfrak{p}} \rangle \geq 0 \quad (1 \leq i \leq m). \quad (12.9)$$

We spell out where balancedness enters. Because 0 is an interior point of the convex hull, the continuous function

$$u \mapsto \max_i (-\langle w_i, u \rangle)$$

is strictly positive on the unit sphere in $\mathbb{R}^r$. Its minimum there is some $c_w > 0$. Hence, for every u ,

$$\max_i (-\langle w_i, u \rangle) \geq c_w \|u\|. \quad (12.10)$$

Let $A_{\mathfrak{p}} = \sum_i a_{i,\mathfrak{p}}$. Inequalities (12.9)–(12.10) give $\|\nu_{\mathfrak{p}}\| = O_w(A_{\mathfrak{p}})$. Thus $\nu_{\mathfrak{p}}$ has at most $O_w((1 + A_{\mathfrak{p}})^r)$ possible integral values, and it is zero when $A_{\mathfrak{p}} = 0$.

Set

$$\mathfrak{a} = \prod_{\mathfrak{p} \notin S} \mathfrak{p}^{A_{\mathfrak{p}}}.$$

The height bounds imply $\log N\mathfrak{a} = O_{K,m,C}(n)$: the left side is the sum of the outside- S positive valuations of the coordinates of x , weighted by $\log N\mathfrak{p}$, and this is one of the terms controlled by their Weil heights. Consequently the number of possible collections $(\nu_{\mathfrak{p}})_{\mathfrak{p} \notin S}$ is at most

$$\prod_{\mathfrak{p}|\mathfrak{a}} O_w((1 + A_{\mathfrak{p}})^r) \leq \tau_K(\mathfrak{a})^{O_w(1)} \leq \exp\left(O\left(\frac{n}{\log n}\right)\right). \quad (12.11)$$

Here $\tau_K(\mathfrak{a})$ is the number of ideal divisors of $\mathfrak{a}$; the last inequality is the standard uniform divisor bound in a fixed number field.

Fix one valuation pattern. The quotient of any two parameters which realize it is an S -unit in each torus coordinate. Choose r linearly independent weights and let $D \neq 0$ be the determinant of their weight matrix. The adjugate-matrix identity expresses each t_j^D as a fixed Laurent monomial in the ratios y_i/x_i . Since $H(y_i/x_i) \leq H(y_i)H(x_i) \leq e^{2Cn}$, it follows that $H(t_j) \leq e^{C'n}$. Dirichlet's S -unit theorem identifies the free part of $\mathcal{O}_{K,S}^\times$ with a lattice under the logarithmic embedding. A box of radius $O(n)$ contains only $n^{O_{K,S,r}(1)}$ lattice points. Multiplying this polynomial factor by (12.11) proves (12.8) when $t \in T(K)$.

Finally suppose only that $y \in T(\overline{K})x \cap K^m$. The same adjugate identity shows that every t_j^D lies in K . Run the valuation argument on these D -th powers, or equivalently on the lattice $D^{-1}\mathbb{Z}^r$. A fixed tuple of powers has at most D^r preimages acting on x , so only the implied constant changes. Zero coordinates are omitted because a torus orbit preserves its coordinate support. $\square$

Here is the invariant-theoretic form. If a torus T acts on an affine variety Y , the affine quotient $Y//T = \text{Spec } K[Y]^T$ records the values of all invariant regular functions. Each quotient fiber contains a unique closed orbit. For a diagonal action, a point has closed orbit and finite stabilizer exactly when the weights on its nonzero coordinates are balanced.

Corollary 12.6 (stable quotient fibers in word balls). *Let G/K be an affine algebraic group, let a K -torus T act on G by conjugation, and let $\pi_T : G \rightarrow G//T$ be the affine quotient. Let $\Delta < G(K)$ be finitely generated. On the locus G^{bal} where the active weights in a fixed faithful equivariant affine embedding are balanced, one has, uniformly in $\xi \in (G//T)(K)$,*

$$\#\{\gamma \in \Delta : |\gamma| \leq n, \gamma \in G^{\text{bal}}, \pi_T(\gamma) = \xi\} \leq \exp\left(O_{G,T,\Delta}\left(\frac{n}{\log n}\right)\right). \quad (12.12)$$

Proof. Pass once to a finite extension which splits T , and choose S so that the fixed generators of Δ and the affine embedding are S -integral. Multiplying at most n fixed generator matrices makes every coordinate height at most $e^{O(n)}$. Balanced points have closed torus orbit and finite stabilizer by the diagonal Hilbert–Mumford criterion. Two such points with the same quotient value therefore lie in the unique closed orbit in that fiber. Theorem 12.5 applies to this orbit. There are only finitely many coordinate supports, so summing over them preserves the bound. $\square$

For SL_d , the quotient functions can be written explicitly as traces of alternating products with a fixed regular semisimple matrix.

Corollary 12.7 (fibers of centralizer invariants in SL_d). *Let $\Delta < \text{SL}_d(K)$ be finitely generated, and let $B \in \Delta$ be split regular semisimple. Write $P_1(B), \dots, P_d(B)$ for its spectral projectors. For every simple directed cycle $I = (i_1, \dots, i_k)$, including loops, put*

$$c_I(A) = \text{tr}(P_{i_1} A P_{i_2} A \cdots P_{i_k} A). \quad (12.13)$$

Let Ψ_B be the finite vector of all these functions, and let U_B be the locus where the directed graph having an edge $i \rightarrow j$ whenever the (i, j) -entry of A is nonzero is strongly connected. Then

$$\max_{\xi} \#\{A \in \Delta : |A| \leq n, A \in U_B, \Psi_B(A) = \xi\} \leq \exp\left(O_{\Delta, B}\left(\frac{n}{\log n}\right)\right). \quad (12.14)$$

Every coordinate of Ψ_B is a fixed linear combination of the alternating traces

$$\mathrm{tr}(B^{e_1} A B^{e_2} A \cdots B^{e_k} A).$$

Proof. In a B -eigenbasis, its connected centralizer modulo the center is the diagonal torus. It acts on the entry a_{ij} with weight $e_i - e_j$. An invariant monomial corresponds to a directed multigraph having equal indegree and outdegree at every vertex. Such an Eulerian multigraph is a union of directed cycles, and every directed cycle decomposes into simple cycles. Thus the diagonal entries and the simple-cycle monomials generate the invariant ring.

The expression in (12.13) is exactly

$$a_{i_1 i_2} a_{i_2 i_3} \cdots a_{i_k i_1}.$$

Strong connectivity says that the active weights span the root lattice. Moreover every active edge lies on a directed cycle; adding these cycles gives a strictly positive linear relation among all active weights. This is precisely balancedness. Corollary 12.6 now gives (12.14). Finally, Lagrange interpolation writes each $P_i(B)$ as a polynomial in B . Expanding (12.13) therefore gives the asserted finite linear combination of alternating traces. $\square$

In rank one the entire quotient collapses to two ordinary traces. If $B = \mathrm{diag}(\lambda, \lambda^{-1})$ and $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, then

$$K[\mathrm{SL}_2]^{Z(B)} = K[a, d, bc]/(ad - bc - 1) = K[a, d] = K[\mathrm{tr} A, \mathrm{tr}(AB)]. \quad (12.15)$$

The last equality holds because $\lambda \neq \lambda^{-1}$, so $(\mathrm{tr} A, \mathrm{tr}(AB))$ is an invertible linear change of (a, d) . The balanced locus is $bc \neq 0$, equivalently $\langle A, B \rangle$ is irreducible.

Thus $\mathrm{tr} A$ and $\mathrm{tr}(AB)$ are literally the affine quotient coordinates for centralizer conjugation in rank one. Theorem 12.2 gives the stronger $\exp(o(n))$ word-ball bound over every finitely generated characteristic-zero coefficient field.

The irreducibility hypothesis is essential for a general matrix group. Indeed, take $B = \mathrm{diag}(2, 1/2)$, $U(t) = \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$, and $\Delta = \langle B, U(1) \rangle < \mathrm{SL}_2(\mathbb{Q})$. Since $BU(t)B^{-1} = U(4t)$, word balls contain exponentially many different upper-unipotent elements, while all of them have $\mathrm{tr} U(t) = 2$ and $\mathrm{tr}(U(t)B) = \mathrm{tr} B$. Cocompact surface groups avoid this horospherical phenomenon because a boundary-point stabilizer is cyclic.

12.3. Trace multiplicity and additive energy. Low trace growth does force large length multiplicities. Let H be a torsion-free finite-index subgroup of a cocompact Fuchsian lattice. Write $m_t(L)$ for the number of primitive hyperbolic H -conjugacy classes of length at most L having absolute trace t .

We keep two statistics separate. The multiplicity second moment

$$M_2(L) = \sum_t m_t(L)^2$$

counts ordered pairs of primitive classes having the same trace. It is not additive energy. For a finite set $S \subset \mathbb{R}$, put

$$r_{S+S}(u) = \#\{(a, b) \in S^2 : a + b = u\}, \quad E_+(S) = \sum_u r_{S+S}(u)^2.$$

Equivalently, $E_+(S)$ is the number of ordered quadruples $(a, b, c, d) \in S^4$ satisfying $a + b = c + d$. This second statistic forgets all geodesic multiplicities and uses each distinct trace value only once.

Proposition 12.8 (trace-multiplicity second moment). *If, for some $\alpha \geq 0$, $\#(T(H) \cap [-X, X]) = O(X^\alpha)$, then*

$$\sum_t m_t(L)^2 \gg \frac{e^{(2-\alpha/2)L}}{L^2}, \quad \max_t m_t(L) \gg \frac{e^{(1-\alpha/2)L}}{L}. \quad (12.16)$$

In particular, linear trace growth gives the exponents 3/2 and 1/2.

Proof. The prime geodesic theorem [34, 13] gives $N_H(L) \asymp e^L/L$ primitive classes. Since $|\operatorname{tr} \gamma| \asymp e^{\ell(\gamma)/2}$, the number $D_H(L)$ of available distinct trace values is $O(e^{\alpha L/2})$. Cauchy–Schwarz gives

$$\sum_t m_t(L)^2 \geq \frac{N_H(L)^2}{D_H(L)},$$

and the bound for the largest fiber follows from $N_H(L) \leq D_H(L) \max_t m_t(L)$. $\square$

This is a second moment of multiplicities, not yet additive energy of the distinct trace support. Fricke does turn an equal-trace pair into an additive quadruple:

$$\operatorname{tr} A = \operatorname{tr} A' \implies \operatorname{tr}(AB) + \operatorname{tr}(AB^{-1}) = \operatorname{tr}(A'B) + \operatorname{tr}(A'B^{-1}). \quad (12.17)$$

To transfer the lower bound in (12.16) to the scalar support, one must prevent too many group-theoretic pairs from collapsing onto the same scalar quadruple in (12.17). The finite-type divisor theorem above gives the required bound on these fibers. We first record the geometric normalization needed to apply it.

Work in the full inverse image $\tilde{H} < \operatorname{SL}_2(\mathbb{R})$, always choosing the positive-trace lift of a primitive class. Fix a hyperbolic $B \in \tilde{H}$, with $y = \operatorname{tr} B \neq 0$. Fix once and for all a compact fundamental set and a deterministic tie-breaking rule, and conjugate each axis to meet that set. Let $\mathcal{P}_X$ be the resulting nested family of representatives of length at most $2 \log X + O(1)$. Then $d(o, Ao) \leq \ell(A) + O_H(1)$, so its elements have norm $O_H(X)$, and $|\mathcal{P}_X| \asymp X^2/\log X$. Define

$$F_B(X) = \max_{x,z} \#\{A \in \mathcal{P}_X : \operatorname{tr} A = x, \operatorname{tr}(AB) = z\}. \quad (12.18)$$

Theorem 12.9 (two trace coordinates on compact Fuchsian lattices). *Every hyperbolic $B \in \tilde{H}$ satisfies*

$$F_B(X) = X^{o(1)}. \quad (12.19)$$

No genericity assumption on B is needed.

Proof. By construction, the axis of every $A \in \mathcal{P}_X$ meets one fixed compact set and $\ell(A) \leq 2 \log X + O(1)$. Hence, for a fixed basepoint o ,

$$d(o, Ao) \leq \ell(A) + O_H(1) = O_H(\log X).$$

The orbit map of the cocompact action is a quasi-isometry (the Milnor–Švarc lemma), so the word length of A in a fixed finite generating set is $O_H(\log X)$.

The field generated over $\mathbb{Q}$ by the entries of a finite generating set of $\tilde{H}$ is finitely generated. On the irreducible locus, Theorem 12.2, with $n = O_H(\log X)$, gives $\exp(o(\log X)) = X^{o(1)}$ possibilities uniformly in the two prescribed traces. On the reducible locus, A and B share a boundary fixed point. A boundary-point stabilizer in a torsion-free cocompact Fuchsian group is cyclic: there are no parabolics, and two hyperbolic elements fixing the same endpoint have the same axis and belong to one maximal cyclic subgroup. That subgroup has only its generator and inverse as primitive elements, so it contributes $O(1)$ representatives. Combining the two loci proves (12.19). $\square$

The fiber estimate for $A \mapsto (\operatorname{tr} A, \operatorname{tr}(AB))$ has the following unconditional consequence, which does not pass through additive energy.

Corollary 12.10 (the sharp universal exponent for distinct lengths). *For every torsion-free cocompact Fuchsian lattice H ,*

$$D_H(X) := \#(T(H) \cap [-X, X]) \geq X^{1-o(1)}.$$

If $N_H^(L)$ denotes the number of distinct primitive closed-geodesic lengths at most L , then*

$$N_H^*(L) \geq \exp((\frac{1}{2} - o(1))L).$$

Thus every closed hyperbolic surface has exponentially many distinct primitive lengths, with the universal exponent $1/2$. This exponent is optimal, because compact arithmetic surfaces have at most $O(e^{L/2})$ distinct lengths up to L .

Proof. The map

$$\mathcal{P}_X \longrightarrow T(H)^2, \quad A \longmapsto (\operatorname{tr} A, \operatorname{tr}(AB))$$

has fibers of size at most $F_B(X) = X^{o(1)}$. Both coordinates have absolute value at most $C_B X$, whereas $|\mathcal{P}_X| \asymp X^2 / \log X$. Therefore

$$D_H(C_B X)^2 \geq \#\{(\operatorname{tr} A, \operatorname{tr}(AB)) : A \in \mathcal{P}_X\} \geq \frac{|\mathcal{P}_X|}{F_B(X)} = X^{2-o(1)},$$

which proves the first assertion after rescaling.

Every nontrivial element of a torsion-free cocompact Fuchsian group is a power of a unique primitive element, up to inversion. If such a power has absolute trace at most X , its translation length is $O(\log X)$. The systole of H is positive, so a fixed primitive element contributes only $O_H(\log X)$ power traces in this range. Hence the number of distinct primitive absolute traces up to X is at least $D_H(X)/O_H(\log X) = X^{1-o(1)}$. Finally $|\operatorname{tr} \gamma| = 2 \cosh(\ell(\gamma)/2)$ converts X to $e^{L/2}$, proving the length statement. The arithmetic upper bound follows from linear trace growth. $\square$

Breuillard–de Cornulier–Lubotzky–Meiri prove exponential growth of the number of characteristic polynomials in word balls of a non-virtually-solvable linear group [9, Theorem 1.2]. In SL_2 the characteristic polynomial is determined by the trace; a cocompact Fuchsian lattice is linear and not virtually solvable, so the theorem applies. Word length is comparable with hyperbolic displacement for a cocompact lattice. Finally, a primitive element of length at most L contributes only $O_H(L)$ powers of length at most L , because the systole is positive. Thus their theorem implies an exponential lower bound for distinct primitive lengths, with a positive constant that may depend on the surface. Corollary 12.10 gives the universal exponent $1/2$ and, through the arithmetic examples, shows that this universal value cannot be increased.

12.4. Trace supports indexed by a finite quotient. The scalar trace support can be replaced by a finite vector of supports in a way which retains the group law. Let $\Delta < \mathrm{SL}_2(\mathbb{R})$ have cocompact, projectively torsion-free image, let $\pi : \Delta \rightarrow Q$ be a finite quotient, and define

$$T_r(X) = \{\operatorname{tr} A : \pi(A) = r, |\operatorname{tr} A| \leq X\} \quad (r \in Q).$$

These supports may overlap as subsets of $\mathbb{R}$; their labels remember the group element, not merely its trace. Fix $o \in \mathbb{H}^2$ and put $\|A\|_o = \exp(d(o, \bar{A}o)/2)$.

Theorem 12.11 (growth of $\operatorname{tr}(\gamma g \gamma^{-1} B)$ on a conjugacy orbit). *Fix noncentral $g, B \in \Delta$, and write*

$$a = \pi(g), \quad b = \pi(B), \quad t = \operatorname{tr} g, \quad y = \operatorname{tr} B.$$

For every $c \in Q$, the set

$$\mathcal{Z}_{g,B,c}(X) = \{\operatorname{tr}(\gamma g \gamma^{-1} B) : \pi(\gamma) = c, d(o, \bar{\gamma}o) \leq \log X\}$$

satisfies

$$\#\mathcal{Z}_{g,B,c}(X) \geq X^{1-o(1)}.$$

For a constant $C = C(g, B, o)$, it also satisfies the two exact component containments

$$\mathcal{Z}_{g,B,c}(X) \subset T_{cac^{-1}b}(CX), \quad ty - \mathcal{Z}_{g,B,c}(X) \subset T_{cac^{-1}b^{-1}}(CX).$$

Proof. Let $\Delta_0 = \ker \pi$. Cocompact orbital counting in each fixed coset gives

$$\#\{\gamma \in \pi^{-1}(c) : d(o, \bar{\gamma}o) \leq R\} \asymp e^R.$$

If two such γ_i give the same conjugate of g , then $\gamma_2^{-1}\gamma_1 \in Z_\Delta(g)$. This centralizer is virtually cyclic and has only $O(R)$ elements in a displacement ball of radius $2R$. With $R = \log X$, we therefore obtain $\gg X/\log X$ distinct conjugates $C = \gamma g \gamma^{-1}$.

The triangle inequality gives $\|C\|_o \ll_g X$, and cocompactness gives word length $O(\log X)$. For prescribed $\text{tr } C = t$ and $\text{tr}(CB) = z$, Theorem 12.2 bounds the irreducible part of the fiber by $X^{o(1)}$. In the reducible part, C fixes an endpoint of B ; the relevant boundary stabilizer is virtually cyclic and contributes only $O(\log X)$ elements. Each z -fiber is therefore $X^{o(1)}$, proving the lower bound.

The image of C in Q is cac^{-1} , which proves the first containment. The Fricke identity

$$\text{tr}(CB) + \text{tr}(CB^{-1}) = \text{tr } C \text{tr } B = ty$$

proves the reflected containment. $\square$

Corollary 12.12 (growth in every quotient component). *For every $r \in Q$, one has $\#T_r(X) \geq X^{1-o(1)}$.*

Proof. Fix any noncentral g , with image $a \in Q$, take $c = 1$, and choose a noncentral $B \in \pi^{-1}(a^{-1}r)$. Such a choice is possible because each coset of the finite-index kernel contains infinitely many elements. The first containment in Theorem 12.11 puts all $X^{1-o(1)}$ values of $\text{tr}(\gamma g \gamma^{-1}B)$ in the component indexed by r . $\square$

For finite $A, B \subset \mathbb{R}$, write

$$E_+(A, B) = \#\{(a_1, b_1, a_2, b_2) \in A \times B \times A \times B : a_1 + b_1 = a_2 + b_2\}.$$

Theorem 12.13 (mixed energy between quotient components). *Assume that $\#(T(\Delta) \cap [-X, X]) = O(X)$. For $q, b \in Q$, choose a noncentral $B \in \pi^{-1}(b)$. Then, for a fixed $C = C(B, o)$,*

$$E_+(T_{qb}(CX), T_{qb^{-1}}(CX)) \geq X^{3-o(1)}.$$

Consequently every pair consisting of one quotient component with itself has mixed energy $X^{3-o(1)}$. If $|Q|$ is odd, the same holds for every ordered pair of quotient components.

Proof. Let

$$\mathcal{B}_q(X) = \{A \in \Delta : \pi(A) = q, \|A\|_o \leq X\}.$$

Finite-index cocompact orbital counting gives $|\mathcal{B}_q(X)| \asymp X^2$: the condition $\pi(A) = q$ selects one coset of the lattice $\ker \pi$, while a displacement ball of radius $2 \log X$ contains $\asymp X^2$ orbit points. For a trace value x , let m_x count the matrices in $\mathcal{B}_q(X)$ with trace x , and let q_x count the distinct values $\text{tr}(AB)$ among them. Cocompactness identifies displacement and word length up to multiplicative and additive constants, so every $A \in \mathcal{B}_q(X)$ has word length $O(\log X)$. For a fixed pair of values x, z , Theorem 12.2 bounds the irreducible matrices with $\text{tr } A = x, \text{tr}(AB) = z$ by $X^{o(1)}$. All reducible matrices lie in the union of the two boundary-point stabilizers determined by B , and these contribute $O(\log X)$ elements in the ball. If m_x is larger than a fixed multiple of $\log X$, the irreducible elements therefore give $q_x \geq m_x X^{-o(1)}$; if $m_x = O(\log X)$, the trivial bound $q_x \geq 1$ gives the same conclusion. Linear trace growth leaves only $O(X)$ possible values of x , because $|\text{tr } A| \ll X$, so

$$\sum_x m_x^2 \geq \frac{(\sum_x m_x)^2}{O(X)} \gg X^3.$$

The element B is noncentral in a projectively torsion-free cocompact group, hence hyperbolic, so $y = \text{tr } B \neq 0$. Every distinct value $z = \text{tr}(AB)$ gives the ordered representation

$$z + \text{tr}(AB^{-1}) = xy.$$

The two summands belong to the components qb and qb^{-1} , respectively. Distinct z 's give distinct representations, and distinct x 's give distinct sums. Therefore

$$E_+(T_{qb}(CX), T_{qb^{-1}}(CX)) \geq \sum_x q_x^2 \geq X^{-o(1)} \sum_x m_x^2 = X^{3-o(1)}.$$

The same averaging shows that for some exact trace $x = x(X)$, the set of values $\text{tr}(AB)$ has size $X^{1-o(1)}$.

For desired components r, s , the displayed pair equals (r, s) exactly when $b^2 = s^{-1}r$, taking $q = rb^{-1}$. The choice $b = 1$ gives every diagonal pair. In a finite group of odd order the squaring map is bijective, so every pair occurs. $\square$

It is therefore useful to retain the entire family of trace sets indexed by Q , rather than to replace it by their union. Fricke's identity couples specified components, and, when Q is abelian, the finite Fourier transform diagonalizes translation in the index Q . This observation does not invoke modular forms. The estimates above have not been shown to be compatible through a tower of quotients associated with a nondistinguished finite place. The arithmeticity criterion itself needs only one component: if the entire set

$$\{g + \gamma g \gamma^{-1} - (\text{tr } g)I : \gamma \text{ lies in a fixed coset of } \ker \pi\}$$

is bounded at all finite places, translation by one coset representative gives a bounded adjoint orbit for the finite-index subgroup $\ker \pi$. Theorem 11.9 makes that subgroup, and hence Δ , arithmetic.

Proposition 12.14 (a lower bound for additive energy). *Let $S_X = T(H) \cap [-C_B X, C_B X]$, with C_B large enough. Then*

$$E_+(S_X) \gg \frac{X^4}{F_B(X)^2 |S_X| (\log X)^2}. \quad (12.20)$$

In particular, Theorem 12.9 gives the unconditional relation $E_+(S_X) |S_X| \geq X^{4-o(1)}$.

Proof. For a trace value x , let m_x count its representatives in $\mathcal{P}_X$, and let q_x count the distinct values $z = \text{tr}(AB)$ among them. Then $q_x \geq m_x / F_B(X)$. Fricke gives, for each such value,

$$z + \text{tr}(AB^{-1}) = xy.$$

Both summands belong to S_X , after enlarging C_B . Distinct z 's give distinct ordered representations of xy ; distinct x 's give distinct sums because $y \neq 0$. Hence

$$E_+(S_X) \geq \sum_x q_x^2 \geq F_B(X)^{-2} \sum_x m_x^2.$$

There are at most $|S_X|$ values of x , while $\sum_x m_x = |\mathcal{P}_X| \asymp X^2 / \log X$. Cauchy-Schwarz therefore gives

$$\sum_x m_x^2 \geq \frac{|\mathcal{P}_X|^2}{|S_X|},$$

which proves (12.20). Insert (12.19) for the final assertion. $\square$

Corollary 12.15 (additive energy under linear trace growth). *For every torsion-free cocompact Fuchsian lattice, linear trace growth implies*

$$E_+(S_X) = X^{3-o(1)}. \quad (12.21)$$

Consequently $|S_X| = X^{1-o(1)}$, and S_X contains subsets of cardinality $X^{1-o(1)}$ and doubling $X^{o(1)}$.

Proof. Linear growth gives $|S_X| = O(X)$. Insert (12.19) into (12.20); the logarithmic factors are $X^{o(1)}$, so the lower bound is $X^{3-o(1)}$. Since $E_+(S_X) \leq |S_X|^3$, the first conclusion follows. Together with the assumed upper bound $|S_X| = O(X)$, this says that the energy is within a factor $X^{o(1)}$ of its maximum. The energy form of the Balog–Szemerédi–Gowers theorem then produces the asserted subset. Here the doubling constant of a finite set A is $|A + A|/|A|$. $\square$

The scalar constraints in (12.12) have a simple geometric description.

Lemma 12.16 (centralizer description of fixed-multiplier fibers). *For hyperbolic B , the pair $(\operatorname{tr} A, \operatorname{tr}(AB))$ determines $A \in \operatorname{SL}_2(\mathbb{R})$, away from the reducible locus, up to at most two orbits under conjugation by $Z_{\operatorname{SL}_2(\mathbb{R})}(B)$.*

Proof. Conjugate B to $\operatorname{diag}(\lambda, \lambda^{-1})$, and write $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. The two specified traces determine a, d , while determinant one determines $bc = ad - 1$. Centralizer conjugation sends (b, c) to $(s^2 b, s^{-2} c)$. For $bc \neq 0$, only the possible real sign of b remains, giving at most two orbits. The excluded locus $bc = 0$ is exactly where A and B share an eigenline. $\square$

The invariant ring explains why commutator identities do not improve this lemma. Over an algebraically closed field of characteristic zero, for a regular semisimple B ,

$$k[\operatorname{SL}_2]^{Z(B)} = k[\operatorname{tr} A, \operatorname{tr}(AB)].$$

Indeed, after diagonalizing B , the centralizer acts on the entries of $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with weights $0, 2, -2, 0$. Its invariant ring is $k[a, d, bc]/(ad - bc - 1) = k[a, d]$, and a, d are recovered linearly from $\operatorname{tr} A, \operatorname{tr}(AB)$. In particular, the trace of every word in A, B , including every iterated commutator, is constant on the same centralizer orbit. Thus no collection of two-generator Fricke identities can bound the fiber in (12.12); one must add a trace against a third fixed matrix whose trace functional is linearly independent of the first two.

Proposition 12.17 (finite reconstruction from three trace coordinates). *Let k have characteristic different from two. Suppose $B, C \in \operatorname{SL}_2(k)$ and the annihilator of $\operatorname{span}_k\{I, B, C\}$ under the trace pairing is kD , where $\det D \neq 0$. Then*

$$\Psi_{B,C} : \operatorname{SL}_2 \longrightarrow \mathbb{A}^3, \quad A \longmapsto (\operatorname{tr} A, \operatorname{tr}(AB), \operatorname{tr}(AC))$$

is finite flat of degree two. Every geometric fiber has length two and at most two points. Its deck involution is the determinant-polar reflection

$$\iota_D(A) = A - \frac{\langle A, D \rangle_{\det}}{\det D} D, \quad \langle A, D \rangle_{\det} = \operatorname{tr}(\operatorname{adj}(A)D).$$

The ramification divisor is $\langle A, D \rangle_{\det} = 0$; the branch divisor is the discriminant of the resulting quadratic equation.

If $\tilde{H} < \operatorname{SL}_2(\mathbb{R})$ is a Fuchsian lattice, one can moreover take $C = hBh^{-1}$ for two hyperbolic elements of the same trace in $\tilde{H}$.

Proof. The three trace functionals have common kernel kD . Choose a linear section $A_0(y)$ of them. Every point over $y \in \mathbb{A}^3$ has the form $A_0(y) + sD$, and

$$\det(A_0 + sD) - 1 = \det D s^2 + \langle A_0, D \rangle_{\det} s + \det A_0 - 1.$$

The leading coefficient is a nonzero constant. After division by it this is a monic quadratic in s , so the coordinate ring of SL_2 is free of rank two over $k[y_1, y_2, y_3]$. The second root above a point A is the displayed $\iota_D(A)$. The polarization identity shows that $\iota_D^2 = 1$ and preserves the determinant. Its fixed locus and the quadratic discriminant give the asserted ramification and branch divisors.

For existence, diagonalize a hyperbolic $B = \text{diag}(\lambda, \lambda^{-1})$. Choose a conjugate $C = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ sharing neither eigenline with B , so $qr \neq 0$. Such a conjugate exists by Zariski density (equivalently, choose h outside the two endpoint stabilizers). Direct calculation gives

$$\text{span}\{I, B, C\}^\perp = k \begin{pmatrix} 0 & q \\ -r & 0 \end{pmatrix}, \quad \det D = qr \neq 0.$$

Conjugation preserves trace, so $\text{tr } C = \text{tr } B$. $\square$

This is a concrete Noether normalization by the three displayed trace coordinates. It has a useful higher-rank analogue. If $\Gamma < \text{SL}_d(k)$ is Zariski dense in characteristic zero, there are $d^2 - 1$ elements $B_j \in \Gamma$ whose trace-pairing annihilator is an invertible line kD . Consequently

$$A \longmapsto (\text{tr}(AB_1), \dots, \text{tr}(AB_{d^2-1}))$$

is finite flat of degree d : on the remaining line its equation is the monic degree- d polynomial $\det(A_0 + sD) - 1$. To see that suitable matrices B_j exist, take $d^2 - 1$ conjugates in SL_d of one noncentral trace-zero element which span $\mathfrak{sl}_d$; the annihilator is then kI . The condition that their trace-pairing annihilator be an invertible line is nonempty and Zariski open, so it meets Γ^{d^2-1} . For a general embedded reductive group, ordinary Noether normalization gives the same principle with sufficiently many matrix coefficients, but the degree and the possibility of choosing all fixed matrices from the group depend on the representation.

The degree-two result also gives an unconditional energy statement involving three trace coordinates. Recall $D_H(X) = \#(T(H) \cap [-X, X])$. For a sufficiently large fixed K , let $S_X = T(H) \cap [-KX, KX]$ and

$$r_X(s) = \#\{(u, v) \in S_X^2 : u + v = s\}.$$

Corollary 12.18 (a fourth moment along multiples of one trace). *There are a nonzero trace y and $K, c > 0$, depending only on H , such that*

$$\sum_t r_X(ty)^2 \geq cX^2, \quad \sum_t r_X(ty)^4 \geq c \frac{X^4}{D_H(KX)}.$$

Thus if $D_H(X) = O(X^\alpha)$, the second lower bound is $\gg X^{4-\alpha}$; under linear growth it is $\gg X^3$. In addition, $D_H(X) \gg X^{2/3}$. At the extremal exponent, if $D_H(X) = X^{2/3+o(1)}$, the sets S_X contain subsets of size $|S_X|^{1-o(1)}$ with doubling $|S_X|^{o(1)}$.

Proof. Use the conjugate matrices B, C from Proposition 12.17, and write $y = \text{tr } B = \text{tr } C$. Let $\mathcal{G}_X = \{A \in \tilde{H} : \|A\| \leq X\}$. Cocompact lattice-point counting gives $\#\mathcal{G}_X \asymp X^2$. If n_t counts its elements of trace t , finite reconstruction says that at least $n_t/2$ distinct pairs

$$(\text{tr}(AB), \text{tr}(AC))$$

occur. Each coordinate is the first term of a Fricke decomposition of the same number ty :

$$\text{tr}(AB) + \text{tr}(AB^{-1}) = ty, \quad \text{tr}(AC) + \text{tr}(AC^{-1}) = ty.$$

All four terms lie in S_X after fixing K . Hence $r_X(ty)^2 \geq n_t/2$. Summing gives the first claim, while squaring and using Cauchy–Schwarz gives

$$\sum_t r_X(ty)^4 \geq \frac{1}{4} \sum_t n_t^2 \geq c \frac{X^4}{D_H(KX)}.$$

Finally, the triple $(\text{tr } A, \text{tr}(AB), \text{tr}(AC)) \in S_X^3$ has fibers of size at most two, so $X^2 \ll D_H(KX)^3$, proving the last assertion after rescaling X . If $D_H(X) = X^{2/3+o(1)}$, the first estimate says

$$E_+(S_X) \geq \sum_t r_X(ty)^2 \gg X^2 = |S_X|^{3-o(1)}.$$

The energy form of Balog–Szemerédi–Gowers gives the final assertion. $\square$

The fourth moment is the strongest conclusion available from finite reconstruction *alone*. It does not by itself yield cubic ordinary additive energy: it only forces $E_+(S_X) \geq \sum_t r_X(ty)^2 \gg X^2$, which is diagonal order under linear growth. Combinatorially, a fiber of n_t reconstructed matrices could map to a balanced $\sqrt{n_t}$ -by- $\sqrt{n_t}$ grid of the two values $\text{tr}(AB)$ and $\text{tr}(AC)$. Theorem 12.2 rules out this grid after either one of these coordinates is fixed. Together with Theorem 12.9, this proves the required incidence multiplicity estimate and hence Corollary 12.15 for compact Fuchsian lattices.

The next proposition limits what can be deduced from a large exact trace fiber. After one independent fixed trace functional is added, the original fiber splits into almost as many new values as its cardinality permits.

Proposition 12.19 (dispersion of a large trace fiber under a fixed multiplier). *Let $\rho: \Pi \rightarrow \text{PSL}_2(\mathbb{R})$ be a faithful cocompact Fuchsian representation whose absolute trace support has linear growth. Let $\phi: \Pi \twoheadrightarrow \mathbb{Z}$, let $H < \ker \phi$ be finitely generated and convex cocompact with critical exponent $\delta_H > 1/2$, and choose $a \in \phi^{-1}(1)$. Fix compatible lifts over a finitely generated characteristic-zero field. If $B \in \text{SL}_2(F)$ satisfies $\text{tr}^2 B \neq 4$, then for every sufficiently large R there are a squared-trace value t_R and a set*

$$E_R \subset \{h \in H : d(o, \rho(h)o) \leq R\}$$

such that

$$|E_R| \geq \exp((\delta_H - \frac{1}{2} - o(1))R), \quad \text{tr}^2 \rho(ah) = t_R \quad (h \in E_R),$$

whereas, uniformly in $s \in F$,

$$\#\{h \in E_R : \text{tr}^2(B\rho(a)\rho(h)) = s\} = \exp(o(R)).$$

Consequently the second squared trace assumes at least $\exp((\delta_H - \frac{1}{2} - o(1))R)$ distinct values on E_R . The same lower bound holds for a vector defined by any fixed finite list of multipliers if at least one multiplier has squared trace different from four.

Proof. Convex-cocompact orbit counting [34] gives $\exp((\delta_H + o(1))R)$ elements in the displayed ball. Every ah has translation length in the original Fuchsian representation at most $R + O(1)$, and hence absolute trace at most $Ce^{R/2}$. Linear trace growth leaves only $O(e^{R/2})$ squared-trace values. Pigeonholing gives t_R and E_R with the asserted cardinality.

Put $A_0 = \rho(a)$ and $A_1 = BA_0$. Then $A_1A_0^{-1} = B$. The simultaneous-fiber assertion of Theorem 13.4, applied directly to the faithful convex-cocompact Fuchsian representation $\rho|_H$, gives

$$\sup_{u,v} \#\{h : d(o, \rho(h)o) \leq R, \text{tr}^2(A_0\rho(h)) = u, \text{tr}^2(A_1\rho(h)) = v\} = \exp(o(R)).$$

Restricting to $u = t_R$ proves the uniform fiber bound, and division gives the asserted number of second-coordinate values. Adding further coordinates can only decrease a joint fiber. $\square$

Thus neither the Fricke identities nor conjugation by a fixed independent multiplier can turn the large first-coordinate fiber into a large simultaneous fiber. The exceptional condition $\text{tr}^2 B = 4$ is exactly where the two-coordinate divisor argument degenerates; central multipliers merely repeat the first coordinate, while noncentral unipotent multipliers require a separate argument. A propagation theorem must therefore use multipliers that vary with the radius or information from a nonarchimedean valuation.

Average additive energy does not control all the pairs needed for a trace gap. Put $H = \Gamma^{(2)}$, and, for $\eta > 0$, consider the set of all small differences

$$\mathcal{C}_\eta(H) = \{x - y : x, y \in T(H), |x - y| < \eta\}.$$

The gap problem asks whether zero is isolated in this set. A family containing a negligible proportion of all pairs can nevertheless contain every sequence of differences tending to zero. Cardinality, additive energy, and Balog–Szemerédi–Gowers extraction do not detect such a family.

Any argument of this kind must therefore control every sufficiently close pair, rather than almost all pairs.

Problem 12.20 (control of every sufficiently small trace difference). Assume that Γ is a cocompact Fuchsian lattice and that $T(\Gamma)$ has bounded clustering or linear growth. Prove, using the group-level Fricke relations and the fiber theorems for $(\operatorname{tr} A, \operatorname{tr}(AB))$, one of the following sufficient conclusions for $H = \Gamma^{(2)}$:

- (a) there are $\eta > 0$ and a finite set $F \subset \mathbb{R}$ such that

$$\mathcal{C}_\eta(H) \subset F + T(H);$$

- (b) there are $\eta > 0$, a number field k with distinguished real embedding v_0 , a fractional ideal $I \subset k$, and constants C_σ such that, under v_0 ,

$$\mathcal{C}_\eta(H) \subset I, \quad |\sigma(d)| \leq C_\sigma \quad (d \in \mathcal{C}_\eta(H))$$

for every nondistinguished archimedean embedding σ .

Conclusion (a) gives a gap by Lemma 11.2; conclusion (b) gives one directly from the product formula, by applying the proof of Lemma 11.3 to $\mathcal{C}_\eta(H)$. The finite-type fiber theorem makes inverse additive theory available, but Theorem 12.4 shows that, after applying the injective coordinate map $A \mapsto (\operatorname{tr}(AB_i))_{i=1}^3$, an exact trace fiber in a word ball of radius n has representation multiplicity $\exp(o(n))$ for every sum other than the forced inverse sum. Thus ordinary Freiman extraction inside that fiber cannot yield either conclusion, both of which must hold for every sufficiently small difference. Problem 12.20 remains one sufficient scalar approach.

A more direct sufficient condition is furnished by Theorem 11.11 and Lemma 11.12. The following theorem quantifies this expansion in terms of traces. Theorem 13.8 attaches to a loxodromic kernel element k an exact cancellation exponent $\mathbf{c}_v(a, k; J)$ and proves

$$h_{\operatorname{tr}}(x) \geq \frac{\delta_J}{1 + \ell_x(k) \mathbf{c}_v(a, k; J) / \ell_v(k)}.$$

Lemma 13.19 shows that an admissible k exists at every unbounded divisor. Thus, when $\delta_J > 1/2$, the remaining finite-place implication is the concrete inequality

$$\frac{\ell_v(k)}{\ell_x(k)} > \frac{\mathbf{c}_v(a, k; J)}{2\delta_J - 1}$$

for at least one choice of J, a, k , or an argument treating the reverse inequality. This statement applies over the function field of a variety on which all trace collisions at the original Fuchsian character persist. It allows the most frequent trace value to vary with the radius and does not presuppose a number-field form.

12.5. Volume of a single trace slab. There is a geometric reason the direct quantitative double-coset approach stalls in the compact case.

Proposition 12.21 (trace-slab volume). *Fix a nonzero $A \in M_2(\mathbb{R})$, and let $\|\cdot\|$ be any matrix norm. With respect to Haar measure on $\operatorname{SL}_2(\mathbb{R})$, as $X \rightarrow \infty$,*

$$\operatorname{vol}\{g : \|g\| \leq X, |\operatorname{tr}(gA)| \leq 1\} \asymp_A \begin{cases} X, & \det A \neq 0, \\ X \log X, & \operatorname{rank} A = 1. \end{cases} \quad (12.22)$$

Proof. Use KAK coordinates $g = k_1 \operatorname{diag}(r, r^{-1}) k_2$, where $r = e^{t/2}$. Haar density is comparable to $\sinh(t) dt dk_1 dk_2 = (r - r^{-3}) dr dk_1 dk_2$, and $\|g\| \asymp r$. Fixing one angular variable and writing $C = Ak_1 = (c_{ij})$, the other angular variable makes $\operatorname{tr}(gA)$ a sinusoid of amplitude D_r , where

$$D_r^2 = r^2 \|Ce_1\|^2 + 2 \det A + r^{-2} \|Ce_2\|^2.$$

Its angular sublevel proportion is exactly

$$\frac{2}{\pi} \arcsin\left(\min\{1, D_r^{-1}\}\right).$$

If A is invertible, $\|Ak_1e_1\|$ is bounded uniformly away from zero, and averaging gives a proportion comparable to r^{-1} . If A has rank one, the first-column factor has simple angular zeros; splitting into $|\cos \theta| \leq r^{-2}$ and dyadic complementary regions gives an averaged proportion comparable to $r^{-1} \log r$. Integration against $(r - r^{-3})dr$, for $r \ll X$, gives X and $X \log X$, respectively. $\square$

For a torsion-free cocompact Fuchsian group, the square-root and secant matrices attached to positive-trace lifts of hyperbolic elements are invertible. This is a continuous Haar-volume calculation, not a lattice-point estimate: standard mixing errors for a ball of volume $\asymp X^2$ may dominate an X -volume slab. Thus even an optimal equidistribution estimate for one such slab would give only the $O(X)$ bound compatible with bounded clustering. A parabolic secant is rank one and gains the logarithm in (12.22), explaining structurally why unipotent amplification is available in the nonuniform proofs.

13. A NOETHERIAN REDUCTION FOR EXCEPTIONAL COMPACT SURFACES

Hao's 2026 theorem proves superlinear trace growth away from a countable union of positive-codimension trace-collision loci in Teichmüller space [19]. At a point in the exceptional union, Noetherianity implies that all trace equalities holding there are consequences of finitely many polynomial equations.

Let $\mathcal{X}$ be an affine $\mathbb{Q}$ -model of the relevant PSL_2 -character component, with its open Teichmüller component $\mathcal{T} \subset \mathcal{X}(\mathbb{R})$. For a group element w , its squared-trace function $\tau_w \in \mathbb{Q}[\mathcal{X}]$ is regular. For $x \in \mathcal{T}$, let J_x be the ideal generated by

$$\tau_u - \tau_v \quad \text{for all pairs with } \tau_u(x) = \tau_v(x), \quad (13.1)$$

and put $V_x = Z(J_x) \subset \mathcal{X}$.

Theorem 13.1 (persistence of trace collisions). *The ideal J_x is generated by finitely many trace-collision equations. Every squared-trace equality holding at x holds identically on V_x .*

If x is a hyperbolic metric with trace-set entropy

$$h_{\mathrm{tr}}(x) = \limsup_{L \rightarrow \infty} \frac{1}{L} \log \#\{\text{distinct primitive absolute traces of length at most } L\} \leq \frac{1}{2}, \quad (13.2)$$

then, for every $\epsilon > 0$, all hyperbolic metrics $y \in V_x$ sufficiently near x satisfy

$$h_{\mathrm{tr}}(y) \leq (1 + \epsilon)h_{\mathrm{tr}}(x) \leq \frac{1 + \epsilon}{2}. \quad (13.3)$$

If the complex algebraic germ of V_x at x is zero-dimensional, the Fricke coordinates of x are algebraic. Since x is a smooth point of the character variety, it is enough for the gradients of finitely many equal-trace equations to span the cotangent space at x . If x has a transcendental coordinate, then $V_x(\mathbb{R}) \cap \mathcal{T}$ contains a nonconstant semialgebraic arc through x .

Proof. The coordinate ring $\mathbb{Q}[\mathcal{X}]$ is Noetherian, so a finite subfamily of (13.1) generates J_x . Every other collision difference belongs to that ideal and vanishes on V_x .

Hyperbolic lengths for two sufficiently close metrics satisfy, uniformly over all closed curves,

$$\ell_x(\gamma) \leq (1 + \epsilon)\ell_y(\gamma)$$

after shrinking the neighborhood. Every collision at x persists at y , so the partition of group elements by trace at y is coarser than the partition at x . Therefore the number of y -trace values represented by curves of y -length at most L is at most the number of x -trace values represented by curves of x -length at most $(1 + \epsilon)L$. Taking logarithmic upper growth rates proves (13.3).

A zero-dimensional complex germ through x places x on a zero-dimensional irreducible component of the $\mathbb{Q}$ -scheme V_x ; its residue field is a finite extension of $\mathbb{Q}$. The Jacobian criterion at the smooth Teichmüller point gives the stated sufficient condition. Finally, every isolated real point of a $\mathbb{Q}$ -semialgebraic set has algebraic coordinates. A transcendental x is therefore not isolated in $V_x(\mathbb{R})$; real curve selection gives a nonconstant semialgebraic arc, and a short segment remains in the open set $\mathcal{T}$. $\square$

Proposition 13.2 (bounded-degree algebraic specializations converging to a transcendental character). *For $x \in \mathcal{T}$, let*

$$I(x) = \{f \in \mathbb{Q}[\mathcal{X}] : f(x) = 0\}, \quad Z_x = V(I(x)).$$

We call Z_x the $\mathbb{Q}$ -Zariski closure, or rational closure, of x . If $\dim Z_x > 0$, there are an integer D and distinct algebraic Fuchsian points $y_j \in Z_x(\mathbb{R})$, all in the Teichmüller component of x , such that

$$y_j \longrightarrow x, \quad [\mathbb{Q}(y_j) : \mathbb{Q}] \leq D. \quad (13.3a)$$

For every finite $E \subset \pi_1(S)$,

$$\#\{\tau_\gamma(y_j) : \gamma \in E\} \leq \#\{\tau_\gamma(x) : \gamma \in E\}. \quad (13.3b)$$

In particular, if the absolute trace set at x has linear growth and

$$\mathcal{B}_R^x = \{\gamma : d_x(o, \gamma o) \leq R\},$$

then, uniformly in j and R ,

$$\#\{\tau_\gamma(y_j) : \gamma \in \mathcal{B}_R^x\} = O_x(e^{R/2}). \quad (13.3c)$$

After discarding finitely many terms, the y_j may be taken nonarithmetic.

Proof. The ideal $I(x)$ is the kernel of evaluation in $\mathbb{R}$, hence is prime and Z_x is irreducible. The point x lies on no proper $\mathbb{Q}$ -defined closed subvariety of Z_x . Since characteristic zero implies generic smoothness, x is therefore a smooth point of Z_x .

Put $d = \dim Z_x$. Choose a rational linear Noether normalization

$$\pi : Z_x \longrightarrow \mathbb{A}^d$$

which is generically étale. Its ramification locus is a proper $\mathbb{Q}$ -closed subset of Z_x , so the $\mathbb{Q}$ -generic point x does not belong to it. After shrinking the target away from the discriminant and nonflat loci, the real inverse-function theorem gives a local inverse branch through x , and every fiber has total degree $\deg \pi$. Choose distinct rational points tending to $\pi(x)$ in this neighborhood and lift them along the branch. The lifts y_j are real algebraic points of degree at most $\deg \pi$, tend to x , and remain in the open Teichmüller component after the branch is restricted.

If $\tau_\gamma(x) = \tau_{\gamma'}(x)$, then $\tau_\gamma - \tau_{\gamma'} \in I(x)$, so the same equality holds at every point of Z_x . Thus the partition at y_j is coarser than the partition at x , proving (13.3b). For $\gamma \in \mathcal{B}_R^x$,

$$|\mathrm{tr}_x \gamma| \leq 2 \cosh(R/2).$$

Linear growth of the absolute trace set at x , together with (13.3b), gives (13.3c).

Finally, the Borel–Prasad theorem gives finitely many conjugacy classes of arithmetic lattices in a fixed semisimple real group whose covolume is at most a prescribed constant [7]. Applying it to $\mathrm{PSL}_2(\mathbb{R})$ with that constant equal to $4\pi(g-1)$ gives, in particular, only finitely many conjugacy classes having the fixed covolume $4\pi(g-1)$. Each unmarked surface gives one mapping-class-group orbit in Teichmüller space, and proper discontinuity makes that orbit meet a relatively compact neighborhood of x in only finitely many points. Hence only finitely many members of the convergent sequence can be arithmetic. $\square$

Theorem 13.3 (relative deformations cannot preserve all collisions). *Let Π be torsion-free, let $H < \Pi$ be a subgroup, and let*

$$\rho_s : \Pi \longrightarrow \mathrm{SL}_2(\mathbb{R}), \quad s \in U,$$

be a continuous family of lifts of faithful Fuchsian representations, where U is connected and contains 0. Suppose that $\rho_s|_H = \rho_0|_H$ for every s , and that $\rho_0(H)$ is convex cocompact. For $a \in \Pi$, put

$$W_a = \mathrm{span}_{\mathbb{R}}\{\rho_s(a) : s \in U\} \subset M_2(\mathbb{R}).$$

Assume that $\dim W_a \geq 3$ and that every collision in the coset aH at $s = 0$ persists throughout the family:

$$\mathrm{tr}^2 \rho_0(ah) = \mathrm{tr}^2 \rho_0(ah') \implies \mathrm{tr}^2 \rho_s(ah) = \mathrm{tr}^2 \rho_s(ah') \quad (s \in U). \quad (13.4)$$

Then every fiber of

$$h \longmapsto \mathrm{tr}^2 \rho_0(ah), \quad h \in H,$$

has at most three elements. If $W_a = M_2(\mathbb{R})$, every fiber has one element.

If, in addition, every ah , $h \in H$, is primitive in Π , then the primitive length-set entropy of $\rho_0(\Pi)$ is at least the critical exponent δ_H of $\rho_0(H)$.

Proof. Fix h_0 in one fiber and write $X_h = \rho_0(h)$. These matrices do not depend on s . The traces in (13.4) never vanish: a nonidentity element in a torsion-free Fuchsian group is hyperbolic, while the identity has trace 2. Connectedness of U therefore gives, for each h in the fiber, a sign $\epsilon_h \in \{\pm 1\}$, independent of s , such that

$$\mathrm{tr}(\rho_s(a)(X_h - \epsilon_h X_{h_0})) = 0 \quad (s \in U).$$

For the nondegenerate pairing $(A, B) \mapsto \mathrm{tr}(AB)$, this says

$$X_h - \epsilon_h X_{h_0} \in W_a^\perp. \quad (13.5)$$

If $\dim W_a = 4$, the right side is zero. Projective faithfulness then gives $h = h_0$.

Suppose $\dim W_a = 3$, and write $W_a^\perp = \mathbb{R}D$ with $D \neq 0$. For a fixed sign, every possible matrix has the form

$$X_h = \epsilon X_{h_0} + cD, \quad \det(\epsilon X_{h_0} + cD) = 1. \quad (13.6)$$

The determinant in (13.6) is a polynomial of degree at most two in c , and $c = 0$ is a root. If $\det D \neq 0$, there is at most one further root for each sign. If $\det D = 0$, the polynomial is linear unless it is identically zero. In the latter case, after writing

$$\epsilon X_{h_0} + cD = \epsilon X_{h_0}(I + cN),$$

the identity of determinants implies $N \neq 0$ and $N^2 = 0$. Two distinct values of c arising from elements of $\rho_0(H)$ would then have a quotient of the form $I + c'N$, a nontrivial unipotent. A convex-cocompact Fuchsian group has no such element. Thus there is again at most one nonzero extra root for each sign. The two roots $c = 0$, one for each sign, represent the same projective element. Consequently the projective fiber has at most three elements.

Finally,

$$\#\{h \in H : d(o, \rho_0(h)o) \leq R\} = e^{(\delta_H + o(1))R}.$$

Moreover $\ell_{\rho_0}(ah) \leq d(o, \rho_0(a)o) + d(o, \rho_0(h)o)$. If all ah are primitive, the uniform fiber bound just proved therefore gives $e^{(\delta_H + o(1))R}$ distinct primitive lengths below $R + O(1)$, proving the entropy assertion. $\square$

For a finitely generated group G , write $\mathfrak{X}(G)$ for its affine $\mathrm{PSL}_2(\mathbb{C})$ -character variety. For $g \in G$, its regular squared-trace coordinate is

$$\tau_g([\sigma]) = \mathrm{tr}^2(\tilde{\sigma}(g)) = \mathrm{tr}(\mathrm{Ad} \sigma(g)) + 1,$$

where either lift of $\sigma(g)$ to $\mathrm{SL}_2(\mathbb{C})$ gives the same square.

Theorem 13.4 (relative fiber bound from two trace evaluations). *Let H be finitely generated, let $F/\mathbb{Q}$ be a finitely generated field of characteristic zero, and let $\rho: H \rightarrow \mathrm{SL}_2(F)$. Assume that there is a faithful torsion-free convex-cocompact Fuchsian character x_H such that*

$$\mathrm{tr}^2 \rho(g) = 4 \text{ in } F \implies \tau_g(x_H) = 4 \quad (g \in H).$$

Thus every squared-trace identity of the form $\mathrm{tr}^2 \rho(g) = 4$ remains valid at x_H . This implication is the only relation between ρ and x_H used below; no homomorphism from the field F to $\mathbb{R}$ is being assumed. Let $A_0, A_1 \in \mathrm{SL}_2(F)$. For the conclusion about a fiber of the first trace coordinate alone, assume in addition that

$$\mathrm{tr}^2(A_0\rho(h)) = \mathrm{tr}^2(A_0\rho(h')) \implies \mathrm{tr}^2(A_1\rho(h)) = \mathrm{tr}^2(A_1\rho(h')). \quad (13.7)$$

Put

$$B = A_1 A_0^{-1} \quad \text{and assume} \quad \mathrm{tr}^2 B \neq 4. \quad (13.8)$$

Then, for every word metric on H , every simultaneous squared-trace fiber satisfies

$$\sup_{c_0, c_1} \#\{h \in H : |h| \leq n, \mathrm{tr}^2(A_i \rho(h)) = c_i \ (i = 0, 1)\} = \exp(o(n)).$$

If (13.7) holds, then also

$$\sup_t \#\{h \in H : |h| \leq n, \mathrm{tr}^2(A_0 \rho(h)) = t\} = \exp(o(n)). \quad (13.9)$$

Condition (13.8) says exactly that the relative matrix $A_1 A_0^{-1}$ has two distinct eigenvalues; no semisimplicity hypothesis is imposed on either representation as a whole.

In particular, the theorem applies when ρ itself is a faithful convex-cocompact Fuchsian representation. If $A_i = \rho_i(a)$ are two evaluations of a deformation fixed on H , all collisions at ρ_0 persist at ρ_1 , and every ah is primitive, then the primitive length-set entropy at ρ_0 is at least δ_H .

Proof. Fix h_* in a nonempty simultaneous squared-trace fiber. After dividing the fiber into at most four parts according to the two possible signs at each trace evaluation, the two quantities

$$\mathrm{tr}(A_0 \rho(h)), \quad \mathrm{tr}(A_1 \rho(h))$$

are fixed. Put $Y = A_0 \rho(h)$. Since $A_1 \rho(h) = BY$, both $\mathrm{tr} Y$ and $\mathrm{tr}(BY)$ are fixed on such a part.

After a finite extension of F , condition (13.8) diagonalizes B with two distinct eigenvalues. Writing

$$Y = \begin{pmatrix} y_{11} & y_{12} \\ y_{21} & y_{22} \end{pmatrix},$$

the two fixed traces determine y_{11} and y_{22} , while $\det Y = 1$ becomes

$$y_{12}y_{21} = y_{11}y_{22} - 1 =: q. \quad (13.10)$$

The entries of a word of length n , and the fixed linear transforms used here, belong to one of the finite-type sets $R_C[n]$ of Section 12. More explicitly, in a nonempty radius- n fiber one may use h_* to define the two diagonal entries and q ; all have finite-type complexity $O(n)$. The fixed diagonalizing matrix is absorbed into the same finite extension and finite-type model. If $q \neq 0$, Theorem 12.1 gives $\exp(o(n))$ possibilities for the ordered factorization in (13.10), uniformly in q .

If $q = 0$, the solutions form two affine lines, according as $y_{12} = 0$ or $y_{21} = 0$. Two distinct matrices Y, Y' on one of these lines satisfy $\det(Y - Y') = 0$. Therefore

$$0 = \det(Y - Y') = 2 - \mathrm{tr}(YY'^{-1}) = 2 - \mathrm{tr}(\rho(h)\rho(h')^{-1}).$$

Thus $\mathrm{tr}^2 \rho(hh'^{-1}) = 4$ in F , and the implication in the statement gives $\tau_{hh'^{-1}}(x_H) = 4$. Faithfulness and convex cocompactness at x_H force $h = h'$. Hence the degenerate part contributes at most two elements for each of the four sign choices. This proves the simultaneous fiber assertion. Under (13.7), every fiber in (13.9) is a simultaneous fiber, so (13.9) follows.

Word length and orbit displacement on the convex-cocompact group H are quasi-isometric. Dividing its $e^{(\delta_H + o(1))R}$ orbit points by the subexponential fiber bound, and using the primitivity of ah , proves the final assertion. $\square$

The preceding proof only uses finite generation to bound the arithmetic complexity of the matrices being counted. We shall need the following version for a subset of an infinitely generated subgroup. Word length is always measured in the fixed ambient group in this statement.

Lemma 13.5 (two trace evaluations on subsets of ambient word balls). *Let Π be finitely generated, let $F/\mathbb{Q}$ be a finitely generated field of characteristic zero, and let $\rho: \Pi \rightarrow \mathrm{SL}_2(F)$. Suppose that x is a faithful, torsion-free Fuchsian character of Π such that*

$$\mathrm{tr}^2 \rho(g) = 4 \text{ in } F \implies \tau_g(x) = 4.$$

Fix a word metric on Π , a constant C , and subsets

$$E_R \subset \{g \in \Pi : |g|_\Pi \leq CR + C\}.$$

If $A_0, A_1 \in \mathrm{SL}_2(F)$ and $\mathrm{tr}^2(A_1 A_0^{-1}) \neq 4$, then, uniformly in c_0, c_1 ,

$$\#\{h \in E_R : \mathrm{tr}^2(A_i \rho(h)) = c_i \ (i = 0, 1)\} = \exp(o(R)).$$

No finite-generation hypothesis is made on the subgroup containing the sets E_R .

Proof. Repeat the proof of Theorem 13.4, dividing into the four possible sign classes. After diagonalizing $A_1 A_0^{-1}$, the two traces determine the diagonal entries of $A_0 \rho(h)$, and the determinant leaves one factorization equation $y_{12} y_{21} = q$. The ambient word bound puts every matrix entry in a finite-type filtration of index $O(R)$, so the divisor theorem gives $\exp(o(R))$ solutions when $q \neq 0$. When $q = 0$, two points on one of the two resulting affine lines have relative squared trace 4 in F . The displayed specialization implication and faithfulness of x make them the same projective matrix and then the same element of Π . The proof therefore does not require a finite generating set for the union of the E_R . $\square$

The preceding subexponential estimate can be sharpened geometrically in a fixed Fuchsian representation. The sharpening also prevents a misleading commensurator argument: an unbounded two-trace fiber can be produced inside every lattice by conjugating with powers of the first multiplier.

Theorem 13.6 (decomposition and uniform count of internal centralizer orbits). *Let $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ be torsion-free and convex cocompact, let $q \in \Gamma$ be hyperbolic, and choose compatible lifts to $\mathrm{SL}_2(\mathbb{R})$. Write*

$$C_\Gamma(q) = \langle z \rangle,$$

where z is primitive. For $u, v \in \mathbb{R}$, put

$$\mathcal{F}_{u,v}(q) = \{g \in \Gamma : \mathrm{tr} g = u, \mathrm{tr}(qg) = v\}.$$

Then every $\mathcal{F}_{u,v}(q)$ is a finite union of $\langle z \rangle$ -conjugacy orbits. Moreover, for any fixed word metric there is $C = C(\Gamma, q)$, independent of u, v and of the orbit, such that

$$\#(\mathcal{O} \cap B_\Gamma(R)) \leq C(R + 1) \tag{13.10a1}$$

for every such orbit $\mathcal{O}$. Consequently, if $N_{u,v}(R)$ is the number of these orbits which meet $B_\Gamma(R)$, then

$$\#(\mathcal{F}_{u,v}(q) \cap B_\Gamma(R)) \leq C(R + 1) N_{u,v}(R). \tag{13.10a2}$$

The same conclusions, after multiplying C by at most four, hold when either trace is prescribed only up to sign and hence for fibers of $(\mathrm{tr}^2 g, \mathrm{tr}^2(qg))$.

Proof. Conjugate so that

$$z = \begin{pmatrix} \theta & 0 \\ 0 & \theta^{-1} \end{pmatrix}, \quad \theta > 1,$$

and write $q = z^m$ projectively. Up to a harmless sign in the chosen lift, put $q = \text{diag}(\lambda, \lambda^{-1})$, $\lambda > 1$. If $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the two trace equations are

$$a + d = u, \quad \lambda a + \lambda^{-1}d = v. \quad (13.10a3)$$

They determine a, d , and the determinant equation becomes

$$bc = ad - 1 =: \kappa. \quad (13.10a4)$$

If $\kappa \neq 0$, conjugation by z^n sends (b, c) to $(\theta^{2n}b, \theta^{-2n}c)$. Every orbit therefore has a unique representative, apart from one boundary convention, for which

$$1 \leq |b/c| < \theta^4. \quad (13.10a5)$$

For fixed u, v , equations (13.10a3)–(13.10a5) put these representatives in a compact subset of $\text{SL}_2(\mathbb{R})$. Discreteness of Γ makes their number finite. If $\kappa = 0$, a nondiagonal point is triangular and its commutator with z is a nonidentity unipotent. A convex-cocompact group has no such element. The degenerate fiber therefore contains at most one point.

It remains to make the orbit count uniform in u, v . Positive systole gives $\epsilon_\Gamma > 0$ such that

$$|\text{tr } \gamma| \geq 2 + \epsilon_\Gamma \quad (1 \neq \gamma \in \Gamma).$$

Direct multiplication gives

$$\text{tr}[z, g] = 2 - bc(\theta - \theta^{-1})^2. \quad (13.10a6)$$

The commutator is nontrivial in the nondegenerate fiber, so

$$|bc| \geq \frac{\epsilon_\Gamma}{(\theta - \theta^{-1})^2} =: \kappa_0. \quad (13.10a7)$$

For the representative in (13.10a5), both off-diagonal entries consequently have absolute value at least $c_0 = \theta^{-2}\sqrt{\kappa_0}$. In a fixed matrix norm,

$$\|z^n g z^{-n}\| \geq c_0 \theta^{2|n|}.$$

On the other hand, $\|\gamma\| \leq AM^{|\gamma|}$ for fixed $A, M > 1$. Membership in $B_\Gamma(R)$ therefore forces $|n| = O_{\Gamma, q}(R + 1)$, proving (13.10a1) and (13.10a2). $\square$

The linear factor is sharp: for noncommuting hyperbolic $q, h \in \Gamma$, all matrices $q^n h q^{-n}$ have the same two traces

$$\text{tr}(q^n h q^{-n}) = \text{tr } h, \quad \text{tr}(q q^n h q^{-n}) = \text{tr}(qh).$$

Thus even a nonarithmetic cocompact lattice has an infinite two-trace fiber with $\asymp R$ points in a radius- R word ball. To deduce arithmeticity, the same transfer parameter must conjugate one fixed finite-index subgroup back into Γ .

Proposition 13.7 (transporting one finite-index subgroup forces arithmeticity). *Let $\Gamma < \text{PSL}_2(\mathbb{R})$ be a lattice, let $q \in \Gamma$ be hyperbolic, put $T = C_{\text{PSL}_2(\mathbb{R})}(q)^\circ$, and put $Z = T \cap \Gamma = \langle z \rangle$. If $\Lambda < \Gamma$ has finite index, define*

$$\mathcal{M}_\Lambda(T) = \{Zt \in Z \backslash T : t\Lambda t^{-1} \subset \Gamma\}.$$

If Γ is nonarithmetic, then $\mathcal{M}_\Lambda(T)$ is finite. Equivalently, infinitely many such transfer parameters force Γ to be arithmetic. More quantitatively, for every proper additive coordinate $s : T \simeq \mathbb{R}$, a nonarithmetic lattice satisfies

$$\#\{t \in T : |s(t)| \leq R, t\Lambda t^{-1} \subset \Gamma\} = O_{\Gamma, q, \Lambda}(R). \quad (13.10a8)$$

Proof. If $t\Lambda t^{-1} \subset \Gamma$, then the subgroup on the left is a lattice and consequently has finite index in the lattice Γ . Since Λ has finite index in Γ , this says precisely that $t\Gamma t^{-1}$ and Γ are commensurable. Thus $t \in \text{Comm}(\Gamma)$. For a nonarithmetic Fuchsian lattice, the commensurator theorem makes $\text{Comm}(\Gamma)$ discrete. Hence $T \cap \text{Comm}(\Gamma)$ is a discrete subgroup of the one-dimensional split torus which contains the lattice Z ; its quotient by Z is finite, and it has only $O(R)$ elements in a coordinate interval of radius R . $\square$

The use of one fixed subgroup Λ in Proposition 13.7 is essential. A point $tg_0t^{-1} \in \Gamma$ transports only one cyclic subgroup. It gives no reason for t to transport a fixed lattice back into Γ , even if many unrelated values of t occur.

Theorem 13.8 (an entropy bound from a loxodromic element at a divisorial valuation). *Let Π be a closed surface group and let $x = [\rho_x]$ be a faithful cocompact Fuchsian character. Let $F/\mathbb{Q}$ be a finitely generated field of characteristic zero and let $\rho: \Pi \rightarrow \text{SL}_2(F)$ be a representation with the following two compatibility properties:*

$$\begin{aligned} \tau_u(x) = \tau_{u'}(x) &\implies \text{tr}^2 \rho(u) = \text{tr}^2 \rho(u') \quad \text{in } F, \\ \text{tr}^2 \rho(g) = 4 \quad \text{in } F &\implies \tau_g(x) = 4. \end{aligned}$$

The first property says that all trace collisions at the Fuchsian character persist in ρ ; the second says that trace identities of ρ specialize back to x .

Let $\phi: \Pi \twoheadrightarrow \mathbb{Z}$, let $J < \ker \phi$ be finitely generated, nonelementary, and convex cocompact under ρ_x , and write δ_J for its critical exponent. Fix $o \in \mathbb{H}^2$, put

$$\mathcal{B}_J(R) = \{h \in J : d(o, \rho_x(h)o) \leq R\},$$

and choose $a \in \phi^{-1}(1)$.

Let v be a divisorial valuation of F . Suppose that for some $k \in \ker \phi$, after a finite field extension and an extension of v , the matrix $D = \rho(k)$ has a spectral decomposition

$$D = \lambda P_+ + \lambda^{-1} P_-, \quad P_+ + P_- = I, \quad |\lambda|_v > 1.$$

Here and below the same letter v denotes the extended absolute value. Set

$$A = \rho(a), \quad L_{\pm}(Y) = \text{tr}(AP_{\pm}Y), \quad \ell_v(k) = 2 \log |\lambda|_v,$$

and let $\ell_x(k)$ be the translation length of $\rho_x(k)$ in $\mathbb{H}^2$. Define

$$\begin{aligned} C_v(R) = \sup \left\{ \log^+ \left| \frac{L_-(\rho(h) - \epsilon \rho(h'))}{L_+(\rho(h) - \epsilon \rho(h'))} \right|_v : h, h' \in \mathcal{B}_J(R), \right. \\ \left. \begin{aligned} &L_+(\rho(h) - \epsilon \rho(h')) \neq 0, \\ &L_-(\rho(h) - \epsilon \rho(h')) \neq 0 \end{aligned} \right\}, \end{aligned}$$

where the supremum of the empty set is zero and $\log^+ t = \max\{0, \log t\}$, and put

$$\mathbf{c}_v(a, k; J) = \limsup_{R \rightarrow \infty} \frac{C_v(R)}{R}.$$

Then $\mathbf{c}_v(a, k; J) < \infty$, and the primitive trace entropy of x satisfies

$$h_{\text{tr}}(x) \geq \frac{\delta_J}{1 + \ell_x(k) \mathbf{c}_v(a, k; J) / \ell_v(k)}.$$

More precisely, if $c > \mathbf{c}_v(a, k; J) / \ell_v(k)$, $n_R = \lceil cR \rceil$, and $w_R = ak^{n_R}$, then

$$\sup_t \# \{h \in \mathcal{B}_J(R) : \tau_{w_R h}(x) = t\} = \exp(o(R)).$$

Proof. We first justify the finiteness assertion. Choose a finite symmetric generating set of J . Convex cocompactness makes word length on J quasi-isometric to the orbit metric, so the word length of every $h \in \mathcal{B}_J(R)$ is $O(R)$. Choose a normal finite-type model of the coefficient field, after the fixed finite extension used to split D , and the filtration $R_C[n]$ of (12.1). Repeated multiplication by the finitely many generator matrices shows that the entries of $\rho(h)$, their differences, and their images under the two fixed linear forms $L_\pm$ belong to $R_{C'}[O(R)]$.

The absolute order at a fixed prime divisor of every nonzero element in this filtration is $O(R)$. This is the valuation estimate used in the proof of Theorem 12.1: horizontal orders are bounded by the degree, vertical orders by the affine coefficient height, and passage through the fixed finite integral extension changes only the constant. Hence the v -absolute value of every nonzero quotient occurring in $C_v(R)$ has logarithm $O(R)$. This proves $\mathbf{c}_v(a, k; J) < \infty$.

Fix $c > \mathbf{c}_v(a, k; J)/\ell_v(k)$. For every sufficiently large R ,

$$C_v(R) < n_R \ell_v(k), \quad n_R = \lceil cR \rceil.$$

Suppose that $h, h' \in \mathcal{B}_J(R)$ satisfy $\tau_{w_R h}(x) = \tau_{w_R h'}(x)$. Persistence turns this into an equality of squared traces in F , so in characteristic zero there is an $\epsilon \in \{1, -1\}$ for which

$$\mathrm{tr}(AD^{n_R}\rho(h)) = \epsilon \mathrm{tr}(AD^{n_R}\rho(h')).$$

With $\Delta = \rho(h) - \epsilon\rho(h')$, the spectral decomposition of D gives

$$\lambda^{n_R} L_+(\Delta) + \lambda^{-n_R} L_-(\Delta) = 0.$$

If both coefficients were nonzero, then

$$\log \left| \frac{L_-(\Delta)}{L_+(\Delta)} \right|_v = 2n_R \log |\lambda|_v = n_R \ell_v(k),$$

contrary to the preceding strict inequality. If one coefficient is zero, the displayed trace equation makes the other zero as well. Thus every collision satisfies

$$L_+(\rho(h)) = \epsilon L_+(\rho(h')), \quad L_-(\rho(h)) = \epsilon L_-(\rho(h')).$$

Choose μ in a further fixed finite extension with $\mu^2 \neq 1$, and put

$$D_\mu = \mu P_+ + \mu^{-1} P_-, \quad A_0 = A, \quad A_1 = AD_\mu.$$

The two traces

$$\mathrm{tr}(A_0 Y) = L_+(Y) + L_-(Y), \quad \mathrm{tr}(A_1 Y) = \mu L_+(Y) + \mu^{-1} L_-(Y)$$

are independent linear combinations of $L_+(Y)$ and $L_-(Y)$. Moreover

$$A_1 A_0^{-1} = AD_\mu A^{-1}, \quad \mathrm{tr}^2(A_1 A_0^{-1}) = (\mu + \mu^{-1})^2 \neq 4.$$

After fixing one element h' in a trace fiber, the two coefficient equalities therefore put the whole fiber in one simultaneous *squared*-trace fiber for the two fixed evaluations

$$h \mapsto (\mathrm{tr}^2(A_0 \rho(h)), \mathrm{tr}^2(A_1 \rho(h))).$$

The sign may vary with h , which is harmless because both coordinates are squared. The specialization hypothesis in Theorem 13.4 is exactly the second compatibility property in the statement. That theorem, and the quasi-isometry between word length and displacement in the Fuchsian representation on J , give the asserted $\exp(o(R))$ fiber estimate.

Orbit counting gives $\#\mathcal{B}_J(R) = \exp((\delta_J + o(1))R)$. Every $w_R h$ is primitive: if $w_R h = u^m$ with $|m| \geq 2$, then $1 = \phi(w_R h) = m\phi(u)$, which is impossible. Finally,

$$\ell_x(w_R h) \leq d(o, \rho_x(a)o) + d(o, \rho_x(k^{n_R})o) + d(o, \rho_x(h)o) \leq (1 + c\ell_x(k) + o(1))R,$$

because $d(o, \rho_x(k^n)o) = n\ell_x(k) + O(1)$. Dividing the orbit count by the subexponential fiber bound, and then letting $c \downarrow \mathbf{c}_v(a, k; J)/\ell_v(k)$, proves the entropy bound. $\square$

Corollary 13.9 (a quantitative criterion excluding an unbounded divisor). *In the setting of Theorem 13.8, suppose that $\delta_J > 1/2$ and*

$$\frac{\ell_v(k)}{\ell_x(k)} > \frac{\mathbf{c}_v(a, k; J)}{2\delta_J - 1}.$$

Then $h_{\text{tr}}(x) > 1/2$. Consequently, if $h_{\text{tr}}(x) \leq 1/2$, no divisor along which all trace collisions at x persist can satisfy this inequality. Equivalently, every unbounded divisor compatible with those collisions, and every admissible v -loxodromic k , must satisfy the exponential-cancellation lower bound

$$\mathbf{c}_v(a, k; J) \geq (2\delta_J - 1) \frac{\ell_v(k)}{\ell_x(k)}.$$

Proof. The displayed inequality says exactly that the lower bound in Theorem 13.8 is strictly greater than $1/2$. $\square$

The preceding theorem controls every possible cancellation by its largest valuation. The collision equation itself is more rigid: it asks whether a particular quotient is an *exact* even power of λ . The first exponent for which no such identity occurs gives a sharper estimate.

Theorem 13.10 (the first exponent not realized by an exact cancellation identity). *Retain the hypotheses and notation of Theorem 13.8. For $R > 0$, let $\mathcal{E}_{v,J}(R) \subset \mathbb{Z}_{\geq 0}$ be the set of integers n for which there are distinct $h, h' \in \mathcal{B}_J(R)$ and $\epsilon \in \{1, -1\}$ such that, for*

$$\Delta = \rho(h) - \epsilon\rho(h'),$$

both $L_+(\Delta)$ and $L_-(\Delta)$ are nonzero and

$$-\frac{L_-(\Delta)}{L_+(\Delta)} = \lambda^{2n}. \quad (13.10a)$$

Define

$$m_{v,J}(R) = \min(\mathbb{Z}_{\geq 0} \setminus \mathcal{E}_{v,J}(R)), \quad \mathbf{m}_v(a, k; J) = \liminf_{R \rightarrow \infty} \frac{m_{v,J}(R)}{R}. \quad (13.10b)$$

Then $m_{v,J}(R) < \infty$,

$$\mathbf{m}_v(a, k; J) \leq \frac{\mathbf{c}_v(a, k; J)}{\ell_v(k)}, \quad (13.10c)$$

and

$$h_{\text{tr}}(x) \geq \frac{\delta_J}{1 + \ell_x(k)\mathbf{m}_v(a, k; J)}. \quad (13.10d)$$

More precisely, if

$$n_R = m_{v,J}(R), \quad w_R = ak^{n_R},$$

then

$$\sup_t \#\{h \in \mathcal{B}_J(R) : \tau_{w_R h}(x) = t\} = \exp(o(R)). \quad (13.10e)$$

Consequently, if $h_{\text{tr}}(x) \leq 1/2$ and $\delta_J > 1/2$, then

$$\mathbf{m}_v(a, k; J) \geq \frac{2\delta_J - 1}{\ell_x(k)}. \quad (13.10f)$$

Equivalently, for every $0 < \eta < 2\delta_J - 1$ and every sufficiently large R , each integer

$$0 \leq n < \frac{(2\delta_J - 1 - \eta)R}{\ell_x(k)} \quad (13.10g)$$

is realized by an exact identity of the form (13.10a).

Proof. For a fixed triple (h, h', ϵ) , equation (13.10a) can hold for at most one n , because $|\lambda|_v > 1$. Thus $\mathcal{E}_{v,J}(R)$ is finite. More quantitatively, if $n \in \mathcal{E}_{v,J}(R)$, then

$$n\ell_v(k) = \log \left| \frac{L_-(\Delta)}{L_+(\Delta)} \right|_v \leq C_v(R).$$

It follows that

$$m_{v,J}(R) \leq 1 + \left\lfloor \frac{C_v(R)}{\ell_v(k)} \right\rfloor,$$

which proves finiteness and (13.10c).

Put $n_R = m_{v,J}(R)$. Suppose that distinct $h, h' \in \mathcal{B}_J(R)$ satisfy $\tau_{w_R h}(x) = \tau_{w_R h'}(x)$. Persistence of trace collisions gives

$$\mathrm{tr}(AD^{n_R}\rho(h)) = \epsilon \mathrm{tr}(AD^{n_R}\rho(h'))$$

for some $\epsilon \in \{1, -1\}$. Hence, with $\Delta = \rho(h) - \epsilon\rho(h')$,

$$\lambda^{n_R}L_+(\Delta) + \lambda^{-n_R}L_-(\Delta) = 0. \quad (13.10h)$$

If both coefficients were nonzero, (13.10h) would say precisely that $n_R \in \mathcal{E}_{v,J}(R)$, contrary to its definition. If either coefficient vanishes, (13.10h) forces the other to vanish as well. Thus every nontrivial collision satisfies

$$L_+(\rho(h)) = \epsilon L_+(\rho(h')), \quad L_-(\rho(h)) = \epsilon L_-(\rho(h')). \quad (13.10i)$$

Choose, in a fixed finite extension, an element μ with $\mu^2 \neq 1$, and put

$$D_\mu = \mu P_+ + \mu^{-1}P_-, \quad A_0 = A, \quad A_1 = AD_\mu.$$

The two traces against A_0, A_1 are independent linear combinations of L_+, L_- , and

$$\mathrm{tr}^2(A_1 A_0^{-1}) = \mathrm{tr}^2(AD_\mu A^{-1}) = (\mu + \mu^{-1})^2 \neq 4.$$

Fixing one member of a trace fiber, (13.10i) places every other member in the same simultaneous squared-trace fiber for A_0, A_1 . Lemma 13.5, applied to $\mathcal{B}_J(R)$, proves (13.10e).

Every $w_R h$ is primitive because $\phi(w_R h) = 1$. Moreover

$$\ell_x(w_R h) \leq R + m_{v,J}(R)\ell_x(k) + O(1). \quad (13.10j)$$

Choose a sequence $R_j \rightarrow \infty$ along which

$$\frac{m_{v,J}(R_j)}{R_j} \rightarrow \mathfrak{m}_v(a, k; J).$$

Convex-cocompact orbit counting gives

$$\#\mathcal{B}_J(R_j) = \exp((\delta_J + o(1))R_j).$$

Dividing by the fiber bound (13.10e), using (13.10j), and taking the limsup in the definition of h_{tr} proves (13.10d). Rearranging (13.10d) under the assumption $h_{\mathrm{tr}}(x) \leq 1/2$ gives (13.10f). Since $m_{v,J}(R)$ is the first absent integer, (13.10g) follows directly from the definition of the liminf. $\square$

Theorem 13.11 (an internal multiplier produces all initial cancellation exponents). *Retain the hypotheses and notation of Theorem 13.8, and suppose in addition that $k \in J$. Assume that there are distinct $u, v \in J$ and a sign $\epsilon \in \{1, -1\}$ such that*

$$\mathrm{tr}(A\rho(u)) = \epsilon \mathrm{tr}(A\rho(v))$$

and, for $\Delta_0 = \rho(u) - \epsilon\rho(v)$,

$$L_+(\Delta_0)L_-(\Delta_0) \neq 0.$$

Then there is a constant C , independent of n and R , such that

$$\left\{ 0, 1, \dots, \left\lfloor \frac{R-C}{\ell_x(k)} \right\rfloor \right\} \subset \mathcal{E}_{v,J}(R) \quad (13.10ab)$$

for all sufficiently large R . In particular,

$$\liminf_{R \rightarrow \infty} \frac{m_{v,J}(R)}{R} \geq \frac{1}{\ell_x(k)}. \quad (13.10ac)$$

If $h_{\text{tr}}(x) < \delta_J$, then such u, v, ϵ necessarily exist. In particular they exist when $h_{\text{tr}}(x) \leq 1/2 < \delta_J$. Consequently, under low trace entropy, the method of Theorem 13.10 cannot prove an entropy bound greater than $1/2$ for a subgroup containing the multiplier k .

Proof. For $n \geq 0$, put

$$h_n = k^{-n}u, \quad h'_n = k^{-n}v.$$

Both elements belong to J . Since $D^{-n} = \lambda^{-n}P_+ + \lambda^n P_-$ and the spectral projectors commute with D , one has

$$\begin{aligned} L_+(\rho(h_n) - \epsilon \rho(h'_n)) &= \lambda^{-n} L_+(\Delta_0), \\ L_-(\rho(h_n) - \epsilon \rho(h'_n)) &= \lambda^n L_-(\Delta_0). \end{aligned}$$

The assumed trace equality is the identity $L_+(\Delta_0) + L_-(\Delta_0) = 0$. The two coefficients are nonzero, so

$$-\frac{L_-(\rho(h_n) - \epsilon \rho(h'_n))}{L_+(\rho(h_n) - \epsilon \rho(h'_n))} = \lambda^{2n}.$$

Moreover

$$d(o, \rho_x(k^{-n}u)o), \quad d(o, \rho_x(k^{-n}v)o) \leq n\ell_x(k) + C,$$

because $d(o, \rho_x(k^n)o) = n\ell_x(k) + O(1)$ and u, v are fixed. This proves (13.10ab) and (13.10ac).

It remains to obtain the fixed collision from low trace entropy. Write $s = h_{\text{tr}}(x) < \delta_J$. Convex-cocompact orbit counting gives

$$\#\mathcal{B}_J(T) = \exp((\delta_J + o(1))T).$$

Every ah , $h \in \mathcal{B}_J(T)$, is primitive because $\phi(ah) = 1$, and its length in the Fuchsian representation is at most $T + O(1)$. Hence these elements assume at most $\exp((s + o(1))T)$ squared-trace values at x . A squared-trace fiber $\mathcal{F}_T$ consequently has size

$$|\mathcal{F}_T| \geq \exp((\delta_J - s - o(1))T). \quad (13.10ad)$$

Fix $v_0 \in \mathcal{F}_T$. Persistence of collisions implies that, for each $u \in \mathcal{F}_T$, one can choose $\epsilon_u \in \{1, -1\}$ with

$$\text{tr}(A\rho(u)) = \epsilon_u \text{tr}(A\rho(v_0)).$$

Choose in a fixed finite extension an element μ with $\mu^2 \neq 1$, and put

$$D_\mu = \mu P_+ + \mu^{-1} P_-, \quad A_0 = A, \quad A_1 = AD_\mu.$$

Suppose, contrary to the assertion, that every $u \in \mathcal{F}_T$ satisfied

$$L_\pm(\rho(u)) = \epsilon_u L_\pm(\rho(v_0)).$$

Then $\mathcal{F}_T$ would lie in one simultaneous squared-trace fiber for A_0, A_1 . Since

$$\text{tr}^2(A_1 A_0^{-1}) = (\mu + \mu^{-1})^2 \neq 4,$$

Lemma 13.5 would give $|\mathcal{F}_T| = \exp(o(T))$, contradicting (13.10ad). Thus for some $u \in \mathcal{F}_T$ at least one of the two coefficient equalities fails. Take $v = v_0$ and $\epsilon = \epsilon_u$. The trace equality says that the sum of the two coefficient differences is zero, so if one is nonzero, then both are nonzero. This completes the proof. $\square$

Remark 13.12 (why the multiplier must lie outside the subgroup). The obstruction is internal multiplication, not specifically conjugacy. To exclude it in a way stable under replacing k by a positive power, one must at least require

$$\langle k \rangle \cap J = \{1\}. \quad (13.10ae)$$

Indeed, if $k^q \in J$, the theorem applies to the equally admissible multiplier k^q , and both its Fuchsian and its v -adic translation length are multiplied by q . Condition (13.10ae) is therefore a necessary elementary separation condition for this argument. By itself, it says nothing about cancellation identities arising from unrelated pairs.

This separation is compatible with a critical exponent as close to one as desired. We record a form which also separates the chosen subgroup from Fuchsian conjugacy relations that could imitate multiplication by k . For $w \in \Gamma$, write

$$\mathcal{L}_w(\Gamma) = \{w^{-1}\gamma w\gamma^{-1} : \gamma \in \Gamma\}.$$

Lemma 13.13 (separated limit sets and twisted commutators). *Let $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ be discrete, let $w \in \Gamma$ be hyperbolic, and let $H < \Gamma$ be nonelementary and convex cocompact. If*

$$\Lambda(H) \cap w^{-1}\Lambda(H) = \emptyset, \quad (13.10af)$$

then $H \cap \mathcal{L}_w(\Gamma)$ is finite.

Proof. Suppose that there are distinct $c_i = w^{-1}\gamma_i w\gamma_i^{-1} \in H$, and put $b_i = wc_i = \gamma_i w\gamma_i^{-1}$. The b_i are distinct hyperbolic elements, all with translation length $\ell(w)$. Their axes leave every compact subset of $\mathbb{H}^2$. Indeed, if the axes met one fixed compact set, the displacement of a fixed basepoint under the b_i would be uniformly bounded; discreteness would then leave only finitely many possibilities for b_i .

Write the endpoints of the axis of b_i as x_i, y_i . After passage to a subsequence they converge to $x, y \in \partial\mathbb{H}^2$. If $x \neq y$, the geodesics with endpoints x_i, y_i meet a common compact neighborhood of the geodesic with endpoints x, y , a contradiction. Thus $x = y = \xi$. A hyperbolic element is determined on the boundary by its two fixed points and its multiplier. Since the multiplier here is fixed, the cross-ratio formula shows that

$$b_i z \longrightarrow \xi, \quad b_i^{-1} z \longrightarrow \xi$$

locally uniformly on $\partial\mathbb{H}^2 \setminus \{\xi\}$. Consequently

$$c_i z = w^{-1}b_i z \longrightarrow w^{-1}\xi, \quad c_i^{-1} z = b_i^{-1}wz \longrightarrow \xi$$

away from the corresponding exceptional points. Applying these limits to points of the closed, H -invariant, nonelementary set $\Lambda(H)$ shows that both ξ and $w^{-1}\xi$ belong to $\Lambda(H)$. Therefore $w^{-1}\xi \in \Lambda(H) \cap w^{-1}\Lambda(H)$, contradicting (13.10af). $\square$

Proposition 13.14 (high-exponent subgroups transverse to a prescribed multiplier). *Let $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ be a torsion-free cocompact lattice, let $\phi : \Gamma \rightarrow \mathbb{Z}$, and put $N = \ker \phi$. Fix $a \in \phi^{-1}(1)$ and a nontrivial $k \in N$, and put $w_n = ak^n$. For every $\varepsilon > 0$ there is a finitely generated nonelementary convex-cocompact subgroup $J < N$ such that*

$$\delta_J > 1 - \varepsilon, \quad \langle k \rangle \cap J = \{1\}, \quad J \cap w_n^{-1}Jw_n = \{1\} \quad (n \geq 0). \quad (13.10ag)$$

It can moreover be chosen so that

$$\tau_q(x) \neq \tau_k(x) \quad (q \in J \setminus \{1\}). \quad (13.10ah)$$

For each fixed n , the intersection $J \cap \mathcal{L}_{w_n}(\Gamma)$ is finite. Given any fixed M , J may additionally be chosen so that these intersections equal $\{1\}$ for every $0 \leq n \leq M$.

Proof. The quotient $\Gamma/N \cong \mathbb{Z}$ is amenable, so the amenable-cover theorem gives $\delta_N = \delta_\Gamma = 1$ [10, 17]. We briefly explain why finitely generated subgroups of N approximate this exponent. On $X = N \backslash \mathbb{H}^2$, amenability gives $\lambda_0(X) = 0$. Choose compactly supported smooth functions with Rayleigh quotients tending to zero, and enclose their supports in connected compact subsurfaces $K_i \subset X$. If $H_i < N$ is the image of $\pi_1(K_i)$, then K_i embeds in the cover $H_i \backslash \mathbb{H}^2$. Extending the lifted test function by zero there gives $\lambda_0(H_i \backslash \mathbb{H}^2) \rightarrow 0$. The groups H_i are eventually nonelementary, are finitely generated, and contain no parabolic elements; hence they are convex cocompact. Sullivan's formula

$$\lambda_0(H_i \backslash \mathbb{H}^2) = \begin{cases} 1/4, & \delta_{H_i} \leq 1/2, \\ \delta_{H_i}(1 - \delta_{H_i}), & \delta_{H_i} > 1/2 \end{cases}$$

then gives $\delta_{H_i} \rightarrow 1$ [46, 8]. Choose one such group $H < N$ with $\delta_H > 1 - \varepsilon$.

We next choose one boundary interval which is separated from all the moving translates needed in (13.10ag). The maps $w_n^{-1} = k^{-n}a^{-1}$ converge locally uniformly, away from $a(k^+)$, to the constant k^- . Choose a point $\xi \in \partial\mathbb{H}^2$ different from k^- , from $a(k^+)$, and from every fixed point of the countable family $\{w_n : n \geq 0\}$. For all sufficiently large n , a small closed interval about ξ is disjoint from its image under w_n^{-1} ; after shrinking it finitely many more times, we obtain an open interval U such that

$$\overline{U} \cap \text{Fix}(k) = \emptyset, \quad \overline{U} \cap w_n^{-1}\overline{U} = \emptyset \quad (n \geq 0). \quad (13.10ai)$$

The normal nonelementary subgroup N has full limit set, and ordered pairs of fixed points of its hyperbolic elements are dense off the diagonal. Choose a hyperbolic $b \in N$ with $b^+ \in U$ and $b^- \notin \Lambda(H)$. North-south dynamics, uniformly on the compact set $\Lambda(H)$, gives $b^L\Lambda(H) \subset U$ for all sufficiently large L . Replace H by $H_1 = b^L H b^{-L}$. Its critical exponent is unchanged and $\Lambda(H_1) \subset U$. The first condition in (13.10ai) implies $H_1 \cap \langle k \rangle = \{1\}$. The second implies $H_1 \cap w_n^{-1}H_1 w_n = \{1\}$: a nontrivial element in this intersection would have both fixed points in the two disjoint limit sets. Lemma 13.13 also makes each $H_1 \cap \mathcal{L}_{w_n}(\Gamma)$ finite.

There are only finitely many H_1 -conjugacy classes whose translation length in the representation x equals $\ell_x(k)$, because H_1 is convex cocompact. The finitely generated linear group H_1 is residually finite. Choose a finite-index subgroup $J \triangleleft H_1$ which excludes one representative of every nontrivial such class. Normality then excludes the whole class and proves (13.10ah). If M was fixed in advance, also exclude the finitely many nonidentity elements in

$$\bigcup_{0 \leq n \leq M} H_1 \cap \mathcal{L}_{w_n}(\Gamma).$$

Passing to finite index preserves the limit set and the critical exponent, so every assertion follows. $\square$

The next elementary rigidity statement explains the usefulness of (13.10ah). It rules out not just propagation by k , but propagation by any fixed semisimple element of J whose translation length at x is different. The formulation allows the exponents of D to lie in an arbitrary four-term arithmetic progression; one multiplication by Q^{-1} is then forced to have the same squared trace as multiplication by D^r , where r is the common difference.

Lemma 13.15 (four cancellation equations determine the multiplier). *Let K be a characteristic-zero valued field. Write*

$$D = \lambda P_+ + \lambda^{-1} P_-, \quad |\lambda| > 1, \quad L_\pm(Y) = \text{tr}(AP_\pm Y).$$

Let $Q \in \text{SL}_2(K)$ have distinct eigenvalues and let $\Delta \in M_2(K)$. Let $n_0 \in \mathbb{Z}$ and $r \geq 1$. Suppose that, for $j = 0, 1, 2, 3$, both $L_+(Q^{-j}\Delta)$ and $L_-(Q^{-j}\Delta)$ are nonzero and

$$-\frac{L_-(Q^{-j}\Delta)}{L_+(Q^{-j}\Delta)} = \lambda^{2(n_0+rj)}. \quad (13.10aj)$$

Then

$$\mathrm{tr}^2 Q = \mathrm{tr}^2(D^r). \quad (13.10ak)$$

More precisely, if $Q = \mu E_+ + \mu^{-1} E_-$, then either $\mu^2 = \lambda^{2r}$ or $\mu^2 = \lambda^{-2r}$; the corresponding two mixed spectral coefficients vanish.

Proof. The assertion may be checked after an algebraic extension over which Q is diagonalizable. Put

$$\alpha = L_+(E_+ \Delta), \quad \beta = L_+(E_- \Delta), \quad \gamma = L_-(E_+ \Delta), \quad \zeta = L_-(E_- \Delta).$$

Set $\Lambda = \lambda^{2r}$. Multiplication of (13.10aj) by μ^j gives

$$\gamma + \zeta(\mu^2)^j + \lambda^{2n_0} \alpha \Lambda^j + \lambda^{2n_0} \beta (\Lambda \mu^2)^j = 0, \quad j = 0, 1, 2, 3. \quad (13.10al)$$

If the four bases $1, \mu^2, \Lambda, \Lambda \mu^2$ are distinct, their Vandermonde determinant is nonzero, so all four coefficients vanish, contrary to the nonvanishing assumption. Since $\Lambda \neq 1$ and Q has distinct eigenvalues, the only possible coincidences are $\mu^2 = \Lambda$ and $\mu^2 = \Lambda^{-1}$. In the first case the three remaining bases in (13.10al) give $\gamma = \beta = 0$ and $\zeta = -\lambda^{2n_0} \alpha \neq 0$. In the second they give $\alpha = \zeta = 0$ and $\gamma = -\lambda^{2n_0} \beta \neq 0$. In either case the squared traces of Q and D^r agree. $\square$

Suppose, under the hypotheses of Theorem 13.10, that one fixed $q \in J \setminus \{1\}$ and one fixed pair $u, v \in J$ were to realize four consecutive exact cancellation equations through $h_n = q^{-n}u$ and $h'_n = q^{-n}v$. Taking $r = 1$, the lemma would give $\mathrm{tr}^2 \rho(q) = \mathrm{tr}^2 \rho(k)$. Persistence and specialization at the original Fuchsian character would then give $\tau_q(x) = \tau_k(x)$, contradicting (13.10ah). Thus, if a long interval of cancellation exponents is realized, the pair realizing the identity must vary with the exponent; one fixed pair and one fixed element of J cannot account for the interval. This conclusion does not bound identities arising from unrelated pairs.

The distinction between a fixed pair and varying pairs is essential. Small sumsets for many geometrically spaced coefficients do not, by themselves, produce even two cancellation equations witnessed by the same pair.

Proposition 13.16 (small sumsets at geometric coefficients need not have a common collision). *Fix an integer $b \geq 2$, and for $q \geq 2$ put $A_q = \{0, 1, \dots, q-1\} \subset \mathbb{Q}$. Whenever $0 \leq n \leq \lfloor \log_b q \rfloor$,*

$$\#(A_q + b^n A_q) = (b^n + 1)(q - 1) + 1 \leq (b^n + 1) \#A_q. \quad (13.10ala)$$

Nevertheless, for two distinct ordered pairs $(x, y), (x', y') \in A_q^2$, the equality

$$x + b^n y = x' + b^n y' \quad (13.10alb)$$

can hold for at most one integer $n \geq 0$. This example occurs inside determinant-one matrices: the map

$$(x, y) \mapsto M(x, y) = \begin{pmatrix} x & 1 \\ xy - 1 & y \end{pmatrix} \quad \text{has} \quad \det M(x, y) = 1,$$

and, over $\mathbb{Q}(\sqrt{b})$, if $D = \mathrm{diag}(b^{-1/2}, b^{1/2})$, then

$$\mathrm{tr}(D^n M(x, y)) = b^{-n/2}(x + b^n y).$$

Thus the obstruction persists for the same trace observables that arise from a diagonal one-parameter subgroup of SL_2 .

Proof. For fixed y , the sums $x + b^n y$ fill the integer interval $[b^n y, b^n y + q - 1]$. Since $b^n \leq q$, the intervals for consecutive values of y overlap or touch, and their union is $[0, (b^n + 1)(q - 1)] \cap \mathbb{Z}$. This proves (13.10ala). If (13.10alb) held for two different integers m, n , subtraction would give $(b^m - b^n)(y - y') = 0$. Hence $y = y'$, and then $x = x'$, contrary to the assumed distinction. The determinant and trace identities follow by direct calculation. $\square$

It is tempting to replace J by the full kernel $\ker \phi$, whose critical exponent is 1 because the quotient is amenable [17]. The following proposition shows that this cannot improve the argument: conjugation alone fills exactly the initial interval that would be needed to cross the exponent $1/2$.

Proposition 13.17 (conjugation realizes the initial cancellation exponents in the full homology kernel). *Retain the preceding hypotheses and put $N = \ker \phi$. Define $\mathcal{E}_{v,N}(R)$ exactly as in (13.10a), but with*

$$\mathcal{B}_N(R) = \{h \in N : d(o, \rho_x(h)o) \leq R\}$$

in place of $\mathcal{B}_J(R)$. Then, for every $\eta \in (0, 1)$ and every sufficiently large R ,

$$\left\{0, 1, \dots, \left\lfloor \frac{(1-\eta)R}{\ell_x(k)} \right\rfloor\right\} \subset \mathcal{E}_{v,N}(R). \quad (13.10k)$$

Consequently, if

$$m_{v,N}(R) = \min(\mathbb{Z}_{\geq 0} \setminus \mathcal{E}_{v,N}(R)),$$

then

$$\liminf_{R \rightarrow \infty} \frac{m_{v,N}(R)}{R} \geq \frac{1}{\ell_x(k)}. \quad (13.10l)$$

Thus replacing the finitely generated subgroup J by the full homology kernel cannot make the argument of Theorem 13.10 prove trace entropy greater than $1/2$.

Proof. Choose once and for all a finitely generated nonelementary convex-cocompact subgroup $J_0 < N$, for example the subgroup J in Theorem 13.10, and let $\delta_0 > 0$ be its critical exponent. For $n \geq 0$, put

$$w_n = ak^n, \quad h_0 = w_n^{-1}a = k^{-n}.$$

For $\gamma \in J_0$, define

$$h_\gamma = w_n^{-1}\gamma a \gamma^{-1}. \quad (13.10m)$$

Both h_0 and h_γ belong to N , and

$$w_n h_0 = a, \quad w_n h_\gamma = \gamma a \gamma^{-1}. \quad (13.10n)$$

Their traces are therefore equal identically in every representation of Π .

There is a constant C , independent of n, R, γ , such that

$$d(o, \rho_x(h_\gamma)o) \leq n\ell_x(k) + 2d(o, \rho_x(\gamma)o) + C. \quad (13.10o)$$

Fix $\eta \in (0, 1)$, and suppose that

$$0 \leq n \leq \frac{(1-\eta)R}{\ell_x(k)}.$$

For large R , put

$$S_{R,n} = \frac{R - n\ell_x(k) - C}{2}.$$

Then $S_{R,n} \geq \eta R/3$. Every $\gamma \in \mathcal{B}_{J_0}(S_{R,n})$ gives $h_\gamma \in \mathcal{B}_N(R)$, and convex-cocompact orbit counting gives

$$\#\mathcal{B}_{J_0}(S_{R,n}) = \exp((\delta_0 + o(1))S_{R,n}) \geq \exp(c_\eta R) \quad (13.10p)$$

for some $c_\eta > 0$, uniformly over the indicated integers n .

The map $\gamma \mapsto h_\gamma$ is injective. Indeed, equality of two values in (13.10m) says that $\gamma_2^{-1}\gamma_1$ centralizes a . Centralizers of nontrivial elements in a closed surface group are cyclic. Moreover $\phi(a) = 1$ implies that a is not a proper power and that

$$C_\Pi(a) \cap N = \{1\}.$$

Since $\gamma_2^{-1}\gamma_1 \in J_0 < N$, one obtains $\gamma_1 = \gamma_2$.

Apply (13.10n) in the representation ρ . With

$$\Delta_\gamma = \rho(h_\gamma) - \rho(h_0),$$

the spectral expansion gives

$$\lambda^n L_+(\Delta_\gamma) + \lambda^{-n} L_-(\Delta_\gamma) = 0. \quad (13.10q)$$

The elements h_γ all have ambient Π -word length $O(R)$. Those for which

$$L_+(\Delta_\gamma) = L_-(\Delta_\gamma) = 0 \quad (13.10r)$$

lie in one simultaneous squared-trace fiber for

$$A_0 = A, \quad A_1 = AD_\mu,$$

where $D_\mu = \mu P_+ + \mu^{-1} P_-$ and $\mu^2 \neq 1$. Lemma 13.5 bounds the number satisfying (13.10r) by $\exp(o(R))$, uniformly in n and in the resulting trace values. This is negligible compared with (13.10p). Hence, for every integer n in the indicated interval, some γ has both coefficients in (13.10q) nonzero. Equation (13.10q) then becomes

$$-\frac{L_-(\Delta_\gamma)}{L_+(\Delta_\gamma)} = \lambda^{2n},$$

so $n \in \mathcal{E}_{v,N}(R)$.

There are only $O(R)$ integers in (13.10k), and all estimates above are uniform in n ; consequently the conclusion holds simultaneously for all of them once R is sufficiently large. This proves (13.10k), and letting $\eta \downarrow 0$ gives (13.10l). $\square$

Remark 13.18. Proposition 13.17 explains why the subgroup in Theorem 13.10 cannot be replaced by the full normal kernel. The conjugacy families (13.10m) in the full kernel realize every exponent up to the limit imposed by their Fuchsian translation length. Thus the limitation comes from an explicit group-theoretic construction, not from the valuation estimate.

Lemma 13.19 (an unbounded representation has a kernel element of positive local translation length). *Let K be a complete valued field of characteristic zero of archimedean or discrete nonarchimedean type, and let $r: \Pi \rightarrow \mathrm{PSL}_2(K)$ be Zariski dense. Suppose that $J < \ker \phi$ has Zariski-dense image, where $\phi: \Pi \twoheadrightarrow \mathbb{Z}$. If $r(\Pi)$ is unbounded, then $r(\ker \phi)$ contains an element which, after a finite extension of K , has eigenvalues λ, λ^{-1} with $|\lambda| > 1$.*

Proof. Put $N = \ker \phi$. Its image is Zariski dense because it contains $r(J)$. The elliptic-subgroup dichotomy for an action on a rank-one symmetric space or a tree says that a subgroup containing no element of positive translation length either has a global fixed point or fixes a point at infinity; finite generation is not required. The second alternative puts its Zariski closure in a proper parabolic and is impossible here. Thus, if $r(N)$ had no such element, its fixed-point set would be nonempty.

This fixed-point set is bounded. Indeed, J is finitely generated and Zariski dense, so it contains two elements with no common eigenline. In rank one, the intersection of the fixed subspaces of two elliptic elements is unbounded only if they have a common endpoint at infinity, equivalently a common eigenline. Hence $\mathrm{Fix}(r(J))$ is bounded, and $\mathrm{Fix}(r(N)) \subset \mathrm{Fix}(r(J))$ is bounded as well. Normality of N makes $\mathrm{Fix}(r(N))$ invariant under $r(\Pi)$. Its circumcenter in the symmetric space, or its center in the tree after subdividing edges if necessary, is consequently fixed by $r(\Pi)$, contrary to the assumed unboundedness. Hence $r(N)$ contains an element of positive translation length. Its two fixed points and eigenvalues are defined after a finite extension, and positive translation length is equivalent to $|\lambda| > 1$. $\square$

Remark 13.20. Both compatibility properties in Theorem 13.8 occur naturally. If x is algebraic and ρ is a lift of its algebraic representation, they follow from injectivity of the distinguished field embedding. If ρ is the generic representation of an irreducible component of the variety V_x on which the trace collisions at x persist, the first property is the definition of V_x . For the second, an identity $\text{tr}^2 \rho(g) = 4$ says that the regular function $\tau_g|_{V_x}$ is identically 4; evaluating that regular identity at x gives $\tau_g(x) = 4$. No homomorphism from the abstract function field to $\mathbb{R}$ is being asserted. Lemma 13.19 shows that every unbounded divisor in either setting admits such an element k ; the remaining condition is now the explicit, scale-invariant inequality in Corollary 13.9.

Replacing k by k^m does not improve that inequality: $\ell_x(k^m)$ and $\ell_v(k^m)$ both multiply by m , while the spectral projectors and $\mathbf{c}_v(a, k; J)$ do not change. Thus the criterion compares the translation length in the original Fuchsian representation, the translation length at v , and the rate of cancellation. Taking a large power changes neither side of this comparison.

For an algebraic character, the same proof works at a nondistinguished archimedean embedding. Use the ordinary absolute value, choose a proximal kernel element with $|\lambda| > 1$, and define $\mathbf{c}_\sigma$ by the same formula. Linear word-height bounds give $\mathbf{c}_\sigma < \infty$, and every subsequent cancellation and two-evaluation step is unchanged. Thus the theorem also gives a quantitative criterion for compactness of a nondistinguished archimedean realization. The divisorial formulation was stated separately because it also applies over function fields of varieties on which a prescribed collection of trace collisions persists.

We now determine when a single scalar trace fiber already admits a second, independent trace functional that is constant on the fiber. The result also identifies the exceptional case in which the group structure or a valuation must provide additional information.

Proposition 13.21 (linear observables constant on one squared-trace fiber). *Let k be a field of characteristic zero, let $A \in \text{SL}_2(k)$, and let $\mathcal{E} \subset \text{SL}_2(k)$ be a nonempty set on which $X \mapsto \text{tr}^2(AX)$ has one nonzero value. Fix $X_0 \in \mathcal{E}$. For each $X \in \mathcal{E}$, choose the unique sign $\epsilon_X \in \{1, -1\}$ for which*

$$\text{tr}(A\hat{X}) = \text{tr}(AX_0), \quad \hat{X} = \epsilon_X X,$$

and take $\epsilon_{X_0} = 1$. Put

$$W_{\mathcal{E}} = \text{span}_k\{\hat{X} - X_0 : X \in \mathcal{E}\} \subset M_2(k).$$

For the nondegenerate trace pairing $\langle B, C \rangle = \text{tr}(BC)$, one has

$$W_{\mathcal{E}} \subset A^\perp, \quad \{B : \text{tr}(B\hat{X}) \text{ is independent of } X \in \mathcal{E}\} = W_{\mathcal{E}}^\perp. \quad (13.10s)$$

Consequently, a linear trace observable independent of the original one exists exactly when $\dim W_{\mathcal{E}} \leq 2$. If $\dim W_{\mathcal{E}} = 3$, then

$$W_{\mathcal{E}} = A^\perp, \quad W_{\mathcal{E}}^\perp = kA,$$

so every constant linear trace observable is a scalar multiple of the original one.

Assume $\dim W_{\mathcal{E}} \leq 2$. Exactly one of the following occurs.

(i) There is $B \in W_{\mathcal{E}}^\perp$ with $\det B \neq 0$ and

$$\Delta(BA^{-1}) := \text{tr}^2(BA^{-1}) - 4\det(BA^{-1}) \neq 0. \quad (13.10t)$$

After adjoining $\sqrt{\det B}$ and rescaling B , the set $\mathcal{E}$ is contained in a simultaneous squared-trace fiber for two matrices in SL_2 whose relative matrix has two distinct eigenvalues.

(ii) One has $\dim W_{\mathcal{E}} = 2$ and

$$\{BA^{-1} : B \in W_{\mathcal{E}}^\perp\} = kI \oplus kN \quad (13.10u)$$

for a nonzero nilpotent matrix N with $N^2 = 0$. This is the only exception.

In case (ii), the matrices $Y_X = A\hat{X}$ have both $\text{tr } Y_X$ and $\text{tr}(NY_X)$ fixed. After extending k , if necessary, and conjugating so that $N = E_{12}$, write $c = \text{tr}(NY_X) = (Y_X)_{21}$. If $c \neq 0$, all the Y_X lie on the single unipotent conjugacy orbit

$$\{(I + sN)Y_{X_0}(I - sN) : s \in k\}. \quad (13.10v)$$

If $c = 0$, the corresponding curve is the union of at most two affine lines; two distinct points on either line have relative trace 2.

Proof. The signs give $\text{tr}(A(\hat{X} - X_0)) = 0$, hence $W_{\mathcal{E}} \subset A^{\perp}$. A matrix B has constant trace on the signed set precisely when it annihilates every difference $\hat{X} - X_0$, which proves (13.10s). Nondegeneracy of the trace pairing gives $\dim W_{\mathcal{E}}^{\perp} = 4 - \dim W_{\mathcal{E}}$, proving the first claims.

Put

$$U = \{BA^{-1} : B \in W_{\mathcal{E}}^{\perp}\} \subset M_2(k).$$

This subspace contains I . The discriminant $\Delta(M) = \text{tr}^2 M - 4 \det M$ is unchanged upon adding a scalar matrix, so it descends to the nondegenerate quadratic form on $M_2(k)/kI \simeq \mathfrak{sl}_2(k)$. A totally isotropic subspace for a nondegenerate quadratic form in three variables has dimension at most one. Thus, if $\dim U \geq 3$, some $M \in U$ has $\Delta(M) \neq 0$. Replacing it by $M + \alpha I$, for a suitable $\alpha \in k$, makes it invertible without changing its discriminant, and $B = MA$ proves (i). If $\dim U = 2$, the same conclusion holds unless U/kI is an isotropic line. In the latter case, subtracting the repeated eigenvalue of a nonscalar generator gives a nonzero matrix N with $N^2 = 0$, proving (13.10u).

In the exceptional case, $NA \in W_{\mathcal{E}}^{\perp}$, so $\text{tr}(NY_X)$ is fixed. With $N = E_{12}$, every such matrix has the form

$$Y(u) = \begin{pmatrix} u & (u(t-u) - 1)/c \\ c & t-u \end{pmatrix}, \quad t = \text{tr } Y_X,$$

when $c \neq 0$. Conjugation by $I + sN$ replaces u by $u + sc$, which proves (13.10v). If $c = 0$, the determinant equation is $u(t-u) = 1$; each of its at most two roots leaves one off-diagonal entry free. Two points on the same resulting line have singular difference, and $\det(P - Q) = 2 - \text{tr}(PQ^{-1})$ gives relative trace 2. $\square$

Theorem 13.22 (subexponential fibers unless the signed differences fill the trace hyperplane). *Let H be finitely generated, let $F/\mathbb{Q}$ be finitely generated of characteristic zero, and let $\rho: H \rightarrow \text{SL}_2(F)$ satisfy the Fuchsian specialization hypothesis of Theorem 13.4. Use the finite-type filtration of Section 12. Let $A_n \in \text{SL}_2(F)$ have complexity $O(n)$, and let $\mathcal{E}_n$ be contained in the word ball of radius n in H and in one nonzero squared-trace fiber of*

$$h \mapsto \text{tr}^2(A_n \rho(h)).$$

Form the signed span $W_{\mathcal{E}_n}$ from the matrices $\rho(h)$ as in Proposition 13.21. If $\dim W_{\mathcal{E}_n} \leq 2$ and alternative (ii) of that proposition does not occur, then

$$|\mathcal{E}_n| = \exp(o(n)),$$

uniformly over the fiber. Hence an exponentially large squared-trace fiber must either satisfy

$$\text{span}_F\{\epsilon_h \rho(h) - \epsilon_{h_0} \rho(h_0) : h \in \mathcal{E}_n\} = A_n^{\perp} \quad (13.10w)$$

or lie in the explicitly described unipotent-conjugation exception.

Proof. Choose at most two signed differences which form a basis of $W_{\mathcal{E}_n}$. A basis of their common annihilator can be written using minors of this fixed-size linear system; no division is needed, and all entries have finite-type complexity $O(n)$. Unless the exceptional alternative holds, the discriminant is not identically zero on this annihilator after right multiplication by A_n^{-1} . One basis vector or the sum of two basis vectors therefore has nonzero discriminant: if the quadratic form

vanished on every basis vector and every such sum, its polarization would vanish on the entire space. We obtain $B_n \in W_{\mathcal{E}_n}^\perp$ such that

$$\Delta(B_n A_n^{-1}) \neq 0$$

and all its entries have complexity $O(n)$.

Put

$$Y = A_n \hat{X}, \quad M = B_n A_n^{-1}, \quad t = \operatorname{tr} Y, \quad u = \operatorname{tr}(MY),$$

and take traceless parts

$$Z = Y - \frac{t}{2}I, \quad S = M - \frac{\operatorname{tr} M}{2}I.$$

Set

$$q = \frac{t^2}{4} - 1, \quad d = -\det S = \frac{\Delta(M)}{4}, \quad r = u - \frac{t \operatorname{tr} M}{2}.$$

Then $d \neq 0$, $Z^2 = qI$, $S^2 = dI$, and $\operatorname{tr}(SZ) = r$. Therefore

$$V = 2dZ - rS$$

satisfies

$$\operatorname{tr}(SV) = 0, \quad \frac{1}{2} \operatorname{tr}(V^2) = d(4dq - r^2) =: \mathcal{N}. \quad (13.10x)$$

All entries and scalars in these formulas have complexity $O(n)$.

Write

$$S = \begin{pmatrix} a & b \\ c & -a \end{pmatrix}, \quad V = \begin{pmatrix} x & y \\ z & -x \end{pmatrix}, \quad d = a^2 + bc.$$

If $b \neq 0$, the equation $\operatorname{tr}(SV) = 0$ and (13.10x) give the injective assignment

$$V \mapsto (bx - ay, y), \quad (bx - ay)^2 - dy^2 = b^2 \mathcal{N}.$$

If $b = 0$ and $c \neq 0$, use instead

$$V \mapsto (cx - az, z), \quad (cx - az)^2 - dz^2 = c^2 \mathcal{N}.$$

If $b = c = 0$, then $x = 0$ and (13.10x) is the factorization equation $yz = \mathcal{N}$. When $\mathcal{N} \neq 0$, Lemma 12.3, or Theorem 12.1 in the last case, gives $\exp(o(n))$ possibilities. When $\mathcal{N} = 0$, after a quadratic extension the solution is the union of at most two affine lines. Two distinct group matrices on one line have singular difference and hence relative squared trace 4. The Fuchsian specialization hypothesis makes them the same projective group element. Thus the degenerate case contributes only $O(1)$ points. The two possible signs change the bound by only a constant factor. $\square$

The full-rank alternative in (13.10w) occurs generically in the scheme defined by the scalar collision equations; it is not an artifact of the proof.

Proposition 13.23 (geometry of the universal scalar-collision scheme). *Let k be a field of characteristic zero. For $m \geq 1$, consider*

$$\mathcal{C}_m = \{(A, X_0, \dots, X_m) \in \operatorname{SL}_2^{m+2} : \operatorname{tr}(AX_i) = \operatorname{tr}(AX_0) \ (1 \leq i \leq m)\}.$$

Then $\mathcal{C}_m$ is an integral complete intersection of codimension m . Its projection to (A, X_0) is flat, and its fiber is the m -fold product of the geometrically integral affine quadric

$$Q_{A,t} = \{X \in \operatorname{SL}_2 : \operatorname{tr}(AX) = t\}, \quad t = \operatorname{tr}(AX_0).$$

The affine span of $Q_{A,t}$ is exactly the hyperplane $\{X \in M_2 : \operatorname{tr}(AX) = t\}$.

For $m \geq 3$, at the generic point of $\mathcal{C}_m$,

$$\operatorname{span}\{X_i - X_0 : 1 \leq i \leq m\} = A^\perp. \quad (13.10y)$$

Consequently, even if B is allowed to have coefficients in the full function field $k(\mathcal{C}_m)$,

$$\mathrm{tr}(BX_i) = \mathrm{tr}(BX_0) \quad (1 \leq i \leq m) \implies B \in k(\mathcal{C}_m)A. \quad (13.10z)$$

The same conclusion holds componentwise for fixed-sign components of the equations $\mathrm{tr}^2(AX_i) = \mathrm{tr}^2(AX_0)$; on the open set where the common trace is nonzero, the signs are unique.

Proof. Left multiplication by A identifies $Q_{A,t}$ with $Q_t = \{Y \in \mathrm{SL}_2 : \mathrm{tr} Y = t\}$. Writing $Y = \begin{pmatrix} u & v \\ w & t-u \end{pmatrix}$ gives

$$k[Q_t] = k[u, v, w]/(u(t-u) - vw - 1).$$

After any field extension, the displayed polynomial is primitive in $k[u, w][v]$, linear and irreducible over $k(u, w)$, so Gauss's lemma proves geometric integrality, including when $t^2 = 4$. Its principal ideal contains no nonzero affine-linear polynomial. Hence Q_t affinely spans the entire trace hyperplane.

The trace morphism $\mathrm{SL}_2 \rightarrow \mathbb{A}^1$ is flat because its coordinate ring is torsion-free over the principal ideal domain $k[t]$. The projection in the statement is its m -fold base change and has geometrically integral fibers over the integral base SL_2^2 ; its total space is therefore integral. Every fiber has dimension $2m$, so $\mathcal{C}_m$ has codimension m in the smooth variety SL_2^{m+2} . The displayed m equations consequently form a regular sequence.

Because the quadric affinely spans its trace hyperplane, three differences from four generic points span its three-dimensional translation space $A^\perp$. Linear independence is a nonempty open condition, which proves (13.10y). The trace pairing is nondegenerate, so the annihilator of this space is the line spanned by A , proving (13.10z). $\square$

The proposition shows that the scalar collision equations alone cannot imply the desired second trace relation. It does not assert that a full-rank fiber of exponential size occurs inside a word ball of a Fuchsian group. To exclude that possibility one must use the group law or arithmetic information not present in the universal collision scheme. In Theorem 13.58, that information comes from reduced divisors and congruence expansion applied to a varying family of group elements. Differentiating a collision gives

$$\mathrm{tr}((dA)(X_i - X_0)) + \mathrm{tr}(A(dX_i - dX_0)) = 0.$$

For a deformation of a representation, the second term depends on the group element and disappears only when the restricted cocycle is a coboundary removed by infinitesimal conjugation, exactly the vertical situation treated in Theorem 13.83. Hasse derivatives leave two free matrix directions at every order on the generic smooth trace quadric. Likewise, an identity in an associated graded or Rees algebra gives a second identity in the original fiber only if it can be lifted or separated by a valuation. Theorem 13.8 provides such a valuation argument.

Finally, the square-root construction itself cannot evade this linear-span alternative by producing a new conjugation-covariant matrix on the same fiber.

Proposition 13.24 (rational conjugation covariants in rank one). *Let k be algebraically closed of characteristic zero. Every rational conjugation-equivariant map*

$$\Phi: \mathrm{SL}_2 \dashrightarrow M_2$$

has the form

$$\Phi(g) = \alpha(\mathrm{tr} g)I + \beta(\mathrm{tr} g)g, \quad \alpha, \beta \in k(t). \quad (13.10aa)$$

The same statement holds after a Kummer extension of the adjoint-quotient field $k(t)$, with α and β in that extension. Consequently, on a fixed trace quadric every such construction is affine-linear in g and preserves the full-affine-span obstruction of Proposition 13.21. In particular,

$$R(g) = \frac{I + g}{\sqrt{2 + \mathrm{tr} g}}$$

does not by itself produce a second linear trace observable on one of its scalar trace fibers.

Proof. For regular semisimple g , equivariance makes $\Phi(g)$ fixed by the centralizer of g . The commutant of that torus in M_2 is the quadratic algebra $k[g] = kI \oplus kg$. Thus $\Phi(g) = \alpha(g)I + \beta(g)g$ on a dense open set. Uniqueness of the two coefficients and equivariance make them conjugation-invariant, while the rational invariant field is $k(\mathrm{SL}_2)^{\mathrm{PGL}_2} = k(\mathrm{tr} g)$. This proves (13.10aa). The same argument after base change proves the Kummer assertion. $\square$

Passing to a finite quotient and a vector-valued representation has a similar limitation. If $q: \Pi \rightarrow Q$ is finite and $\sigma: Q \rightarrow \mathrm{GL}(V)$, then

$$\mathrm{tr}((\rho \otimes \sigma)(g)) = \mathrm{tr} \rho(g) \chi_\sigma(q(g)).$$

On a fixed q -coset this is only a fixed scalar multiple of the original trace. The partial trace over the two-dimensional factor is $\mathrm{tr} \rho(g) \sigma(q(g))$, again a fixed matrix multiple on that coset. Thus deck-character twists, Kummer sign characters, and the regular representation of a finite quotient do not by themselves produce a second left multiplier. To obtain one, a vector-valued argument must use either a different representation of the subgroup or the valuation separation used above.

13.1. The trace recurrence and its height at all places. There is another way to retain more than one scalar trace without adding a new representation. Fix one hyperbolic element k , and record the two coordinates

$$(\mathrm{tr} g, \mathrm{tr}(kg)).$$

All later coordinates $\mathrm{tr}(k^n g)$ are obtained from these two by one linear recurrence. The expansion rate of that recurrence at all places is a global height, and not merely the translation length at the distinguished real place.

Lemma 13.25 (the two-trace recurrence). *Let $k, g \in \mathrm{SL}_2(\mathbb{C})$ and put $s = \mathrm{tr} k$. Define*

$$q_0 = 0, \quad q_1 = 1, \quad q_{n+1} = sq_n - q_{n-1}.$$

Then, for every $n \geq 1$,

$$k^n = q_n(s)k - q_{n-1}(s)I, \quad \mathrm{tr}(k^n g) = q_n(s) \mathrm{tr}(kg) - q_{n-1}(s) \mathrm{tr}(g). \quad (13.10am)$$

If

$$\alpha_n = -\frac{q_{n-1}(s)}{q_n(s)}, \quad \beta_n = -\alpha_n^2, \quad (13.10an)$$

then $\beta_n \in \mathbb{Q}(s^2)$. Moreover

$$q_n(s)^2 - q_{n-1}(s)q_{n+1}(s) = 1. \quad (13.10ao)$$

Proof. Cayley–Hamilton gives $k^2 - sk + I = 0$. Induction proves the matrix identity in (13.10am), and taking the trace after multiplication by g proves the scalar identity. The polynomial q_n has the parity of $n - 1$, so the square of q_{n-1}/q_n is rational in s^2 . The left side of (13.10ao) is constant in n by the recurrence and is equal to one at $n = 1$. $\square$

Proposition 13.26 (all cyclic trace vectors have one determinantal modulus). *Put $q_0(T) = 0$, $q_1(T) = 1$, and $q_{j+1}(T) = Tq_j(T) - q_{j-1}(T)$ for every $j \in \mathbb{Z}$; thus $q_{-j} = -q_j$. Let $J \subset \mathbb{Z}$ be finite with at least two elements, and put*

$$C_J(T) = (-q_{j-1}(T), q_j(T))_{j \in J} \in M_{J \times 2}(\mathbb{Z}[T]).$$

If

$$g_J = \gcd\{|i - j| : i, j \in J, i \neq j\},$$

then the ideal generated by the maximal minors of $C_J(T)$ is exactly

$$I_2(C_J) = (q_{g_J}(T)) \subset \mathbb{Z}[T]. \quad (13.10ao1)$$

This equality survives every specialization $T \mapsto s$ to a commutative ring. In particular, if K is a commutative ring, $A \in \mathrm{SL}_2(K)$, $s = \mathrm{tr} A$, and $X \in M_2(K)$, then

$$\mathrm{tr}(A^j X) = -q_{j-1}(s) \mathrm{tr} X + q_j(s) \mathrm{tr}(AX), \quad (13.10\mathrm{ao}2)$$

and the common determinantal divisor of any finite vector of such trace coordinates is the single Chebyshev term $q_{g_J}(s)$.

Proof. Cayley–Hamilton gives $A^j = q_j(s)A - q_{j-1}(s)I$ for $j \geq 0$. The identity $A^{-1} = sI - A$, or the same recurrence run backwards, proves this for every $j \in \mathbb{Z}$ and gives $q_{-j} = -q_j$. For any $i, j \in \mathbb{Z}$, the corresponding minor is

$$q_i q_{j-1} - q_{i-1} q_j = q_{j-i}. \quad (13.10\mathrm{ao}3)$$

The addition and subtraction identities

$$q_{u+v} = q_u q_{v+1} - q_{u-1} q_v, \quad q_{u-v} = q_u q_{v+1} - q_{u+1} q_v \quad (13.10\mathrm{ao}4)$$

hold in $\mathbb{Z}[T]$. Consequently $(q_u, q_v) = (q_v, q_{u-v})$ for $u \geq v > 0$. The Euclidean algorithm on the indices gives

$$(q_u, q_v) = (q_{\gcd(u,v)})$$

without localization or exceptional primes. Apply this repeatedly to the differences $i - j$ in (13.10ao3). Since the resulting ideal equality is already over $\mathbb{Z}[T]$, evaluation in any commutative ring preserves it. $\square$

Thus adding finitely many positions in one orbit of the cyclic group $\langle A \rangle$ never creates independent recurrence moduli. If the index differences generate $\mathbb{Z}$, the determinantal ideal is the unit ideal. If the indices form a primitive congruence pattern modulo n , meaning that the greatest common divisor of their differences is exactly n , the common modulus is exactly the same $q_n(s)$. Finite-character twists multiply these rows by scalars from a fixed finite set. After zero coordinates are discarded and the fixed coefficient field and bad-prime set are enlarged, they introduce only factors independent of the recurrence indices; they do not produce a new growing Chebyshev modulus. Hence a finite vector of cyclic trace coordinates cannot replace the radical of one Lucas–Chebyshev term by a product of independent factors. To obtain independent growing moduli, one must use either a noncommuting group element or congruence information modulo the full prime powers.

Proposition 13.27 (the first noncommuting recurrence determinant). *Let K have characteristic zero, let $a, b \in \mathrm{SL}_2(K)$, and assume that a is regular semisimple and that a, b have no common invariant line over $\overline{K}$. Then*

$$\{I, a, b, ab\}$$

is a basis of $M_2(K)$. Put

$$Q_m = q_m(\mathrm{tr} a), \quad R_n = q_n(\mathrm{tr} b).$$

Relative to any fixed coordinates on $M_2(K)$,

$$\det(I, a^m, b^n, a^m b^n) = Q_m^2 R_n^2 \det(I, a, b, ab). \quad (13.10\mathrm{ao}5)$$

Equivalently, the four trace functionals

$$X \mapsto (\mathrm{tr} X, \mathrm{tr}(a^m X), \mathrm{tr}(b^n X), \mathrm{tr}(a^m b^n X))$$

have determinant equal to a fixed nonzero factor times $Q_m^2 R_n^2$.

Proof. Write $P_m = q_{m-1}(\mathrm{tr} a)$ and $S_n = q_{n-1}(\mathrm{tr} b)$. Cayley–Hamilton gives

$$a^m = Q_m a - P_m I, \quad b^n = R_n b - S_n I.$$

In the ordered basis (I, a, b, ab) , the change-of-basis matrix to $(I, a^m, b^n, a^m b^n)$ is triangular, with diagonal

$$1, \quad Q_m, \quad R_n, \quad Q_m R_n.$$

This proves (13.10a5). To see that the fixed determinant is nonzero, diagonalize a over $\overline{K}$, say $a = \text{diag}(\lambda, \lambda^{-1})$, and write the off-diagonal entries of b as u, v . Direct expansion gives the fixed determinant, up to the sign imposed by the coordinate order, as

$$uv(\lambda - \lambda^{-1})^2 = 2 - \text{tr}[a, b].$$

Regularity makes the last eigenvalue factor nonzero, and absence of a common invariant line makes $uv \neq 0$. Finally, the trace pairing on $M_2(K)$ is nondegenerate, so the determinant of the four trace functionals differs from the matrix-vector determinant by one fixed nonzero factor. $\square$

Proposition 13.27 identifies the first point at which two independent recurrence moduli occur. The determinant identity alone does not give a useful count. One group ball has $e^{R+o(R)}$ elements, whereas four scalar trace supports of size $e^{R/2+o(R)}$ form an ambient grid of size $e^{2R+o(R)}$. A direct Cartesian-support comparison therefore loses a factor $e^{R-o(R)}$. Thus a direct comparison with the Cartesian product is too weak by a factor of $e^{R-o(R)}$. Any application of the determinant identity must use additional incidence relations among the four trace coordinates.

Lemma 13.28 (the recurrence dilates eventually generate the squared-trace field). *Let s be algebraic, suppose that one real embedding of s^2 is greater than 4, and put*

$$K = \mathbb{Q}(s^2), \quad \beta_n = -\frac{q_{n-1}(s)^2}{q_n(s)^2} \quad (n \geq 2).$$

Then

$$\mathbb{Q}(\beta_n) = K$$

for all but finitely many n .

Proof. Put $u = s^2$. If $z + z^{-1} = u - 2$, the closed formula for the recurrence gives

$$\beta_n = f_n(u) = -z \left(\frac{z^{n-1} - 1}{z^n - 1} \right)^2. \quad (13.10\text{ap})$$

The expression is invariant under $z \mapsto z^{-1}$, and is therefore rational in u .

A number field has only finitely many subfields. If the assertion failed, one proper subfield $F \subsetneq K$ would contain $f_n(u)$ for infinitely many n . Fix the distinguished embedding $\sigma_0 : K \hookrightarrow \mathbb{C}$, for which $\sigma_0(u) > 4$. The restriction $\sigma_0|_F$ has two distinct extensions $\sigma_0, \tau : K \hookrightarrow \mathbb{C}$. Hence

$$f_n(\sigma_0 u) = f_n(\tau u) \quad (13.10\text{aq})$$

for infinitely many n .

Choose $z_0 > 1$ with $z_0 + z_0^{-1} = \sigma_0 u - 2$, and choose w with $|w| \geq 1$ and $w + w^{-1} = \tau u - 2$. No denominator in (13.10ap) vanishes. Indeed, $q_n(s)$ is a nonzero algebraic number because its distinguished conjugate is positive. Substitution in (13.10aq) and clearing denominators gives the exponential polynomial

$$z_0(z_0^{n-1} - 1)^2(w^n - 1)^2 - w(w^{n-1} - 1)^2(z_0^n - 1)^2. \quad (13.10\text{ar})$$

By the Skolem–Mahler–Lech theorem [24], infinitely many zeros force (13.10ar) to vanish for every n in an infinite arithmetic progression.

If $|w| > 1$, passage to the limit along that progression gives $-z_0^{-1} = -w^{-1}$. Thus $w = z_0$ and $\tau(u) = \sigma_0(u)$. Since u generates K , this contradicts $\tau \neq \sigma_0$.

It remains to consider $|w| = 1$. If $w \neq 1$ is a root of unity, then $w^m = 1$ for some m , and the conjugate $q_m(\tau s)$ vanishes, again contradicting $q_m(s) \neq 0$. If $w = 1$, then $\tau(u) = 4$; irreducibility of the minimal polynomial of u would force $u = 4$, contrary to $\sigma_0(u) > 4$. Thus w is not a root

of unity. On a progression $n = a + mj$, the powers w^{a+mj} are dense in the unit circle, whereas $f_{a+mj}(\sigma_0 u) \rightarrow -z_0^{-1}$. But

$$f_n(\tau u) = -\frac{1}{w} \left(\frac{w^n - w}{w^n - 1} \right)^2.$$

Subsequences converging away from the pole at 1 would make the nonconstant rational function

$$t \mapsto -\frac{1}{w} \left(\frac{t - w}{t - 1} \right)^2$$

constant on the unit circle, a contradiction. $\square$

For an algebraic number ξ , put $F = \mathbb{Q}(\xi)$ and define the Conlon–Lim height

$$H(\xi) = N_{F/\mathbb{Q}}(\mathfrak{D}_{\xi;F}) \prod_{\sigma:F \hookrightarrow \mathbb{C}} (1 + |\sigma\xi|), \quad \mathfrak{D}_{\xi;F} = \{a \in \mathcal{O}_F : a\xi \in \mathcal{O}_F\}. \quad (13.10as)$$

Every complex embedding occurs separately in the product. Thus, for $\xi = p/q \in \mathbb{Q}$ in lowest terms, $H(\xi) = |p| + |q|$. Conlon and Lim prove that every finite $A \subset \mathbb{C}$ satisfies

$$|A + \xi A| \geq (H(\xi) - o_\xi(1))|A| \quad (|A| \rightarrow \infty). \quad (13.10at)$$

Their theorem is sharp [16].

We now compute this height for the recurrence dilates. For a place v of $K = \mathbb{Q}(s^2)$, choose a root z_v of

$$z_v + z_v^{-1} = s^2 - 2$$

in an algebraic closure of K_v , and set

$$\chi_v(s) = \frac{1}{2} \log \max\{|z_v|_v, |z_v|_v^{-1}\}, \quad \mathcal{J}_K(s) = \sum_{v \in M_K} [K_v : \mathbb{Q}_v] \chi_v(s). \quad (13.10au)$$

These quantities are independent of the choice $z_v \leftrightarrow z_v^{-1}$. At the distinguished real place, $\chi_{v_0}(s) = \ell(k)/2$. The sum $\mathcal{J}_K(s)$ measures the growth, over all places, of the recurrence determined by $s^2 - 2$; equivalently, it is its Mahler measure in logarithmic form. The distinguished translation length accounts for only one of its nonnegative summands.

Lemma 13.29 (global height of the recurrence dilates). *With the preceding notation,*

$$h(\alpha_n) = \frac{n}{[K : \mathbb{Q}]} \mathcal{J}_K(s) + O_s(\log n). \quad (13.10av)$$

Whenever $\mathbb{Q}(\beta_n) = K$,

$$\log H(\beta_n) = 2n \mathcal{J}_K(s) + O_s(\log n). \quad (13.10aw)$$

Proof. Let $L = \mathbb{Q}(s)$. For each place w of L , choose $\lambda_w + \lambda_w^{-1} = s$. The closed formula

$$q_n(s) = \frac{\lambda_w^n - \lambda_w^{-n}}{\lambda_w - \lambda_w^{-1}}$$

and (13.10ao) give

$$\log \max\{|q_{n-1}(s)|_w, |q_n(s)|_w\} = n \chi_w(s) + O_{s,w}(\log n). \quad (13.10ax)$$

At a nonarchimedean place, if $|s|_w > 1$, then $|q_n(s)|_w = |s|_w^{n-1}$. If $|s|_w \leq 1$, the recurrence bounds both adjacent terms by one and (13.10ao) prevents both from having norm less than one. At an archimedean place the closed formula proves (13.10ax); a logarithmic error is needed only at a repeated root.

The projective-height formula, followed by summation of local degrees over the places of L above a place of K , proves (13.10av). Also $h(\beta_n) = 2h(\alpha_n)$. If $K = \mathbb{Q}(\beta)$, then

$$\log H(\beta) = \sum_{v \nmid \infty} [K_v : \mathbb{Q}_v] \log^+ |\beta|_v + \sum_{v \mid \infty} [K_v : \mathbb{Q}_v] \log(1 + |\beta|_v),$$

and hence

$$0 \leq \log \mathbf{H}(\beta) - [K : \mathbb{Q}]h(\beta) \leq [K : \mathbb{Q}] \log 2.$$

Together with (13.10av), this proves (13.10aw). $\square$

The recurrence becomes useful only after one deals with a combinatorial point which is absent from the usual sum-of-dilates problem. The pairs $(\operatorname{tr} g, \operatorname{tr}(kg))$ need not contain a large Cartesian product. The next two lemmas show that a dense bipartite graph of such pairs is already enough. We include the details because this is where an unjustified “dense graph contains a large rectangle” assertion would otherwise enter.

Lemma 13.30 (a difference-set bound on a bipartite graph). *Let U, V be finite subsets of a field, let $G \subset U \times V$, and let $\alpha \neq 0$. For $u \in U$, write*

$$N_G(u) = \{v \in V : (u, v) \in G\}.$$

For every $d \in U - \alpha^2 U$, choose one representation $d = u_d - \alpha^2 u'_d$, and put $b(d) = \#(N_G(u_d) \cap N_G(u'_d))$. Then

$$\sum_{d \in U - \alpha^2 U} b(d) \leq \#\{u + \alpha v : (u, v) \in G\} \#\{v + \alpha u : (u, v) \in G\}. \quad (13.10ay)$$

In particular, if every two vertices of U , including two equal vertices, have at least r common neighbours, then

$$\#(U - \alpha^2 U) \leq \frac{1}{r} \#\{u + \alpha v : (u, v) \in G\} \#\{v + \alpha u : (u, v) \in G\}. \quad (13.10az)$$

Proof. For each d and each $v \in N_G(u_d) \cap N_G(u'_d)$, consider

$$(d, v) \mapsto (u_d + \alpha v, v + \alpha u'_d).$$

This map is injective. Indeed, its image (x, y) first recovers $d = x - \alpha y$; the chosen representative (u_d, u'_d) is then known, and $v = (x - u_d)/\alpha$. Counting its domain and codomain proves (13.10ay), and (13.10az) follows immediately. $\square$

Lemma 13.31 (dependent random choice with an explicit loss). *Let $G \subset U \times V$ be a bipartite graph, where $\#U, \#V \leq N$, and write*

$$\#G = N^2 e^{-a}, \quad a \geq 0.$$

For every positive integer t , there is $W \subset U$ such that

$$\#W \geq \frac{1}{2} N e^{-at} \quad (13.10ba)$$

and every two vertices of W , including two equal vertices, have at least

$$r = \left\lfloor e^{-a} N (2N)^{-1/t} \right\rfloor \quad (13.10bb)$$

common neighbours.

Proof. Pad the two vertex classes by isolated vertices so that both have size N . Choose t vertices of V , independently and with replacement, and let X be their common neighbourhood. Convexity gives

$$\mathbb{E} \#X = \sum_{u \in U} \left(\frac{\deg u}{N} \right)^t \geq N \left(\frac{\#G}{N^2} \right)^t = N e^{-at}.$$

Call an unordered pair, allowing a repeated vertex, bad if its common neighbourhood has size less than r . The expected number B of bad pairs contained in X is at most

$$N^2 (r/N)^t \leq \frac{1}{2} N e^{-at}.$$

Some choice therefore has $\#X - B \geq Ne^{-at}/2$. Repeatedly remove one vertex from each remaining bad pair. At most B vertices are removed, and the surviving set W has (13.10ba) and (13.10bb). $\square$

Theorem 13.32 (large trace sets with simultaneous recurrence control). *Let $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ be a torsion-free cocompact lattice and suppose that its signed trace set has linear growth. Choose a lift $\hat{\Gamma} < \mathrm{SL}_2(\mathbb{R})$, a hyperbolic $k \in \hat{\Gamma}$ with $s = \mathrm{tr} k > 2$, and a basepoint on the axis of k . Let $\ell = \ell(k)$, and let β_n be as in (13.10an). There are finite sets $W_R \subset \mathbb{R}$ and numbers $\Delta_R = o(R)$, depending on k but not on n , such that*

$$\#W_R = \exp((\tfrac{1}{2} - o(1))R) \quad (13.10bc)$$

and, simultaneously for every integer $n \geq 2$,

$$\#(W_R + \beta_n W_R) \leq \exp(n\ell + \Delta_R) \#W_R. \quad (13.10bd)$$

More explicitly, define the simple bipartite graph

$$G_R = \{(\mathrm{tr} g, \mathrm{tr}(kg)) : g \in \hat{\Gamma}, d(o, go) \leq R\}. \quad (13.10be)$$

For a sufficiently large fixed C , put

$$N_R = \lceil Ce^{R/2} \rceil, \quad a_R = \log \frac{N_R^2}{\#G_R}, \quad L_R = \log(2N_R).$$

Then $a_R = o(R)$. If

$$\bar{a}_R = \max\{a_R, L_R^{-1}\}, \quad t_R = \left\lceil \sqrt{L_R / \bar{a}_R} \right\rceil, \quad (13.10bf)$$

one may take

$$\Delta_R \leq a_R + a_R t_R + L_R / t_R + O_{\Gamma,k}(1) = O(1 + a_R + \sqrt{a_R R}) = o(R). \quad (13.10bg)$$

Proof. The two coordinate projections of G_R have at most N_R elements: matrix norm estimates put both traces in an interval of size $O(e^{R/2})$, and the assumed linear trace growth bounds the number of values in that interval. Cocompact orbit counting gives $e^{R+O(1)}$ group elements in the ball. Lemma 13.5, with $A_0 = I$ and $A_1 = k$, bounds every fiber of the two trace coordinates by $e^{o(R)}$. Hence

$$\#G_R = e^{R-o(R)}, \quad a_R = o(R). \quad (13.10bh)$$

Apply Lemma 13.31 with $t = t_R$, and call the resulting left vertex set W_R . Formula (13.10ba), together with $a_R t_R = o(R)$, proves (13.10bc). The quantity inside the floor in (13.10bb) tends to infinity; consequently, after changing an absolute constant,

$$r_R \geq \tfrac{1}{2} e^{-a_R} N_R (2N_R)^{-1/t_R}. \quad (13.10bi)$$

Fix $n \geq 2$, and put $\alpha_n = -q_{n-1}(s)/q_n(s)$. The recurrence and its inverse give

$$\begin{aligned} q_n(s)(\mathrm{tr} g + \alpha_n \mathrm{tr}(kg)) &= \mathrm{tr}(k^{-(n-1)}g), \\ q_n(s)(\mathrm{tr}(kg) + \alpha_n \mathrm{tr} g) &= \mathrm{tr}(k^n g). \end{aligned} \quad (13.10bj)$$

The basepoint lies on the axis of k , so the elements on the right have displacement at most $R + n\ell$. Linear trace growth therefore bounds each of the two graph projections in Lemma 13.30 by

$$O_{\Gamma,k}(e^{(R+n\ell)/2}). \quad (13.10bk)$$

Since $\beta_n = -\alpha_n^2$, Lemma 13.30, (13.10ba), and (13.10bi) now give

$$\frac{\#(W_R + \beta_n W_R)}{\#W_R} \leq \exp(n\ell + a_R + a_R t_R + L_R / t_R + O_{\Gamma,k}(1)).$$

This is (13.10bd)–(13.10bg). Notice that W_R and Δ_R were chosen before n ; the estimate holds simultaneously for all recurrence indices. $\square$

Theorem 13.32 requires no large Cartesian product inside the trace-pair graph. It gives one large set on which every recurrence dilate has a controlled sumset, with a loss of only $\exp(o(R))$.

Theorem 13.33 (Chebyshev recurrence and Chinese-remainder uncertainty on a curve). *Let C/k be a smooth projective geometrically integral curve over a field of characteristic zero, let $s \in k(C)$ be nonconstant, and let D_0 be an effective divisor containing $(s)_\infty$. Put*

$$q_0(T) = 0, \quad q_1(T) = 1, \quad q_{n+1}(T) = Tq_n(T) - q_{n-1}(T), \quad \beta_n = -\frac{q_{n-1}(s)^2}{q_n(s)^2}.$$

For $n \geq 2$, let A_n be the part of the zero divisor $(q_n(s)^2)_0$ supported outside $|D_0|$. Let I be a finite set of pairwise coprime integers at least 2. Suppose that

$$U \subset \Gamma(C \setminus |D_0|, \mathcal{O}_C)$$

is finite and nonempty and that an effective divisor D , supported on D_0 , satisfies

$$(u - u')_\infty \leq D \quad (u, u' \in U). \quad (13.10bh1)$$

If

$$\sum_{n \in I} \deg A_n > \deg D, \quad (13.10bh2)$$

then

$$\#U \leq \prod_{n \in I} \frac{\#(U + \beta_n U)}{\#U}. \quad (13.10bh3)$$

Writing $d = \deg(s)_\infty$, the divisors A_n , $n \in I$, have pairwise disjoint supports and

$$2(n-1)d - C_{D_0, s} \leq \deg A_n \leq 2(n-1)d, \quad (13.10bh4)$$

where the constant is independent of n . One may take

$$C_{D_0, s} = 2 \sum_{\substack{P \in |D_0| \\ s(P) \neq \infty}} e_P(s) \deg P,$$

after extending the constant field if necessary; degrees are unchanged by that extension.

Proof. As polynomials,

$$q_n(T) = U_{n-1}(T/2), \quad \gcd(q_m, q_n) = q_{\gcd(m, n)}. \quad (13.10bh5)$$

Thus pairwise coprime m, n give coprime polynomials q_m, q_n , and their zero divisors after substitution $T = s$ are disjoint. The polynomial q_n is monic of degree $n-1$, so the full zero divisor of $q_n(s)^2$ has degree $2(n-1)d$. All roots of q_n are simple in characteristic zero. At a fixed point $P \in |D_0|$ where $s(P)$ is finite, a zero therefore has order $2 \operatorname{ord}_P(s - s(P))$, independently of n ; at a pole of s there is no zero. Removing the fixed set $|D_0|$ consequently costs at most a constant, proving (13.10bh4).

Let π_n denote reduction modulo A_n , and choose $T_n \subset U$ containing one representative of each class in $\pi_n(U)$. Cassini's identity makes $q_{n-1}(s)$ a unit modulo A_n . The map

$$T_n \times U \longrightarrow q_n(s)^2(U + \beta_n U), \quad (t, u) \longmapsto q_n(s)^2 u - q_{n-1}(s)^2 t \quad (13.10bh6)$$

is injective. Indeed, reduction modulo A_n first recovers the class of t , hence t itself, and then equality in the function field recovers u . Therefore

$$\frac{\#(U + \beta_n U)}{\#U} \geq \#\pi_n(U). \quad (13.10bh7)$$

The Chinese remainder theorem applies to the pairwise disjoint divisors A_n . The joint reduction map

$$U \longrightarrow \prod_{n \in I} \Gamma(A_n, \mathcal{O}_{A_n})$$

is injective: otherwise a nonzero difference $u - u'$ would have at least $\sum_{n \in I} \deg A_n$ zeros, counted with multiplicity, but by (13.10bh1) at most $\deg D$ poles. Hence $\#U \leq \prod_{n \in I} \#\pi_n(U)$, and multiplication of (13.10bh7) proves (13.10bh3). $\square$

Corollary 13.34 (entropy bound on an algebraic curve). *In the setting of Theorem 13.33, suppose that finite sets U_R satisfy, for constants $\sigma, c, H > 0$,*

$$\log \#U_R \geq (\sigma - o(1))R, \quad (13.10bh8)$$

that every difference of two elements of U_R has a common pole bound of degree at most $(H + o(1))R$, and, uniformly for $n = o(R)$,

$$\log \frac{\#(U_R + \beta_n U_R)}{\#U_R} \leq cn + \Delta_R, \quad \Delta_R = o(R). \quad (13.10bh9)$$

Then

$$\sigma \leq \frac{cH}{2 \deg(s)_\infty}. \quad (13.10bh10)$$

Proof. Choose $N_R = o(R)$ so that

$$N_R \gg \Delta_R, \quad N_R^2 \gg R \log N_R.$$

Write the degree of the common pole bound as at most $HR + e_R$, where $e_R = o(R)$. Choose $\eta_R \downarrow 0$ so slowly that

$$\eta_R R \gg |e_R| + R/N_R + N_R.$$

Greedily select primes in $[N_R, 2N_R]$ until

$$\sum_{n \in I_R} n \geq \frac{HR + \eta_R R}{2 \deg(s)_\infty}. \quad (13.10bh11)$$

The overshoot is at most $2N_R = o(R)$. The prime number theorem gives enough primes: the required number is $O(R/N_R)$, while the interval contains $\gg N_R / \log N_R$ primes and its total prime sum is $\gg N_R^2 / \log N_R \gg R$. Moreover $\#I_R = O(R/N_R)$, so (13.10bh4) gives

$$\sum_{n \in I_R} \deg A_n \geq 2 \deg(s)_\infty \sum_{n \in I_R} n - O(R/N_R) > HR + e_R.$$

Thus (13.10bh2) holds. Apply (13.10bh3) and (13.10bh9). Since $\#I_R = O(R/N_R)$,

$$(\sigma - o(1))R \leq c \sum_{n \in I_R} n + \#I_R \Delta_R = \frac{cHR}{2 \deg(s)_\infty} + o(R).$$

Division by R proves (13.10bh10). $\square$

Lemma 13.35 (sharpness of the function-field radical bound). *In the notation of Theorem 13.33, let R_s be the ramification divisor of the morphism $s : C \rightarrow \mathbb{P}^1$, and let B_n be the reduced part, outside $|D_0|$, of $(q_n(s))_0$. Then*

$$(n-1)d - \deg R_s - \deg(D_0)_{\text{red}} \leq \deg B_n \leq (n-1)d, \quad d = \deg(s)_\infty. \quad (13.10bha)$$

The leading constant is best possible: for $C = \mathbb{P}^1$, $s = T$, and $D_0 = [\infty]$, one has $\deg B_n = n-1$ for every $n \geq 2$.

Proof. After extending the constant field, write

$$(q_n(s))_0 = \sum_{q_n(a)=0} s^*[a].$$

The $n-1$ roots of q_n are distinct, so this divisor has degree $(n-1)d$. Passing to its reduced divisor loses $\sum (e_P(s) - 1) \deg P$ over only some ramification points and hence at most $\deg R_s$. Deleting the fixed support of D_0 loses at most $\deg (D_0)_{\text{red}}$. Degrees are unchanged by the constant-field extension. In the stated example s is unramified and $q_n(T)$ is square-free. $\square$

This is the leading term given by Mason–Stothers. It cannot sharpen (13.10bh10): Theorem 13.33 already uses the nonreduced divisor of $q_n(s)^2$, whose degree is the exact maximum $2(n-1)d$. Replacing it by the radical gives only $(n-1)d + O(1)$, and the example in the lemma rules out a universal strict improvement. Thus the possible loss on a curve comes from the coefficients and from the alignment of their poles, not from a weaker-than-necessary function-field radical estimate.

For the sets given by Theorem 13.32, the natural values are $\sigma = 1/2$ and $c = \ell_x(k)$. The corollary then gives the sharp necessary inequality

$$\deg(s)_\infty \leq \ell_x(k)H. \quad (13.10bh12)$$

This is a product-formula estimate on a curve along which the prescribed trace collisions persist, but it is not by itself a contradiction: the common pole rate H can be governed by different boundary valuations for different words. A strict improvement of the coefficient c , together with one word nearly maximizing $\deg(s)_\infty / \ell_x(k)$ relative to that same pole bound, would exclude the curve. The near-maximizing condition is additional and is not implied merely by taking a supremum over unrelated boundary valuations.

Theorem 13.36 (a lower sumset bound would eliminate every nondistinguished expanding place). *Retain the hypotheses and sets of Theorem 13.32. Suppose that s is algebraic, put $K = \mathbb{Q}(s^2)$, and choose integers $n_R \rightarrow \infty$ such that*

$$n_R = o(R), \quad \Delta_R = o(n_R). \quad (13.10bl)$$

Such a choice always exists; for example one may take the integer part of $\sqrt{R(1 + \Delta_R)}$, after an immaterial monotone adjustment. Suppose that the following lower bound holds along this sequence:

$$\log \frac{\#(W_R + \beta_{n_R} W_R)}{\#W_R} \geq \log H(\beta_{n_R}) - o(n_R). \quad (13.10bm)$$

Then

$$\mathcal{J}_K(s) = \frac{\ell(k)}{2}. \quad (13.10bn)$$

Equivalently, $\chi_v(s) = 0$ at every place of K other than the distinguished real place. In particular, s^2 is an algebraic integer and every nonidentity conjugate of s^2 lies in $[0, 4]$.

Proof. Lemma 13.28 shows that $\mathbb{Q}(\beta_{n_R}) = K$ for all sufficiently large R . Combining (13.10bd), (13.10bm), and Lemma 13.29 gives

$$2n_R \mathcal{J}_K(s) + o(n_R) \leq n_R \ell(k) + \Delta_R.$$

Thus $\mathcal{J}_K(s) \leq \ell(k)/2$. The distinguished real place alone contributes $\ell(k)/2$, and every local term is nonnegative, proving (13.10bn). At a finite place, $\chi_v(s) = 0$ says that the two roots of $z + z^{-1} = s^2 - 2$ are units; hence s^2 is integral there. At an archimedean place it says that those roots have modulus one, so the corresponding conjugate of s^2 is real and belongs to $[0, 4]$. $\square$

One can instead count the distribution of the two trace coordinates modulo the recurrence ideal. This avoids both the loss from dependent random choice and the need for a general sum-of-dilates theorem.

Lemma 13.37 (counting trace pairs by congruence classes). *Let R be a Dedekind domain, let $P, Q \in R \setminus \{0\}$ generate coprime ideals, and let $G \subset U \times V$, where $U, V \subset R$ are finite. Put*

$$X_- = \{Qu - Pv : (u, v) \in G\},$$

$$X_+ = \{Qv - Pu : (u, v) \in G\},$$

and

$$M_{(PQ)}(V) = \max_{c \in R/(PQ)} \#(V \cap (c + (PQ))).$$

Then

$$\#X_- \#X_+ \geq \frac{1}{M_{(PQ)}(V)} \sum_{v \in V} \deg_G(v)^2 \geq \frac{(\#G)^2}{\#V M_{(PQ)}(V)}. \quad (13.10bp0)$$

Proof. A wedge is a triple (u, v, u') with $(u, v), (u', v) \in G$. Map it to

$$(x, y) = (Qu - Pv, Qv - Pu') \in X_- \times X_+.$$

For fixed (x, y) , reduction of these equations gives

$$v \equiv -P^{-1}x \pmod{(Q)}, \quad v \equiv Q^{-1}y \pmod{(P)}.$$

The displayed inverses exist because $(P) + (Q) = R$. The Chinese remainder theorem confines v to one class modulo (PQ) . Once v, x, y are fixed, the two equations determine u, u' uniquely in the fraction field. Thus every fiber of the wedge map has size at most $M_{(PQ)}(V)$. There are $\sum_v \deg_G(v)^2$ wedges, and Cauchy–Schwarz proves the second inequality. $\square$

For $P = q_{n-1}(s)$, $Q = q_n(s)$, the two sets in this lemma are not formal sumsets. Equation (13.10bj) identifies them, up to signs, with actual trace supports

$$\{\mathrm{tr}(k^{-(n-1)}g) : (\mathrm{tr} g, \mathrm{tr}(kg)) \in G\}, \quad \{\mathrm{tr}(k^n g) : (\mathrm{tr} g, \mathrm{tr}(kg)) \in G\}. \quad (13.10bp1)$$

Thus linear trace growth bounds their product by $\exp(R + n\ell(k) + O(1))$ when $G \subset G_R$.

There is an essential normalization when the coefficients are not integral. Let F be a number field containing s and the trace vertices, put $K = \mathbb{Q}(s^2)$, and set $\mathcal{J}_F(s) = [F : K]\mathcal{J}_K(s)$. For

$$P = q_{n-1}(s), \quad Q = q_n(s),$$

define the integral ideal

$$\mathfrak{m}_n = \prod_{\mathfrak{p}} \mathfrak{p}^{|v_{\mathfrak{p}}(P) - v_{\mathfrak{p}}(Q)|}. \quad (13.10bp2)$$

Thus $\mathfrak{m}_n$ is the product of the coprime numerator and denominator ideals of P/Q . If Λ is a fractional $\mathcal{O}_F$ -ideal containing a common rescaling of the vertex sets, the two congruences in the proof of Lemma 13.37 confine the middle vertex to one coset of $\Lambda\mathfrak{m}_n$. Indeed, locally they say

$$v_{\mathfrak{p}}(v - v') - v_{\mathfrak{p}}(\Lambda) \geq \max\{v_{\mathfrak{p}}(P) - v_{\mathfrak{p}}(Q), v_{\mathfrak{p}}(Q) - v_{\mathfrak{p}}(P)\}.$$

This formulation includes primes where the original representation is not integral and is independent of the common rescaling.

At an archimedean embedding, choose an algebraic characteristic root λ with $\lambda + \lambda^{-1} = s$, and put $z = \lambda^2$. If $|\lambda| \neq 1$, then P/Q and its inverse tend to nonzero limits. If $z = 1$, the repeated-root formula $q_n(\pm 2) = (\pm 1)^{n-1}n$ gives polynomial growth. Otherwise z is not a root of unity: if $z^m = 1$, then the conjugate of $q_m(s)$ at this embedding would vanish, although its distinguished real conjugate is positive. Baker's lower bound for linear forms in logarithms gives a constant $C_z > 0$ such that

$$|z^j - 1| \geq j^{-C_z} \quad (j \geq 2)$$

[1]. Applying this to the closed formulas for $q_{n-1}(s)$ and $q_n(s)$ shows that P/Q and its inverse are $n^{O_s(1)}$. Formula (13.10av) and the product formula therefore give

$$\log N_{F/\mathbb{Q}} \mathbf{m}_n = 2n\mathcal{J}_F(s) + O_s(\log n). \quad (13.10bp2a)$$

This is the correct exponent. Taking an unnormalized full ideal norm in the quadratic sign extension can incorrectly double it.

Theorem 13.38 (a uniform congruence-count criterion). *Let G_R be the trace-pair graph (13.10be), with vertex classes U_R, V_R . Write*

$$\#G_R = e^{R-\eta_R}, \quad \eta_R = o(R).$$

Choose integers $n_R \rightarrow \infty$ such that

$$n_R = O(R), \quad \eta_R = o(n_R). \quad (13.10bp3)$$

Suppose that, after a common rescaling into a fractional ideal Λ_R ,

$$\max_c \#(V_R \cap (c + \Lambda_R \mathbf{m}_{n_R})) \leq \frac{\#V_R}{N\mathbf{m}_{n_R}} e^{o(n_R)}. \quad (13.10bp4)$$

Then

$$\mathcal{J}_F(s) = \frac{\ell(k)}{2}.$$

Consequently $F = K$, $\mathcal{J}_K(s) = \ell(k)/2$, and all nondistinguished local expansion exponents vanish.

Proof. The two vertex classes have size at most $e^{R/2+O(1)}$. Apply Lemma 13.37, in the fractional-lattice form just proved, with $n = n_R$. Equations (13.10bp2a)–(13.10bp4) give

$$\begin{aligned} \log(\#X_- \#X_+) &\geq 2\log \#G_R - 2\log \#V_R + \log N\mathbf{m}_{n_R} - o(n_R) \\ &\geq R + 2n_R\mathcal{J}_F(s) - o(n_R). \end{aligned}$$

On the other hand, (13.10bp1) and linear trace growth give

$$\log(\#X_- \#X_+) \leq R + n_R\ell(k) + O(1).$$

Divide by n_R . This yields $2\mathcal{J}_F(s) \leq \ell(k)$; the distinguished real place already contributes $\ell(k)/2$, so equality holds and every other nonnegative local term is zero. Finally, $\mathcal{J}_F(s) = [F : K]\mathcal{J}_K(s)$ and $\mathcal{J}_K(s) \geq \ell(k)/2$, so equality forces $[F : K] = 1$. $\square$

Condition (13.10bp4) asks for a uniform upper bound on the number of trace values in each residue class modulo the recurrence ideal. It can be deduced from a corresponding count of group elements. Remove from V_R vertices of degree less than $\#G_R/(4\#V_R)$, and make the analogous removal on the other side. A positive proportion of the edges remains, while every surviving trace value is represented by at least $\#G_R/(4\#V_R)$ group elements. Consequently a uniform estimate

$$\max_c \#\{g : d(o, go) \leq R + O(1), \operatorname{tr} g \equiv c \pmod{\Lambda_R \mathbf{m}_{n_R}}\} \leq \frac{e^{R+o(n_R)}}{N\mathbf{m}_{n_R}} \quad (13.10bp5)$$

implies (13.10bp4) for the remaining graph and hence gives the theorem. Property τ , together with uniform lattice-point counting, is the natural source of (13.10bp5) [18].

13.2. The congruence criterion over the full trace field. We now put the preceding implication in the form needed for the compact conjectures. This also removes two possible ambiguities. The coefficient field must be the full trace field of the subgroup generated by squares, and the finitely many places at which a matrix model has denominators must be deleted before ideal norms are taken.

Let $H = \Gamma^{(2)}$, let $\tilde{H}$ be its inverse image in $\mathrm{SL}_2(\mathbb{R})$, and suppose for the moment that

$$k = \mathbb{Q}(\mathrm{tr} \tilde{H}) \quad (13.10bpa)$$

is a number field. The k -algebra generated by $\tilde{H}$ is its invariant quaternion algebra B/k . Choose a finite set S of finite places which contains the ramified finite places of B and clears the denominators of a finite generating set. Thus, for an $\mathcal{O}_{k,S}$ -order $\mathcal{B} \subset B$,

$$\tilde{H} \subset \mathrm{SL}_1(\mathcal{B})(\mathcal{O}_{k,S}). \quad (13.10bpb)$$

Enlarge S , once and for all, so that it is saturated over a finite set of rational primes containing the denominators and the bad-reduction and model discriminants of the restriction-of-scalars realization used below: if one place above such a rational prime is removed, all places above it are removed. This harmless enlargement is needed when a number-field ideal is compared with a rational congruence modulus. Norms of ideals of $\mathcal{O}_{k,S}$ will mean norms of their prime-to- S parts.

Lemma 13.39 (the full trace field gives the full restriction-of-scalars closure). *The connected $\mathbb{Q}$ -Zariski closure of the restriction-of-scalars image of $\tilde{H}$ is*

$$\mathrm{Res}_{k/\mathbb{Q}} \mathrm{SL}_1(B). \quad (13.10bpba)$$

Proof. The group $\tilde{H}$ is non-elementary and is Zariski dense in $\mathrm{SL}_1(B)$ over k . After extension to $\overline{\mathbb{Q}}$, the ambient restriction of scalars is a product of SL_2 -factors indexed by the embeddings $\sigma : k \hookrightarrow \overline{\mathbb{Q}}$, and the connected closure projects onto every factor.

A connected subdirect algebraic subgroup of a product of simple A_1 -groups is a product of diagonal blocks. This follows, for example, by applying the Lie-algebra version of Goursat's lemma successively: a surjective subalgebra of a product of simple Lie algebras either contains a factor or identifies it with another factor by an isomorphism. Suppose a block joined two different embeddings $\sigma \neq \tau$. After passing to the finite-index subgroup $\tilde{H}_0$ lying in the connected closure, the two projections would be identified by an automorphism of SL_2 . Every such automorphism preserves trace, so

$$\sigma(\mathrm{tr} h) = \tau(\mathrm{tr} h) \quad (h \in \tilde{H}_0). \quad (13.10bpbb)$$

The invariant trace field of a non-elementary Fuchsian group is unchanged on passage to finite index [26, Chapter 3]. Since $H = \Gamma^{(2)}$, this invariant trace field is precisely the full field k in (13.10bpa). Equation (13.10bpbb), also applied to squares, would therefore make σ and τ agree on k , a contradiction. Every diagonal block is a singleton, proving (13.10bpba). $\square$

Fix $a \in \tilde{H}$ with $s = \mathrm{tr} a > 2$, and put $\ell = \ell(a)$. For the recurrence in Lemma 13.25, set

$$P_n = q_{n-1}(s), \quad Q_n = q_n(s), \quad \mathbf{n}_n = (P_n Q_n) \mathcal{O}_{k,S} \quad (n \geq 2). \quad (13.10bpc)$$

The two factors in (13.10bpc) are coprime in $\mathcal{O}_{k,S}$, by Cassini's identity (13.10ao).

Lemma 13.40 (norm of the prime-to- S recurrence modulus). *With the preceding notation,*

$$\log N \mathbf{n}_n = 2n \mathcal{J}_k(s) + O_{s,S}(\log n). \quad (13.10bpd)$$

Before S is chosen, the intrinsic version is obtained by dividing the two principal fractional ideals by their common fractional-ideal factor.

Proof. Put $\alpha_n = P_n/Q_n$. The local calculation in the proof of Lemma 13.29, now summed over the places of k , gives

$$[k : \mathbb{Q}]h(\alpha_n) = n\mathcal{J}_k(s) + O_s(\log n). \quad (13.10bpe)$$

We explain why deleting S loses only the displayed logarithmic error. At a fixed archimedean place, choose an algebraic characteristic root λ and set $z = \lambda^2$. If the two characteristic roots have different moduli, the quotient of consecutive recurrence terms converges to a nonzero limit. If $z = 1$, use $q_n(\pm 2) = (\pm 1)^{n-1}n$. In the remaining unit-circle case, z is not a root of unity: otherwise $q_m(s)$ would vanish under this embedding for some m , and injectivity of the embedding would give $q_m(s) = 0$, contrary to its positive distinguished conjugate. Baker's linear-forms theorem gives $|z^n - 1| \geq n^{-C_z}$ [1]. The closed formula for q_n now shows that both $\log^+ |\alpha_n|_v$ and $\log^+ |\alpha_n^{-1}|_v$ are $O_s(\log n)$.

The same assertion holds at every fixed nonarchimedean place in S . If $|s|_v > 1$, then $|q_n(s)|_v = |s|_v^{n-1}$, so the valuation of the quotient of two consecutive terms is constant. Suppose that $|s|_v \leq 1$, and pass to a finite extension containing a characteristic root λ . Then λ and $z = \lambda^2$ are units, and, unless $z = 1$,

$$v(q_n(s)) = v(z^n - 1) - v(z - 1). \quad (13.10bpe-local)$$

We use the following elementary local valuation lemma. If $E/\mathbb{Q}_p$ is finite and $z \in \mathcal{O}_E^\times$ is not a root of unity, then

$$v_E(z^n - 1) \leq C_z + v_E(n). \quad (13.10bpe-lte)$$

Indeed, let r be the order of the residue class of z . Choose j so that $w = z^{rp^j}$ lies in a principal-unit neighborhood on which the p -adic logarithm preserves valuations. If $r \nmid n$, the left side is zero. If $n = rp^a u$, with $p \nmid u$, then for $a < j$ it is bounded by the maximum over the finitely many values $a = 0, \dots, j-1$, because a binomial expansion gives $v_E((z^{rp^a})^u - 1) = v_E(z^{rp^a} - 1)$. For $a \geq j$, the logarithm gives

$$v_E(z^n - 1) = v_E(w - 1) + v_E(p^{a-j}u) = v_E(w - 1) + (a - j)v_E(p).$$

This proves (13.10bpe-lte). Equation (13.10bpe-local) therefore gives $v(q_n(s)) = O_{s,v}(\log n)$. When $z = 1$, the repeated-root formula gives the same bound directly. No other root of unity can occur, since $z^m = 1$, $z \neq 1$, would force $q_m(s) = 0$. Thus the archimedean and S -adic parts of the heights of both α_n and α_n^{-1} are $O_{s,S}(\log n)$.

Outside S , coprimality says that the denominator ideals of α_n and α_n^{-1} are respectively (Q_n) and (P_n) . The ideal-height formula applied to (13.10bpe) in the two directions gives

$$\log N(Q_n) = n\mathcal{J}_k(s) + O_{s,S}(\log n), \quad \log N(P_n) = n\mathcal{J}_k(s) + O_{s,S}(\log n).$$

Adding proves (13.10bpd). In particular, even a place at which the original representation is unbounded costs only $O(\log n)$ after the fixed bad set has been deleted. $\square$

Choose a basepoint o on the axis of a , put

$$\mathcal{B}_R = \{g \in \tilde{H} : d(o, \bar{g}o) \leq R\},$$

and form the simple graph

$$\mathcal{G}_R = \{(\text{tr } g, \text{tr}(ag)) : g \in \mathcal{B}_R\}. \quad (13.10bpf)$$

Write $\#\mathcal{G}_R = e^{R-\eta_R}$. Under linear trace growth, orbit counting and Lemma 13.5 give $\eta_R = o(R)$, while both vertex classes have size at most $e^{R/2+O_a(1)}$.

Theorem 13.41 (full-trace-field congruence criterion). *Suppose that the signed trace set of Γ has linear growth. Let $n_R \rightarrow \infty$ satisfy*

$$n_R = O(R), \quad \eta_R = o(n_R). \quad (13.10bpg)$$

Suppose, uniformly in $c \in \mathcal{O}_{k,S}/\mathfrak{n}_{n_R}$, that

$$\#\{g \in \mathcal{B}_R : \text{tr}(ag) \equiv c \pmod{\mathfrak{n}_{n_R}}\} \leq \frac{\exp(R + \epsilon_R)}{N\mathfrak{n}_{n_R}}, \quad \eta_R + \epsilon_R = o(n_R). \quad (13.10bph)$$

Then

$$\mathcal{J}_k(s) = \frac{\ell(a)}{2}. \quad (13.10bpi)$$

Consequently s is an algebraic integer and every nondistinguished embedding $\sigma : k \hookrightarrow \mathbb{C}$ satisfies $\sigma(s) \in [-2, 2]$.

Proof. Let U_R, V_R be the two vertex classes of (13.10bpf). Delete only those vertices $v \in V_R$ for which

$$\deg(v) < d_R := \frac{\#\mathcal{G}_R}{4\#V_R}.$$

Fewer than one quarter of the edges are deleted. If $\mathcal{G}'_R \subset U_R \times V'_R$ is the remaining graph, then

$$\#\mathcal{G}'_R \geq \frac{3}{4}\#\mathcal{G}_R, \quad \#V'_R \geq \frac{\#\mathcal{G}'_R}{\#U_R} = \exp(R/2 - \eta_R - O_a(1)). \quad (13.10bpj)$$

Choose one group element representing each edge. Every retained right vertex occurs for at least d_R of these distinct representatives, so (13.10bph) implies

$$\max_c \#(V'_R \cap (c + \mathfrak{n}_{n_R})) \leq \frac{\#V'_R}{N\mathfrak{n}_{n_R}} \exp(2\eta_R + \epsilon_R + O_a(1)). \quad (13.10bpk)$$

This pruning is necessary: equidistribution of group elements is a weighted statement and does not directly imply flatness of the unweighted trace support.

Apply Lemma 13.37 to $\mathcal{G}'_R$, with $P = P_{n_R}$ and $Q = Q_{n_R}$. The two images are actual trace supports, because

$$\begin{aligned} Q \text{tr } g - P \text{tr}(ag) &= \text{tr}(a^{-(n_R-1)}g), \\ Q \text{tr}(ag) - P \text{tr } g &= \text{tr}(a^{n_R}g). \end{aligned} \quad (13.10bpl)$$

Linear trace growth therefore gives

$$\#X_- \#X_+ \leq \exp(R + n_R\ell + O_a(1)). \quad (13.10bpm)$$

On the other hand, (13.10bpj), (13.10bpk), and the wedge injection give

$$\#X_- \#X_+ \geq \exp(R - o(n_R))N\mathfrak{n}_{n_R}. \quad (13.10bpn)$$

Lemma 13.40, followed by division by n_R , now gives $2\mathcal{J}_k(s) \leq \ell$. The distinguished real place alone contributes $\ell/2$, and all local terms are nonnegative. Equality follows and every other local term vanishes. At a finite place this says $|s|_v \leq 1$; at an archimedean place it says that the roots of $X^2 - \sigma(s)X + 1$ have modulus one, whence $\sigma(s) \in [-2, 2]$. $\square$

Corollary 13.42 (the algebraic compact conjectures from recurrence counting). *Let $\Gamma < \text{PSL}_2(\mathbb{R})$ be torsion-free and cocompact. Assume that its invariant traces are algebraic and that its trace set has linear growth. If (13.10bpg)–(13.10bph) hold for every hyperbolic element of $\widetilde{\Gamma^{(2)}}$, then Γ is arithmetic. Thus this one congruence-counting statement implies both the algebraic compact Schmutz conjecture and, since bounded clustering implies linear growth, the algebraic compact Sarnak conjecture.*

Proof. A finite set of character coordinates generates all traces integrally. Their algebraicity makes k in (13.10bpa) a number field. Apply Theorem 13.41 to every nonidentity element of $\widetilde{\Gamma^{(2)}}$. All its traces are algebraic integers, and every nondistinguished embedding sends every trace into $[-2, 2]$; in particular k is totally real. Takeuchi's criterion [47] makes $\Gamma^{(2)}$, and therefore its finite-index overgroup Γ , arithmetic. $\square$

Every count in this proof is initially a count of group elements: this is true for the graph (13.10bpf), for cocompact orbit counting, and for the congruence estimate. Sparsity concerns only the set of trace values. In a torsion-free Fuchsian group, every nonidentity element is a power of a unique primitive element up to inversion, and $|\operatorname{tr} g| = 2 \cosh(\ell(g)/2)$. This is exactly the trace/primitive-length conversion in the standard formulations. The two choices of lift change cardinalities by at most two.

Lemma 13.43 (quotient mixing implies the required trace count). *Let $\mathfrak{n}$ be prime to S . Suppose strong approximation in the form of Weisfeiler's theorem [52] gives a bounded-index image $Q_{\mathfrak{n}} \subset \operatorname{SL}_1(\mathcal{B})(\mathcal{O}_{k,S}/\mathfrak{n})$, uniformly in $\mathfrak{n}$, and suppose*

$$\max_{x \in Q_{\mathfrak{n}}} \#\{g \in \mathcal{B}_R : \pi_{\mathfrak{n}}(g) = x\} \leq \frac{\exp(R + o(R))}{\#Q_{\mathfrak{n}}}. \quad (13.10bpo)$$

Then, uniformly in c ,

$$\#\{g \in \mathcal{B}_R : \operatorname{tr}(ag) \equiv c \pmod{\mathfrak{n}}\} \leq \frac{\exp(R + o(R))}{N\mathfrak{n}}. \quad (13.10bpp)$$

The errors are uniform in any family for which $\log N\mathfrak{n} = O(R)$ and (13.10bpo) is uniform.

Proof. Outside S , the group scheme is SL_2 . If a prime $\mathfrak{p}$ has residue cardinality q , then, for $A = \mathcal{O}_k/\mathfrak{p}^e$,

$$\max_t \#\{x \in \operatorname{SL}_2(A) : \operatorname{tr} x = t\} \leq (e+1)q^{2e}. \quad (13.10bpq)$$

Indeed, after choosing the upper-left entry u , the determinant equation is $bc = u(t-u)-1$. Splitting b according to its valuation gives at most $(e+1)q^e$ pairs (b, c) . Since $\#\operatorname{SL}_2(A) = q^{3e}(1-q^{-2})$, the Chinese remainder theorem gives

$$\frac{\max_t \#\{x : \operatorname{tr} x = t\}}{\#\operatorname{SL}_1(\mathcal{B})(\mathcal{O}_{k,S}/\mathfrak{n})} \leq \frac{(N\mathfrak{n})^{o(1)}}{N\mathfrak{n}}. \quad (13.10bpr)$$

A bounded-index image changes only the implied constant. Multiplication by a permutes that image, so (13.10bpo)–(13.10bpr) prove (13.10bpp). $\square$

A uniform spectral gap for all the congruence quotients, combined with uniform lattice-point counting on the corresponding compact covers, gives (13.10bpo) in a range $\log N\mathfrak{n} \leq cR$: more precisely the count in a quotient coset has a main term $e^R/\#Q_{\mathfrak{n}}$ and an error $O(e^{(1-\delta)R})$, with $\delta > 0$ independent of the modulus [11, 12, 18]. Equations (13.10bpq)–(13.10bpr) absorb the error after $c > 0$ is chosen sufficiently small. Therefore full arbitrary-ideal superapproximation for the group in (13.10bpb) proves (13.10bph), with $n_R = \lfloor \vartheta R \rfloor$ and $\vartheta > 0$ sufficiently small, and hence proves the algebraic compact Sarnak and Schmutz conjectures.

This last spectral estimate is not currently available in the required generality. Salehi Golsefidy proves expansion for moduli m^E , where m is square-free and the exponent E is fixed in advance [37]. The claimed arbitrary-prime-power extension in [36, 37] has an unresolved issue, recorded explicitly in [48, Remark 2.8], so no such extension is used here. Varjú treats square-free ideals in the split SL_2 case [50]. He and de Saxcé prove all integer moduli for integral groups with absolutely simple Zariski closure [20]; their published theorem therefore does not include $\operatorname{Res}_{k/\mathbb{Q}} \operatorname{SL}_1(B)$ when $[k : \mathbb{Q}] > 1$, because that group becomes a product over $\overline{\mathbb{Q}}$, and it also does not state the needed S -integer version. Tang and Zhang prove all integer moduli, including fixed denominators, for the split non-simple cases $\operatorname{SL}_2 \times \operatorname{SL}_2$ and ASL_2 [48]; their theorem does not cover a general twisted restriction-of-scalars form. The general S -integer arbitrary-modulus assertion remains open. The induction arguments of [38] transfer expansion in certain extensions, but do not prove the twisted restriction-of-scalars assertion required here. Thus Corollary 13.42 is a complete implication theorem, not a claim that this external spectral problem has already been settled.

Proposition 13.44 (the congruence criterion on balls from another Fuchsian metric). *Let x be a faithful cocompact Fuchsian character of a closed surface group Π , and assume that its signed trace set has linear growth. Let $Z \subset V_x$ be a $\mathbb{Q}$ -subvariety through x , and let $y \in Z(\overline{\mathbb{Q}})$ also be a faithful cocompact Fuchsian character. Fix a lift $a \in \widetilde{\Pi^{(2)}}$ of a nonidentity element of $\Pi^{(2)}$, put*

$$s_y = \text{tr}_y(a), \quad \mathcal{B}_R^x = \{g \in \widetilde{\Pi^{(2)}} : d_x(o, \bar{g}o) \leq R\},$$

and choose o on the x -axis of a . Form

$$\mathcal{G}_{R,y} = \{(\text{tr}_y g, \text{tr}_y(ag)) : g \in \mathcal{B}_R^x\}, \quad \#\mathcal{G}_{R,y} = e^{R-\eta_{R,y}}. \quad (13.10bpr1)$$

Let k_y be the full trace field of the specialized subgroup generated by squares, choose its finite bad set S_y , and let $\mathfrak{n}_{n,y}$ be the recurrence ideal (13.10bpc) formed from s_y . Suppose that, for $n_R = \lfloor \vartheta R \rfloor$, where $\vartheta > 0$ is fixed, one has uniformly in c

$$\#\{g \in \mathcal{B}_R^x : \text{tr}_y(ag) \equiv c \pmod{\mathfrak{n}_{n_R,y}}\} \leq \frac{\exp(R + \epsilon_R)}{N\mathfrak{n}_{n_R,y}}, \quad \eta_{R,y} + \epsilon_R = o(R). \quad (13.10bpr2)$$

Then

$$2\mathcal{J}_{k_y}(s_y) \leq \ell_x(a). \quad (13.10bpr3)$$

Proof. Every equality of squared traces at x holds on V_x , hence at y . An equality of signed traces at x can split into at most the two choices of sign at y . Linear trace growth at x therefore bounds both coordinate projections of (13.10bpr1) by $e^{R/2+O_a(1)}$.

The representation at y is faithful Fuchsian. The two-trace finite-type estimate, Lemma 13.5, applied to the fixed x -ball, bounds every fiber of the two coordinates by $e^{o(R)}$; word length and x -displacement are comparable. Cocompact orbit counting in the metric x now gives

$$\#\mathcal{G}_{R,y} = e^{R-o(R)}, \quad \eta_{R,y} = o(R). \quad (13.10bpr4)$$

The recurrence identities hold in the representation y :

$$\begin{aligned} q_n(s_y) \text{tr}_y g - q_{n-1}(s_y) \text{tr}_y(ag) &= \text{tr}_y(a^{-(n-1)}g), \\ q_n(s_y) \text{tr}_y(ag) - q_{n-1}(s_y) \text{tr}_y g &= \text{tr}_y(a^n g). \end{aligned} \quad (13.10bpr5)$$

The elements on the right lie in the x -ball of radius $R + n\ell_x(a) + O_a(1)$. Persistence again bounds their y -trace supports, up to the two signs, by the corresponding x -trace supports. Consequently

$$\#X_- \#X_+ \leq \exp(R + n\ell_x(a) + O_a(1)). \quad (13.10bpr6)$$

The pruning and congruence-class counting proof of Theorem 13.41, using (13.10bpr2), is otherwise unchanged and gives

$$\#X_- \#X_+ \geq \exp(R - o(n_R))N\mathfrak{n}_{n_R,y} = \exp(R + 2n_R\mathcal{J}_{k_y}(s_y) - o(n_R)). \quad (13.10bpr7)$$

Comparison of (13.10bpr6) and (13.10bpr7), followed by division by n_R , proves (13.10bpr3). $\square$

There is no change in the spectral interpretation of the hypothesis in Proposition 13.44. A congruence kernel defined by the algebraic representation y is a finite-index abstract subgroup of Π , hence also defines a cover of the fixed surface (S, x) . Expansion of the y -congruence quotients gives a uniform spectral gap on these x -covers by the Brooks–Burger transfer [11, 12], and uniform lattice counting then proves (13.10bpr2) for sufficiently small ϑ . No displacement ball for the metric y is used.

Lemma 13.45 (local expansion and marked Fuchsian rigidity). *Let x and y be faithful cocompact Fuchsian characters of the same closed surface group Π , lying in the same Fuchsian component, and assume that y is algebraic. Let k_y be the full trace field of the subgroup generated by squares. Suppose that every nonidentity $b \in \Pi^{(2)}$ satisfies*

$$\mathcal{J}_{k_y}(\mathrm{tr}_y b) \leq \frac{\ell_x(b)}{2}. \quad (13.10\mathrm{bpr}9)$$

Then $y = x$.

Proof. For $\gamma \in \Pi \setminus \{1\}$, choose a lift b of γ^2 . Since $\mathcal{J}_{k_y}(-t) = \mathcal{J}_{k_y}(t)$, the choice of lift does not matter. At the distinguished real embedding,

$$|\mathrm{tr}_y b| = 2 \cosh\left(\frac{\ell_y(\gamma^2)}{2}\right),$$

so this place contributes $\ell_y(\gamma^2)/2$ to the nonnegative sum $\mathcal{J}_{k_y}(\mathrm{tr}_y b)$. Hence

$$\ell_y(\gamma) = \frac{\ell_y(\gamma^2)}{2} \leq \mathcal{J}_{k_y}(\mathrm{tr}_y b) \leq \frac{\ell_x(\gamma^2)}{2} = \ell_x(\gamma).$$

Thurston's minimal-stretch theorem identifies the logarithm of the least Lipschitz constant in the marked homotopy class with

$$d_{\mathrm{Th}}(x, y) = \log \sup_{\gamma \neq 1} \frac{\ell_y(\gamma)}{\ell_x(\gamma)}.$$

The common Fuchsian component identifies x and y with marked points of the same Teichmüller space. The displayed formula defines a nonnegative separating asymmetric metric on that space [49]. The preceding length domination gives $d_{\mathrm{Th}}(x, y) \leq 0$. Therefore $d_{\mathrm{Th}}(x, y) = 0$, and separation gives $y = x$. $\square$

For comparison, we formulate a general superapproximation hypothesis that would also imply both compact conjectures, including the case of a transcendental character. Let $\mathrm{SA}_{\mathrm{all}}$ denote the following assertion: for every number field k , quaternion algebra B/k , finite set S , and finitely generated Zariski-dense subgroup

$$\Delta < \mathrm{SL}_1(B)(\mathcal{O}_{k,S})$$

whose full trace field is k , the Cayley graphs of its reduction images modulo all ideals prime to S have a uniform spectral gap.

Theorem 13.46 (arbitrary-ideal superapproximation implies both compact conjectures). *Assume $\mathrm{SA}_{\mathrm{all}}$. Then every torsion-free cocompact Fuchsian lattice whose signed trace set has linear growth is arithmetic. Consequently both the compact Schmutz linear-growth conjecture and the compact Sarnak bounded-clustering conjecture hold.*

Proof. Let x be the Fuchsian character. If x is algebraic, the expansion hypothesis, Lemma 13.43, and Corollary 13.42 prove arithmeticity.

Suppose that x is transcendental. Its rational Zariski closure has positive dimension. Proposition 13.2 gives an algebraic Fuchsian point $y \neq x$, arbitrarily close to x in the same Fuchsian component, at which all trace collisions at x persist. The hypothesis $\mathrm{SA}_{\mathrm{all}}$ and the preceding spectral interpretation verify (13.10bpr2). Thus, for every nonidentity element of $\Pi^{(2)}$ and either lift $a \in \Pi^{(2)}$,

$$\mathcal{J}_{k_y}(\mathrm{tr}_y a) \leq \frac{1}{2} \ell_x(a). \quad (13.10\mathrm{bpr}8)$$

Lemma 13.45 now gives $y = x$, contrary to the choice of y .

Thus x is algebraic and the first paragraph applies. Finally, bounded clustering implies linear trace growth, so the Sarnak implication follows from the Schmutz implication. $\square$

Nor can estimates for the prime-power factors simply be multiplied. The following finite model records the exact obstruction.

Proposition 13.47 (flat marginals need not be flat modulo a product). *Let p be prime and let*

$$V = \{(t, t) : t \in \mathbb{F}_p\} \subset \mathbb{F}_p \times \mathbb{F}_p.$$

Each coordinate projection is perfectly flat: every residue class has exactly

$$\frac{\#V}{p} = 1$$

preimage. But the joint reduction occupies only p of the p^2 classes, and its largest fiber is 1, rather than the formal product prediction $\#V/p^2 = 1/p$. Equivalently, if

$$E_i(V) = \sum_{a \in \mathbb{F}_p} \#\{v \in V : v_i = a\}^2,$$

then $E_i(V) = \#V^2/p = p$ for each factor, whereas the joint collision energy is $\#V = p$, not $\#V^2/p^2 = 1$.

Proof. Both coordinate maps restrict to bijections on V , while the joint map is the inclusion of the diagonal. The formulas follow immediately. $\square$

If $\log p$ is proportional to the recurrence index, the discrepancy in Proposition 13.47 is exponential. Therefore one-prime or one-prime-power trace estimates do not imply (13.10bph) for a mixed recurrence ideal. One would need either a spectral theorem directly for the product quotient or an estimate that rules out correlations between the different prime-power coordinates.

The available expansion theorems nevertheless give an unconditional restriction on the argument with a fixed group element whenever a nondistinguished place contributes positive recurrence height. For a fixed integer $D \geq 1$, write

$$\mathfrak{c}_{n,D} = \prod_{\mathfrak{p} \notin S} \mathfrak{p}^{\min\{v_{\mathfrak{p}}(\mathfrak{n}_n), D\}}, \quad \mathfrak{h}_{n,D} = \mathfrak{n}_n \mathfrak{c}_{n,D}^{-1}. \quad (13.10bps)$$

Thus $\mathfrak{c}_{n,D}$ is obtained by truncating every local exponent at D , while $\mathfrak{h}_{n,D}$ contains the remaining powers.

Theorem 13.48 (large prime-power contributions from a nondistinguished expanding place). *Under the algebraic and linear-growth hypotheses above, for every fixed hyperbolic $a \in \tilde{H}$ and every fixed D ,*

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log N \mathfrak{c}_{n,D} \leq \ell(a). \quad (13.10bpt)$$

Consequently

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log N \mathfrak{h}_{n,D} \geq 2\mathcal{J}_k(s) - \ell(a). \quad (13.10bpu)$$

If a has any nondistinguished local expansion, the right side of (13.10bpu) is positive. Hence, for every fixed cutoff D , a linear amount of the recurrence height is forced into prime powers of exponent strictly greater than D .

Proof. After restriction of scalars, (13.10bpb) embeds the finitely generated group in $\mathrm{GL}_N(\mathbb{Z}[1/q_0])$ for one fixed q_0 . If m is the square-free product of the rational primes below $\mathfrak{c}_{n,D}$, then

$$m^D \mathcal{O}_{k,S} \subset \mathfrak{c}_{n,D}. \quad (13.10bpv)$$

Indeed, at a prime $\mathfrak{p} \mid p$, the exponent on the left is $De(\mathfrak{p}/p)$, which is at least $\min\{v_{\mathfrak{p}}(\mathfrak{n}_n), D\}$; recall that larger ideal exponents mean smaller ideals. The rational prime p is prime to q_0 by the saturation convention on S . Reduction modulo $\mathfrak{c}_{n,D}$ is therefore a quotient of reduction modulo the bounded power m^D .

By Lemma 13.39, the connected $\mathbb{Q}$ -Zariski closure is the full group $\text{Res}_{k/\mathbb{Q}} \text{SL}_1(B)$. It is semisimple and therefore perfect; moreover, it is connected and simply connected, since both properties are preserved by restriction of scalars. Salehi Golsefidy's theorem for square-free moduli raised to a fixed power [37] applies; absolute simplicity is not required in this theorem. Weisfeiler's strong-approximation theorem [52], applied after restriction of scalars and after enlarging the fixed bad set, gives a uniformly bounded-index congruence image and hence $\#Q_c \gg (Nc)^3$ for $c = \mathfrak{c}_{n,D}$. The spectral gap descends from modulus m^D to its quotient at modulus c .

For completeness, uniform lattice-point counting on the congruence covers gives, for some $\delta_D > 0$, a count per quotient element bounded by

$$\frac{e^{R+o(R)}}{\#Q_c} + O(e^{(1-\delta_D)R}).$$

The local trace-fiber estimate (13.10bpq), the Chinese remainder theorem, and the ideal-divisor bound $\prod_{\mathfrak{p} \in c} (e+1) = (Nc)^{o(1)}$ therefore give

$$\#\{g \in \mathcal{B}_R : \text{tr}(ag) \equiv t \pmod{c}\} \ll (Nc)^{o(1)} \left\{ \frac{e^R}{Nc} + (Nc)^2 e^{(1-\delta_D)R} \right\}. \quad (13.10bpw)$$

Lemma 13.40 gives $\log Nc \leq 2\mathcal{J}_k(s)n + O(\log n)$. Take $R = C_{D,a,\Gamma}n$, with the fixed constant so large that $\delta_D C_{D,a,\Gamma} > 6\mathcal{J}_k(s)$. Then the second term in (13.10bpw) is at most $e^{R+o(n)}/Nc$, and $\eta_R = o(n)$. Repeat the proof of Theorem 13.41, with $\mathfrak{c}_{n,D}$ in place of $\mathfrak{n}_n$. Comparison of (13.10bpm) and (13.10bpn) gives

$$\log N\mathfrak{c}_{n,D} \leq n\ell(a) + o(n),$$

which proves (13.10bpt). Subtract this from Lemma 13.40 to obtain (13.10bpu). $\square$

Theorem 13.49 (a one-term Chebyshev radical bound). *Retain the algebraic and linear-trace-growth hypotheses of Theorem 13.48. Fix a hyperbolic $a \in \tilde{H}$, choose its lift so that $s = \text{tr } a > 2$, and put*

$$\ell = \ell(a), \quad P_n = q_{n-1}(s), \quad Q_n = q_n(s).$$

For $n \geq 2$, define the prime-to- S ideals

$$\mathfrak{r}_n = \prod_{\substack{\mathfrak{p} \notin S \\ v_{\mathfrak{p}}(Q_n) > 0}} \mathfrak{p}, \quad \mathfrak{d}_n = \mathfrak{r}_n^2. \quad (13.10bpw1)$$

Thus $\mathfrak{d}_n$ is the ideal square of the radical of the single recurrence term $q_n(s)$; it is not the radical of $q_n(s)^2$. Then

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log N\mathfrak{r}_n \leq \frac{\ell(a)}{2}. \quad (13.10bpw2)$$

Consequently, if $\mathfrak{q}_n = (Q_n)\mathfrak{r}_n^{-1}$ is the part of the principal prime-to- S ideal left after one copy of each prime divisor is removed, then

$$\liminf_{n \rightarrow \infty} \frac{1}{n} \log N\mathfrak{q}_n \geq \mathcal{J}_k(s) - \frac{\ell(a)}{2}. \quad (13.10bpw3)$$

In particular, any nondistinguished expanding place forces a positive linear amount of norm into repeated prime factors of every individual term $q_n(s)$.

Proof. Let m_n be the square-free product of the rational primes lying below the prime ideals occurring in $\mathfrak{r}_n$. The rational saturation of S gives

$$m_n^2 \mathcal{O}_{k,S} \subseteq \mathfrak{d}_n. \quad (13.10bpw4)$$

To check the direction of this inclusion, fix $\mathfrak{p} \mid p$. The exponent of $\mathfrak{p}$ on the left is $2e(\mathfrak{p}/p)$. It is at least 2 if $\mathfrak{p} \mid \mathfrak{d}_n$, while the required exponent is zero at the other primes. Since larger ideal exponents give smaller ideals, (13.10bpw4) follows. Hence

$$\mathcal{O}_{k,S}/m_n^2 \mathcal{O}_{k,S} \twoheadrightarrow \mathcal{O}_{k,S}/\mathfrak{d}_n,$$

so a spectral gap modulo the bounded power m_n^2 descends to the quotient modulo $\mathfrak{d}_n$.

After restriction of scalars, the connected Zariski closure is the perfect semisimple group $\text{Res}_{k/\mathbb{Q}} \text{SL}_1(B)$, by Lemma 13.39. Salehi Golsefidy's theorem for fixed powers of square-free rational moduli, used here with the square, therefore applies [37]. Combining it with uniform lattice counting and the local trace-fiber estimate (13.10bpq) gives, for some fixed $\delta_2 > 0$,

$$\max_c \# \{g \in \mathcal{B}_R : \text{tr } g \equiv c \pmod{\mathfrak{d}_n}\} \ll (N\mathfrak{d}_n)^{o(1)} \left\{ \frac{e^R}{N\mathfrak{d}_n} + (N\mathfrak{d}_n)^2 e^{(1-\delta_2)R} \right\}. \quad (13.10bpw5)$$

The factor $(N\mathfrak{d}_n)^{o(1)}$ comes from the ideal-divisor factor in the local trace count. Since $\log N\mathfrak{d}_n = O(n)$, it is $\exp(o(n))$; the lattice-counting error denoted by $o(R)$ below is also $o(n)$ after $R = Cn$. Since $(Q_n^2) \subseteq \mathfrak{d}_n$, the one-factor norm estimate in the proof of Lemma 13.40 gives

$$\log N\mathfrak{d}_n \leq 2n\mathcal{J}_k(s) + O_{s,S}(\log n). \quad (13.10bpw6)$$

Choose a fixed $C > 6\mathcal{J}_k(s)/\delta_2$ and put $R = Cn$. The ratio of the error term in braces in (13.10bpw5) to its main term is

$$(N\mathfrak{d}_n)^{3+o(1)} e^{-\delta_2 R} \leq \exp((6\mathcal{J}_k(s) - \delta_2 C)n + o(n)) = o(1).$$

Thus

$$\max_c \# \{g \in \mathcal{B}_R : \text{tr } g \equiv c \pmod{\mathfrak{d}_n}\} \leq \frac{e^{R+o(n)}}{N\mathfrak{d}_n}. \quad (13.10bpw7)$$

Use the set W_R produced in the proof of Theorem 13.32, as a subset of the left vertex class $\{\text{tr } g : g \in \mathcal{B}_R\}$. Lemma 13.31 allows the two selected vertices to be equal. Applied to the repeated pair (u, u) , its common-neighbour conclusion says that every $u \in W_R$ has degree at least r_R . Equations (13.10ba) and (13.10bi) consequently give

$$\#W_R r_R \geq \frac{1}{4} N_R^2 \exp(-a_R(t_R + 1) - L_R/t_R) = \exp(R - o(R)). \quad (13.10bpw8)$$

Choose one group element representing each edge incident to W_R ; different edges have different representatives. If W_R meets $M_{R,n}$ residue classes modulo $\mathfrak{d}_n$, all these representatives lie in the union of those $M_{R,n}$ residue classes. In particular,

$$M_{R,n} \max_c \# \{g \in \mathcal{B}_R : \text{tr } g \equiv c \pmod{\mathfrak{d}_n}\} \geq \#W_R r_R.$$

Equations (13.10bpw7)–(13.10bpw8), and $R = Cn$, now give the explicit support estimate

$$M_{R,n} \geq \frac{\#W_R r_R}{e^{R+o(n)}/N\mathfrak{d}_n} \geq N\mathfrak{d}_n e^{-o(n)}. \quad (13.10bpw9)$$

No second application of dependent random choice and no further degree pruning are needed.

Let $T_R \subset W_R$ contain exactly one representative from each occupied residue class modulo $\mathfrak{d}_n$, and recall that $\beta_n = -P_n^2/Q_n^2$. The map

$$T_R \times W_R \longrightarrow Q_n^2(W_R + \beta_n W_R), \quad (t, u) \longmapsto Q_n^2 u - P_n^2 t \quad (13.10bpw10)$$

is injective. Indeed, reduction of an equality of two images modulo $\mathfrak{d}_n$ kills the Q_n^2 -terms. Because $\mathfrak{d}_n$ is supported on primes dividing Q_n , Cassini's identity makes P_n a unit modulo $\mathfrak{d}_n$, so it recovers the residue class of t . The definition of T_R then gives $t = t'$, and equality in the number field gives $u = u'$. Therefore (13.10bpw9) implies

$$\frac{\#(W_R + \beta_n W_R)}{\#W_R} \geq N\mathfrak{d}_n e^{-o(n)}. \quad (13.10bpw11)$$

The simultaneous upper bound (13.10bd), at $R = Cn$, is

$$\frac{\#(W_R + \beta_n W_R)}{\#W_R} \leq \exp(n\ell + o(n)).$$

Comparison gives $\log N\mathfrak{d}_n \leq n\ell + o(n)$. Since $N\mathfrak{d}_n = (N\mathfrak{r}_n)^2$, this proves (13.10bpw2). Finally,

$$\log N((Q_n)) = n\mathcal{J}_k(s) + O_{s,S}(\log n) = \log N\mathfrak{r}_n + \log N\mathfrak{q}_n,$$

which proves (13.10bpw3). $\square$

Proposition 13.50 (residue-class deficit modulo a complete recurrence term). *Retain the hypotheses and notation of Theorem 13.49. Let W_R and Δ_R be given by Theorem 13.32, and put*

$$\mathfrak{m}_n = (q_n(s)^2)\mathcal{O}_{k,S}, \quad M_{R,n} = \#\mathrm{im}(W_R \longrightarrow \mathcal{O}_{k,S}/\mathfrak{m}_n). \quad (13.10bpw12)$$

Then, for every R and $n \geq 2$,

$$M_{R,n} \leq \frac{\#(W_R + \beta_n W_R)}{\#W_R} \leq \exp(n\ell(a) + \Delta_R). \quad (13.10bpw13)$$

Consequently, for every fixed $C > 0$,

$$\log \frac{N\mathfrak{m}_n}{M_{Cn,n}} \geq (2\mathcal{J}_k(s) - \ell(a))n - o(n). \quad (13.10bpw14)$$

In particular, if along an unbounded sequence

$$M_{Cn,n} \geq N\mathfrak{m}_n e^{-o(n)}, \quad (13.10bpw15)$$

then $2\mathcal{J}_k(s) \leq \ell(a)$. The contribution of the distinguished real place gives the reverse inequality, so equality holds and every nondistinguished local expansion exponent vanishes.

Proof. Write $P_n = q_{n-1}(s)$, $Q_n = q_n(s)$, and choose $T_{R,n} \subset W_R$ with one representative of every class modulo $\mathfrak{m}_n$. The map

$$T_{R,n} \times W_R \longrightarrow Q_n^2(W_R + \beta_n W_R), \quad (t, u) \longmapsto Q_n^2 u - P_n^2 t,$$

is injective. Reducing an equality modulo (Q_n^2) recovers the residue class of t , because Cassini's identity makes P_n a unit modulo (Q_n^2) . The choice of representatives then gives $t = t'$, and equality in the number field gives $u = u'$. This proves the first inequality in (13.10bpw13); the second is (13.10bd). Finally,

$$\log N\mathfrak{m}_n = 2n\mathcal{J}_k(s) + O_{s,S}(\log n),$$

and $\Delta_{Cn} = o(n)$, which proves (13.10bpw14) and the final assertion. $\square$

This criterion asks less than superapproximation for arbitrary ideals. It requires only that the particular set W_{Cn} , obtained by dependent random choice, occupy almost all residue classes modulo the particular principal ideal $(q_n(s)^2)$. Conversely, (13.10bpw14) says that a nondistinguished place with positive recurrence height forces this set to miss an exponentially large proportion of that quotient. Bounded-power expansion proves the required estimate only after $(q_n(s)^2)$ is replaced by the square of its radical. Theorem 13.49 measures exactly the resulting loss.

The same bounded-power estimate gives a square-divisor sieve when the group element, rather than the recurrence index, is allowed to vary. We first record the exact local calculation. It also explains why divisibility by $\mathfrak{p}^2$ is not a condition of codimension two modulo $\mathfrak{p}$: it is a second-order congruence along a smooth divisor.

Lemma 13.51 (local density of a Chebyshev power divisor). *Fix $m \geq 2$. Let $\mathfrak{p}$ be a prime ideal at which the group scheme is split, put $q = N\mathfrak{p}$, and assume that the residue characteristic does not divide $2m$. Define*

$$r_+(m, q) = \frac{\gcd(2m, q-1) - 2}{2}, \quad r_-(m, q) = \frac{\gcd(2m, q+1) - 2}{2}. \quad (13.10bpw16)$$

For every $e \geq 1$,

$$\frac{\#\{x \in \mathrm{SL}_2(\mathcal{O}_k/\mathfrak{p}^e) : q_m(\mathrm{tr} x) = 0\}}{\#\mathrm{SL}_2(\mathcal{O}_k/\mathfrak{p}^e)} = \frac{r_+(m, q)(q+1) + r_-(m, q)(q-1)}{q^{e-1}(q^2-1)}. \quad (13.10bpw17)$$

In particular, this density is

$$\frac{r_+(m, q) + r_-(m, q)}{q^e} + O_m(q^{-e-1}), \quad (13.10bpw18)$$

and the set counted in the numerator on the left of (13.10bpw17) has $O_m(q^{2e})$ elements in $\mathrm{SL}_2(\mathcal{O}_k/\mathfrak{p}^e)$.

Moreover, for $x \in \mathrm{SL}_2(\mathcal{O}_k/\mathfrak{p}^e)$,

$$q_m(\mathrm{tr} x) = 0 \implies x^m \in \{I, -I\}. \quad (13.10bpw19)$$

Conversely, if the reduction of x modulo $\mathfrak{p}$ is not central, then x^m scalar implies $q_m(\mathrm{tr} x) = 0$. On the two central residue discs, a scalar m -th power occurs only for $x = I$ and $x = -I$. Thus the scalar-power locus consists of the divisor counted in (13.10bpw17), together with these two central points.

Proof. Write $t = \lambda + \lambda^{-1}$. Away from $t = \pm 2$,

$$q_m(t) = \frac{\lambda^m - \lambda^{-m}}{\lambda - \lambda^{-1}}. \quad (13.10bpw20)$$

Hence a split root over $\mathbb{F}_q$ is obtained from an element $\lambda \in \mathbb{F}_q^\times$ satisfying $\lambda^{2m} = 1$, with $\lambda = \pm 1$ deleted and λ identified with λ^{-1} . This gives $r_+(m, q)$ roots. The nonsplit roots are obtained in the same way from the norm-one subgroup of $\mathbb{F}_{q^2}^\times$, whose order is $q+1$, and their number is $r_-(m, q)$. The assumption on the residue characteristic makes all these roots simple. It also excludes ± 2 , since

$$q_m(2) = m, \quad q_m(-2) = (-1)^{m-1}m.$$

For $t \in \mathbb{F}_q$, direct counting in the determinant equation gives

$$\#\{x \in \mathrm{SL}_2(\mathbb{F}_q) : \mathrm{tr} x = t\} = q(q + \chi(t^2 - 4)), \quad (13.10bpw21)$$

where χ is the quadratic character. Indeed, after choosing the upper-left entry a , one counts pairs (b, c) with $bc = a(t - a) - 1$. A split root in (13.10bpw20) has $\chi(t^2 - 4) = 1$, and a nonsplit root has character -1 . Every root of q_m has one Hensel lift modulo $\mathfrak{p}^e$. The trace morphism on SL_2 is smooth above a regular trace, so every point of its fibre modulo $\mathfrak{p}$ has $q^{2(e-1)}$ lifts in the fibre of a prescribed lifted trace. Since

$$\#\mathrm{SL}_2(\mathcal{O}_k/\mathfrak{p}^e) = q^{3(e-1)}q(q^2 - 1),$$

equation (13.10bpw17) follows.

Cayley–Hamilton gives

$$x^m = q_m(\mathrm{tr} x)x - q_{m-1}(\mathrm{tr} x)I. \quad (13.10bpw22)$$

If the first coefficient vanishes, Cassini's identity gives $q_{m-1}(\mathrm{tr} x)^2 = 1$, which proves (13.10bpw19). Conversely, if x^m is scalar and x is noncentral modulo $\mathfrak{p}$, an off-diagonal entry or the difference of the two diagonal entries in (13.10bpw22) shows that $q_m(\mathrm{tr} x) = 0$ modulo $\mathfrak{p}^e$. Finally, write an element in a central residue disc as $x = \epsilon u$, where $\epsilon \in \{I, -I\}$ and $u \equiv I \pmod{\mathfrak{p}}$. If x^m is scalar, then u^m is a scalar congruent to I modulo $\mathfrak{p}$. Its determinant is one, so it equals I , because the

residue characteristic is odd. The kernel of reduction is a finite group of residue-characteristic power order, whereas that characteristic does not divide m ; hence $u = I$. $\square$

We now state the precise quotient-counting hypothesis used in the sieve. For each fixed integer $E \geq 2$, suppose that there are $\delta_E > 0$ and $C_E > 0$ such that, for every good prime ideal $\mathfrak{p}$, every $1 \leq e \leq E$, and every residue class x in the congruence image,

$$\frac{\#\{g \in \mathcal{B}_R : g \equiv x \pmod{\mathfrak{p}^e}\}}{\#\mathcal{B}_R} \leq \frac{C_0}{\#\mathrm{SL}_2(\mathcal{O}_k/\mathfrak{p}^e)} + C_E e^{-\delta_E R}. \quad (13.10\mathrm{bpw}23)$$

Here C_0 is independent of E , e , and $\mathfrak{p}$, and $\#\mathcal{B}_R = e^{R+O(1)}$. A uniformly bounded congruence-image index gives the constant C_0 . In the present setting, (13.10bpw23) follows, after enlarging the fixed set of bad primes, from superapproximation for these fixed powers and uniform lattice-point counting [37, 18]. Only a fixed value of E is used at any one time. In particular, no assertion about an unbounded prime-power exponent is being invoked.

Theorem 13.52 (localization of repeated prime factors from bounded powers). *Fix $m \geq 2$, enlarge S by the primes below $2m$, and put*

$$A_m(g) = q_m(\mathrm{tr} g).$$

Assume (13.10bpw23). For a fixed $D \geq 2$, define

$$\mathcal{E}_{m,z}^{(D)}(g) = \sum_{\substack{\mathfrak{p} \notin S \\ N\mathfrak{p} \leq z}} \min\{(v_{\mathfrak{p}}(A_m(g)) - 1)_+, D - 1\} \log N\mathfrak{p}. \quad (13.10\mathrm{bpw}24)$$

If $0 < \alpha < \delta_D/(2D + 1)$, then

$$\frac{1}{\#\mathcal{B}_R} \sum_{g \in \mathcal{B}_R} \mathcal{E}_{m,e^{\alpha R}}^{(D)}(g) = O_{m,D,k}(1). \quad (13.10\mathrm{bpw}25)$$

The average is over group elements, counted with multiplicity; it is not an average over distinct trace values. It follows that a proportion $1 - o(1)$ of the elements satisfy

$$\mathcal{E}_{m,e^{\alpha R}}^{(D)}(g) = o(R). \quad (13.10\mathrm{bpw}26)$$

Suppose in addition that the standard height estimate

$$\sum_{\mathfrak{p} \notin S} v_{\mathfrak{p}}(A_m(g)) \log N\mathfrak{p} \leq H_m R + O_m(1) \quad (g \in \mathcal{B}_R) \quad (13.10\mathrm{bpw}27)$$

holds. This estimate is automatic for a fixed polynomial in the matrix entries of a finitely generated S -integral group. Then, for every $\epsilon > 0$, there is a fixed $D = D(\epsilon, m, k)$, followed by a fixed $\alpha = \alpha(\epsilon, m, k, \Gamma) > 0$, such that

$$\limsup_{R \rightarrow \infty} \frac{1}{R \#\mathcal{B}_R} \sum_{g \in \mathcal{B}_R} \sum_{\substack{\mathfrak{p} \notin S \\ N\mathfrak{p} \leq e^{\alpha R}}} (v_{\mathfrak{p}}(A_m(g)) - 1)_+ \log N\mathfrak{p} < \epsilon. \quad (13.10\mathrm{bpw}28)$$

Consequently, after first making the right side smaller than ϵ^2 , Markov's inequality shows that a proportion at least $1 - \epsilon - o(1)$ of the group elements have repeated-prime-factor contribution at most ϵR from primes of norm at most $e^{\alpha R}$.

Proof. Lemma 13.51 and (13.10bpw23) give, uniformly for $2 \leq j \leq D$,

$$\frac{\#\{g \in \mathcal{B}_R : v_{\mathfrak{p}}(A_m(g)) \geq j\}}{\#\mathcal{B}_R} \leq C_0 C_m (N\mathfrak{p})^{-j} + C_{m,D} (N\mathfrak{p})^{2j} e^{-\delta_D R}, \quad (13.10\mathrm{bpw}29)$$

where C_m is independent of D and j . The elementary identity

$$\min\{(v-1)_+, D-1\} = \sum_{j=2}^D \mathbf{1}_{v \geq j}$$

therefore reduces the average in (13.10bpw25) to a sum of (13.10bpw29). The main term is bounded because

$$\sum_{\mathfrak{p}} \frac{\log N\mathfrak{p}}{(N\mathfrak{p})^j} < \infty \quad (j \geq 2),$$

and prime-ideal counting bounds the total error by

$$O_{m,D,k}(e^{-\delta_D R} z^{2D+1}).$$

This proves (13.10bpw25). Taking the exceptional threshold to be $\sqrt{R}$, for example, proves (13.10bpw26) by Markov's inequality.

For the second assertion use (13.10bpw23) only through the fixed depth $D+1$. The part above the cutoff in (13.10bpw24) is bounded pointwise, using (13.10bpw27), by

$$(H_m R + O_m(1)) \mathbf{1}_{\{\exists \mathfrak{p}: N\mathfrak{p} \leq z, v_{\mathfrak{p}}(A_m(g)) \geq D+1\}}. \quad (13.10bpw30)$$

The union bound and (13.10bpw29), now at depth $D+1$, show that if $0 < \alpha < \delta_{D+1}/(2D+3)$, then

$$\begin{aligned} & \frac{\#\{g \in \mathcal{B}_R : \exists \mathfrak{p}, N\mathfrak{p} \leq e^{\alpha R}, v_{\mathfrak{p}}(A_m(g)) \geq D+1\}}{\#\mathcal{B}_R} \\ & \leq C_0 C_m \sum_{\mathfrak{p} \notin S} (N\mathfrak{p})^{-(D+1)} + o(1). \end{aligned} \quad (13.10bpw31)$$

The prime-ideal sum tends to zero as the fixed integer D tends to infinity. Choose D first so that (13.10bpw30)–(13.10bpw31) contribute less than ϵR on average, then choose α , and only then let R tend to infinity. The truncated sum is $o(R)$ on average by (13.10bpw25). This proves (13.10bpw28). Notice that no uniform lower bound for δ_D , and no choice $D = D(R)$, is required. $\square$

The order of quantifiers in Theorem 13.52 is essential. The exponent α may decrease when D is increased. Moreover, (13.10bpw27) only confines a prime which occurs to exponent at least two to

$$N\mathfrak{p} \leq \exp(H_m R/2 + O_m(1)).$$

Thus the theorem leaves the interval

$$e^{\alpha R} < N\mathfrak{p} \leq \exp(H_m R/2 + O_m(1)) \quad (13.10bpw32)$$

untouched. A single prime in this interval can account for a positive multiple of R in the difference between the logarithmic norm and the logarithm of the radical.

The following elementary family shows that the uncontrolled interval cannot be removed by a formal refinement of the preceding estimate. Fix

$$0 < \alpha < \beta < H/2, \quad 0 < h < \beta.$$

For all sufficiently large R , the prime number theorem gives $\lfloor e^{hR} \rfloor$ distinct primes $p_i \in [e^{\beta R}, 2e^{\beta R}]$. The integers $a_i = p_i^2$ satisfy

$$\log a_i \leq HR + O(1), \quad \log \frac{a_i}{\text{rad}(a_i)} = \beta R + O(1),$$

but no prime at most $e^{\alpha R}$ square-divides any a_i , and the repeated-prime supports of distinct a_i are disjoint. Hence even perfect information below $e^{\alpha R}$, together with all estimates for pairs or larger tuples sharing the same repeated prime, does not force one member of such a family to have a nearly full radical. If one also requires compatibility with the Chebyshev congruence, choose $p_i \equiv 1$

(mod $2m$); equation (13.10bpw20) and Hensel lifting then give regular classes $t_i \pmod{p_i^2}$ for which $p_i^2 \mid q_m(t_i)$. Removing the interval (13.10bpw32) therefore requires an additional large-prime argument, not a further consequence of superapproximation for fixed powers of square-free moduli.

We avoid this large-prime problem by fixing the recurrence index and varying the group element whose trace enters the recurrence. Roth's theorem then gives a lower bound for the reduced divisor. Expansion modulo squares is enough to choose the varying element so that no prime occurs to a large power. Thus the available expansion theorem for fixed powers suffices.

We begin with the precise Diophantine statement. Absolute values on a number field k are normalized to extend the standard absolute values on $\mathbb{Q}$, and $n_v = [k_v : \mathbb{Q}_v]$. Thus $\sum_v n_v \log |a|_v = 0$, and

$$[k : \mathbb{Q}]h(a) = \sum_v n_v \log^+ |a|_v = \frac{1}{2} \sum_v n_v |\log |a|_v|.$$

Lemma 13.53 (Roth's theorem for a reduced numerator–denominator divisor). *Let k be a number field, let S be a finite set of finite places, and let $f \in k(T)$ have degree $d \geq 2$. Suppose that every zero and every pole of f on $\mathbb{P}_k^1$ is simple. For $t \in k$ which is neither a zero nor a pole, put*

$$\mathfrak{m}_{f,S}(t) = \prod_{\mathfrak{p} \notin S} \mathfrak{p}^{|\text{ord}_{\mathfrak{p}} f(t)|}.$$

For every $\epsilon > 0$, outside a finite set of $t \in k$,

$$\log N\mathfrak{m}_{f,S}(t) \geq (2d - 2 - \epsilon)[k : \mathbb{Q}]h(t) - O_{f,S,\epsilon}(1). \quad (13.10mov1)$$

The exceptional set and the implied constant may depend on f, S, ϵ , which remain fixed while t varies.

Proof. Let T consist of S and all archimedean places. The product formula and height functoriality give

$$\sum_{v \in M_k} n_v |\log |f(t)|_v| = 2[k : \mathbb{Q}]h(f(t)) = 2d[k : \mathbb{Q}]h(t) + O_f(1). \quad (13.10mov2)$$

At a place $v \in T$, the quantity $|\log |f(t)|_v|$, up to a bounded local term, is a Weil proximity function to one member of the finite set consisting of the zeros and poles of f . Simplicity is important here: every target occurs with multiplicity one. Pass once to a fixed finite extension L/k which contains every zero and pole, and sum over all extensions $w \mid v$ of each $v \in T$. Roth's theorem over L , divided by $[L : k]$, retains the coefficient 2 with the base-field normalization: $[L : \mathbb{Q}]h(t)/[L : k] = [k : \mathbb{Q}]h(t)$ for $t \in k$, and the local degrees above each v sum to $[L : k]n_v$. There are only finitely many ways to choose one such target at each place of L above T . The number-field Roth theorem, applied to each choice, gives

$$\sum_{v \in T} n_v |\log |f(t)|_v| \leq (2 + \epsilon)[k : \mathbb{Q}]h(t) + O_{f,S,\epsilon}(1) \quad (13.10mov3)$$

outside a finite set. We use here the simultaneous-place form of Roth's theorem, for example [4, Theorem 6.4.1]. The finite-place part outside S of the left side of (13.10mov2) is exactly $\log N\mathfrak{m}_{f,S}(t)$. Subtracting (13.10mov3) from (13.10mov2), and replacing ϵ by a smaller positive number if necessary, proves (13.10mov1). $\square$

For $N \geq 3$, put

$$P_N(T) = q_{N-1}(T), \quad Q_N(T) = q_N(T), \quad F_N(T) = P_N(T)Q_N(T).$$

The rational function P_N/Q_N has degree $N - 1$. Its finite zeros and poles are simple and disjoint, and infinity is one further simple zero. Indeed, the Chebyshev polynomials q_j are square-free in characteristic zero, while Cassini's identity makes two consecutive terms coprime.

Corollary 13.54 (the reduced divisor of two consecutive recurrence terms). *Let k, S, N be fixed, with $N \geq 3$, and enlarge S so that the arguments under consideration are S -integral. For every $\epsilon > 0$, outside a finite set of $t \in k$,*

$$\log N(F_N(t)\mathcal{O}_{k,S}) \geq (2N - 4 - \epsilon)\mathcal{J}_k(t) - O_{N,S,\epsilon}(1). \quad (13.10\text{mov4})$$

Here and below the norm of a principal S -ideal means the norm of its prime-to- S part.

Proof. Cassini's identity says that $P_N(t)$ and $Q_N(t)$ generate the unit ideal of $\mathcal{O}_{k,S}$. Consequently $\mathfrak{m}_{P_N/Q_N,S}(t) = F_N(t)\mathcal{O}_{k,S}$. Apply Lemma 13.53 with $d = N - 1$. The local definition (13.10au) gives, uniformly in $t \in k$,

$$|\mathcal{J}_k(t) - [k : \mathbb{Q}]h(t)| \leq [k : \mathbb{Q}] \log 2 :$$

at a nonarchimedean place the two sides have the same local contribution, and at each archimedean place their difference is bounded. This converts (13.10mov1) into (13.10mov4). $\square$

We next use expansion modulo squares to sieve the group elements. The order of quantifiers is important. Let x be a faithful cocompact Fuchsian character of a closed surface group Π , and let y be an algebraic faithful cocompact Fuchsian character in the rational closure of x . Thus every squared-trace equality at x also holds at y . Write $\Delta = \Pi^{(2)}$, let k_y be the full trace field of the representation y , and use a fixed finite generating set of Δ for word length.

Lemma 13.55 (selection of elements with bounded recurrence exponents). *Fix a noncentral $b \in \Delta$, an integer $N \geq 3$, and a sufficiently small fixed $\theta > 0$. There are a rationally saturated finite set $S = S(N, \theta, b, y)$ of finite places of k_y , an integer $D = D(N, \theta, b, y)$, and, for every sufficiently large m , an element*

$$a_m = b^m h_m, \quad |h_m| \leq \theta m, \quad (13.10\text{mov5})$$

with the following properties. If $s_m = \text{tr}_y(a_m)$, then

$$\mathcal{J}_{k_y}(s_m) \geq m\mathcal{J}_{k_y}(\text{tr}_y b) - O_{b,y}(\theta m) - O_{b,y}(1), \quad (13.10\text{mov6})$$

and, for a fixed basepoint o on the x -axis of b ,

$$D_x(a_m) := d_x(o, a_m o) \leq m\ell_x(b) + O_{b,x}(\theta m) + O_{b,x}(1). \quad (13.10\text{mov7})$$

Moreover, $F_N(s_m) \neq 0$, every exponent in the prime-to- S ideal

$$\mathfrak{c}_m = F_N(s_m)\mathcal{O}_{k_y,S} \quad (13.10\text{mov8})$$

is at most D , and

$$\log N\mathfrak{c}_m = O_{N,b,y}(m). \quad (13.10\text{mov9})$$

Only expansion modulo the square of a rational prime is used in choosing a_m . In particular, neither D -th-power expansion nor a lower bound for a spectral gap depending on D is used.

Proof. First enlarge a rationally saturated set S_0 so that the representation is integral, the restriction-of-scalars model has good reduction outside S_0 , and the primes below $2N(N - 1)$ are removed. Let h be sampled from the distribution of the lazy random walk on Δ after $L = \lfloor \theta m \rfloor$ steps, and consider $a = b^m h$.

There are only finitely many places v of k_y for which $\chi_v(\text{tr}_y b) > 0$. Over one fixed finite extension which splits the quaternion algebra and the characteristic polynomial of b , choose at each such place the eigenvalue λ_v with $|\lambda_v|_v > 1$. Then

$$\text{tr}_y(b^m h) = \lambda_v^m L_{v,+}(h) + \lambda_v^{-m} L_{v,-}(h), \quad (13.10\text{mov10})$$

where $L_{v,+}$ and $L_{v,-}$ are fixed regular linear functions of the matrix entries of h , and $L_{v,+}(I) = 1$. Take the product of the Galois norms of all the $L_{v,+}$. After clearing a fixed denominator this is a nonzero regular function Φ , defined over $\mathbb{Q}$, on the restriction-of-scalars group.

Choose one sufficiently large good rational prime r . The Lang–Weil estimate [23] and the bounded index of the congruence image make the proportion of its reduction on which $\Phi = 0$ smaller than $1/8$. Enlarge S_0 by all places above r . Mixing on this one fixed quotient then gives

$$\Pr(\Phi(h) = 0) < \frac{1}{4} \quad (13.10\text{mov}11)$$

for all sufficiently large m . If $\Phi(h) \neq 0$, the height of each nonzero $L_{v,+}(h)$, and the height of $L_{v,-}(h)$, is $O(L)$. The product formula therefore bounds the negative logarithm of the former and the positive logarithm of the latter at v by $O(L)$. For θ sufficiently small, the first term of (13.10mov10) dominates the second and gives

$$\chi_v(\text{tr}_y(b^m h)) \geq m\chi_v(\text{tr}_y b) - O_{b,y}(L) - O_{b,y}(1).$$

Summing over the finitely many expanding places proves (13.10mov6). The triangle inequality and the bounded displacement of the fixed generators give (13.10mov7).

It remains to impose bounded exponents without destroying (13.10mov11). For a value of h with $\Phi(h) \neq 0$, formula (13.10mov6) already makes $\mathcal{J}_{k_y}(\text{tr}_y(b^m h))$ grow linearly. Thus, for large m , its trace avoids the fixed zero set of F_N . The word complexity of $b^m h$ is $O(m)$, so the usual height estimate for the fixed polynomial $F_N(\text{tr}_y(\cdot))$ gives a constant A_N such that

$$\log N(F_N(\text{tr}_y(b^m h))\mathcal{O}_{k_y, S_0}) \leq A_N m + O_N(1) \quad (13.10\text{mov}12)$$

whenever the value is nonzero. Choose a large constant Z , and enlarge S_0 by every place above a rational prime at most Z ; call the result S .

Outside S , the polynomial $F_N = P_N Q_N$ is square-free. The local calculation of Lemma 13.51, applied to its two disjoint factors and followed by Hensel lifting, shows that for a good rational prime p

$$\Pr(\mathfrak{p}^2 \mid F_N(\text{tr}_y(b^m h)) \text{ for some } \mathfrak{p} \mid p) \leq C_N p^{-2} + p^B e^{-\delta L}. \quad (13.10\text{mov}13)$$

Here the union over all prime ideals above p is taken inside the single restriction-of-scalars quotient modulo p^2 . More explicitly, $p^2 \mathcal{O}_{k_y} \subseteq \mathfrak{p}^2$, so reduction modulo rational p^2 surjects onto reduction modulo every $\mathfrak{p}^2$ above it. The constants $B, C_N, \delta > 0$ are fixed. Salehi Golsefidy’s expansion theorem for moduli p^2 applies to that quotient, starting the walk at the arbitrary point b^m ; summing the pointwise mixing error over the local divisor costs only the fixed power p^B , and the bounded index of the congruence image is absorbed into C_N . Lemma 13.39 establishes the required connected perfect Zariski closure. No modulus p^e with $e > 2$ occurs here.

If an exponent in (13.10mov8) is greater than D , (13.10mov12) shows that the rational prime below it is at most

$$Y_m = \exp\left(\frac{A_N m + O_N(1)}{D + 1}\right).$$

Indeed, $(D + 1) \log N\mathfrak{p} \leq \log N\mathfrak{c}_m \leq A_N m + O_N(1)$, and the rational prime p below $\mathfrak{p}$ satisfies $p \leq N\mathfrak{p} \leq Y_m$. The union bound in (13.10mov13) gives

$$\begin{aligned} & \Pr(\text{some prime-to-}S \text{ exponent exceeds } D) \\ & \leq C_N \sum_{p > Z} p^{-2} + \exp\left(\left(\frac{(B + 1)A_N}{D + 1} - \delta\theta\right)m + o(m)\right). \end{aligned} \quad (13.10\text{mov}14)$$

Choose Z first so that the first term is less than $1/8$, and then choose the fixed integer D so that the exponential term tends to zero. For large m , (13.10mov14) is less than $1/4$. Its union with the event in (13.10mov11) still has probability less than one, so there is a value h_m satisfying all the assertions. Formula (13.10mov6) tends to infinity linearly; hence s_m eventually avoids the roots of F_N and any prescribed finite set. Finally (13.10mov9) follows from (13.10mov12). $\square$

The next lemma gives a fiber bound uniform in the varying multiplier a_m . Its proof avoids diagonalizing a_m over a field that depends on m .

Lemma 13.56 (uniform fibers for a varying pair of trace coordinates). *Retain x, y, Δ as above. Suppose that $A_m \in \Delta$ has word complexity $O(m)$ and is noncentral. For each fixed $C > 0$, uniformly in $\xi, \zeta \in k_y$,*

$$\#\{H \in \Delta : d_x(o, Ho) \leq Cm, \operatorname{tr}_y H = \xi, \operatorname{tr}_y(A_m H) = \zeta\} = \exp(o_C(m)). \quad (13.10\text{mov15})$$

The same conclusion holds for squared traces, with only the four possible choices of signs added.

Proof. Pass once to a fixed finite splitting field for the invariant quaternion algebra and use the fixed arithmetic filtration of Section 12. Write

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad H = \begin{pmatrix} r & s \\ u & \xi - r \end{pmatrix}, \quad w = \zeta - d\xi,$$

and put

$$\Delta_A = (a + d)^2 - 4, \quad y_0 = 2br - (a - d)s, \quad K = b\{\xi(a - d) - 2w\}.$$

Since A is a noncentral element of a cocompact Fuchsian group, $\Delta_A \neq 0$. If $b \neq 0$, direct elimination using $\det H = 1$ and $\operatorname{tr}(AH) = \zeta$ gives

$$(\Delta_A s + K)^2 - \Delta_A(y_0 - b\xi)^2 = K^2 - \Delta_A b^2(\xi^2 - 4). \quad (13.10\text{mov16})$$

For fixed A, ξ, ζ , the map

$$H \mapsto (Y, V) = (y_0 - b\xi, \Delta_A s + K)$$

is injective: V recovers s , Y recovers r , and the trace equation recovers u . All coefficients and all possible solutions in (13.10mov16) have complexity $O_C(m)$.

The right side of (13.10mov16) is

$$4b^2(\operatorname{tr}[A, H] - 2), \quad (13.10\text{mov17})$$

by the Fricke identity. If it is nonzero, Lemma 12.3 gives $\exp(o_C(m))$ possible pairs (Y, V) , and hence as many H . If it is zero, then $\operatorname{tr}[A, H] = 2$. The faithful cocompact Fuchsian image has no nontrivial parabolic or unipotent element, so $[A, H] = I$. The centralizer of a nonidentity element of a surface group is cyclic; fixing $\operatorname{tr} H = \xi$ leaves $O(1)$ choices.

If $b = 0$ and $c \neq 0$, transpose the calculation. If $b = c = 0$, the two trace equations determine both diagonal entries of H , and $\det H = 1$ fixes the product su . A nonzero product has $\exp(o_C(m))$ factorizations by Theorem 12.1. A zero product would make the commutator unipotent; cocompactness again forces commutation and leaves only $O(1)$ choices. This proves (13.10mov15). Replacing two squared trace conditions by their four signed versions proves the last assertion. $\square$

We now combine the bounded exponent condition with the congruence count in Lemma 13.37. The basepoint remains fixed throughout; neither the axis nor the conjugation varies.

Theorem 13.57 (upper bound for bounded-exponent recurrence ideals). *Retain x, y, Δ , and suppose that the signed trace set at x has linear growth. Fix $N \geq 3$. Let $a_m \in \Delta$ have word complexity $O(m)$, assume that a_m is noncentral for all sufficiently large m , put $s_m = \operatorname{tr}_y(a_m)$, and suppose that*

$$D_x(a_m) = d_x(o, a_m o) = O(m).$$

Let S and D be fixed, with S rationally saturated and containing all bad and ramified primes, and assume that the nonzero ideals

$$\mathfrak{c}_m = (q_{N-1}(s_m)q_N(s_m))\mathcal{O}_{k_y, S}$$

have every prime exponent at most D and satisfy $\log N\mathfrak{c}_m = O(m)$. Then

$$\log N\mathfrak{c}_m \leq \frac{2N-1}{2} D_x(a_m) + o(m). \quad (13.10\text{mov}18)$$

The error may depend on N, S, D, x, y and on fixed implied constants in the word-complexity, displacement, and ideal-norm bounds for a_m . All of these are fixed before $m \rightarrow \infty$.

Proof. Let M_m be the square-free product of the rational primes below $\mathfrak{c}_m$. Rational saturation and the removal of ramified primes give

$$M_m^D \mathcal{O}_{k_y, S} \subseteq \mathfrak{c}_m. \quad (13.10\text{mov}19)$$

Thus the quotient modulo $\mathfrak{c}_m$ is a quotient of the reduction modulo the fixed power M_m^D of a square-free rational modulus. Lemma 13.39 and bounded-power superapproximation [37], together with the Brooks–Burger transfer and uniform lattice counting used after Proposition 13.44, imply the following. There is a sufficiently large fixed C , depending also on the fixed implied complexity and norm bounds for the sequence a_m , such that, for $R = Cm$,

$$\max_c \#\{g \in \mathcal{B}_R^x : \text{tr}_y g \equiv c \pmod{\mathfrak{c}_m}\} \leq \frac{e^{R+o(m)}}{N\mathfrak{c}_m}. \quad (13.10\text{mov}20)$$

Here is the quantitative point in this invocation. The local trace fiber has relative size $(N\mathfrak{c}_m)^{-1+o(1)}$, because all local exponents are at most D . The expander error is at most a fixed power of $N(M_m^D \mathcal{O}_{k_y, S})$ times $e^{-\delta_D R}$. Both this norm and $N\mathfrak{c}_m$ have logarithm $O_D(m)$, so choosing C after D absorbs the error. No uniformity of δ_D as $D \rightarrow \infty$ is asserted or needed.

Form the simple bipartite graph, with the trace involving a_m in the first coordinate,

$$\mathcal{G}_m = \{(u, v) = (\text{tr}_y(a_m g), \text{tr}_y g) : g \in \mathcal{B}_R^x\}. \quad (13.10\text{mov}21)$$

Cocompact orbit counting gives $e^{R+O(1)}$ elements in the ball, and Lemma 13.56 bounds every fiber of the two coordinates by $e^{o(m)}$. Hence

$$\#\mathcal{G}_m = e^{R-o(m)}. \quad (13.10\text{mov}22)$$

Let V_m be the second vertex class. Persistence of squared-trace collisions from x to y , with the two possible signs restored, and linear trace growth at x give

$$\#V_m \leq e^{R/2+O(1)}. \quad (13.10\text{mov}23)$$

Delete the vertices of V_m whose degree is less than $\#\mathcal{G}_m/(4\#V_m)$, and let $\mathcal{G}'_m$ be the remaining graph. At most one quarter of the edges are deleted. Choose one group element representing each remaining edge. Equation (13.10mov20) then implies

$$M_{\mathfrak{c}_m}(V'_m) := \max_c \#\{v \in V'_m : v \equiv c \pmod{\mathfrak{c}_m}\} \leq \frac{e^{R/2+o(m)}}{N\mathfrak{c}_m}. \quad (13.10\text{mov}24)$$

Indeed, every retained v occurs in at least $\#\mathcal{G}_m/(4\#V_m) = e^{R/2-o(m)}$ distinct chosen representatives to the same congruence class.

Apply Lemma 13.37 over $\mathcal{O}_{k_y, S}$ to $\mathcal{G}'_m$, with

$$P = q_{N-1}(s_m), \quad Q = q_N(s_m).$$

Cassini's identity makes P, Q coprime. Equations (13.10mov22)–(13.10mov24) give

$$\#X_- \#X_+ \geq \frac{(\#\mathcal{G}'_m)^2}{\#V'_m M_{\mathfrak{c}_m}(V'_m)} \geq e^{R-o(m)} N\mathfrak{c}_m. \quad (13.10\text{mov}25)$$

The recurrence identities identify the two images with actual traces:

$$\begin{aligned} Q \operatorname{tr}_y(a_m g) - P \operatorname{tr}_y g &= \operatorname{tr}_y(a_m^N g), \\ Q \operatorname{tr}_y g - P \operatorname{tr}_y(a_m g) &= \operatorname{tr}_y(a_m^{-(N-1)} g). \end{aligned} \quad (13.10\text{mov}26)$$

Since the basepoint is fixed,

$$d_x(o, a_m^j g o) \leq R + j D_x(a_m) \quad (j \geq 0),$$

and the analogous inequality holds for negative powers. Persistence and linear trace growth therefore give

$$\#X_- - \#X_+ \leq \exp\left(R + \frac{2N-1}{2} D_x(a_m) + O(1)\right). \quad (13.10\text{mov}27)$$

Comparison of (13.10mov25) and (13.10mov27) proves (13.10mov18). $\square$

As $N \rightarrow \infty$, the coefficient in Roth's lower bound and the coefficient in the geometric upper bound have the same leading term. Their comparison gives the following unconditional estimate.

Theorem 13.58 (recurrence height at algebraic specializations preserving trace equalities). *Let x be a faithful cocompact Fuchsian character of a closed surface group Π , and suppose that its signed trace set has linear growth. Let y be any algebraic faithful cocompact Fuchsian character in the rational closure of x , and let k_y be the full trace field of the subgroup generated by squares. Then every nonidentity $b \in \widehat{\Pi^{(2)}}$ satisfies*

$$\mathcal{J}_{k_y}(\operatorname{tr}_y b) \leq \frac{\ell_x(b)}{2}. \quad (13.10\text{mov}28)$$

Proof. If b is central, then $b = \pm I$, $\mathcal{J}_{k_y}(\operatorname{tr}_y b) = 0 = \ell_x(b)$, and there is nothing to prove. Assume henceforth that b is noncentral. Fix in this order an integer $N \geq 3$, positive numbers ϵ, θ , with θ sufficiently small, and then the set S , exponent D , and sequence a_m given by Lemma 13.55. Finally choose the constant C in Theorem 13.57. All of these data remain fixed while $m \rightarrow \infty$. Formula (13.10mov6) makes s_m leave every finite set, so Corollary 13.54 applies for all sufficiently large m . Together with (13.10mov18) it gives

$$(2N-4-\epsilon)(m\mathcal{J}_{k_y}(\operatorname{tr}_y b) - O(\theta m) - O(1)) \leq \frac{2N-1}{2}(m\ell_x(b) + O(\theta m) + O(1)) + o(m). \quad (13.10\text{mov}29)$$

Let $m \rightarrow \infty$, then $\theta \rightarrow 0$, then $\epsilon \rightarrow 0$, and finally $N \rightarrow \infty$. This proves (13.10mov28). Equivalently, the complete quantifier order is: fix N, ϵ, θ ; choose the fixed sieve cutoff, S, D, C ; let $m \rightarrow \infty$; then let $\theta, \epsilon \rightarrow 0$; and only then let $N \rightarrow \infty$. $\square$

Theorem 13.59 (trace sparsity forces arithmeticity). *Let $\Gamma < \operatorname{PSL}_2(\mathbb{R})$ be torsion-free and cocompact. If its signed trace set has linear growth, then Γ is arithmetic. Consequently the compact Schmutz linear-growth conjecture and the compact Sarnak bounded-clustering conjecture hold.*

Proof. Suppose first that the Fuchsian character x is algebraic and take $y = x$. Theorem 13.58 gives the upper bound (13.10mov28), while the distinguished real place contributes $\ell_x(b)/2$ and every other local contribution is nonnegative. Thus equality holds, all finite-place contributions vanish, and all nondistinguished archimedean contributions vanish. Apply this to every nonidentity element of $\widehat{\Gamma^{(2)}}$. All its traces are algebraic integers, its full trace field is totally real, and every nondistinguished embedding sends every trace into $[-2, 2]$. Takeuchi's criterion [47] makes $\Gamma^{(2)}$, and hence Γ , arithmetic.

Suppose now that x is transcendental. Proposition 13.2 gives an algebraic faithful Fuchsian character $y \neq x$, arbitrarily close to x in the same Fuchsian component, on which all squared-trace

collisions at x persist. Theorem 13.58 gives, for every nonidentity $b \in \widetilde{\Pi^{(2)}}$,

$$\mathcal{J}_{k_y}(\mathrm{tr}_y b) \leq \frac{\ell_x(b)}{2}. \quad (13.10\mathrm{mov}30)$$

Lemma 13.45 therefore gives $y = x$, contradicting the choice of y . Thus the transcendental case cannot occur, and the algebraic argument proves arithmeticity. Bounded clustering implies linear growth, which gives the Sarnak conclusion from the Schmutz conclusion. $\square$

Remark 13.60 (the independent pointwise radical problem). Theorem 13.59 does not prove the universal pointwise assertion that every fixed Chebyshev sequence has the full radical rate. That assertion still contains the Mersenne and non-Wieferich problem described below. The proof instead uses the group family in an essential way: for each fixed recurrence index, it chooses a nearby multiplier $b^m h_m$ whose reduced recurrence divisor has bounded exponents. The pointwise radical results and examples below therefore remain separate results, but the proof of the compact trace-sparsity conjectures does not depend on them.

Primitive-divisor theorems for Lucas sequences guarantee new prime factors, but they do not show that the bounded-exponent part in (13.10bps) captures $e^{(2\mathcal{J}_k(s)-o(1))n}$ of the norm. Exponential control of the powerful part is closely related to unresolved radical and Wieferich-type questions. Theorem 13.48 gives additional information about this problem for the recurrence of a fixed element without assuming a stronger Lucas-sequence estimate.

The recurrence permits a further sharpening. Except at finitely many places, divisibility acquired after a prime first appears is accounted for by the ordinary prime divisors of the index. Those later contributions have only logarithmic total norm. Any exponential powerful part must therefore come from primes which already divide the recurrence term to unusually high order when they first appear.

Lemma 13.61 (local valuation formula and the Lucas–Wieferich excess). *Let $s \in \mathcal{O}_{k,S}$, let $q_0 = 0, q_1 = 1$, and let $q_{m+1} = sq_m - q_{m-1}$. Enlarge S by the prime ideals dividing $2(s^2 - 4)$. For a prime ideal $\mathfrak{p} \notin S$ which divides some $q_m(s)$, let $r_{\mathfrak{p}}$ be the least positive integer r for which $\mathfrak{p} \mid q_r(s)$, and put*

$$w_{\mathfrak{p}} = v_{\mathfrak{p}}(q_{r_{\mathfrak{p}}}(s)).$$

Then

$$\mathfrak{p} \mid q_m(s) \iff r_{\mathfrak{p}} \mid m, \quad v_{\mathfrak{p}}(q_m(s)) = w_{\mathfrak{p}} + v_{\mathfrak{p}}(m/r_{\mathfrak{p}}). \quad (13.10\mathrm{bpx})$$

For $D \geq 1$, define

$$\mathfrak{W}_{m,D} := \prod_{\substack{\mathfrak{p} \notin S \\ r_{\mathfrak{p}} \mid m}} \mathfrak{p}^{\max\{w_{\mathfrak{p}} - D, 0\}}, \quad \mathfrak{h}_{m,D}^{(q)} := \prod_{\mathfrak{p} \notin S} \mathfrak{p}^{\max\{v_{\mathfrak{p}}(q_m(s)) - D, 0\}}.$$

Then

$$\log N \mathfrak{h}_{m,D}^{(q)} \leq \log N \mathfrak{W}_{m,D} + [k : \mathbb{Q}] \log m + O_{s,S}(1). \quad (13.10\mathrm{bpy})$$

If the finitely many newly excluded places are retained instead, their total contribution is $O_{s,S}(\log m)$.

Proof. Let $L = k(\lambda)$, where $\lambda + \lambda^{-1} = s$. At a place outside S , the extension L/k is unramified, and λ and $\lambda^2 - 1$ are units. The closed formula is

$$q_m(s) = \lambda^{-(m-1)} \frac{\lambda^{2m} - 1}{\lambda^2 - 1}. \quad (13.10\mathrm{bpz})$$

Thus $r_{\mathfrak{p}}$ is the multiplicative order of the reduction of λ^2 at any place of L above $\mathfrak{p}$. In the split case the two reductions are inverse and have the same order. The local lifting-the-exponent formula, in odd residue characteristic, applied after writing $m = r_{\mathfrak{p}}u$, proves (13.10bpx).

For every prime in the product,

$$\max\{w_{\mathfrak{p}} + v_{\mathfrak{p}}(u) - D, 0\} \leq \max\{w_{\mathfrak{p}} - D, 0\} + v_{\mathfrak{p}}(m).$$

Summing against $\log N\mathfrak{p}$ proves (13.10bpy), since $\log N(m\mathcal{O}_k) = [k : \mathbb{Q}] \log m$. At each of the finitely many excluded places, (13.10bpz) and the corresponding local lifting-the-exponent estimate give $v_{\mathfrak{p}}(q_m(s)) = O_{s,\mathfrak{p}}(\log m)$. $\square$

Theorem 13.62 (a nondistinguished expanding place forces large prime-power multiplicities). *Use the notation of Theorem 13.48, and fix $D \geq 1$. If the signed trace set has linear growth, then every hyperbolic a , with $s = \text{tr } a$, satisfies*

$$\liminf_{n \rightarrow \infty} \frac{1}{n} (\log N\mathfrak{W}_{n,D} + \log N\mathfrak{W}_{n-1,D}) \geq 2\mathcal{J}_k(s) - \ell(a). \quad (13.10bpza)$$

In particular, if a has a nondistinguished expanding place, then, for every fixed D , the primes whose initial multiplicities $w_{\mathfrak{p}}$ exceed D carry positive exponential norm. Conversely, if for some fixed D

$$\log N\mathfrak{W}_{n,D} + \log N\mathfrak{W}_{n-1,D} = o(n) \quad (13.10bpzb)$$

along an unbounded sequence, then every nondistinguished local expansion exponent of s vanishes.

Proof. Cassini's identity makes $(q_n(s))$ and $(q_{n-1}(s))$ coprime. Apply Lemma 13.61 to both factors. It gives

$$\log N\mathfrak{h}_{n,D} \leq \log N\mathfrak{W}_{n,D} + \log N\mathfrak{W}_{n-1,D} + O_{s,S}(\log n).$$

The lower bound (13.10bpu) now proves (13.10bpza). Conversely, (13.10bpzb) and the same estimate show that $\log N\mathfrak{h}_{n,D} = o(n)$. Hence

$$\log N\mathfrak{c}_{n,D} = 2n\mathcal{J}_k(s) - o(n)$$

along the chosen sequence. Comparing this with (13.10bpt) yields $2\mathcal{J}_k(s) \leq \ell(a)$. The contribution of the distinguished real place gives the reverse inequality, so equality holds and every other nonnegative local term is zero. $\square$

Proposition 13.63 (the one-term radical bound at an algebraic specialization). *Let x be a faithful cocompact Fuchsian character whose signed trace set has linear growth, and let y be an algebraic Fuchsian point in V_x . For a nonidentity element $\bar{a} \in \Pi^{(2)}$, choose the lift $a \in \Pi^{(2)}$ for which*

$$s_y = \text{tr}_y(a) > 2, \quad Q_{n,y} = q_n(s_y).$$

Let k_y be the full trace field at y , choose a rationally saturated bad set S_y , and set

$$\mathfrak{r}_{n,y} = \prod_{\substack{\mathfrak{p} \notin S_y \\ v_{\mathfrak{p}}(Q_{n,y}) > 0}} \mathfrak{p}. \quad (13.10bpzc)$$

Then

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log N\mathfrak{r}_{n,y} \leq \frac{\ell_x(\bar{a})}{2}. \quad (13.10bpzcb)$$

All constants may depend on the fixed specialization y .

Proof. Use the graph $\mathcal{G}_{R,y}$ of Proposition 13.44, with a as chosen above. Persistence of squared-trace collisions at x bounds each coordinate set at y by $e^{R/2+O_a(1)}$: a signed trace value at x can split into at most its two signs at y . The two-trace finite-type estimate at y , together with cocompact orbit counting in the metric x , gives

$$\#\mathcal{G}_{R,y} = e^{R-o(R)}. \quad (13.10bpzcc1)$$

Apply Lemma 13.31 to this graph with the parameters used in the proof of Theorem 13.32. It produces a set

$$W_{R,y} \subset \{\text{tr}_y g : g \in \mathcal{B}_R^x\}$$

of $e^{R/2-o(R)}$ left vertices, each incident to $e^{R/2-o(R)}$ edges. Indeed, repeated pairs are allowed in Lemma 13.31, and for the pair (u, u) the common neighbours are exactly the neighbours of u . If this minimum degree is denoted by $r_{R,y}$, then

$$\#W_{R,y} r_{R,y} = e^{R-o(R)}. \quad (13.10bpzcc1a)$$

Repeating the graph–Ruzsa calculation with the recurrence identities at y gives, simultaneously for every $n \geq 2$,

$$\frac{\#(W_{R,y} + \beta_{n,y}W_{R,y})}{\#W_{R,y}} \leq \exp(n\ell_x(\bar{a}) + o(R)), \quad \beta_{n,y} = -\frac{q_{n-1}(s_y)^2}{q_n(s_y)^2}. \quad (13.10bpzcc2)$$

Indeed, the recurrence images are y -traces of elements in x -balls of radius $R + n\ell_x(\bar{a}) + O_a(1)$, and persistence transfers the linear trace-support bound from x to y . This reconstructs the dependent-random-choice argument in the fixed metric x ; it does not silently change the metric in Theorem 13.32.

Put $\mathfrak{d}_{n,y} = \mathfrak{r}_{n,y}^2$, and let $m_{n,y}$ be the square-free product of the rational primes below $\mathfrak{r}_{n,y}$. Rational saturation gives

$$m_{n,y}^2 \mathcal{O}_{k_y, s_y} \subseteq \mathfrak{d}_{n,y}. \quad (13.10bpzcc3)$$

Thus reduction modulo $\mathfrak{d}_{n,y}$ is a quotient of reduction modulo the fixed square of the square-free rational modulus $m_{n,y}$. Salehi Golsefidy’s theorem for fixed powers of square-free rational moduli applies to the algebraic representation at y [37]. Its spectral gap transfers to the corresponding covers of the fixed surface (S, x) by the Brooks–Burger comparison [11, 12]. Uniform lattice counting in the metric x therefore gives, for some $\delta_y > 0$,

$$\max_c \#\{g \in \mathcal{B}_R^x : \mathrm{tr}_y g \equiv c \pmod{\mathfrak{d}_{n,y}}\} \ll (N\mathfrak{d}_{n,y})^{o(1)} \left\{ \frac{e^R}{N\mathfrak{d}_{n,y}} + (N\mathfrak{d}_{n,y})^2 e^{(1-\delta_y)R} \right\}. \quad (13.10bpzcc4)$$

The one-factor recurrence norm estimate gives

$$\log N\mathfrak{d}_{n,y} \leq 2n\mathcal{J}_{k_y}(s_y) + O_y(\log n).$$

Choose a fixed C_y with $\delta_y C_y > 6\mathcal{J}_{k_y}(s_y)$, and put $R = C_y n$. The second term in (13.10bpzcc4) is then absorbed by the first.

Choose one group representative for every edge incident to $W_{R,y}$. Representatives of distinct graph edges are distinct. If this set of left vertices occupies $M_{R,n,y}$ residue classes modulo $\mathfrak{d}_{n,y}$, then (13.10bpzcc1a) and (13.10bpzcc4) give

$$M_{R,n,y} \max_c \#\{g \in \mathcal{B}_R^x : \mathrm{tr}_y g \equiv c \pmod{\mathfrak{d}_{n,y}}\} \geq \#W_{R,y} r_{R,y}.$$

Since $R = C_y n$, it follows that

$$M_{R,n,y} \geq N\mathfrak{d}_{n,y} e^{-o(n)}. \quad (13.10bpzcc5)$$

Let $T_{R,n,y} \subset W_{R,y}$ contain one representative from each occupied class. Exactly as in (13.10bpw10), the map

$$T_{R,n,y} \times W_{R,y} \longrightarrow Q_{n,y}^2(W_{R,y} + \beta_{n,y}W_{R,y}), \quad (t, u) \longmapsto Q_{n,y}^2 u - q_{n-1}(s_y)^2 t$$

is injective. The ideal $\mathfrak{d}_{n,y}$ is supported on primes dividing $Q_{n,y}$, so $Q_{n,y}^2$ vanishes modulo it while Cassini’s identity makes $q_{n-1}(s_y)$ a unit there. Consequently

$$\frac{\#(W_{R,y} + \beta_{n,y}W_{R,y})}{\#W_{R,y}} \geq N\mathfrak{d}_{n,y} e^{-o(n)}.$$

Comparison with (13.10bpzcc2) gives

$$\log N\mathfrak{d}_{n,y} \leq n\ell_x(\bar{a}) + o(n).$$

Since $\mathfrak{d}_{n,y} = \mathfrak{r}_{n,y}^2$, this is (13.10bpzcb). $\square$

We next record the exact consequence of the number-field *abc* conjecture. For a number field L , a finite set S_L of finite places, and an S_L -integer α , write

$$\text{rad}_{S_L}(\alpha) = \prod_{\substack{\mathfrak{P} \notin S_L \\ v_{\mathfrak{P}}(\alpha) > 0}} \mathfrak{P}.$$

We use the standard number-field *abc* assertion in the following fixed-field normalization: for every $\epsilon > 0$ and every nonzero solution $A + B = C$,

$$[L : \mathbb{Q}]h(A : B : C) \leq (1 + \epsilon) \sum_{\substack{\mathfrak{P} \\ v_{\mathfrak{P}}(A), v_{\mathfrak{P}}(B), v_{\mathfrak{P}}(C) \text{ are not all equal}}} \log N\mathfrak{P} + C_{L,\epsilon}. \quad (13.10bpzcc)$$

Deleting a fixed finite set S_L changes only the constant. This is the usual fixed-number-field form of the Masser–Oesterlé–Vojta conjecture [51, p. 84]; a discriminant term in a field-uniform formulation is constant here.

Lemma 13.64 (*abc* gives the full Chebyshev radical). *Let k be a number field, let $s \in \mathcal{O}_{k,S}$ have a distinguished real embedding with $s > 2$, and let $q_{n+1} = sq_n - q_{n-1}$, $q_0 = 0, q_1 = 1$. Assume the number-field *abc* conjecture for $L = k(\lambda)$, where $\lambda + \lambda^{-1} = s$. After enlarging S by a fixed finite set,*

$$\log N_k \text{rad}_S(q_n(s)) = n\mathcal{J}_k(s) + o_s(n). \quad (13.10bpzd)$$

Equivalently, the quotient of the principal ideal $(q_n(s))$ by its radical has logarithmic norm $o_s(n)$.

Proof. Put $z = \lambda^2$. The distinguished embedding shows that z is not a root of unity. The product formula and the definition (13.10au) give the global identity

$$[L : \mathbb{Q}]h(z) = [L : k]\mathcal{J}_k(s). \quad (13.10bpze)$$

When $z \notin k$, the two roots z, z^{-1} give this identity place by place after extension to L ; when $z \in k$, it follows after summing because the positive and negative logarithms have equal total mass.

Enlarge S so that $\lambda, z - 1$, and their inverses are units above its complement, and so that every prime ramified in L/k is removed. In the group application this enlargement is made after adjoining the model and denominator primes and is then saturated over rational primes. Removing this fixed set changes every logarithmic radical by only $O_{s,S}(1)$. Apply *abc* over L to

$$z^n + (-1) = z^n - 1.$$

Outside the enlarged fixed set, only $z^n - 1$ contributes to the radical, whereas

$$h(z^n : -1 : z^n - 1) = nh(z) + O_z(1).$$

Thus, for every $\epsilon > 0$,

$$\log N_L \text{rad}_{S_L}(z^n - 1) \geq \frac{n[L : k]\mathcal{J}_k(s)}{1 + \epsilon} - O_{s,S,\epsilon}(1). \quad (13.10bpzff)$$

The closed formula

$$q_n(s) = \lambda^{-(n-1)} \frac{z^n - 1}{z - 1}$$

identifies the two radicals outside S_L . Since $q_n(s) \in k$ and L/k is unramified there, every prime above a prime divisor of $q_n(s)$ occurs, and

$$\log N_L \text{rad}_{S_L}(q_n(s)\mathcal{O}_L) = [L : k] \log N_k \text{rad}_S(q_n(s)). \quad (13.10bpzg)$$

Combine (13.10bpzff)–(13.10bpzg), then let $\epsilon \downarrow 0$, to obtain the lower bound in (13.10bpzd). The local norm calculation in Lemma 13.40, applied to one recurrence factor, gives

$$\log N_k(q_n(s)) = n\mathcal{J}_k(s) + O_{s,S}(\log n).$$

The radical divides this principal ideal, which gives the matching upper bound and the assertion about the quotient. $\square$

Conjecture 13.65 (the one-term Chebyshev radical hypothesis). Let k be a number field, let S be a finite set of finite places, and let $s \in \mathcal{O}_{k,S}$ have a real embedding at which $s > 2$. Then

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log N_k \operatorname{rad}_S(q_n(s)) \geq \mathcal{J}_k(s). \quad (\text{CR}_{\text{Ch}})$$

The reverse inequality follows from the one-factor recurrence norm formula

$$\log N_k((q_n(s))) = n\mathcal{J}_k(s) + O_{s,S}(\log n),$$

so the assertion is equivalently equality of the two exponential rates. It is unchanged by enlarging S by a fixed finite set.

Theorem 13.66 (the one-term radical hypothesis implies the compact conjectures). *Assume Conjecture 13.65. Then every torsion-free cocompact Fuchsian lattice whose signed trace set has linear growth is arithmetic. Consequently the compact Schmutz linear-growth conjecture and the compact Sarnak bounded-clustering conjecture hold.*

Proof. First suppose that the Fuchsian character x is algebraic. For every nontrivial projective element of $\Gamma^{(2)}$, choose its positive-trace lift a , put $s = \operatorname{tr} a > 2$, and use the full trace field k . Conjecture 13.65 and Theorem 13.49 give

$$\mathcal{J}_k(s) \leq \limsup_{n \rightarrow \infty} \frac{1}{n} \log N_k \operatorname{rad}_S(q_n(s)) \leq \frac{\ell(a)}{2}. \quad (13.10\text{rad}1)$$

The distinguished real place already contributes $\ell(a)/2$. Therefore equality holds and every other local expansion exponent vanishes. This holds for every projectively nontrivial element of $\Gamma^{(2)}$. Takeuchi's criterion [47] makes $\Gamma^{(2)}$ arithmetic, and arithmeticity passes to its finite-index overgroup Γ .

Suppose next that x is transcendental. Let $y_j \rightarrow x$ be the bounded-degree algebraic Fuchsian specializations given by Proposition 13.2. Since $Z_x \subset V_x$, the squared-trace equalities needed below persist at every y_j ; the proposition also places all y_j in the same Teichmüller component as x . For each fixed j , Proposition 13.63 and Conjecture 13.65 give, for every nonidentity $\bar{a} \in \Pi^{(2)}$,

$$\mathcal{J}_{k_{y_j}}(\operatorname{tr}_{y_j} a) \leq \frac{\ell_x(\bar{a})}{2}, \quad (13.10\text{rad}2)$$

where the lift is chosen to have positive trace at y_j . No constants need be uniform in j , because the limit in n is taken with j fixed. Because $\mathcal{J}_k(-s) = \mathcal{J}_k(s)$, the choice of lift in (13.10rad2) is immaterial. Taking any $y_j \neq x$, the inequality holds for every nonidentity element of the subgroup generated by squares. Lemma 13.45 gives $y_j = x$, a contradiction. Hence only the algebraic case occurs. Finally, bounded clustering implies linear growth. $\square$

Proposition 13.67 (specialization to Mersenne numbers). *Let*

$$k = \mathbb{Q}(\sqrt{2}), \quad s = \frac{3}{\sqrt{2}},$$

and let S contain the unique prime of k above 2. Then

$$q_n(s) = 2^{-(n-1)/2}(2^n - 1) \quad (13.10\text{mer}1)$$

and

$$N_k \operatorname{rad}_S(q_n(s)) = \operatorname{rad}(2^n - 1)^2, \quad \mathcal{J}_k(s) = 2 \log 2. \quad (13.10\text{mer}2)$$

Consequently Conjecture 13.65 at this single value of s is exactly

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log \operatorname{rad}(2^n - 1) = \log 2. \quad (13.10\text{mer}3)$$

In particular, (13.10mer3) implies that there are infinitely many non-Wieferich primes to base 2.

Proof. Take $\lambda = \sqrt{2}$, so that $\lambda + \lambda^{-1} = 3/\sqrt{2}$. The closed recurrence formula gives (13.10mer1). The power of 2 is an S -unit. Every odd rational prime is unramified in k , and the product of the norms of all primes above it is its square. This proves the first identity in (13.10mer2). Moreover, for the prime-to- S principal ideal,

$$\log N_k((q_n(s))) = 2 \log(2^n - 1) = 2n \log 2 + O(1).$$

The one-factor recurrence norm formula therefore gives $\mathcal{J}_k(s) = 2 \log 2$, proving the equivalence with (13.10mer3).

Suppose that only finitely many odd primes E are non-Wieferich to base 2. If $p \notin E$ divides $2^n - 1$, put $r_p = \text{ord}_p(2)$. Then $r_p \mid n$ and $r_p \mid p - 1$. Since $p \nmid (p - 1)/r_p$, the lifting-the-exponent formula gives

$$v_p(2^{r_p} - 1) = v_p(2^{p-1} - 1) \geq 2.$$

A second application, using $r_p \mid n$, gives $v_p(2^n - 1) \geq 2$. Hence

$$\log \text{rad}(2^n - 1) \leq \frac{1}{2} \log(2^n - 1) + O_E(1),$$

contradicting (13.10mer3). Thus this special case already implies the classical infinitude of non-Wieferich primes; number-field abc implies it by Lemma 13.64, in agreement with Silverman's theorem [42]. $\square$

Known unconditional recurrence results are far below the exponential rate in Conjecture 13.65. In the classical integral setting, Stewart's lower bound for a simple nondegenerate recurrence with a dominant root has the shape

$$\text{rad}(u_n) > n^{C \log_2 n / \log_3 n},$$

where $\log_2 n = \log \log n$ and $\log_3 n = \log \log \log n$ [44]. Its logarithm is still $o(n)$. His later lower bound for the greatest prime divisor of a Lucas-Lehmer cyclotomic factor is likewise subexponential on the scale needed here [45]. Primitive-divisor theorems guarantee a new prime at large indices, but one new prime per index does not give a linear lower bound for the logarithm of the radical. Thus the Mersenne specialization is not hiding a known Lucas-sequence theorem.

Proposition 13.68 (a nondistinguished expanding place contradicts fixed-field abc). *Assume the algebraic linear-trace-growth hypotheses of Theorem 13.48. Fix a hyperbolic element with positive-trace lift a , put $s = \text{tr } a > 2$, choose $\lambda + \lambda^{-1} = s$, and set*

$$L = k(\lambda), \quad d = [L : k], \quad z = \lambda^2.$$

For $m \geq 1$, define the height and radical of the abc triple $z^m + (-1) = z^m - 1$ by

$$\begin{aligned} H_m &= [L : \mathbb{Q}] h(z^m : -1 : z^m - 1), \\ R_m &= \sum_{\substack{\mathfrak{p} \subset \mathcal{O}_L \\ v_{\mathfrak{p}}(z^m), v_{\mathfrak{p}}(-1), v_{\mathfrak{p}}(z^m - 1) \text{ are not all equal}}} \log N \mathfrak{p}. \end{aligned} \tag{13.10bpzj}$$

With the convention $H_m/R_m = +\infty$ when $R_m = 0$, one has

$$\liminf_{m \rightarrow \infty} \frac{H_m}{R_m} \geq \frac{2\mathcal{J}_k(s)}{\ell(a)}. \tag{13.10bpzk}$$

Consequently, if a has a nondistinguished expanding place, then the fixed number field L and the explicit sequence $(z^m, -1, z^m - 1)$ contradict the fixed-field abc conjecture (13.10bpzcc).

Proof. Use the same enlarged set S as in Lemma 13.64: above its complement, z and $z - 1$ are units and L/k is unramified. At every place outside S_L ,

$$z^m - 1 = \lambda^{m-1}(z - 1)q_m(s)$$

shows that the prime support of the triple is exactly that of $q_m(s)$. Because $q_m(s) \in k$ and the extension is unramified there,

$$\text{rad}_{S_L}(q_m(s)\mathcal{O}_L) = \text{rad}_S(q_m(s))\mathcal{O}_L.$$

The finitely many places in S_L contribute only $O_{s,S}(1)$ to a radical, regardless of the sizes of their valuations. Therefore

$$R_m = d \log N_k \text{rad}_S(q_m(s)) + O_{s,S}(1). \quad (13.10\text{bpzl})$$

Likewise, the ultrametric triangle inequality at the finite places and the ordinary triangle inequality at the archimedean places give

$$H_m = [L : \mathbb{Q}]h(z^m) + O_L(1) = dm\mathcal{J}_k(s) + O_{s,L}(1), \quad (13.10\text{bpzm})$$

where the last equality is (13.10bpze).

Write $r_m = \log N_k \text{rad}_S(q_m(s))$. Theorem 13.49 gives directly

$$r_m \leq \frac{m\ell(a)}{2} + o(m). \quad (13.10\text{bpzn})$$

Together with (13.10bpzl)–(13.10bpzm), this gives

$$\frac{H_m}{R_m} \geq \frac{dm\mathcal{J}_k(s) + O(1)}{dm\ell(a)/2 + o(m)}$$

whenever $R_m > 0$; if $R_m = 0$, the asserted inequality holds by convention. Taking the limit inferior proves (13.10bpzk) for every individual recurrence term, without averaging consecutive indices.

Finally suppose $2\mathcal{J}_k(s)/\ell(a) > 1$, and choose

$$0 < \epsilon < \frac{2\mathcal{J}_k(s)}{\ell(a)} - 1.$$

There are an unbounded sequence of indices m and a fixed $\delta > 0$ for which $H_m \geq (1 + \epsilon + \delta)R_m$. If R_m is unbounded on a subsequence, then $H_m - (1 + \epsilon)R_m \rightarrow \infty$ there; if it is bounded on a subsequence, the same follows from (13.10bpzm). Thus no constant $C_{L,\epsilon}$ can make (13.10bpzcc) hold for these triples. This is a counterexample to the fixed-field *abc* conjecture over L . $\square$

Theorem 13.69 (number-field *abc* implies compact Schmutz and Sarnak). *Assume the number-field *abc* conjecture for every number field. Then every torsion-free cocompact Fuchsian lattice whose signed trace set has linear growth is arithmetic. Consequently the compact Schmutz linear-growth conjecture and the compact Sarnak bounded-clustering conjecture both hold.*

Proof. For every fixed k, S, s , Lemma 13.64 shows that number-field *abc* over $k(\lambda)$, where $\lambda + \lambda^{-1} = s$, implies Conjecture 13.65. The assumed family of fixed-field *abc* conjectures therefore establishes that hypothesis for every algebraic specialization needed here. Apply Theorem 13.66. $\square$

There is a coefficient-uniform way to prove the lower bound (13.10bm) if the trace set is sufficiently spread out in residue classes. Unlike the general sum-of-dilates theorem, it has no error depending on the size of the coefficients.

Lemma 13.70 (coprime dilates and residue classes). *Let L be a number field, let $A \subset \mathcal{O}_L$ be finite, and let $a, b \in \mathcal{O}_L \setminus \{0\}$ generate coprime ideals. Denote by $\nu_{\mathfrak{m}}(A)$ the number of residue classes modulo $\mathfrak{m}$ met by A . Then*

$$\#(aA + bA) \geq \max\{\nu_{(a)}(A), \nu_{(b)}(A)\} \#A. \quad (13.10\text{bp})$$

More generally, put

$$E_{\mathfrak{m}}(A) = \sum_{\xi \in \mathcal{O}_L/\mathfrak{m}} \#(A \cap (\xi + \mathfrak{m}))^2.$$

Then

$$\#(aA + bA) \geq \frac{(\#A)^3}{\min\{E_{(a)}(A), E_{(b)}(A)\}}. \quad (13.10bq)$$

The same statements hold for arbitrary $a, b \in L^\times$: divide their principal fractional ideals by their greatest common ideal and multiply the sumset by a common scalar.

Proof. Choose one element $u \in A$ from each residue class modulo (a) . The sets $aA + bu$ are pairwise disjoint. Indeed, a collision would give $b(u - u') \in (a)$, and coprimality would give $u - u' \in (a)$. This proves the first bound using (a) ; interchange a, b for the other one.

For the second bound, let $\mathcal{E}$ count solutions of $ax + by = ax' + by'$ in A^4 . Coprimality implies $x \equiv x' \pmod{(b)}$ and $y \equiv y' \pmod{(a)}$. Once (x, x', y) is fixed, there is at most one y' , and symmetrically. Hence

$$\mathcal{E} \leq \#A \min\{E_{(a)}(A), E_{(b)}(A)\}.$$

Cauchy–Schwarz gives $\#(aA + bA) \geq (\#A)^4/\mathcal{E}$. $\square$

For the recurrence, this residue modulus has exactly the required exponential size. Work in $K = \mathbb{Q}(s^2)$, and divide the two principal fractional ideals generated by $q_{n-1}(s)^2$ and $q_n(s)^2$ by their greatest common ideal. Denote the resulting coprime integral ideals by $\mathfrak{a}_n$ and $\mathfrak{b}_n$, respectively. Cassini’s identity (13.10ao) is the algebraic reason for this coprimality. If $\mathbb{Q}(\beta_n) = K$, then

$$\log N\mathfrak{a}_n = \log N\mathfrak{b}_n = 2n\mathcal{J}_K(s) + O_s(\log n). \quad (13.10br)$$

To see this, use (13.10ap) at every archimedean embedding. When the corresponding algebraic root z has modulus different from one, β_n tends to a nonzero limit. If $|z| = 1$, then z is not a root of unity by Lemma 13.28, and Baker’s linear-forms theorem supplies $C_z > 0$ such that $|z^n - 1| \geq n^{-C_z}$ for $n \geq 2$ [1]. Thus the archimedean factors in the heights of both β_n and β_n^{-1} are $e^{O_s(\log n)}$. Lemma 13.29 then puts the full exponential height into the denominator ideals of β_n and its inverse, which are $\mathfrak{b}_n$ and $\mathfrak{a}_n$.

We state a sufficient group-theoretic hypothesis for applying Lemma 13.70. This formulation permits a finite extension L/K , since the trace values need not themselves lie in K .

Proposition 13.71 (congruence spreading is sufficient for recurrence expansion). *Retain the algebraic hypotheses of Theorem 13.36. Suppose all trace values under consideration lie in a fixed number field $L \supset K$. Fix a sufficiently small $\vartheta > 0$, put $n_R = \lfloor \vartheta R \rfloor$, and let $\mathfrak{m}_R$ be either $\mathfrak{a}_{n_R}\mathcal{O}_L$ or $\mathfrak{b}_{n_R}\mathcal{O}_L$. After multiplying W_R by a common scalar to make it integral, assume that*

$$\nu_{\mathfrak{m}_R}(W_R) \geq \exp(2n_R\mathcal{J}_K(s) - o(R)). \quad (13.10bs)$$

Then $\mathcal{J}_K(s) = \ell(k)/2$, so every nondistinguished local expansion of the recurrence vanishes.

It is enough, more concretely, to have the following uniform counting estimate for one of the two ideals. Let $G_R(W_R)$ be the edges of (13.10be) whose first coordinate belongs to W_R , and choose one group element representing each such edge. If, uniformly in residue classes c ,

$$\#\{g \text{ among these representatives} : \text{tr } g \equiv c \pmod{\mathfrak{m}_R}\} \leq \frac{\exp(R + o(R))}{\exp(2n_R\mathcal{J}_K(s))}, \quad (13.10bt)$$

then (13.10bs) follows.

Proof. Since $\beta_n = -q_{n-1}(s)^2/q_n(s)^2$, multiplication by a common scalar identifies $W_R + \beta_n W_R$ with a sum of the two coprime dilates defining $\mathfrak{a}_n, \mathfrak{b}_n$. Equations (13.10bp), (13.10br), and (13.10bs) give

$$\#(W_R + \beta_{n_R} W_R) \geq \exp(2n_R\mathcal{J}_K(s) - o(R))\#W_R.$$

The upper bound (13.10bd) is $\exp(n_R \ell(k) + o(R)) \#W_R$. Divide logarithms by $n_R \sim \vartheta R$ to obtain $2\mathcal{J}_K(s) \leq \ell(k)$; the distinguished-place contribution gives the reverse inequality.

For the final assertion, the dependent-random-choice construction gives minimum degree at least r_R , so (13.10ba) and (13.10bi) imply

$$\#G_R(W_R) \geq \#W_R r_R = \exp(R - o(R)).$$

If W_R met M residue classes, all chosen edge representatives would lie in the union of the corresponding M trace congruence classes. Estimate (13.10bt) therefore forces $M \geq \exp(2n_R \mathcal{J}_K(s) - o(R))$, which is (13.10bs). $\square$

This proposition makes the remaining arithmetic issue concrete. Expansion theorems give estimates of the shape (13.10bt) for several important families of congruence moduli: square-free ideals, bounded powers of square-free rational moduli, and special groups for which arbitrary-modulus theorems are known, subject to the usual good-reduction hypotheses. The ideals $\mathfrak{a}_n, \mathfrak{b}_n$, however, can have simultaneously many prime factors and unbounded prime-power exponents. Moreover, at a finite place where the original representation is unbounded, there is no fixed integral reduction map. Neither issue may be suppressed: one needs either an arbitrary-ideal trace equidistribution theorem in the range $\log N\mathfrak{m} = O(R)$, or a recurrence-specific substitute which retains the full norm in (13.10br).

The pointwise theorem of Conlon and Lim, (13.10at), does not prove (13.10bm): its error is uniform in the size of the finite set only after the algebraic dilate has been fixed, whereas here $n_R \rightarrow \infty$. This order of quantifiers is substantive. The desired estimate would have to be uniform for algebraic dilates of bounded degree whose heights grow with R , for the particular sets W_R furnished by the trace graph. It may be written in the more informative equivalent form

$$\#(W_R + \beta_{n_R} W_R) \geq \exp(2n_R \mathcal{J}_K(s) - o(n_R)) \#W_R. \quad (13.10bo)$$

Corollary 13.72 (the arithmetic consequence of uniform recurrence expansion). *Let $\Gamma < \mathrm{PSL}_2(\mathbb{R})$ be cocompact with linearly growing trace set. Assume that all squared traces of Γ are algebraic and that (13.10bm) holds for every hyperbolic element of the subgroup generated by squares, with the corresponding sets furnished by Theorem 13.32. Then Γ is arithmetic. Consequently, the stated uniform recurrence-expansion estimate, together with the exclusion of transcendental exceptional characters, is another sufficient condition for both compact arithmeticity conclusions.*

Proof. The invariant trace field is a number field because a finite collection of character coordinates generates it and those coordinates are algebraic. Theorem 13.36, applied to every hyperbolic element of $\Gamma^{(2)}$, says that every squared trace is an algebraic integer and that all of its nondistinguished conjugates lie in $[0, 4]$. Takeuchi's arithmeticity criterion [47] now applies to $\Gamma^{(2)}$, and arithmeticity passes to its finite-index overgroup Γ . Bounded clustering implies linear growth, so the same implication handles both hypotheses. $\square$

Proposition 13.73 (coefficient extraction for a pure cyclic twist). *Let k be a field of characteristic zero, let*

$$D(z) = \begin{pmatrix} z & 0 \\ 0 & z^{-1} \end{pmatrix}, \quad B = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(k),$$

and assume $bc \neq 0$ and $(a, d) \neq (0, 0)$. Let $\Delta < \mathrm{SL}_2(k)$ contain no nonidentity unipotent. Then every fiber on Δ of

$$X \mapsto \mathrm{tr}^2(D(z)BD(z)^{-1}X) \in k[z, z^{-1}]$$

has at most two projective elements, and hence at most four elements of Δ .

Consequently, suppose that ρ_z is an algebraic pure-twist family and that, for some $w \in \Pi$ and $H < \Pi$, lifts can be chosen so that

$$\tilde{\rho}_z(wh) = D(z)BD(z)^{-1}\tilde{\rho}_1(h) \quad (h \in H).$$

Assume that $\rho_1|_H$ is faithful and convex cocompact and that every squared-trace collision among the elements wh , $h \in H$, at $z = 1$ persists identically in z . Then every fiber of $h \mapsto \text{tr}^2 \tilde{\rho}_1(wh)$ has at most two projective elements. If, moreover, $H \subset \ker \phi$ and $\phi(w) = 1$ for a homomorphism $\phi: \Pi \rightarrow \mathbb{Z}$, then all wh are primitive and

$$h_{\text{tr}}(\rho_1) \geq \delta_H.$$

Proof. For $X = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$, direct multiplication gives

$$\text{tr}(D(z)BD(z)^{-1}X) = ap + ds + z^2br + z^{-2}cq.$$

If the squares of this Laurent polynomial for X and X_0 agree, the two polynomials differ by one sign $\epsilon \in \{1, -1\}$. Comparing coefficients gives

$$q = \epsilon q_0, \quad r = \epsilon r_0, \quad ap + ds = \epsilon(ap_0 + ds_0).$$

After replacing X_0 by ϵX_0 , fix q, r and put $P = 1 + qr$. The remaining equations are

$$ps = P, \quad ap + ds = L.$$

If $ad \neq 0$, elimination gives a quadratic equation and hence at most two solutions. If exactly one of a, d is nonzero, the equations give a unique solution except when their solution set is an affine line. Two distinct matrices on that exceptional line have a nonidentity upper or lower unipotent quotient. The line therefore meets Δ in at most one point. Negating X exchanges the two signs without changing its projective class, proving the first assertion.

A faithful convex-cocompact Fuchsian group has no nonidentity unipotent. Persistence turns every trace fiber at $z = 1$ into a fiber of the Laurent polynomial map just considered. Finally, $\phi(wh) = 1$, whereas the ϕ -value of a proper m -th power is divisible by $|m| \geq 2$. Thus every wh is primitive. Orbit counting in H , together with $\ell(wh) \leq d(o, \rho_1(w)o) + d(o, \rho_1(h)o)$, gives the entropy bound. $\square$

Proposition 13.74 (coefficients of one trace functional do not determine a matrix jet). *Let k be algebraically closed of characteristic zero, let H be finitely generated, and let*

$$\rho: H \longrightarrow \text{SL}_2(k[[t]])$$

have Zariski-dense reduction $\bar{\rho}: H \rightarrow \text{SL}_2(k)$. Write

$$\rho(h) = (I + tz(h) + O(t^2))\bar{\rho}(h),$$

and assume that the class of z in $H^1(H, \mathfrak{sl}_{2\text{Ad } \bar{\rho}})$ is nonzero. For every $r \geq 1$, the Zariski closure of the reduction of $\rho(H)$ in

$$J_r \text{SL}_2 = \text{SL}_2(k[t]/(t^{r+1}))$$

is the whole jet group.

For fixed $A(t) \in M_2(k[[t]])$, consider

$$\Phi_r: J_r \text{SL}_2 \longrightarrow \mathbb{A}^{r+1}, \quad X(t) \longmapsto ([t^j] \text{tr}(A(t)X(t)))_{0 \leq j \leq r}.$$

Every irreducible component of every nonempty fiber of Φ_r has dimension at least

$$3(r+1) - (r+1) = 2(r+1).$$

The same conclusion holds for a Laurent matrix after multiplication by a fixed power of t and reindexing its coefficients.

This loss of two thirds of the dimensions is already exact at the matrix level. Put $R = k[[t]]$ and $A_m = \text{diag}(t^{-m}, t^m)$, where $m \geq 1$. Fix $p_0, s_0 \in R$. For arbitrary $s \in R$ and $q \in R^\times$, set

$$p = p_0 + t^{2m}(s_0 - s), \quad r = \frac{ps - 1}{q}, \quad X_{q,s} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}.$$

Then $X_{q,s} \in \mathrm{SL}_2(R)$ and

$$\mathrm{tr}(A_m X_{q,s}) = t^{-m} p_0 + t^m s_0$$

for every (q, s) , while their reductions fill a two-dimensional open subset of the surface $\bar{p} = \bar{p}_0$ in $\mathrm{SL}_2(k)$.

Proof. Let L_1 be the Zariski closure of the first-jet image. Its projection to SL_2 is onto. The kernel is an $\mathrm{Ad} \mathrm{SL}_2$ -invariant subspace of $\mathfrak{sl}_2$, hence is either zero or all of $\mathfrak{sl}_2$. In the first case L_1 is the graph of an algebraic 1-cocycle on SL_2 . The vanishing of algebraic $H^1(\mathrm{SL}_2, \mathfrak{sl}_2)$ makes that cocycle a coboundary, contrary to the hypothesis on z . Hence $L_1 = J_1 \mathrm{SL}_2$.

Let L_r be the closure at order r . Truncation sends L_r onto $L_1 = J_1 \mathrm{SL}_2$, because the image of a Zariski closure under the surjective homomorphism of jet groups contains the dense first-jet image. Its differential is therefore onto. In the Lie algebra of L_r , choose lifts of the degree-one vectors tX , $X \in \mathfrak{sl}_2$. Brackets of two such lifts have initial term $t^2[X, Y]$. Since $[\mathfrak{sl}_2, \mathfrak{sl}_2] = \mathfrak{sl}_2$, all degree-two congruence directions occur. Bracketing a degree-one lift with one whose initial term has degree j produces every direction of degree $j+1$. Induction gives all congruence layers through degree r , so $\mathrm{Lie} L_r = \mathrm{Lie} J_r \mathrm{SL}_2$. The jet group is connected, and therefore $L_r = J_r \mathrm{SL}_2$.

The jet group has dimension $3(r+1)$, while a fiber of Φ_r is cut out by $r+1$ scalar equations. Krull's principal ideal theorem gives the stated lower bound. Finally, the displayed $X_{q,s}$ has determinant one, direct substitution gives the fixed Laurent trace, and reduction modulo t leaves the two free parameters $(\bar{q}, \bar{s})$, with $\bar{q} \neq 0$. $\square$

The last proposition concerns dimension counting; it does not construct a large fiber inside a discrete group. A Zariski-dense subset need not be dense in any specified fiber. Deforming the representation introduces new jet variables at the same rate as new trace coefficients. A discrete estimate must therefore use the distribution of words, as in Theorem 13.8, or a second independent trace observable.

Proposition 13.75 (the limit of pigeonholing with two trace observables). *Let $\rho_0: \Pi \rightarrow \mathrm{PSL}_2(\mathbb{R})$ be Fuchsian, let $\phi: \Pi \twoheadrightarrow \mathbb{Z}$, and let $J < \ker \phi$ be finitely generated and convex cocompact, with critical exponent δ_J . Fix $o \in \mathbb{H}^2$ and put*

$$\mathcal{B}_J(R) = \{h \in J : d(o, \rho_0(h)o) \leq R\}.$$

Suppose that $h_{\mathrm{tr}}(\rho_0) \leq s$. For $j = 1, 2$, let $w_{j,R} \in \phi^{-1}(1)$ and assume

$$d(o, \rho_0(w_{j,R})o) \leq (a_j + o(1))R, \quad a_j \geq 0.$$

Then the map

$$h \mapsto (\mathrm{tr}^2 \rho_0(w_{1,R}h), \mathrm{tr}^2 \rho_0(w_{2,R}h)), \quad h \in \mathcal{B}_J(R),$$

has at most

$$\exp((s(2 + a_1 + a_2) + o(1))R)$$

values. Consequently, pigeonholing alone guarantees a joint fiber only of size

$$\exp((\delta_J - s(2 + a_1 + a_2) - o(1))R).$$

In particular, when $s = 1/2$, this exponent is nonpositive because $\delta_J \leq 1$. Scalar trace sparsity, even in finite covers for which δ_J tends to 1, therefore cannot by itself force a collision of two independent trace observables.

Proof. The equality $\phi(w_{j,R}h) = 1$ shows that every $w_{j,R}h$ is primitive. Moreover,

$$\ell_{\rho_0}(w_{j,R}h) \leq d(o, \rho_0(w_{j,R})o) + d(o, \rho_0(h)o) \leq (1 + a_j + o(1))R.$$

The definition of h_{tr} , with an arbitrarily small fixed enlargement of s , bounds the number of values of the j -th coordinate by $\exp((s(1 + a_j) + o(1))R)$. Multiplying the two bounds proves the first assertion. Convex-cocompact orbit counting gives $\#\mathcal{B}_J(R) = \exp((\delta_J + o(1))R)$, and division gives

the joint fiber bound. Finally, a convex-cocompact subgroup of $\mathrm{PSL}_2(\mathbb{R})$ has critical exponent at most the ambient exponent 1. $\square$

Remark 13.76 (a sharp finite model). For a prime power q , put $U_q = \mathbb{F}_q^2$, and define

$$f_t(x, y) = x + ty \quad (t \in \mathbb{F}_q), \quad f_\infty(x, y) = y.$$

Every f_t has exactly q values and every fiber has q elements, whereas $(f_s, f_t): U_q \rightarrow \mathbb{F}_q^2$ is a bijection whenever $s \neq t$. With $R = 2 \log q$, this gives $\#U_q = e^R$ and scalar supports of size $e^{R/2}$, but no nontrivial simultaneous fiber for any two independent observables. Thus even an arbitrarily large collection of individually large fibers need not contain two aligned partitions.

Proposition 13.77 (the valuation cost of extracting several Cartan coefficients). *Let K be a nonarchimedean field of characteristic zero, let $\rho_v: \Pi \rightarrow \mathrm{SL}_2(K)$, fix $a \in \Pi$, and let $k_1, \dots, k_m \in \ker \phi$ be split regular semisimple elements. Write*

$$A = \rho_v(a), \quad \rho_v(k_i) = \lambda_i P_i^+ + \lambda_i^{-1} P_i^-, \quad |\lambda_i|_v > 1,$$

and put $\ell_v(k_i) = 2 \log |\lambda_i|_v$. For $\epsilon = (\epsilon_1, \dots, \epsilon_m) \in \{+, -\}^m$, set

$$L_\epsilon(Y) = \mathrm{tr}(A P_1^{\epsilon_1} \cdots P_m^{\epsilon_m} Y), \quad \alpha_\epsilon(\mathbf{n}) = \prod_{i=1}^m \lambda_i^{\epsilon_i n_i},$$

where a sign in the exponent means $+1$ or -1 . Then

$$\mathrm{tr}(A \rho_v(k_1)^{n_1} \cdots \rho_v(k_m)^{n_m} Y) = \sum_{\epsilon \in \{+, -\}^m} \alpha_\epsilon(\mathbf{n}) L_\epsilon(Y).$$

In particular, regardless of m , the functionals in this expansion span a space of dimension at most four.

Let $\mathcal{L} = (L_1, \dots, L_q)$ be q nonzero linearly independent functionals remaining after only coefficient-independent identifications among these terms, with coefficients $\alpha_1, \dots, \alpha_q$. Thus each $|\alpha_r|_v$ is one of the displayed monomial absolute values; no additional valuation separation caused by cancellation of equal-size monomials is used. Necessarily $q \leq 4$. For

$$\mathcal{D}_R = \{\rho_v(h) - \eta \rho_v(h') : h, h' \in \mathcal{B}_J(R), \eta \in \{1, -1\}\},$$

define

$$C_{\mathcal{L}}(R) = \sup_{\substack{\Delta \in \mathcal{D}_R, \\ L_i(\Delta) L_j(\Delta) \neq 0}} \left| \log \left| \frac{L_i(\Delta)}{L_j(\Delta)} \right|_v \right|.$$

Order the coefficients so that $b_1 > \cdots > b_q$, where $b_r = \log |\alpha_r|_v$. If

$$b_r - b_{r+1} > C_{\mathcal{L}}(R) \quad (1 \leq r < q),$$

then, uniformly for $\Delta \in \mathcal{D}_R$,

$$\sum_{r=1}^q \alpha_r L_r(\Delta) = 0 \implies L_1(\Delta) = \cdots = L_q(\Delta) = 0.$$

Every choice satisfying these coefficient-separation inequalities obeys the necessary valuation budget

$$\sum_{i=1}^m n_i \ell_v(k_i) > (q-1) C_{\mathcal{L}}(R).$$

This is a necessary cost for this uniform leading-term argument, not a claim about every possible arithmetic proof of the same conclusion.

Proof. Multiplying the spectral decompositions proves the expansion. All the L_ϵ lie in the four-dimensional space $M_2(K)^*$.

Suppose that the displayed sum vanishes and not every $L_r(\Delta)$ is zero. Let r be the least index with $L_r(\Delta) \neq 0$. For every $j > r$ with $L_j(\Delta) \neq 0$, the definitions give

$$\log |\alpha_r L_r(\Delta)|_v - \log |\alpha_j L_j(\Delta)|_v \geq b_r - b_j - C_{\mathcal{L}}(R) > 0.$$

Thus $\alpha_r L_r(\Delta)$ is the unique term of largest absolute value, contradicting the ultrametric triangle inequality. This proves the coefficient extraction.

Adding the $q - 1$ separation inequalities gives $b_1 - b_q > (q - 1)C_{\mathcal{L}}(R)$. Since the coefficients are pure Cartan weights,

$$b_1 - b_q \leq 2 \sum_i n_i \log |\lambda_i|_v = \sum_i n_i \ell_v(k_i),$$

which proves the cost bound. $\square$

Lemma 13.78 (Fuchsian displacement of a product of large powers). *Retain the preceding notation and suppose that $\rho_0: \Pi \rightarrow \mathrm{PSL}_2(\mathbb{R})$ is faithful, with torsion-free discrete image. Let $Q_i^\pm$ be the spectral projectors of $\rho_0(k_i)$. If*

$$Q_i^+ Q_{i+1}^+ \neq 0 \quad (1 \leq i < m),$$

then, as $\min_i n_i \rightarrow \infty$,

$$d(o, \rho_0(k_1^{n_1} \cdots k_m^{n_m})o) = \sum_{i=1}^m n_i \ell_0(k_i) + O(1).$$

If the projector condition fails, the two adjacent elements share a boundary fixed point. They are then commensurable, and combining their powers shows that they do not provide two independent spectral directions under any representation of Π .

Proof. Write

$$\rho_0(k_i)^{n_i} = \mu_i^{n_i} Q_i^+ + \mu_i^{-n_i} Q_i^-, \quad |\mu_i| = e^{\ell_0(k_i)/2}.$$

The projector assumption makes the leading product $Q_1^+ \cdots Q_m^+$ nonzero. Every other term is smaller by an exponential factor as $\min_i n_i \rightarrow \infty$, and hence

$$\|\rho_0(k_1^{n_1} \cdots k_m^{n_m})\| \asymp \prod_i |\mu_i|^{n_i}.$$

The standard comparison $d(o, go) = 2 \log \|g\| + O(1)$ proves the formula.

The equality $Q_i^+ Q_{i+1}^+ = 0$ says that the attracting fixed point of k_{i+1} is the repelling fixed point of k_i . In a torsion-free discrete Fuchsian group, two hyperbolic elements sharing one endpoint share both and belong to the same cyclic subgroup. Their abstract commensurability persists under every representation, proving the last assertion. $\square$

Corollary 13.79 (the cost of extracting four coefficients from two Cartan factors). *For two factors, suppose that all four functionals $L_{\epsilon_1, \epsilon_2}$ are linearly independent. Put $x_i = n_i \ell_v(k_i)$, and assume $x_1 \geq x_2$. The four logarithmic weights, in decreasing order, are*

$$\frac{x_1 + x_2}{2}, \quad \frac{x_1 - x_2}{2}, \quad \frac{-x_1 + x_2}{2}, \quad \frac{-x_1 - x_2}{2}.$$

Their consecutive gaps are $x_2, x_1 - x_2, x_2$. Uniform extraction with the bound $C_{\mathcal{L}}(R)$ therefore requires

$$x_2 > C_{\mathcal{L}}(R), \quad x_1 - x_2 > C_{\mathcal{L}}(R),$$

and in particular

$$x_1 + x_2 > 3C_{\mathcal{L}}(R).$$

If the corresponding word in the Fuchsian representation satisfies Lemma 13.78 and

$$r_* = \min_i \frac{\ell_0(k_i)}{\ell_v(k_i)},$$

the Fuchsian displacement contributed by the prefix is greater than $3r_*C_{\mathcal{L}}(R) + O(1)$.

More generally, if q extracted coordinates have subexponential common fibers and $c_{\mathcal{L}} = \limsup C_{\mathcal{L}}(R)/R$, this particular leading-term scheme can certify no more than the entropy exponent

$$\frac{\delta_J}{1 + (q-1)r_*c_{\mathcal{L}}}.$$

This is a bound on the method, not an upper bound for the actual trace entropy. Extracting three or four coordinates improves on the two-coordinate argument only if the corresponding cancellation exponent drops by more than a factor of two or three. A product of many Cartan powers therefore does not generically amortize cancellation.

Remark 13.80 (the exceptional two-dimensional family). Suppose $W = \text{span}_{\mathbb{R}}\{A_s : s \in U\}$ has dimension two. If the determinant quadratic form restricted to W is nondegenerate, any nonconstant connected subfamily contains a pair satisfying (13.8), so Theorem 13.4 applies. If it is degenerate, write $W = \text{span}\{A_0, D\}$. Infinitely many points of W have determinant one, so the polynomial $\det(A_0 + cD)$ is identically one along the relevant affine line. Consequently

$$A_s = A_0(I + c(s)N), \quad N \neq 0, \quad N^2 = 0. \quad (13.11)$$

Thus (13.11) is the only two-dimensional family of matrices A_s not covered by the two theorems. Relative Fenchel–Nielsen twists are motions in the semisimple centralizer of a hyperbolic boundary element and are not of the form (13.11).

Theorem 13.81 (constant restriction on a locus preserving trace collisions forces trace entropy at least δ_H). *Let Π be a closed surface group, let $\rho_0 : \Pi \rightarrow \text{PSL}_2(\mathbb{R})$ be a faithful cocompact Fuchsian representation, and let $H < \Pi$ be nonelementary and convex cocompact under ρ_0 . Suppose that $H \subset \ker \phi$ for an epimorphism $\phi : \Pi \rightarrow \mathbb{Z}$, and write δ_H for the critical exponent of $\rho_0(H)$.*

Let $\mathcal{C}$ be an irreducible algebraic subvariety, through $[\rho_0]$, of the $\text{PSL}_2(\mathbb{C})$ -character component containing the Fuchsian characters. Assume that:

- (a) *the restriction map $\mathcal{C} \rightarrow \mathfrak{X}(H)$ is constant;*
- (b) *every equality between two squared traces at ρ_0 holds identically on $\mathcal{C}$.*

If $\dim \mathcal{C} > 0$, then the primitive trace entropy at ρ_0 is at least δ_H . In particular, if $\delta_H > 1/2$ and that entropy is at most $1/2$, then $\mathcal{C}$ is zero-dimensional.

Proof. Suppose that $\dim \mathcal{C} > 0$, and choose $[\rho_1] \in \mathcal{C}$ distinct from $[\rho_0]$. Because $\rho_0|_H$ is absolutely irreducible, equality of the restricted characters means conjugacy of the restricted representations. Conjugate ρ_1 so that its projective restriction to H equals that of ρ_0 :

$$\rho_1|_H = \rho_0|_H. \quad (13.12)$$

Here equality is in $\text{PSL}_2(\mathbb{C})$. Since H is finitely generated and lies in $\ker \phi$, it has infinite index in Π . The covering surface corresponding to H is noncompact, and finite generation gives a compact core with nonempty boundary. Such a surface retracts onto a finite graph, so H is free (of rank at least two because it is nonelementary). We may therefore choose one lift $\tilde{\rho}_H : H \rightarrow \text{SL}_2(\mathbb{R})$ of this common projective restriction.

Consider

$$(\rho_1, \rho_0) : \Pi \longrightarrow \text{PSL}_2(\mathbb{C}) \times \text{PSL}_2(\mathbb{C}).$$

Both projections are Zariski dense because their restrictions to H are conjugate to the nonelementary Fuchsian group $\rho_0(H)$. Algebraic Goursat’s lemma says that the Zariski closure of the paired

image is either the full product or the graph of an algebraic automorphism of PSL_2 . In the graph case, (13.12) forces that automorphism to fix the Zariski-dense subgroup $\rho_0(H)$ pointwise; it is therefore the identity. This would make the two characters equal, contrary to the choice of ρ_1 . Hence the paired image is Zariski dense in the product.

Put $K = \ker \phi$. It contains H , so the projections of the Zariski closure of $(\rho_1, \rho_0)(K)$ are both PSL_2 . That closure is normal in the full product. Since PSL_2 is adjoint and simple, the only such normal algebraic subgroup is the full product. Every coset of K , in particular $\phi^{-1}(1)$, consequently has Zariski-dense image under (ρ_1, ρ_0) .

The condition

$$\mathrm{tr}^2(\rho_1(w)\rho_0(w)^{-1}) \neq 4 \quad (13.13)$$

is a nonempty Zariski-open condition on the product. Choose $w \in \phi^{-1}(1)$ satisfying (13.13). Choose arbitrary lifts $A_i \in \mathrm{SL}_2(\mathbb{C})$ of $\rho_i(w)$, with $A_0 \in \mathrm{SL}_2(\mathbb{R})$. The squared trace in (13.13) is independent of these two sign choices. The entries of A_0, A_1 and of the images of a finite free basis of H generate a finitely generated field $F/\mathbb{Q}$. Hypothesis (b), restricted to the elements wh , $h \in H$, gives implication (13.7) in Theorem 13.4, with $\rho = \tilde{\rho}_H$; (13.13) gives its condition (13.8). That theorem gives subexponential fibers for

$$h \mapsto \mathrm{tr}^2 \rho_0(wh)$$

in word balls of H .

Every wh is primitive in Π . Indeed, if $wh = u^m$ for an integer $|m| \geq 2$, then $1 = \phi(wh) = m\phi(u)$, which is impossible in $\mathbb{Z}$. Convex-cocompact orbit counting and the subexponential fiber bound therefore give primitive trace entropy at least δ_H . The final assertion follows. $\square$

Corollary 13.82 (fibers of restriction to Hao's complementary subgroup). *Let x be a closed hyperbolic surface of genus $g \geq 6$ with $h_{\mathrm{tr}}(x) \leq 1/2$, and let H be the complementary subgroup in the surface decomposition used in Hao's proof of [19, Theorem A, Proposition 3.2, and (5.1)]. Thus $\delta_H > 1/2$, and H lies in the kernel of a homomorphism from the surface group onto $\mathbb{Z}$. For the variety V_x , on which all trace collisions at x persist, the fiber through x of the restriction map*

$$r_H: V_x \longrightarrow \mathfrak{X}(H), \quad [\rho] \longmapsto [\rho|_H],$$

is zero-dimensional at x . In particular, every positive-dimensional algebraic germ in V_x through x must vary the character of this complementary subgroup.

Proof. The required homomorphism is explicit in the standard generators of the splitting. In the one-holed-torus case set $\phi(a_1) = 1$ and set ϕ equal to zero on the remaining standard generators, which generate H . In the pair-of-pants case set $\phi(a_2) = 1$ and set it equal to zero on the displayed generators of H (and on a_1). Thus $H < \ker \phi$ in both cases.

If the fiber had positive dimension at x , take a reduced irreducible component through x of

$$(V_x \cap r_H^{-1}(r_H(x)))_{\mathrm{red}}.$$

Every squared-trace equality at x would persist on that component, while its restricted H -character would be constant. Theorem 13.81 rules this out. $\square$

The preceding argument excludes an algebraic family contained in a restriction fiber. Persistence in fact excludes even an infinitesimal vertical direction. This distinction matters at a ramification point: a morphism can have a zero-dimensional fiber and nevertheless have a noninjective differential there.

Theorem 13.83 (restriction has injective differential on a trace-collision locus). *Let Π be a closed surface group, let $x = [\rho]$ be a faithful cocompact Fuchsian character in the liftable $\mathrm{PSL}_2(\mathbb{C})$ -character component, and suppose that $h_{\mathrm{tr}}(x) \leq 1/2$. Let $\phi: \Pi \twoheadrightarrow \mathbb{Z}$, and let $H < \ker \phi$ be finitely generated, nonelementary, and convex cocompact under ρ , with critical exponent $\delta_H > 1/2$.*

If Z is any reduced algebraic subvariety of V_x , where all trace collisions at x persist, and Z contains x , then restriction has injective differential at x :

$$d(r_H)_x: T_x Z \longrightarrow T_{[\rho|_H]} \mathfrak{X}(H) \quad \text{is injective.} \quad (13.14)$$

In particular, this conclusion also rules out vertical tangent vectors at a singular point of Z ; no integrability assumption on the tangent vector is needed. If H is free of rank $r \geq 2$, then the restricted Fuchsian character is a smooth point of $\mathfrak{X}(H)$, of dimension $3r - 3$. Consequently

$$\dim_{\mathbb{C}} T_x Z \leq 3r - 3.$$

Proof. We first translate a putative tangent vector into a cocycle. The Fuchsian representation ρ is irreducible and its centralizer in $\mathrm{PSL}_2(\mathbb{C})$ is trivial. The usual tangent-space description of the character variety at such a smooth surface-group character is

$$T_x \mathfrak{X}(\Pi) = H^1(\Pi, \mathfrak{sl}_2(\mathbb{C})_{\mathrm{Ad} \rho}). \quad (13.15)$$

For clarity, a representative z of a class on the right satisfies

$$z(uv) = z(u) + \mathrm{Ad} \rho(u)z(v). \quad (13.16)$$

It gives the dual-number deformation

$$\rho_\epsilon(u) = (I + \epsilon z(u))\rho(u), \quad \epsilon^2 = 0. \quad (13.17)$$

Coboundaries are precisely infinitesimal conjugations. Formula (13.15) can also be obtained directly by differentiating the surface relation; the vanishing of the infinitesimal centralizer identifies the quotient by infinitesimal conjugation with H^1 .

Suppose that $0 \neq v \in T_x Z$ belongs to the kernel in (13.14), and let z represent its nonzero cohomology class. The vanishing of the restricted tangent class says that $z|_H$ is a coboundary. Subtracting the corresponding coboundary on all of Π , we may and shall assume

$$z(h) = 0 \quad (h \in H). \quad (13.18)$$

The resulting cocycle is not a coboundary on Π , because $v \neq 0$.

We next find a group element whose infinitesimal displacement is regular semisimple. Put

$$G = \mathrm{PSL}_2(\mathbb{C}), \quad \mathfrak{g} = \mathfrak{sl}_2(\mathbb{C}),$$

and give $G \ltimes \mathfrak{g}$ the multiplication

$$(g, U)(g', U') = (gg', U + \mathrm{Ad}(g)U').$$

Equation (13.16) says that

$$\widehat{\rho}: \Pi \longrightarrow G \ltimes \mathfrak{g}, \quad \widehat{\rho}(u) = (\rho(u), z(u)), \quad (13.19)$$

is a homomorphism. Let L be the Zariski closure of its image. Its projection to G is onto, since the Fuchsian group $\rho(\Pi)$ is Zariski dense. The subgroup $L \cap \mathfrak{g}$ is a vector subspace invariant under the adjoint action of G . The adjoint module of PSL_2 is irreducible, so this intersection is either 0 or all of $\mathfrak{g}$.

If $L \cap \mathfrak{g} = 0$, projection identifies L with G , and L is the graph of an algebraic cocycle $c: G \rightarrow \mathfrak{g}$. Such a cocycle is a coboundary. Indeed, its differential at the identity is a Lie-algebra cocycle $\mathfrak{g} \rightarrow \mathfrak{g}$; Whitehead's lemma makes that differential a coboundary. After subtracting the corresponding algebraic coboundary, the cocycle has zero differential. Since G is connected in characteristic zero, it is then constant, and its value at the identity is zero. It follows that z is a coboundary on the Zariski-dense subgroup $\rho(\Pi)$, contrary to the choice of v . Consequently

$$L = G \ltimes \mathfrak{g}. \quad (13.20)$$

Here we used that a subgroup which contains the full translation group $\mathfrak{g}$ and projects onto G is the whole semidirect product: multiply any lift of $g \in G$ by a suitable translation to obtain $(g, 0)$.

Let $K = \ker \phi$, and let M be the Zariski closure of $\widehat{\rho}(K)$. Since K is normal in Π , (13.20) makes M a normal algebraic subgroup of $G \ltimes \mathfrak{g}$. Its projection to G is onto: it contains $(\rho(h), 0)$ for $h \in H$, and a nonelementary subgroup of PSL_2 is Zariski dense. As above, $M \cap \mathfrak{g}$ is either 0 or $\mathfrak{g}$. In the first case M is the graph of an algebraic cocycle on G . That cocycle vanishes on the Zariski-dense set $\rho(H)$, by (13.18), and hence vanishes identically. Thus the first case would give

$$M = G \times \{0\}, \quad z|_K = 0. \quad (13.21)$$

Choose $w_0 \in \Pi$ with $\phi(w_0) = 1$. For every $k \in K$, the cocycle identity and (13.21) give

$$0 = z(w_0 k w_0^{-1}) = z(w_0) - \mathrm{Ad} \rho(w_0 k w_0^{-1}) z(w_0).$$

The group $\rho(K)$ is Zariski dense because it contains $\rho(H)$, and the adjoint representation has no nonzero invariant vector. Hence $z(w_0) = 0$. Since K together with w_0 generates Π , this would make $z = 0$, again a contradiction. We have therefore proved

$$M = G \ltimes \mathfrak{g}. \quad (13.22)$$

Every coset of K , and in particular $\phi^{-1}(1)$, now has Zariski-dense image under $\widehat{\rho}$. The complement of the nilpotent cone

$$\{U \in \mathfrak{sl}_2 : \det U = 0\}$$

is nonempty and Zariski open. We can consequently choose

$$w \in \phi^{-1}(1) \quad \text{such that} \quad N := z(w) \text{ satisfies } \det N \neq 0. \quad (13.23)$$

Since $\mathrm{tr} N = 0$, the matrix N has two distinct eigenvalues.

We use the two first-order trace coordinates in (13.23). Fix a lift of ρ to $\mathrm{SL}_2(\mathbb{R})$, use the same notation for this lift, and put

$$Y_h = \rho(w)\rho(h), \quad h \in H.$$

If $h, h' \in H$ satisfy $\mathrm{tr}^2 Y_h = \mathrm{tr}^2 Y_{h'}$, then $\tau_{wh} - \tau_{wh'}$ is one of the defining collision functions of V_x . It vanishes on Z , so every tangent vector to Z at x , including v , annihilates its differential. This is true even when v is only a dual-number tangent and is not tangent to an algebraic or analytic arc. By (13.18),

$$z(wh) = z(w), \quad \left. \frac{d}{d\epsilon} \right|_{\epsilon=0} \mathrm{tr} \rho_\epsilon(wh) = \mathrm{tr}(NY_h). \quad (13.24)$$

Fix h_* in one squared-trace fiber. Every wh is nontrivial—in fact it is primitive by the argument below—so its trace at the Fuchsian point is nonzero. There is therefore a sign $\varepsilon_h \in \{\pm 1\}$ such that

$$\mathrm{tr} Y_h = \varepsilon_h \mathrm{tr} Y_{h_*}.$$

Differentiating $\mathrm{tr}^2 Y_h = \mathrm{tr}^2 Y_{h_*}$ along v , and using (13.24), gives

$$\mathrm{tr}(NY_h) = \varepsilon_h \mathrm{tr}(NY_{h_*}). \quad (13.25)$$

Thus, after dividing the fiber into its two sign classes, both $\mathrm{tr} Y_h$ and $\mathrm{tr}(NY_h)$ are fixed.

Diagonalize N over a finite extension of the field generated by its entries:

$$N = \begin{pmatrix} \lambda & 0 \\ 0 & -\lambda \end{pmatrix}, \quad \lambda \neq 0.$$

Writing $Y_h = (y_{ij})$ in this fixed basis, the two quantities in (13.25) determine y_{11} and y_{22} , while determinant one gives

$$y_{12}y_{21} = y_{11}y_{22} - 1 =: q. \quad (13.26)$$

We spell out why Theorem 12.1 applies here despite the use of a tangent vector. Let $F/\mathbb{Q}$ be the field generated by the entries of ρ on a finite generating set of Π , the finitely many entries of $N = z(w)$, and the entries of the fixed diagonalizing matrix. This is a finitely generated field of characteristic zero. Choose for F a normal finite-type model and a filtration $R_C[n]$ as in (12.1). If

h has word length at most n , repeated multiplication by the fixed generator matrices shows that every entry of Y_h , after the fixed change of basis, belongs to $R_C[n]$ for one constant C independent of h and n . The entries of the dual-number cocycle do not grow here: only the single fixed matrix $N = z(w)$ occurs in (13.24). After increasing C , the fixed diagonal entries and the right side q of (13.26) belong to the same filtration. If $q \neq 0$, the finite-type divisor bound therefore leaves only $\exp(o(n))$ possibilities for the ordered pair (y_{12}, y_{21}) , uniformly in the fixed trace fiber.

If $q = 0$, the solutions in (13.26) are the two triangular affine lines $y_{12} = 0$ and $y_{21} = 0$. Each line contains at most one matrix Y_h arising from H . Indeed, two distinct such matrices on the same line would satisfy $\det(Y_h - Y_{h'}) = 0$. For determinant-one matrices,

$$\det(U - V) = 2 - \operatorname{tr}(UV^{-1}),$$

and hence

$$2 = \operatorname{tr}(Y_h Y_{h'}^{-1}) = \operatorname{tr} \rho(h h'^{-1}).$$

Faithfulness and torsion-free convex cocompactness make this impossible when $h \neq h'$. Combining the two signs and the two triangular lines, we conclude that every fiber of

$$h \mapsto \operatorname{tr}^2 \rho(wh) \tag{13.27}$$

in an H -word ball of radius n has size $\exp(o(n))$. Word length and orbit displacement are quasi-isometric on the convex-cocompact group H , so the same statement is $\exp(o(R))$ in a displacement ball of radius R .

Finally, orbit counting gives $\exp((\delta_H + o(1))R)$ elements $h \in H$ with $d(o, \rho(h)o) \leq R$. Every wh is primitive: if $wh = u^m$ with $|m| \geq 2$, then $1 = \phi(wh) = m\phi(u)$, which is impossible. Their lengths are at most $R + O(1)$. The hypothesis $h_{\operatorname{tr}}(x) \leq 1/2$ says that they assume at most $\exp((1/2 + o(1))R)$ distinct absolute-trace values. Pigeonholing therefore produces a fiber of (13.27) of size

$$\exp((\delta_H - \tfrac{1}{2} - o(1))R),$$

contradicting the subexponential bound because $\delta_H > 1/2$. The kernel in (13.14) is consequently zero. $\square$

Theorem 13.84 (restriction to the subgroup has degree one). *Let Π be a closed surface group and let $x = [\rho_x]$ be a faithful cocompact Fuchsian character. Let $\phi: \Pi \rightarrow \mathbb{Z}$, and let $H < \ker \phi$ be a finitely generated subgroup for which $\rho_x(H)$ is nonelementary and convex cocompact with critical exponent $\delta_H > 1/2$. Assume $h_{\operatorname{tr}}(x) \leq 1/2$. Let Z be an irreducible positive-dimensional $\mathbb{Q}$ -subvariety of V_x , on which all trace collisions at x persist, and let*

$$r_H: Z \rightarrow \mathfrak{X}(H)$$

be restriction of characters. If r_H is generically finite onto its image $Y = \overline{r_H(Z)}$, then it has degree one:

$$\mathbb{Q}(Z) = \mathbb{Q}(Y).$$

Proof. Put $K = \mathbb{Q}(Z)$ and $L = \mathbb{Q}(Y)$. Suppose $[K : L] > 1$. Characteristic zero makes this extension separable, so there is a nonidentity L -embedding $\sigma: K \rightarrow \bar{K}$.

The generic absolutely irreducible PSL_2 -character of Z can be represented after a finite extension of K : equivalently, split its degree-two central simple algebra and choose matrices for a finite generating set. Extending σ to this finite coefficient field gives two generic representations

$$\rho_0, \rho_1: \Pi \rightarrow \operatorname{PSL}_2(\Omega),$$

where Ω is an algebraically closed extension of a finitely generated characteristic-zero field. They correspond to the identity and σ embeddings of K . Since σ fixes L , their restrictions to H have the same character. Their generic restricted character is Zariski dense. Indeed, it is faithful. For each fixed nonidentity $h \in H$, the regular function $\tau_h - 4$ on Z is nonzero at x , because ρ_x is

faithful and torsion-free Fuchsian. It is therefore nonzero in the function field K , so h is not the identity under the generic projective representation. This holds simultaneously for all $h \neq 1$: the function-field point evaluates every nonzero element of the coordinate ring as a nonzero element, and no intersection of countably many Zariski-open sets is being asserted. Now H is a nonabelian free group, whereas every proper algebraic subgroup of $\mathrm{PSL}_2(\mathbb{C})$ is virtually solvable. A faithful image of H cannot lie in such a subgroup. Hence the generic restricted image is Zariski dense. In particular the restriction is absolutely irreducible, so we may conjugate ρ_1 and arrange the projective equality

$$\rho_1|_H = \rho_0|_H \quad (13.28)$$

in $\mathrm{PSL}_2(\Omega)$. The group H is a finitely generated infinite-index subgroup of a closed surface group and is therefore free by the same compact-core argument used above. Choose one common lift

$$r: H \longrightarrow \mathrm{SL}_2(\Omega)$$

of the projective representation in (13.28).

If ρ_0 and ρ_1 were projectively equal on Π , then σ would fix every squared-trace coordinate on Z . Those coordinates generate K : in the adjoint representation, $\mathrm{tr}(\mathrm{Ad} \rho(g)) = \mathrm{tr}^2(\bar{\rho}(g)) - 1$, and traces of words in a faithful matrix representation generate the character function field. Thus σ would be the identity. Hence the representations are not projectively equal. Algebraic Goursat's lemma, followed by the normal-subgroup argument in the proof of Theorem 13.81, shows that

$$(\rho_1, \rho_0)(\ker \phi)$$

is Zariski dense in $\mathrm{PSL}_2 \times \mathrm{PSL}_2$. We may consequently choose $w \in \phi^{-1}(1)$ and arbitrary lifts $A_i \in \mathrm{SL}_2(\Omega)$ of $\rho_i(w)$ such that

$$B = A_1 A_0^{-1} \quad \text{has} \quad \mathrm{tr}^2 B \neq 4. \quad (13.29)$$

There are $e^{(\delta_H + o(1))R}$ elements $h \in H$ with $d(o, \rho_x(h)o) \leq R$, and every wh is primitive because $\phi(wh) = 1$. Their x -lengths are at most $R + O(1)$, so for every $\epsilon > 0$, the hypothesis $h_{\mathrm{tr}}(x) \leq 1/2$ leaves at most $e^{(1/2+\epsilon)R}$ squared-trace values for all sufficiently large R . Therefore one value is assumed by at least

$$e^{(\delta_H - 1/2 - \epsilon - o(1))R} \quad (13.30)$$

of these elements.

All collisions at x persist on Z . Thus the squared-trace functions of the elements in (13.30) are equal as elements of K . Evaluating these identities first at the identity generic point and then at its σ -conjugate shows that the elements h form one simultaneous squared-trace fiber for

$$h \longmapsto (\mathrm{tr}^2(A_0 r(h)), \mathrm{tr}^2(A_1 r(h))).$$

The coefficient field generated by the finitely many matrices involved is finitely generated over $\mathbb{Q}$. For each $g \in H$, the element $\mathrm{tr}^2 r(g)$ is the regular squared-trace coordinate τ_g at the generic point of Y . If $\mathrm{tr}^2 r(g) = 4$, then $\tau_g - 4$ is identically zero on Y , and hence it also vanishes at $x|_H$. Thus the common H -representation and the Fuchsian character $x|_H$ satisfy the exact implication in Theorem 13.4. That theorem and (13.29) bound this simultaneous fiber by $e^{o(R)}$. Here convex cocompactness gives $|h|_H = O(R)$, and the fixed matrices and diagonalizer are absorbed into one finite extension and finite-type model. Choosing $0 < \epsilon < \delta_H - 1/2$ contradicts (13.30). Hence $[K : L] = 1$. $\square$

Theorem 13.85 (the adjoint trace field is already generated on the complementary subgroup). *Let Π be a closed surface group and let $x = [\rho_x]$ be a faithful cocompact Fuchsian character which is an algebraic point of the liftable $\mathrm{PSL}_2(\mathbb{C})$ -character component. Let $\phi: \Pi \twoheadrightarrow \mathbb{Z}$, and let $H < \ker \phi$ be finitely generated, nonelementary, and convex cocompact under ρ_x , with critical*

exponent δ_H . Suppose that $h_{\text{tr}}(x) < \delta_H$. Define the adjoint trace field and the corresponding field for the restriction to H by

$$K_x = \mathbb{Q}(\tau_g(x) : g \in \Pi), \quad L_H = \mathbb{Q}(\tau_h(x) : h \in H).$$

Then

$$K_x = L_H.$$

Equivalently, no nonidentity embedding of the full adjoint trace field can fix all squared traces on H .

Proof. Both fields are number fields and $L_H \subset K_x$. Suppose that the inclusion is proper. Separability in characteristic zero gives a nonidentity L_H -embedding $\sigma : K_x \hookrightarrow \overline{\mathbb{Q}}$. Applying σ to the character coordinates of x gives an algebraic character x^σ . It differs from x , since the squared-trace coordinates generate K_x , but its restriction to H has exactly the same character as $x|_H$. The latter is absolutely irreducible. After realizing the two characters over $\overline{\mathbb{Q}}$ and conjugating the second realization, we may therefore write

$$\rho^\sigma|_H = \rho_x|_H \quad \text{in } \text{PSL}_2(\overline{\mathbb{Q}}).$$

The cover corresponding to the finitely generated infinite-index subgroup H is noncompact and has a compact core with nonempty boundary. That core retracts onto a finite graph, so H is free. Choose a common lift $r : H \rightarrow \text{SL}_2(\overline{\mathbb{Q}})$.

The paired representation (ρ^σ, ρ_x) is Zariski dense in $\text{PSL}_2 \times \text{PSL}_2$. Indeed, both factors are Zariski dense. If its closure were the graph of an automorphism, equality on the Zariski-dense subgroup $r(H)$ would make that automorphism the identity, and the two global characters would coincide, contrary to $x^\sigma \neq x$. The normal-subgroup argument from Theorem 13.81 then shows that the paired image of $\ker \phi$ is also Zariski dense in the product. Hence one can choose $w \in \phi^{-1}(1)$ and lifts $A_0, A_1 \in \text{SL}_2(\overline{\mathbb{Q}})$ of $\rho_x(w), \rho^\sigma(w)$, respectively, such that

$$\text{tr}^2(A_1 A_0^{-1}) \neq 4.$$

There are $\exp((\delta_H + o(1))R)$ elements $h \in H$ whose displacement in the Fuchsian representation is at most R , and every wh is primitive because $\phi(wh) = 1$. Choose $\epsilon > 0$ with $h_{\text{tr}}(x) + \epsilon < \delta_H$. By the definition of trace entropy, some squared trace is shared by

$$\exp((\delta_H - h_{\text{tr}}(x) - \epsilon - o(1))R)$$

of these elements. Equality of their squared traces at x remains an equality after applying σ . They consequently lie in one simultaneous fiber of

$$h \longmapsto (\text{tr}^2(A_0 r(h)), \text{tr}^2(A_1 r(h))).$$

Theorem 13.4 bounds this simultaneous fiber by $\exp(o(R))$. (Its specialization hypothesis is immediate here because r is the faithful convex-cocompact Fuchsian restriction itself.) This contradicts $\delta_H - h_{\text{tr}}(x) - \epsilon > 0$, and proves $K_x = L_H$. $\square$

Corollary 13.86 (the invariant quaternion algebra is generated on the complementary subgroup). *Under the hypotheses of Theorem 13.85, let $\mathcal{A}_x$ be the invariant quaternion algebra of $\rho_x(\Pi)$ over K_x : after choosing lifts, this is the K_x -subalgebra generated by $\rho_x(\Pi^{(2)})$. Then the K_x -subalgebra generated by $\rho_x(H^{(2)})$ is all of $\mathcal{A}_x$.*

Proof. The subgroup $H^{(2)}$ has finite index in the finitely generated group H , because the quotient is the finite elementary abelian group $H_1(H, \mathbb{F}_2)$. It is therefore nonelementary, and its two-dimensional representation is absolutely irreducible. Its image lies in $\mathcal{A}_x$, since $H^{(2)} < \Pi^{(2)}$. The K_x -subalgebra it generates becomes all of M_2 after extending scalars to an algebraic closure, by Burnside's theorem. It consequently has dimension four over K_x . The ambient quaternion algebra $\mathcal{A}_x$ also has dimension four over K_x , so the inclusion is an equality. $\square$

Corollary 13.87 (birational determination by the complementary subgroup). *Under the hypotheses of Corollary 13.82, restriction to H is birational onto its image on every irreducible positive-dimensional $\mathbb{Q}$ -subvariety of V_x through x . Its differential at x is injective on the tangent space of every reduced such subvariety.*

Proof. Corollary 13.82 says that the fiber dimension at x is zero. Upper semicontinuity of fiber dimension therefore makes the restriction morphism generically finite on Z . Equivalently, the induced extension of function fields is finite. Theorem 13.84 makes its degree one. The final assertion is Theorem 13.83. $\square$

For completeness, the following direct coordinate calculation verifies the same conclusion on the specific twist subspaces used in Hao's proof.

Corollary 13.88 (explicit calculation in Hao's Fenchel–Nielsen coordinates). *Let x be a closed hyperbolic surface of genus $g \geq 6$ with $h_{\text{tr}}(x) \leq 1/2$, and let V_x be the variety on which all trace collisions at x persist. For the one-holed-torus or pair-of-pants surface decomposition used in Hao's proof of [19, Theorem A, Proposition 3.2, and (5.1)], the complementary subgroup H has $\delta_H > 1/2$. The real germ of V_x at x has zero-dimensional intersection with the associated Fenchel–Nielsen twist line in the one-holed-torus case, and with the full two-twist plane in the pair-of-pants case.*

Proof. We give the matrix calculation because it is where the algebraic locus preserving the trace collisions meets the geometry of the splitting. In the one-holed-torus case, the deformation fixes H and has

$$\rho_s(a_1) = D(s)\rho_0(a_1)D(-s), \quad D(s) = \text{diag}(\kappa^s, \kappa^{-s}).$$

Hao's normal form has $a_{22} = -\kappa a_{11}$ and $a_{11}a_{12}a_{21} \neq 0$. The diagonal part and the two off-diagonal matrix units therefore show that $\text{span}\{\rho_s(a_1)\}$ has dimension three. The homology class of a_1h has a_1 -coordinate one, so a_1h cannot be a proper power. Theorem 13.3 would force primitive length entropy at least $\delta_H > 1/2$ on any nonconstant piece of V_x in this twist line, contrary to $h_{\text{tr}}(x) \leq 1/2$.

In the pair-of-pants case, use twist parameters s, t , put $u = e^s$, $v = e^t$, and choose the standard matrices so that

$$\rho_{s,t}(a_1^{-1}a_2) = E(-s)NE(t)P^{-1}, \quad E(r) = \text{diag}(e^r, e^{-r}),$$

where every entry n_{ij} of N is nonzero. Before the fixed right factor P^{-1} , the four entries are

$$n_{11}\frac{v}{u}, \quad \frac{n_{12}}{uv}, \quad n_{21}uv, \quad n_{22}\frac{u}{v}. \quad (13.31)$$

After projectivizing and multiplying all coordinates by uv , (13.31) becomes the diagonally rescaled Segre map

$$(u^2, v^2) \mapsto [n_{11}v^2 : n_{12} : n_{21}u^2v^2 : n_{22}u^2].$$

Writing these four homogeneous coordinates as $[X_{11} : X_{12} : X_{21} : X_{22}]$, direct multiplication gives the exact equation

$$n_{12}n_{21}X_{11}X_{22} = n_{11}n_{22}X_{12}X_{21}.$$

All n_{ij} are nonzero, so this is a smooth quadric in $\mathbb{P}^3$. Its two families of projective lines are obtained by holding u^2 constant or holding v^2 constant. Hence, on any irreducible curve on which both s and t vary, the matrices in (13.31) span at least three dimensions: otherwise the projective image would lie on a line, forcing s or t to be constant. Every $a_1^{-1}a_2h$ is primitive because its two indicated homology coordinates are $(-1, 1)$. Theorem 13.3 excludes such a curve from V_x .

It remains to treat the two rulings. If $t = t_0$, choose $Z \in H$ so that

$$M = \rho_0(a_1^{-1}Za_1)$$

has two nonzero off-diagonal entries and a nonzero diagonal part. Such Z exists by Zariski density of the non-elementary group $\rho_0(H)$. For $a_Z = a_1^{-1} Z a_1 a_2^{-1}$, the varying matrix is

$$\rho_{s,t_0}(a_Z) = E(-s)ME(s)B_2(-t_0)\rho_0(a_2)^{-1}.$$

Its constant diagonal part and its two independently scaled off-diagonal parts span three dimensions. Also $a_Z h$ has a_2 -homology coordinate -1 , so it is primitive. The other ruling is identical after interchanging the two boundary generators and using the B_2 -eigenbasis. Thus no nonconstant real-analytic curve in the two-twist plane can lie in V_x . Real curve selection now gives the claimed zero-dimensional real germs. $\square$

Theorem 13.89 (trace functions forced to coincide in every finite cover). *Let x be the point of Teichmüller space represented by a closed hyperbolic surface S , let $\Gamma = \pi_1(S)$, and let ρ be its holonomy representation. Count distinct absolute traces by*

$$D_x(T) = \#\{|\operatorname{tr} \tilde{\rho}(\gamma)| : \gamma \in \Gamma, |\operatorname{tr} \tilde{\rho}(\gamma)| \leq T\},$$

where the choice of lift to $\operatorname{SL}_2(\mathbb{R})$ is immaterial. Assume $D_x(T) \leq C(1+T)$.

Let $S_G \rightarrow S$ be any connected finite cover of genus $G \geq 6$, and write x_G for the restricted Fuchsian character. Let $\mathfrak{X}_G$ be the $\mathbb{Q}$ -defined PSL_2 -character component containing its Teichmüller space. Define

$$I(x_G) = \{f \in \mathbb{Q}[\mathfrak{X}_G] : f(x_G) = 0\}, \quad Z_G = V(I(x_G)) = \overline{\{x_G\}}^{\mathbb{Q}}.$$

Then the surface decomposition in Hao's proof of [19, Theorem A, Proposition 3.2, and (5.1)] gives a convex-cocompact subgroup $H = \pi_1(\Sigma')$, an element $A \in \pi_1(S_G)$, and a number $\delta_H \geq \delta(G) > 1/2$ with the following property. There are constants $c > 0$ and R_0 , depending on this cover and metric, such that, for every $R \geq R_0$, there are $h_1, \dots, h_M \in H$ for which

$$M \geq c \exp((\delta_H - \tfrac{1}{2})R), \tag{13.32}$$

and:

- (i) $d(o, \rho(h_i)o) \leq R$, and every Ah_i is primitive;
- (ii) the squared-trace functions

$$\tau_{Ah_i} : \mathfrak{X}_G \longrightarrow \mathbb{A}^1, \quad \tau_w([\sigma]) = \operatorname{tr}^2 \sigma(w),$$

are pairwise distinct regular functions;

- (iii) all restrictions $\tau_{Ah_i}|_{Z_G}$ are the same regular function on Z_G .

Proof. Convex-cocompact orbit counting gives

$$\#\{h \in H : d(o, \rho(h)o) \leq R\} \geq c_0 e^{\delta_H R}$$

for all large R . Hao's primitivity lemma for this decomposition says that every Ah , $h \in H$, is primitive [19, Lemma 4.7]. If $D_A = d(o, \rho(A)o)$, then

$$\ell_x(Ah) \leq D_A + d(o, \rho(h)o) \leq D_A + R.$$

Since $|\operatorname{tr} \tilde{\rho}(w)| = 2 \cosh(\ell_x(w)/2)$, all these elements have absolute trace at most $C_A e^{R/2}$. The assumed linear bound on the base trace set applies because the cover group is a subgroup of the base surface group. It therefore leaves at most $C_1 e^{R/2}$ possible absolute traces. Pigeonholing produces a subset $E_R \subset H$ of cardinality

$$|E_R| \geq (c_0/C_1) e^{(\delta_H - 1/2)R}$$

on which all squared traces agree at x_G .

For $u, v \in E_R$, $\tau_{Au}(x_G) - \tau_{Av}(x_G) = 0$; hence $\tau_{Au} - \tau_{Av} \in I(x_G)$. By the definition of Z_G , their restrictions to Z_G are equal. It remains to ensure that the functions on the full component are distinct. Partition E_R by equality of the regular functions τ_{Ah} on $\mathfrak{X}_G$. Teichmüller space

is Zariski dense in this component, so these are precisely the universal trace-equivalence classes considered in Hao's proof. His matrix calculation bounds every such class by three in the one-holed-torus decomposition and by one in the pair-of-pants decomposition [19, Lemmas 4.2–4.3 and Corollary 4.6]. Choosing one representative from each class loses a factor of at most three and proves (13.32) and (i)–(iii). $\square$

The lower exponent in (13.32) can approach $1/2$. More explicitly, Hao obtains

$$\delta(G) = \frac{1}{2} \left(1 + \sqrt{1 - \frac{2.728 H(G)}{(4G - 6)\pi}} \right) > \frac{1}{2},$$

where

$$H(G) = 4 \log G + 5.362 - 4 \log 2 + 4 \log \left(1 + \frac{4(G - 1)\pi}{2 \log G + 2.681} \right).$$

For all sufficiently large G ,

$$\delta(G) > 1 - \frac{5.456 \log G}{(2G - 3)\pi},$$

so $\delta(G) \rightarrow 1$.

Lemma 13.90 (transcendence degree is unchanged by a finite cover). *Let $\Gamma' < \Gamma$ have finite index, and for a character x put*

$$K_\Gamma = \mathbb{Q}(\tau_\gamma(x) : \gamma \in \Gamma), \quad K_{\Gamma'} = \mathbb{Q}(\tau_h(x) : h \in \Gamma').$$

Then $K_\Gamma/K_{\Gamma'}$ is algebraic. Consequently the rational Zariski closures of x in the full character components for Γ and Γ' have the same dimension.

Proof. For every $\gamma \in \Gamma$, left multiplication by $\langle \gamma \rangle$ on the finite set of left cosets Γ/Γ' gives an integer $1 \leq n \leq [\Gamma : \Gamma']$ with $\gamma^n \in \Gamma'$. The Chebyshev trace identity has the form

$$\tau_{\gamma^n} = Q_n(\tau_\gamma)$$

for a nonconstant polynomial $Q_n \in \mathbb{Z}[T]$. Thus every $\tau_\gamma(x)$ is algebraic over $K_{\Gamma'}$, proving the first assertion. A character-coordinate field is algebraic over its squared-trace field: choose finitely many trace generators and adjoin their square roots. Its transcendence degree, which is the dimension of the rational Zariski closure of the point, is therefore unchanged. $\square$

Corollary 13.91 (at least $e^{(1/2-\eta)R}$ distinct trace functions agree after restriction). *If the field generated by the squared-trace coordinates of x in Theorem 13.89 has positive transcendence degree over $\mathbb{Q}$, then Z_G is positive-dimensional in every finite cover. For every $\eta > 0$, one can choose one cyclic cover and a constant $c_\eta > 0$ such that, for every sufficiently large R , at least*

$$c_\eta e^{(1/2-\eta)R}$$

pairwise distinct squared-trace functions on the full character component restrict to one and the same function on Z_G .

Proof. A cyclic cover of degree m has genus $G = 1 + m(g - 1)$, which tends to infinity with m . Choose m so that $\delta(G) > 1 - \eta$, and apply (13.32). Positivity of $\dim Z_G$ follows from Lemma 13.90. $\square$

The cover is chosen before R tends to infinity, and the constants c, R_0, D_A may depend on that cover. The pairwise distinct functions live on the full character variety of the cover; its twist deformations need not descend to the base surface. Consequently the theorem does not by itself rule out a transcendental character. It shows that any transcendental character with linear trace growth would produce, in every sufficiently large cover, almost $e^{R/2}$ distinct trace polynomials that agree identically on one positive-dimensional algebraic locus.

Proposition 13.92 (arbitrary nonconstant loci can have trace-function fibers of size $e^{(1/2-o(1))R}$). *For every $\eta > 0$, there are a closed arithmetic hyperbolic surface (S, x) , a positive-dimensional nonconstant algebraic locus W through x in its character variety, and constants $c > 0$ and R_0 such that, for every $R \geq R_0$, at least*

$$ce^{(1/2-\eta)R}$$

pairwise distinct squared-trace functions of primitive elements of length at most $R + O(1)$ restrict to one and the same constant function on W . Consequently there is no cover- and locus-independent estimate $e^{(1/2-\epsilon)R+o(R)}$, with fixed $\epsilon > 0$, for fibers of trace functions on arbitrary nonconstant loci.

Proof. Start with a torsion-free compact arithmetic surface S_0 of sufficiently large genus. Choose disjoint embedded one-holed tori T_1, T_2 , with geodesic boundary, such that

$$C = \overline{S_0} \setminus (\overline{T_1 \cup T_2})$$

is connected and has positive genus; one may first pass to a finite cover of an arithmetic surface to arrange this topology. Choose an epimorphism $\psi: \pi_1(S_0) \rightarrow \mathbb{Z}$ which vanishes on both torus subgroups and whose restriction to $\pi_1(C)$ is still onto. For example, use a primitive cohomology class dual to a nonseparating curve contained in the interior of C . Let $S_m \rightarrow S_0$ be the cyclic cover defined by the reduction of ψ modulo m , equip it with the lifted hyperbolic metric, and denote its Fuchsian character by x_m . The inverse image of C is connected because $\psi(\pi_1(C)) = \mathbb{Z}$, whereas each T_i has m isometric lifts. Choose one lift $\tilde{T}_{i,m}$ of each torus and set

$$Y_m = S_m \setminus (\text{int } \tilde{T}_{1,m} \cup \text{int } \tilde{T}_{2,m}), \quad K_m = \pi_1(Y_m).$$

The surface Y_m is connected. It is the convex core of the complete convex-cocompact surface

$$X_m = K_m \backslash \mathbb{H}^2.$$

Moreover

$$\text{area}(Y_m) = m \text{area}(S_0) - \text{area}(T_1) - \text{area}(T_2),$$

whereas $\text{length}(\partial Y_m)$ is independent of m . Extend the function equal to one on Y_m by a fixed compactly supported cutoff across the two funnels of X_m . Its Dirichlet energy is bounded by a constant times $\text{length}(\partial Y_m)$, while its squared L^2 -norm is at least $\text{area}(Y_m)$. The Rayleigh quotient therefore gives $\lambda_0(X_m) \rightarrow 0$. For a convex-cocompact hyperbolic surface, once $\lambda_0(X_m) < 1/4$,

$$\lambda_0(X_m) = \delta_m(1 - \delta_m),$$

where δ_m is the critical exponent of K_m [8]. It follows that

$$\delta_m \rightarrow 1.$$

Put $H_m = \pi_1(S_m \setminus \text{int } \tilde{T}_{1,m})$. Keep the H_m -character fixed and vary the hyperbolic structure on $\tilde{T}_{1,m}$ with its boundary holonomy and gluing fixed. A one-holed torus with fixed boundary has positive-dimensional Teichmüller space. The Zariski closure of this deformation lies in a positive-dimensional component W_m of the algebraic restriction fiber, and every squared-trace function τ_w , $w \in H_m$, is constant on W_m .

Choose a standard generator A_m of the other chosen one-holed torus $\tilde{T}_{2,m}$, and choose a symplectic homology basis of S_m adapted to that torus. Every loop in Y_m has zero A_m -coordinate in this basis. Hence, for $h \in K_m$, the element $A_m h$ lies in H_m and has A_m -coordinate one. If $A_m h = u^q$ with $|q| \geq 2$, applying this integral homology coordinate would give $1 = q[u]_{A_m}$, which is impossible. Thus every $A_m h$ is primitive. Convex-cocompact orbit counting gives $c'_m e^{\delta_m R}$ such elements of length at most $R + O_m(1)$. The surface S_m is arithmetic, so its trace set has linear growth; these elements therefore assume at most $C_m e^{R/2}$ squared-trace values. One value is assumed at least

$$c''_m e^{(\delta_m - 1/2)R}$$

times.

It remains to discard possible identities between trace functions on the full character variety. Twist the chosen torus $\tilde{T}_{2,m}$ along its separating boundary while fixing the representation of its complement, and hence fixing K_m . In an eigenbasis for the boundary holonomy this deformation has

$$\rho_s(A_m) = D(s)\rho_0(A_m)D(-s), \quad D(s) = \text{diag}(e^s, e^{-s}).$$

Write $\rho_0(A_m) = (a_{ij})$. Both off-diagonal entries are nonzero: if one vanished, A_m and the boundary holonomy would share a fixed point, which in a discrete torsion-free Fuchsian group would force them to have the same two fixed points, contrary to the topology of the one-holed torus. Its diagonal part is also nonzero, since a determinant-one matrix with zero diagonal has projective order two. Consequently the constant diagonal part and the two independently scaled off-diagonal matrix units span a three-dimensional subspace of $M_2(\mathbb{R})$.

If two squared-trace functions $\tau_{A_m h}$ agree on the full character component, their equality persists along this twist. Applying Theorem 13.3 with $H = K_m$ shows that each such class contains at most three elements. Choosing one representative from each class loses only this factor. All remaining functions are globally distinct, while their restrictions to W_m are the same constant because the elements $A_m h$ lie in H_m . Finally choose m with $\delta_m > 1 - \eta$, take $c = c''_m/3$, and relabel (S_m, x_m, W_m) as (S, x, W) . The constants are then fixed before R tends to infinity, as required. $\square$

The loci W_m in Proposition 13.92 need not preserve collisions involving the torus that moves. The proposition therefore does not weaken the results above concerning trace equalities that hold identically on a fixed locus. It shows instead that any argument through these loci must exploit the full collision ideal, or another condition that forces the varying part of the representation to be seen by the chosen trace functions.

Although these results are not used in the recurrence proof of arithmeticity, they give a concrete dichotomy for a locus V_x with low trace entropy: either the Fricke coordinates are algebraic, or a real semialgebraic deformation preserves all trace equalities that hold at x . Classifying the latter loci means determining when exponentially many equalities between individual closed-curve length functions can hold identically on one algebraic locus; it is a structural question about the exceptional locus, not a remaining step in the compact conjectures.

14. QUANTITATIVE SEPARATION FROM THE PRODUCT FORMULA

The product formula converts upper bounds at all nondistinguished places into a lower bound at the distinguished place.

Theorem 14.1 (adelic separation). *Let k be a number field, v_0 a distinguished archimedean place of local degree $n_0 \in \{1, 2\}$. Write $n_v = [k_v : \mathbb{Q}_v]$ for the usual local degree (so a real place has degree one and a complex place degree two), and normalize absolute values by $\prod_v |z|_v^{n_v} = 1$. Let $S_T \subset k$ be a family such that for every $T \geq 2$ and every distinct $x, y \in S_T$, with $z = x - y$,*

$$|z|_v \leq C_v T^{a_v} \quad (v \neq v_0), \quad (14.1)$$

where $C_v \geq 1$, $a_v \geq 0$, and $C_v = 1, a_v = 0$ outside a fixed finite set. Put

$$A = \sum_{v \neq v_0} n_v a_v, \quad C = \prod_{v \neq v_0} C_v^{n_v}.$$

Then

$$|x - y|_{v_0} \geq C^{-1/n_0} T^{-A/n_0}. \quad (14.2)$$

Consequently a real unit interval, or a complex unit disk, contains $O(T^A)$ points of S_T . Explicitly, the total counts are

$$\#(S_T \cap [-T, T]) = O(T^{1+A}), \quad \#\{s \in S_T : |s| \leq T\} = O(T^{2+A}) \quad (14.3)$$

at a real and a complex distinguished place, respectively.

Proof. The product formula and (14.1) give

$$1 \leq |z|_{v_0}^{n_0} C T^A,$$

which is (14.2). One-dimensional interval packing and two-dimensional disk packing give the local bounds; covering the radius- T region by $O(T^{n_0})$ unit regions gives (14.3). $\square$

The theorem gives an upper bound controlled by the *sum* of the growth exponents at the nondistinguished places; it does not say that this bound is sharp. In a derived arithmetic group, all nondistinguished archimedean coordinates are bounded and all finite-place denominators are bounded, so $A = 0$. For semi-arithmetic groups, modular embeddings control archimedean escape and lead to the recent weak clustering results of Belolipetsky–Cosac–Dória–Teixeira Paula [2]. For quasi-arithmetic or S -arithmetic behavior, a separate estimate relating Fuchsian translation length to Bruhat–Tits displacement is indispensable; the size of the real trace alone does not control denominators.

15. WHY ITERATED SQUARE ROOTS DO NOT PROVE THE COMPACT CONJECTURES

Apply Theorem 7.5 to the finite-index lattice $H = \Gamma^{(2)}$. Since H is normal in Γ , every $\gamma \in \Gamma$ commensurates H , and $\gamma^2 \in H$. Among primitive H -classes represented by such literal squares and having length at most L , there are at most, up to bounded finite-index class splitting, as many as Γ -classes of length at most $L/2$, namely only

$$e^{hL/2+o(L)}$$

classes. This is exactly the critical exponent allowed for a nonarithmetic H when $m = 2$.

There is no further iteration inside this fixed subgroup: for $r \geq 2$, the element γ^{2^r} is already a proper power in H , so it is not primitive there. One may instead define the characteristic tower

$$H_0 = \Gamma, \quad H_{r+1} = H_r^{(2)}.$$

Then $\gamma^{2^r} \in H_r$, while $\gamma \in \text{Comm}_G(H_r)$, and the number of possibly primitive H_r -classes obtained this way below length L is at most $e^{hL/2^r+o(L)}$. Theorem 7.5 with $m = 2^r$ permits exactly that exponent for a nonarithmetic lattice. Thus the iterated tower stays at, rather than crosses, the critical threshold.

Nor can an arbitrary element of $\Gamma^{(2)}$ be replaced by a bounded product of literal squares. For a closed surface group the square word is proper, and its verbal subgroup has infinite width by Myasnikov–Nikolaev [29]. At the algebro-geometric level, the Kummer obstruction $\delta_2(h) \in H^1(k, \mathcal{J}_h[2])$ lives in a centralizer that varies with h , and has no canonical multiplicative law across products of noncommuting elements. Therefore one cannot prove (9.1) simply by taking square roots after passing to the subgroup generated by squares.

16. THE MAIN ARGUMENT AND THE AUXILIARY RESULTS

We summarize the logical role of the principal results.

1. *Incidence estimates and universal growth.* Theorems 12.2 and 12.9 give the sharp universal bounds of Corollary 12.10. Theorem 12.11 strengthens this to every component of every finite quotient, and Theorem 12.13 gives the corresponding mixed correlations under linear growth. No algebraicity assumption on the lattice is used.

2. *Why ordinary additive closure is insufficient.* Theorem 12.4 proves that, after recording $(\text{tr}(AB_i))_{i=1}^3$ with fixed B_i whose trace functionals recover A , every matrix sum except the forced inverse sum has representation multiplicity $\exp(o(n))$ on a word ball of radius n . Thus the energy estimates cannot by themselves make the full trace fiber bounded at every finite place. This is a proved obstruction to that method, not an omitted estimate.
3. *Simultaneous control of the recurrence dilates.* The two trace coordinates $(\text{tr } g, \text{tr}(ag))$ form a graph with $e^{R-o(R)}$ edges and two vertex classes of size $O(e^{R/2})$. Theorem 13.32 extracts one set of $e^{(1/2-o(1))R}$ vertices on which all Chebyshev recurrence coefficients are controlled simultaneously. On a nonconstant curve, Theorem 13.33 combines the coprime recurrence zero divisors by the Chinese remainder theorem and gives the exact bound (13.10bh10).
4. *The congruence criterion for a fixed element.* Lemma 13.37 does not require a Cartesian product in the trace-pair graph. It requires only a uniform bound for the number of trace values in each residue class modulo

$$\mathfrak{n}_n = (q_{n-1}(\text{tr } a)q_n(\text{tr } a)).$$

Theorem 13.41 turns that estimate into $\mathcal{J}_k(\text{tr } a) = \ell(a)/2$. If it holds for every $a \in \widetilde{\Gamma(2)}$, Takeuchi's theorem [47] proves arithmeticity. Lemma 13.43 derives the estimate from uniform expansion and lattice counting in all these mixed congruence quotients.

5. *The proof using varying group elements.* Fix the recurrence index N , instead of sending it to infinity along one fixed element. Corollary 13.54 gives a $(2N - 4 - \epsilon)$ -multiple of the global trace height in the reduced divisor of two consecutive recurrence terms. Lemma 13.55 uses only expansion modulo rational p^2 to choose $a_m = b^m h_m$ for which that divisor has bounded prime exponents. Lemma 13.56 is uniform in this moving multiplier, and Theorem 13.57 uses bounded-power superapproximation to give the geometric upper coefficient $(2N - 1)/2$. Letting $N \rightarrow \infty$ proves Theorem 13.58:

$$\mathcal{J}_{k_y}(\text{tr}_y b) \leq \frac{\ell_x(b)}{2}$$

at every algebraic faithful cocompact Fuchsian character y in the rational closure of x .

6. *Arithmeticity and transcendental characters.* At an algebraic original character, the distinguished real place gives the reverse inequality, and Takeuchi's theorem [47] gives arithmeticity. If the original character were transcendental, Proposition 13.2 would give a distinct nearby algebraic Fuchsian character in the same component and in its rational closure. The preceding inequality makes every marked length at the specialization at most the corresponding original marked length. Lemma 13.45, which applies Thurston's asymmetric metric, then forces the two characters to coincide. This contradiction proves Theorem 13.59, and hence both compact conjectures, without arbitrary-ideal expansion or abc .
7. *Separate pointwise results for a fixed recurrence.* Theorem 13.49 gives

$$\limsup_n \frac{1}{n} \log N \text{rad}_S(q_n(\text{tr } a)) \leq \frac{\ell(a)}{2}$$

for every hyperbolic a . Thus a nondistinguished place with positive local expansion forces the repeated prime factors in every individual recurrence term to have positive linear logarithmic norm, not merely on average over two consecutive terms. Theorem 13.48 also forces every hypothetical nondistinguished expanding place in the fixed-element argument to put positive linear logarithmic norm above each fixed prime-power cutoff, and Theorem 13.62 locates the whole possible loss in exceptional initial Lucas–Wieferich multiplicity. The matching lower radical rate is isolated as Conjecture 13.65; Theorem 13.66 shows that it implies both compact conjectures. Number-field abc gives the lower rate, whereas Proposition 13.67 shows why it is not currently an unconditional Lucas-sequence theorem. Proposition 13.47 proves that flatness at the separate prime-power factors does not formally glue to flatness modulo their product.

These statements describe the stronger problem for a fixed recurrence. The proof above, which varies the group element, does not resolve the Mersenne radical problem and does not require it.

8. *Valuative results.* The proper-model descent theorem constructs a central simple algebra and an order from invariant lattices. The entropy estimate at a divisorial valuation excludes every unbounded divisor for which the local translation length exceeds the explicit cancellation rate. The recurrence argument proves the more general local expansion theorem without having to establish the reverse inequality in the remaining case.

The logical dependence of the principal ingredients is:

$$\begin{array}{lcl}
 \begin{array}{l} \text{finite-type divisor bounds} \\ + \text{quadratic norm identity} \end{array} & \longrightarrow & \text{uniform moving two-trace fibers,} \\
 \begin{array}{l} \text{reduced-divisor Roth} \\ + \text{expansion modulo rational } p^2 \end{array} & \longrightarrow & \begin{array}{l} \text{moving recurrence ideals} \\ \text{with bounded exponents,} \end{array} \\
 \begin{array}{l} \text{algebraic Fuchsian specialization} \\ + \text{persistent trace equalities} \\ + \text{bounded-power expansion} \\ + \text{congruence-class count} \\ + \text{linear trace growth} \end{array} & \longrightarrow & \mathcal{J}_{k_y}(\text{tr}_y b) \leq \ell_x(b)/2, \\
 \begin{array}{l} \text{nearby algebraic specialization} \\ + \text{the local height inequality} \end{array} & \longrightarrow & \ell_y(\gamma) \leq \ell_x(\gamma) \quad (\gamma \neq 1), \\
 \begin{array}{l} \ell_y \leq \ell_x \\ + \text{Thurston's asymmetric metric} \end{array} & \longrightarrow & \text{the original character is algebraic,} \\
 \begin{array}{l} \text{local height inequality at } x \\ + \text{distinguished real contribution} \\ + \text{nonnegative local decomposition} \\ + \text{Takeuchi and commensurability} \end{array} & \longrightarrow & \text{compact Schmutz and compact Sarnak,} \\
 \text{independently:} & \begin{array}{l} \text{one-term radical lower rate} \\ + \text{proved one-term upper rate} \end{array} & \longrightarrow \text{the same arithmeticity conclusion,} \\
 \begin{array}{l} \text{number-field } abc \\ + \text{superapproximation modulo rational squares} \end{array} & \longrightarrow & \text{the stronger pointwise radical equality.}
 \end{array} \tag{16.1}$$

17. COMPARISON WITH KNOWN RESULTS

The following table places the results beside the closest results in the literature.

Statement	Known result
Positive trace gap $\Rightarrow$ derived, Fuchsian/Kleinian	Bogachev’s manuscript; earlier nonuniform work established related arithmeticity and growth implications, but not this full statement in this form.
Bounded clustering $\Rightarrow$ arithmetic, nonuniform Fuchsian	Geninska–Leuzinger.
Linear trace growth $\Rightarrow$ arithmetic, nonuniform Fuchsian	Hao, in the stronger $o(T \log \log \log T)$ form.
Bounded clustering or linear growth, cocompact Fuchsian	Proved here in Theorem 13.59. The more general Theorem 13.58 gives the required local height inequality at every algebraic specialization preserving the trace equalities at the original character.
Same compact conjectures under the one-term Chebyshev radical hypothesis	Also proved here in Theorem 13.66. The hypothesis has an exact Mersenne specialization and remains open as a stronger pointwise statement.
Same compact conjectures under number-field abc	Theorem 13.69 is retained as an independent conditional implication through the stronger pointwise radical equality.
Same compact conjectures under arbitrary-ideal superapproximation	Theorem 13.46 is an independent conditional implication. The general expansion theorem remains open but is not needed for the unconditional result.
Every closed hyperbolic surface	Corollary 12.10 gives the sharp universal exponent $N_H^*(L) \geq e^{(1/2-o(1))L}$, and arithmetic surfaces show that $1/2$ is optimal. The preceding word-length comparison and removal of proper powers turn [9, Theorem 1.2] into exponential primitive-length growth with an unspecified surface-dependent positive exponent. We do not know an earlier result with this universal exponent.
Generic compact surfaces of genus at least six	Hao proves distinct primitive-length exponent $> 1/2$ outside a countable union of positive-codimension algebraic loci [19].
Semi-arithmetic surfaces with modular embeddings	Weak clustering and exponential mean multiplicity are known in arithmetic dimension at most two; higher-dimensional results currently require a strong contraction hypothesis [2, 53].

Three natural analytic approaches do not apply under the present hypotheses. First, the nonarchimedean Anosov correlation theory does not apply to a closed surface group. Sambarino observes that a ϑ -Anosov representation over a nonarchimedean local field would give continuous transverse boundary maps, forcing the Gromov boundary to be a Cantor set and the group to be virtually free [39, Remark 5.2.6].

Second, replacing a nondistinguished representation by its Bruhat–Tits translation length loses essential information. Cantrell–Reyes prove that the orbit pseudometric of an action on an $\mathbb{R}$ -tree

without a global fixed point lies in their Manhattan boundary exactly when the action has bounded backtracking [14, Proposition 6.7]. No such bounded-backtracking theorem is known for the actions needed here. Chen–He instead show that the nonarchimedean harmonic map factors through a folding $T_q \rightarrow T_F$, with q Jenkins–Strebel; over a locally compact field, an unbounded reductive surface-group representation has a non-small Bruhat–Tits action [15, Theorems 1.2–1.3 and Proposition 2.15]. Thus the resulting tree is generally a folded quotient, not simply the dual tree of a measured foliation. Moreover its integral translation length has only $O(R)$ possible values on an R -ball for the Fuchsian metric, so at least one exact length level may retain the full exponential growth rate. A useful estimate must distinguish traces modulo high powers, rather than recording only their valuations.

Third, ordinary congruence equidistribution does not provide enough independent congruence conditions. A single trace condition in $\mathrm{SL}_2(\mathbb{F}_q)$ has density $\asymp q^{-1}$, whereas prescribing $(\mathrm{tr} A, \mathrm{tr}(AB))$ has generic density $\asymp q^{-2}$. An exact real trace fiber does not prescribe the second coordinate, so summing over it loses that extra factor. Existing uniform congruence counting, such as Magee–Oh–Winter’s theorem for Schottky semigroups in $\mathrm{SL}_2(\mathbb{Z})$, carries a q^C loss and is not a closed-surface periodic-orbit theorem at the growing-modulus scale required here [27]. Their vector-valued transfer operator on the regular representation of the congruence quotient is nevertheless the relevant model for a possible vector-valued argument. Such an argument would require matrix-coefficient bounds against trace-fiber indicators; the trace values themselves do not form a small right-translation module.

The classification results in Sections 5–7 use the positive trace-gap theorem, while the divisor estimates and the universal exponent in Section 12 do not assume arithmeticity or algebraicity of the surface group. Theorem 13.59 proves the compact bounded-clustering and linear-growth implications. The conditional and pointwise results concern stronger arithmetic statements and explain which parts of an argument based on one fixed recurrence remain open.

18. CONCLUSION

The square-root formula is best understood as a special case of division on the regular centralizer, whose branches are governed by Kummer theory and Galois monodromy. Type A_1 is exceptional: projectivization turns this multivalued correspondence into the rational map $[1 + g]$, and the Cayley–Hamilton identity expresses every trace $\mathrm{tr}(g^n h)$ in terms of two initial trace coordinates. In higher type A , polynomial secants remain rational even when roots do not; their failure to descend to a projective unitary group distinguishes outer forms from inner forms.

For compact Fuchsian groups, the divisor bounds over finitely generated fields yield the sharp universal lower exponents for distinct traces and primitive lengths. The same estimates hold separately in the components indexed by a finite quotient, and linear trace growth gives mixed additive energy $X^{3-o(1)}$ between components. Theorem 12.4 also proves that, apart from the forced inverse sum, applying the injective coordinate map $A \mapsto (\mathrm{tr}(AB_i))_{i=1}^3$ to an exact trace fiber in a word ball of radius n gives additive representation multiplicities $\exp(o(n))$. These facts explain why an argument using only additive energy or small doubling cannot prove arithmeticity.

The arithmeticity proof fixes the recurrence index N and varies the group element. Roth’s theorem gives a lower bound for the reduced divisor of $q_{N-1}(\mathrm{tr} a_m)q_N(\mathrm{tr} a_m)$. A random-walk choice $a_m = b^m h_m$, using expansion only modulo rational p^2 , keeps the global recurrence height close to that of b^m while ensuring that every prime occurs to uniformly bounded order in this divisor. The explicit quadratic norm identity (13.10mov16) gives a fiber estimate uniform in a_m :

$$g \longmapsto (\mathrm{tr}_y(a_m g), \mathrm{tr}_y g).$$

Known superapproximation for fixed powers of square-free moduli then bounds the number of traces in each residue class modulo the recurrence ideal. Lemma 13.37 converts this bound into

the required upper estimate. Comparing the lower coefficient $2N - 4 - \epsilon$ with the upper coefficient $(2N - 1)/2$, and then letting $N \rightarrow \infty$, gives the factor $1/2$ in the height inequality below.

The result is the inequality

$$\mathcal{J}_{k_y}(\mathrm{tr}_y b) \leq \frac{\ell_x(b)}{2} \quad (17.1)$$

for every algebraic faithful Fuchsian character y in the rational closure of a linear-growth character x . This is the general theorem from which the compact conjectures follow. When $y = x$ is algebraic, the contribution of the distinguished real place gives equality, and Takeuchi's theorem [47] gives arithmeticity. If x were transcendental, a distinct nearby algebraic Fuchsian specialization y in the same component would preserve all of its trace collisions. Inequality (17.1) would then give $\ell_y(\gamma) \leq \ell_x(\gamma)$ for every nontrivial marked closed curve. Thurston's asymmetric metric [49] forces $y = x$, a contradiction. Thus

$$\begin{array}{l} \text{Roth reduced divisors} \\ + \text{selection using expansion modulo } p^2 \\ + \text{bounded-power congruence counting} \end{array} \implies (17.1) \implies \text{compact Schmutz and compact Sarnak}$$

is unconditional.

The arbitrary-ideal and one-term radical theorems give stronger statements, but they are not prerequisites for this conclusion. In particular,

$$\limsup_n \frac{1}{n} \log N \mathrm{rad}_S(q_n(\mathrm{tr} a)) \leq \frac{\ell(a)}{2}$$

still gives a pointwise restriction on every fixed recurrence. Its matching universal lower bound follows from number-field abc , while the specialization $s = 3/\sqrt{2}$ identifies that stronger lower bound with full exponential radical growth of $2^n - 1$. By varying the group element, the proof avoids this Mersenne and non-Wieferich problem; it does not solve that stronger pointwise problem.

The Lucas–Wieferich localization theorem, the Chebyshev Chinese-remainder theorem on algebraic curves, proper-model descent, and the entropy theorem at a divisorial valuation are additional results. They describe the limitations of arguments based on one fixed recurrence or on a single valuation. Inequality (17.1), proved from the varying family a_m , is what completes the arithmeticity argument.

REFERENCES

- [1] A. BAKER, *Linear forms in the logarithms of algebraic numbers*, Mathematika, 13 (1966), pp. 204–216. doi:10.1112/S0025579300003971.
- [2] M. BELOLIPETSKY, G. COSAC, C. DÓRIA, AND G. T. PAULA, *On multiplicities in length spectra of semi-arithmetic hyperbolic surfaces*, Comm. Math. Phys., 407 (2026). Article 76; doi:10.1007/s00220-026-05581-w; arXiv:2507.00211.
- [3] N. BOGACHEV, *Proof of the positive trace gap conjecture*. Manuscript, version dated 20 September 2026, 2026.
- [4] E. BOMBIERI AND W. GUBLER, *Heights in Diophantine Geometry*, vol. 4 of New Mathematical Monographs, Cambridge University Press, Cambridge, 2006. doi:10.1017/CBO9780511542879.
- [5] A. BOREL, *Density properties for certain subgroups of semi-simple groups without compact components*, Ann. of Math. (2), 72 (1960), pp. 179–188. doi:10.2307/1970150.
- [6] A. BOREL AND HARISH-CHANDRA, *Arithmetic subgroups of algebraic groups*, Ann. of Math. (2), 75 (1962), pp. 485–535. doi:10.2307/1970210.
- [7] A. BOREL AND G. PRASAD, *Finiteness theorems for discrete subgroups of bounded covolume in semi-simple groups*, Publ. Math. Inst. Hautes Études Sci., 69 (1989), pp. 119–171. doi:10.1007/BF02698843.
- [8] D. BORTHWICK, *Spectral Theory of Infinite-Area Hyperbolic Surfaces*, vol. 318 of Progress in Mathematics, Birkhäuser, 2 ed., 2016. doi:10.1007/978-3-319-33877-4.
- [9] E. BREUILLARD, Y. DE CORNULIER, A. LUBOTZKY, AND C. MEIRI, *On conjugacy growth of linear groups*, Math. Proc. Cambridge Philos. Soc., 154 (2013), pp. 261–277. arXiv:1106.4773.
- [10] R. BROOKS, *The bottom of the spectrum of a Riemannian covering*, J. Reine Angew. Math., 357 (1985), pp. 101–114. doi:10.1515/crll.1985.357.101.
- [11] ———, *The spectral geometry of a tower of coverings*, J. Differential Geom., 23 (1986), pp. 97–107. doi:10.4310/jdg/1214439903.

- [12] M. BURGER, *Spectre du Laplacien, graphes et topologie de Fell*, Comment. Math. Helv., 63 (1988), pp. 226–252. doi:10.1007/BF02566764.
- [13] P. BUSER, *Geometry and Spectra of Compact Riemann Surfaces*, vol. 106 of Progress in Mathematics, Birkhäuser, 1992. doi:10.1007/978-0-8176-4992-0.
- [14] S. CANTRELL AND E. REYES, *Manhattan geodesics and the boundary of the space of metric structures on hyperbolic groups*, Comment. Math. Helv., 100 (2025), pp. 11–59. doi:10.4171/CMH/579.
- [15] J. CHEN AND S. HE, *On the non-archimedean Hitchin map for $SL_2(F)$* , 2025. arXiv:2511.03469 [math.DG].
- [16] D. CONLON AND J. LIM, *Sums of algebraic dilates*, 2025. arXiv:2508.18586 [math.CO].
- [17] R. DOUGALL AND R. SHARP, *Amenability, critical exponents of subgroups and growth of closed geodesics*, Math. Ann., 365 (2016), pp. 1359–1377. doi:10.1007/s00208-015-1338-1; arXiv:1411.6817.
- [18] A. GORODNIK AND A. NEVO, *Counting lattice points*, J. Reine Angew. Math., 663 (2012), pp. 127–176. doi:10.1515/CRELLE.2011.096; arXiv:0903.1515.
- [19] Y. HAO, *On the length sets of closed hyperbolic surfaces*, 2026. arXiv:2607.20808 [math.GT].
- [20] W. HE AND N. DE SAXCÉ, *Trou spectral dans les groupes simples*, Enseign. Math., 70 (2024), pp. 425–477. doi:10.4171/LEM/1078; arXiv:2103.06679.
- [21] R. D. HOROWITZ, *Characters of free groups represented in the two-dimensional special linear group*, Comm. Pure Appl. Math., 25 (1972), pp. 635–649. doi:10.1002/cpa.3160250602.
- [22] K. KRUPIŃSKI AND S. MACHADO, *On the structure of approximate rings*, 2026. arXiv:2604.04581 [math.RA].
- [23] S. LANG AND A. WEIL, *Number of points of varieties in finite fields*, Amer. J. Math., 76 (1954), pp. 819–827. doi:10.2307/2372655.
- [24] C. LECH, *A note on recurring series*, Ark. Mat., 2 (1953), pp. 417–421. doi:10.1007/BF02590997.
- [25] N. LEMIRE, V. L. POPOV, AND Z. REICHSTEIN, *Cayley groups*, J. Amer. Math. Soc., 19 (2006), pp. 921–967. arXiv:math/0409004.
- [26] C. MACLACHLAN AND A. W. REID, *The Arithmetic of Hyperbolic 3-Manifolds*, vol. 219 of Graduate Texts in Mathematics, Springer-Verlag, New York, 2003. doi:10.1007/978-1-4757-6720-9.
- [27] M. MAGEE, H. OH, AND D. WINTER, *Uniform congruence counting for Schottky semigroups in $SL_2(\mathbb{Z})$* , J. Reine Angew. Math., 753 (2019), pp. 89–135. doi:10.1515/crelle-2016-0072.
- [28] G. A. MARGULIS, *Discrete Subgroups of Semisimple Lie Groups*, vol. 17 of Ergebnisse der Mathematik und ihrer Grenzgebiete (3), Springer-Verlag, Berlin, 1991.
- [29] A. MYASNIKOV AND A. NIKOLAEV, *Verbal subgroups of hyperbolic groups have infinite width*, J. Lond. Math. Soc. (2), 90 (2014), pp. 573–591. doi:10.1112/jlms/jdu034; arXiv:1107.3719.
- [30] J. NEUKIRCH, *Algebraic Number Theory*, vol. 322 of Grundlehren der mathematischen Wissenschaften, Springer-Verlag, Berlin, 1999. doi:10.1007/978-3-662-03983-0.
- [31] R. S. PIERCE, *Associative Algebras*, vol. 88 of Graduate Texts in Mathematics, Springer-Verlag, New York, 1982. doi:10.1007/978-1-4757-0163-0.
- [32] V. PLATONOV AND A. RAPINCHUK, *Algebraic Groups and Number Theory*, vol. 139 of Pure and Applied Mathematics, Academic Press, Boston, 1994.
- [33] V. L. POPOV, *Cross-sections, quotients, and representation rings of semisimple algebraic groups*, Transform. Groups, 16 (2011), pp. 827–856. doi:10.1007/s00031-011-9137-6; arXiv:0908.0826.
- [34] T. ROBLIN, *Ergodicité et équidistribution en courbure négative*, Mém. Soc. Math. Fr. (N.S.), (2003). doi:10.24033/msmf.408.
- [35] M. ROSENBLICHT, *Toroidal algebraic groups*, Proc. Amer. Math. Soc., 12 (1961), pp. 984–988.
- [36] A. SALEHI GOLSEFIDY, *Super-approximation, I: p -adic semisimple case*, Int. Math. Res. Not. IMRN, (2017), pp. 7190–7263. doi:10.1093/imrn/rnw208.
- [37] ———, *Super-approximation, II: the p -adic case and the case of bounded powers of square-free integers*, J. Eur. Math. Soc., 21 (2019), pp. 2163–2232. doi:10.4171/JEMS/883; arXiv:1602.00409.
- [38] A. SALEHI GOLSEFIDY AND X. ZHANG, *Inducing super-approximation*, Int. Math. Res. Not. IMRN, (2021), pp. 10442–10468. doi:10.1093/imrn/rnz115; arXiv:1802.03561.
- [39] A. SAMBARINO, *A report on an ergodic dichotomy*, Ergodic Theory Dynam. Systems, 44 (2024), pp. 236–289. doi:10.1017/etds.2023.13.
- [40] P. SAMUEL, *À propos du théorème des unités*, Bull. Sci. Math. (2), 90 (1966), pp. 89–96.
- [41] N. A. SHAH, *Invariant measures and orbit closures on homogeneous spaces for actions of subgroups generated by unipotent elements*, in Lie Groups and Ergodic Theory (Mumbai, 1996), vol. 14 of Tata Inst. Fund. Res. Stud. Math., Tata Institute of Fundamental Research, Bombay, 1998, pp. 229–271. arXiv:math/0002183.
- [42] J. H. SILVERMAN, *Wieferich’s criterion and the abc-conjecture*, J. Number Theory, 30 (1988), pp. 226–237. doi:10.1016/0022-314X(88)90019-4.
- [43] R. STEINBERG, *On power maps in algebraic groups*, Math. Res. Lett., 10 (2003), pp. 621–624. doi:10.4310/MRL.2003.v10.n5.a5.

- [44] C. L. STEWART, *On the greatest square-free factor of terms of a linear recurrence sequence*, in Diophantine Equations, N. Saradha, ed., vol. 20 of Tata Inst. Fund. Res. Stud. Math., Narosa Publishing House, New Delhi, 2008, pp. 257–264.
- [45] ———, *On divisors of Lucas and Lehmer numbers*, Acta Math., 211 (2013), pp. 291–314. doi:10.1007/s11511-013-0105-y; arXiv:1008.1274.
- [46] D. SULLIVAN, *Related aspects of positivity in Riemannian geometry*, J. Differential Geom., 25 (1987), pp. 327–351. doi:10.4310/jdg/1214440985.
- [47] K. TAKEUCHI, *A characterization of arithmetic Fuchsian groups*, J. Math. Soc. Japan, 27 (1975), pp. 600–612. doi:10.2969/jmsj/02740600.
- [48] J. TANG AND X. ZHANG, *Super approximation for $SL_2 \times SL_2$ and ASL_2* , 2023. Version 5, revised 2 May 2026; arXiv:2308.09982 [math.GR].
- [49] W. P. THURSTON, *Minimal stretch maps between hyperbolic surfaces*, 1986. 1986 preprint; arXiv:math/9801039.
- [50] P. P. VARJÚ, *Expansion in $SL_d(\mathcal{O}_K/I)$, I square-free*, J. Eur. Math. Soc., 14 (2012), pp. 273–305. doi:10.4171/JEMS/302.
- [51] P. VOJTA, *Diophantine Approximations and Value Distribution Theory*, vol. 1239 of Lecture Notes in Mathematics, Springer-Verlag, Berlin, 1987.
- [52] B. WEISFEILER, *Strong approximation for Zariski-dense subgroups of semi-simple algebraic groups*, Ann. of Math. (2), 120 (1984), pp. 271–315. doi:10.2307/2006943.
- [53] A. ZUEVSKY, *Exponential growth of mean multiplicities in length spectra of semi-arithmetic surfaces of arbitrary arithmetic dimension*, 2026. arXiv:2608.17604 [math.FA].